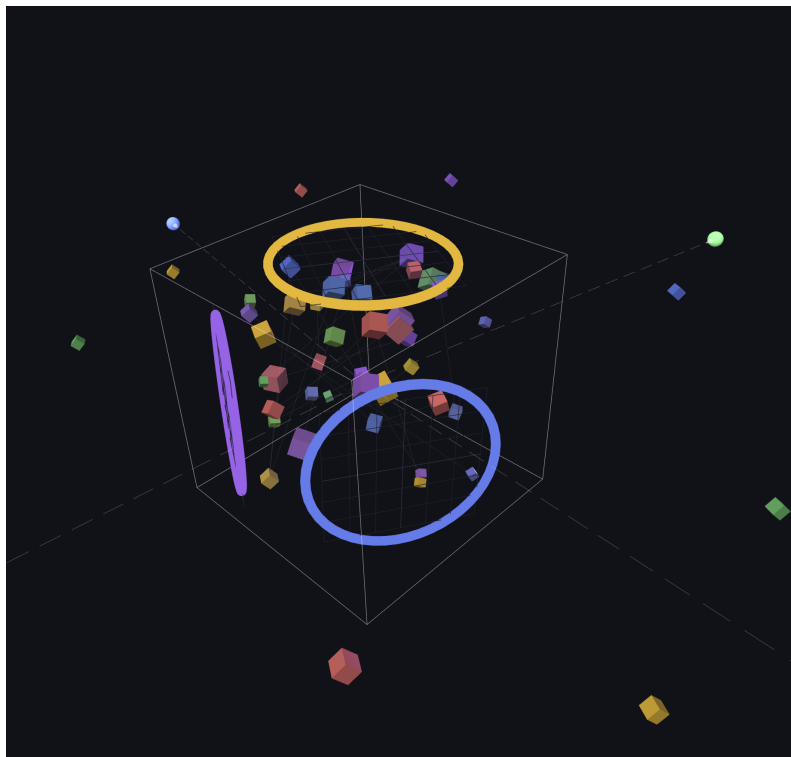


# Towards Multimodal Cartography of Role Constellations in Sustainability Transitions



*Babajide Alamu Owoyele*

# Towards Multimodal Cartography of Role Constellations in Sustainability Transitions

A dissertation submitted to the  
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*Babajide Alamu Owoyele*

**Promotor:** Prof. dr. Derk Loorbach  
**Co-promotor:** Prof.dr. Gerard de Melo  
**Co-promotor:** Dr. Jonathan Antonion Edleman

## Doctoral Committee

*Internal Examiner:* [Name], *Erasmus University Rotterdam*  
*External Examiner:* [Name], *[University]*  
*External Examiner:* [Name], *[University]*

## Public Defence

[Date TBD]

[Time TBD]

Senaatszaal, Erasmus Building  
Erasmus University Rotterdam

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# Abstract

This dissertation develops a multimodal cartography of role constellations in sustainability transitions, with empirical focus on intermediary organizations within the European Institute of Innovation and Technology (EIT) ecosystem. Drawing on Complex Adaptive Systems theory and Design Science Research methodology, we introduce **ROLEBOX**—a computational toolkit for mapping how intermediaries position themselves and are positioned by others across multiple data modalities.

The research addresses a persistent methodological divide in transitions research: qualitative approaches offer rich insight but limited scale, while computational methods achieve scale but often miss theoretical depth. We propose multimodal cartography as a bridge—systematically integrating website texts, news media coverage, investment records, social media presence, and visual materials to triangulate role positions. These intermediaries connect to thousands of ecosystem actors; **ROLEBOX** provides the infrastructure to scale such analysis across these networks.

Five empirical studies examine the role constellations of EIT intermediaries across these modalities. We find that interstitial positioning—intermediaries claiming to bridge separate communities—is substantially more common in self-presentation than in external positioning. Specifically, 34% of intermediary actors claim connector roles, but only 12% show cross-modal evidence for actually performing them. This “interstitial gap” reveals how role aspirations exceed role realities.

The dissertation contributes: (1) a theoretical framework for understanding role constellations as dynamic, multi-dimensional configurations; (2) **ROLEBOX**, an open-source toolkit for multimodal role mapping that can scale to broader ecosystem analysis; (3) empirical findings on role dynamics among intermediaries in Europe’s largest climate innovation ecosystem; and (4) methodological guidance for bridging qualitative and computational approaches in transitions research.

## Open Science Infrastructure

All code, data pipelines, and interactive tools developed for this dissertation are publicly available:

- ▶ **RoleBox** (umbrella repository): [github.com/babajideowoyele/rolebox](https://github.com/babajideowoyele/rolebox)
- ▶ **RoleSim** (agent-based simulation): [github.com/babajideowoyele/rolesim](https://github.com/babajideowoyele/rolesim)
- ▶ **RoleXray** (role extraction & NLP): [github.com/babajideowoyele/rolexray](https://github.com/babajideowoyele/rolexray)
- ▶ **Data repositories**: Wikipedia/Wikidata, websites, Twitter, news, Crunchbase, images, video

Interactive data explorers with Carbon Design System UIs accompany each empirical chapter.

**Keywords:** role constellations, innovation ecosystems, sustainability transitions, multimodal analysis, complex adaptive systems, EIT, intermediaries, open science



# Samenvatting

Dit proefschrift ontwikkelt een multimodale cartografie van rolconstellaties in duurzaamheidstransities, met empirische focus op intermediaire organisaties binnen het ecosysteem van het Europees Instituut voor Innovatie en Technologie (EIT). Voortbouwend op de theorie van Complexe Adaptieve Systemen en Design Science Research methodologie, introduceren wij ROLEBOX—een computationele toolkit voor het in kaart brengen van hoe intermediairs zichzelf positioneren en door anderen worden gepositioneerd over meerdere datamodaliteiten.

Het onderzoek adresseert een hardnekkige methodologische kloof in transitieonderzoek: kwalitatieve benaderingen bieden rijke inzichten maar beperkte schaal, terwijl computationele methoden schaal bereiken maar vaak theoretische diepgang missen. Wij stellen multimodale cartografie voor als brug—het systematisch integreren van website teksten, nieuwsmediaberichtgeving, investeringsgegevens, sociale media-aanwezigheid en visueel materiaal om rolposities te trianguleren. Deze intermediairs zijn verbonden met duizenden ecosysteemactoren; ROLEBOX biedt de infrastructuur om dergelijke analyse over deze netwerken te schalen.

Vijf empirische studies onderzoeken de rolconstellaties van EIT-intermediairs over deze modaliteiten. Wij vinden dat interstitiële positionering—intermediairs die beweren afzonderlijke gemeenschappen te verbinden—substantieel vaker voorkomt in zelfpresentatie dan in externe positionering. Specifiek claimen 34% van de intermediaire actoren verbindingsrollen, maar slechts 12% toont cross-modale evidentie voor het daadwerkelijk uitvoeren ervan. Deze “interstitiële kloof” onthult hoe rolaspiraties rolrealiteiten overtreffen.

Het proefschrift draagt bij: (1) een theoretisch kader voor het begrijpen van rolconstellaties als dynamische, multidimensionale configuraties; (2) ROLEBOX, een open-source toolkit voor multimodale rolkartering die kan schalen naar bredere ecosysteemanalyse; (3) empirische bevindingen over roldynamiek onder intermediairs in Europa’s grootste klimaatinnovatie-ecosysteem; en (4) methodologische richtlijnen voor het overbruggen van kwalitatieve en computationele benaderingen in transitieonderzoek.

**Trefwoorden:** rolconstellaties, innovatie-ecosystemen, duurzaamheidstransities, multimodale analyse, complexe adaptieve systemen, EIT, intermediairs

# Zusammenfassung

Diese Dissertation entwickelt eine multimodale Kartographie von Rollenkonstellationen in Nachhaltigkeitstransitionen, mit empirischem Fokus auf Intermediärorganisationen innerhalb des Ökosystems des Europäischen Instituts für Innovation und Technologie (EIT). Aufbauend auf der Theorie komplexer adaptiver Systeme und der Design Science Research Methodologie führen wir ROLEBOX ein—ein computergestütztes Toolkit zur Erfassung, wie sich Intermediäre selbst positionieren und von anderen über mehrere Datenmodalitäten hinweg positioniert werden.

Die Forschung adressiert eine anhaltende methodische Kluft in der Transitionsforschung: Qualitative Ansätze bieten reichhaltige Einblicke, aber begrenzte Reichweite, während computergestützte Methoden Reichweite erzielen, aber oft theoretische Tiefe vermissen lassen. Wir schlagen multimodale Kartographie als Brücke vor—die systematische Integration von Website-Texten, Nachrichtenmedienberichterstattung, Investitionsdaten, Social-Media-Präsenz und visuellem Material zur Triangulation von Rollenpositionen. Diese Intermediäre sind mit Tausenden von Ökosystem-Akteuren verbunden; ROLEBOX bietet die Infrastruktur, um solche Analysen über diese Netzwerke zu skalieren.

Fünf empirische Studien untersuchen die Rollenkonstellationen von EIT-Intermediären über diese Modalitäten hinweg. Wir stellen fest, dass interstitielle Positionierung—Intermediäre, die behaupten, getrennte Gemeinschaften zu verbinden—in der Selbstdarstellung wesentlich häufiger vorkommt als in der externen Positionierung. Konkret beanspruchen 34% der Intermediär-Akteure Verbindungsrollen, aber nur 12% zeigen modalitätsübergreifende Evidenz für deren tatsächliche Ausübung. Diese “interstitielle Lücke” zeigt, wie Rollenaspirationen die Rollenrealitäten übersteigen.

Die Dissertation leistet folgende Beiträge: (1) einen theoretischen Rahmen zum Verständnis von Rollenkonstellationen als dynamische, mehrdimensionale Konfigurationen; (2) ROLEBOX, ein Open-Source-Toolkit für multimodale Rollenkartierung, das auf breitere Ökosystemanalysen skaliert werden kann; (3) empirische Befunde zur Rollendynamik unter Intermediären in Europas größtem Klimainnovations-Ökosystem; und (4) methodische Leitlinien zur Überbrückung qualitativer und computergestützter Ansätze in der Transitionsforschung.

**Schlüsselwörter:** Rollenkonstellationen, Innovationsökosysteme, Nachhaltigkeitstransitionen, multimodale Analyse, komplexe adaptive Systeme, EIT, Intermediäre

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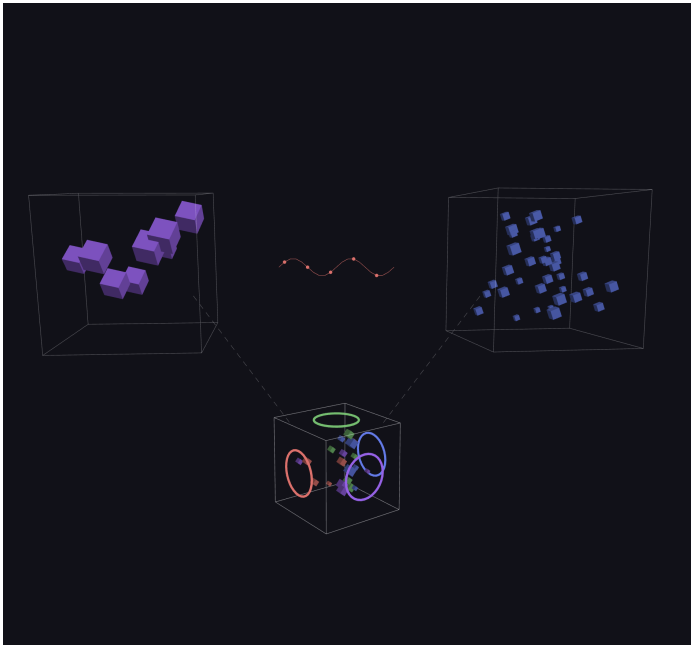
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Part I

## FOUNDATIONS



# Introduction1

Why do some sustainability transitions take off while others stall? This dissertation argues that part of the answer lies in understanding role constellations—the dynamic configurations of roles that intermediary organizations perform within transition ecosystems. But studying these constellations presents a methodological puzzle: the rich theoretical frameworks we have do not scale, and the computational methods that do scale often miss what matters most.

This chapter introduces **multimodal cartography** as a response: a systematic approach to mapping role constellations by integrating multiple data modalities within a Design Science framework. It establishes the problem, makes the case for treating methodological integration as a design challenge, introduces **ROLEBox** as the platform for this cartography, and maps the dissertation’s journey ahead.

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## 1.1 The Urgency of Navigating Transitions

We are living through what historians may one day call the Great Reckoning. Climate change has outpaced many scientific predictions [?].<sup>1</sup> Biodiversity loss continues at what some scientists characterize as a sixth mass extinction [?]. Social inequalities have widened, exacerbated by technological disruption and pandemic aftershocks. These crises share a common origin: the unsustainability of systems that have dominated since the industrial revolution.

Addressing these challenges requires more than incremental fixes. We need *transitions*<sup>2</sup>—fundamental shifts in how societies produce energy, grow food, move people, and organize economic life. The shift from coal to renewables. From industrial to regenerative agriculture. From private cars to shared mobility. Each touches every dimension of socio-technical systems.

Over three decades, sustainability transitions research has developed sophisticated frameworks for understanding these transformations. The Multi-Level Perspective sees transitions emerging from interactions between niche innovations, dominant regimes, and broader landscape pressures [1].<sup>3</sup> Strategic Niche Management explores how protected spaces help radical innovations mature [2]. Transition Management investigates governance approaches for steering complex systems toward sustainability [3].

These frameworks have generated valuable insights. But the field faces a persistent problem—one that motivates this entire dissertation.

1: **420 ppm**—CO<sub>2</sub> levels in 2023, unseen for three million years.

2: **Transitions**—Deep structural changes in technologies, institutions, practices, and power relations.

[1]: Geels (2002), *Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study*

3: The MLP remains the field’s most influential framework, though it has faced critiques for underspecifying actor agency and micro-level dynamics.





Figure 1.1: The depth-breadth trade-off in transitions research.

## 1.2 The Methodological Divide

### 1.2.1 The Qualitative Tradition

Most transitions research is qualitative and case-based. Researchers spend years embedded in transition arenas, conducting interviews, analyzing documents, observing how actors negotiate visions and navigate institutional barriers. This work excels at capturing complexity, contingency, and context. It reveals how actors make sense of change, how meanings shift, how power operates through discourse.

Consider the exemplary studies of the Dutch energy transition [?],<sup>4</sup> or the longitudinal analyses tracing organic agriculture's evolution from countercultural movement to mainstream practice [?]. These provide the theoretical depth that makes transitions research valuable.

But this tradition has limits.<sup>5</sup> Case studies typically examine thirty to fifty actors—enough for rich understanding, not enough for pattern identification. By the time a transition is documented, it has largely already happened. Cross-case comparison remains challenging when contexts differ and operationalization varies.

4: **The Dutch example**—Years of embedded research yielding rich insights, but examining only dozens of actors.

5: Qualitative strengths: deep context, rich meaning, process tracing, actor voice. Limits: small samples, hard to generalize, retrospective, hard to compare.

### 1.2.2 The Computational Turn

Meanwhile, computational social science has developed powerful methods for analyzing large-scale social phenomena [?].<sup>6</sup> Network analysis reveals hidden collaborative structures. Natural language processing extracts themes from millions of documents. Machine learning identifies patterns invisible to human observation.

Applied to transitions, these methods promise to overcome scale limitations. Researchers have mapped innovation networks through

6: **Scale possibilities**—Studies now analyze thousands of patents, millions of tweets, and entire startup ecosystems.

patent analysis, tracked climate discourse dynamics on Twitter, and analyzed startup ecosystems at population level [4, 5]. The potential is substantial.

Yet computational approaches face their own problems—particularly regarding theory.<sup>7</sup> Many studies begin with available data rather than theoretical questions, leading to sophisticated analysis of what *can* be measured rather than what *matters*. The rich concepts animating transitions research—intentionality, power, meaning, legitimacy, institutional work—often resist straightforward operationalization.

7: Computational strengths: population scale, pattern detection, replicability, high-frequency. Limits: theory-light, validation hard, context-blind, digital only.

### 1.2.3 Why Bridging Requires Design

The divide between these approaches is not absolute—some researchers combine methods, some studies integrate data types.<sup>8</sup> But such integration remains ad hoc rather than systematic, opportunistic rather than principled.

8: **The design metaphor**—Like bridge engineering, methodological bridging requires attention to materials and constraints.

What is missing is explicit methodological work on *how* to bridge these approaches while preserving each tradition's strengths. This dissertation argues that bridging requires treating integration as a **design problem**. Given particular materials (concepts, data, methods, expertise) and constraints (validity, resources, ethics), how can we construct artifacts enabling analysis that neither approach achieves alone?

This is where Design Science Research enters the picture. But before introducing that framework, we need to understand why role constellations make such a compelling test case for bridging work.

## 1.3 Role Constellations: Central Yet Elusive

### 1.3.1 What Are Role Constellations?

Transitions unfold through the actions of diverse actors occupying different positions and performing different functions. Some develop radical innovations in protected niches. Others maintain incumbent regimes. Still others broker connections, build coalitions, or provide legitimacy. Understanding these differentiated positions—what we call *roles*—is essential for explaining why some transitions accelerate while others stall.<sup>9</sup>

9: **Role**—A position or function enacted through relationships and activities, not fixed identity.

But roles do not exist in isolation. A niche innovation needs more than just entrepreneurs. It requires financiers willing to bet on uncertainty, brokers who can translate between worlds, legitimators who confer credibility, and sometimes even sympathetic regime insiders. Transitions depend not only on what actors do individually, but on how diverse roles configure together.

#### Role Constellation

A **role constellation** is the dynamic configuration of roles that an actor or set of actors performs, including how these roles relate to each other, evolve over time, and combine to enable (or

constrain) transition processes.

The concept operates at two levels.<sup>10</sup> At the *actor level*, a role constellation describes the portfolio of roles a single organization performs—how a university might simultaneously act as knowledge producer, network convenor, and policy advisor. At the *system level*, it describes how different actors' roles combine into functional configurations that support innovation and change.

10: **Two levels**—(1) Multiple roles one actor performs, or (2) roles configured across a network. We focus on the first.

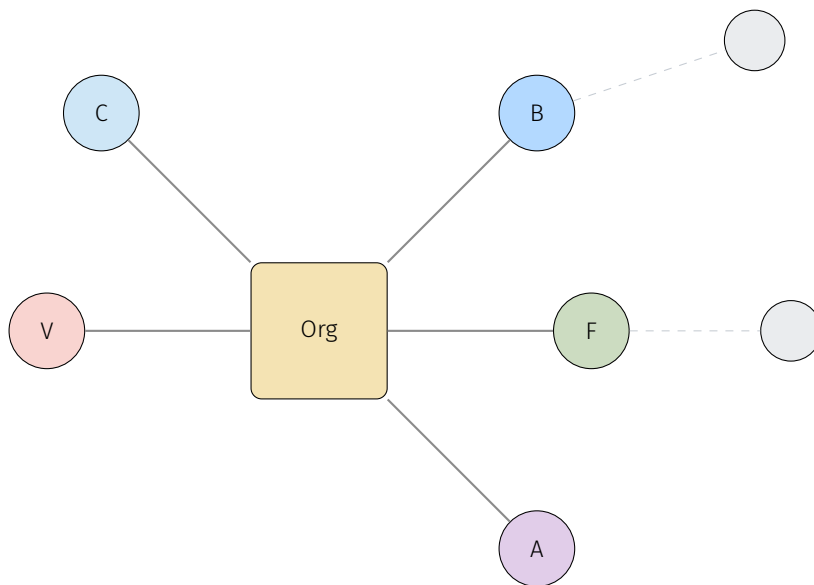


Figure 1.2: Role constellation (colored) and network connections (gray).

### 1.3.2 Scope: Intermediaries and Their Roles

Intermediary organizations occupy a special place in transitions [?].<sup>11</sup> Positioned between innovators and regimes, between sectors, between policy and practice, they perform critical bridging functions: facilitating connections, translating between logics, providing resources, conferring legitimacy, and helping innovations navigate institutional barriers.

11: **Intermediaries**—Organizations bridging innovators and regimes, translating between logics, navigating barriers.

This dissertation focuses specifically on **intermediary organizations and their role constellations**. We ask: what roles do intermediaries perform? How do these roles combine and evolve? What explains variation across intermediaries and over time?

**Scope and scalability.** While our empirical focus is intermediary organizations and the roles they perform, the methodological approach we develop—the analytical platform, integration protocols, and theoretical framework—scales to studying the broader networks that intermediaries mediate. We demonstrate feasibility at the intermediary level; extension to full ecosystem mapping is a logical next step.

This scoping choice is strategic. Intermediaries are theoretically central to transitions yet understudied empirically. They leave substantial digital traces amenable to computational analysis. And their bounded number (dozens to hundreds, not thousands) makes comprehensive analysis feasible while still requiring methods that go beyond traditional case studies.

### 1.3.3 Six Problems in Current Research

Despite the theoretical importance of roles, existing research faces six interconnected problems. These are not fatal flaws but productive tensions pointing toward opportunities for development.

1. **Definitional ambiguity.** The term “role” means different things to different scholars: position, function, identity, category, or behavior. Are roles what actors *do*, who actors *are*, or where actors are *positioned*? This conceptual slippage makes comparison and accumulation difficult.
2. **Static categorization.** Most typologies present roles as fixed boxes—entrepreneur, incumbent, intermediary—that actors occupy permanently. Yet empirical observation suggests otherwise. Actors shift roles over time, occupy multiple roles simultaneously, and vary their role performance by context. Start-ups become incumbents. Regime actors launch niche innovations. Static typologies cannot capture this dynamism.
3. **Actor-centric bias.** Roles are typically treated as attributes of individual actors: this organization *is* an intermediary, that firm *is* an incumbent. But relational approaches suggest roles emerge through interactions and network positions. An intermediary is only an intermediary *in relation to* others it connects.
4. **Normative loading.** Role concepts often carry implicit value judgments—intermediaries presumed beneficial, incumbents framed as barriers. These judgments may sometimes be warranted, but they should be examined explicitly rather than smuggled in through terminology.
5. **Methodological constraints.** The small-N qualitative methods dominating transitions research enable rich understanding but struggle with population-level questions. How common is role multiplicity? Do particular constellations correlate with transition success? How do constellations evolve across transition phases?
6. **Theoretical fragmentation.** Different transitions frameworks emphasize different roles. The Multi-Level Perspective focuses on niche-regime dynamics. Strategic Niche Management highlights innovation champions. Transition Management emphasizes frontrunners and change agents. These perspectives complement rather than compete, yet the field lacks an integrative framework.

### 1.3.4 The Promise of Systematic Integration

These six problems point toward an opportunity. If roles could be analyzed computationally at population scale *while* retaining sensitivity to their emergent, relational, and dynamic properties, new insights might emerge that neither tradition could generate alone. Patterns invisible

in small samples could become apparent. Hypotheses could be tested systematically across contexts. The fluidity and multiplicity of roles could be tracked over time rather than frozen in static typologies.<sup>12</sup>

This dissertation proposes **multimodal cartography** as a methodological response. The cartographic metaphor is deliberate. Maps do not merely represent pre-existing territory; they render complex landscapes legible, navigable, and actionable. They combine multiple information sources—satellite imagery, survey data, historical records—into integrated representations that serve particular purposes. Similarly, mapping role constellations requires integrating multiple data modalities—what organizations say on their websites, how they interact on social media, how media portrays them, what their visual communications convey—into representations that make transition ecosystems comprehensible.

### Multimodal Cartography

**Multimodal cartography** is a methodological approach that integrates multiple data modalities (text, networks, images, video, structured databases) to construct systematic representations of complex social phenomena—in this case, the role constellations of intermediary organizations in sustainability transitions.

The “multimodal” dimension is essential.<sup>13</sup> Roles are not singular phenomena observable through any single lens. An intermediary may claim a convening role on its website, perform a brokering role through its Twitter interactions, and be perceived as an advocate in news coverage. These are not contradictions to be resolved but facets to be mapped. Single-modality analysis—whether text-only, network-only, or interview-only—inevitably misses dimensions that other modalities would reveal.

This dissertation develops ROLEBOX as the platform for multimodal cartography, instantiating six data modalities in an integrated analytical environment. But before detailing the artifacts, we must establish the methodological framework that guides their construction: Design Science Research.

## 1.4 Design Science Research as Framework

### 1.4.1 Why Design?

The challenge of bridging methodological divides is not primarily a matter of applying existing methods more skillfully or collecting more data. Rather, it requires *creating* new methodological artifacts—conceptual frameworks, analytical tools, integration protocols, validation procedures—that enable forms of analysis not previously possible.<sup>14</sup>

This is fundamentally a design problem. Engineers designing a bridge must understand materials, forces, and contexts. Similarly, bridging methodological divides requires systematic attention to what each approach offers, where tensions arise, and how artifacts can be constructed that enable productive integration. The question shifts from

12: **Cartography**—Map-making renders landscapes legible, navigable, and actionable.

13: **Why multimodal?**—Organizations *claim* roles on websites, *perform* them on Twitter, and are *perceived* differently in media.

14: **Design problem**—Given materials and constraints, how can artifacts enable new analytical capabilities?

“which method is better?” to “what can we build that serves both traditions?”

## 1.4.2 Design Science in Sustainability Transitions: Uncharted Territory

Design Science Research (DSR) provides a systematic framework for this kind of methodological innovation.<sup>15</sup> [6] Originating in information systems research [?], DSR treats the creation and evaluation of novel artifacts as legitimate scientific contribution [7]. Unlike explanatory research that seeks to understand existing phenomena, or critical research that interrogates power and assumptions, design research asks: *what new capabilities become possible through systematic artifact creation?*

Yet DSR remains largely uncharted territory in sustainability transitions research.<sup>16</sup> The field has produced sophisticated theoretical frameworks—the Multi-Level Perspective, Strategic Niche Management, Transition Management—but has devoted far less attention to the *artifacts* that enable empirical analysis.

- **Analytical platforms** with documented codebases that others can deploy
- **Curated datasets** with clear provenance and quality documentation
- **Computational workflows** specifying reproducible analytical steps
- **User interfaces** enabling researchers without programming expertise to conduct analyses
- **Integration protocols** detailing how to combine different data sources and methods
- **Methodological guidelines** translating tacit expertise into transferable procedures

This absence is consequential. Without such artifacts, each research team starts from scratch. Findings cannot be replicated or extended. Methodological innovations remain locked in individual projects rather than accumulating as shared infrastructure. The field advances through theoretical refinement but lacks the material foundations for cumulative empirical progress.<sup>17</sup>

This dissertation imports the Design Science paradigm from information systems to address this gap.<sup>18</sup> The contribution is not merely to *use* computational methods but to *design, document, and share* the artifacts that make multimodal cartography accessible and reproducible.

## 1.4.3 Four Meta-Design Requirements

Applying DSR to the bridging challenge, this dissertation develops four Meta-Design Requirements (MDRs) that any systematic integration effort must address. These are not specific to role constellation analysis but represent general principles for combining computational and qualitative approaches.

15: **Design Science Research**—Paradigm treating artifact creation and evaluation as scientific contribution.

[6]: Hevner et al. (2004), *Design science in information systems research*

16: **The artifact gap**—Transitions research excels at theory but rarely produces reusable tools or datasets.

17: Artifacts developed: ROLEBox (platform), ROLESIM (simulation), 6 data modalities (datasets), pipelines (workflows), integration protocols, and replication guidelines.

18: **Importing from IS**—This dissertation adapts DSR from information systems for social science methodology.

**MDR-1: Theory-Data Reconciliation.** Bridging requires bidirectional translation between theoretical constructs and data-based measures. This means not only operationalizing theory as measurable variables (the typical direction) but also theorizing what data patterns reveal, acknowledging what resists operationalization, and being explicit about gaps. Reconciliation is ongoing negotiation, not one-time mapping.

**MDR-2: Hypothesis-Driven Data Selection.** Data collection must be guided by theoretical questions rather than opportunistic availability. Computational approaches risk analyzing whatever data exist rather than collecting data needed to test specific propositions. Each data source and modality requires theoretical justification: what does it help examine, and what are its limitations?

**MDR-3: Computational-Human Complementarity.** Neither full automation nor purely manual analysis suffices. Some tasks benefit from computational scale and consistency (pattern detection, network analysis). Others require human judgment (ambiguity resolution, context sensitivity, meaning interpretation). The design challenge is determining appropriate division of labor.

**MDR-4: Knowledge Accumulation Infrastructure.** Bridging work must produce not only results but reusable artifacts enabling others to replicate, validate, extend, and adapt. This requires open-source code, documented data, transparent procedures, and explicit limitations. Infrastructure enables cumulative progress rather than isolated studies.

These requirements are operationalized throughout this dissertation. Each chapter addresses one or more MDRs, and the limitations chapter evaluates how well the dissertation meets them. They provide evaluative criteria for assessing whether the bridging work succeeds by its own standards.

## 1.5 Research Questions and Structure

### 1.5.1 Guiding Questions

The overarching question guiding this dissertation is:

*How can we systematically develop multimodal cartography for mapping role constellations of intermediary organizations in sustainability transitions, and what do such maps reveal?*

This question combines methodological and empirical ambitions.<sup>19</sup> It asks not only *how* to construct the cartographic apparatus but also *what* we learn by using it. Four sub-questions structure the inquiry:

#### Theoretical Foundation

How can role constellations be reconceptualized to enable computational operationalization while preserving theoretical depth?

19: **Two dimensions**—Methodological (how to map) and empirical (what maps reveal).

**Methodological Artifacts**

What artifacts and design principles enable systematic multimodal cartography of role constellations?

**Empirical Demonstration**

What patterns and dynamics of role constellations emerge when applying multimodal cartography across multiple transition domains?

**Infrastructure and Accumulation**

How can the artifacts and findings be documented to enable replication, validation, and adaptation by others?

## 1.6 Contributions in Brief

Before embarking on the journey, it is worth previewing what lies at the destination. This dissertation makes contributions across four dimensions, each corresponding to a different audience and purpose.

**Theoretical contributions** advance how we conceptualize role constellations in sustainability transitions.<sup>20</sup> The CAS reconceptualization offers mechanisms—emergence, self-organization, co-evolution, non-linearity—that explain constellation dynamics while enabling operationalization.

20: CAS mechanisms; 15 propositions; universal patterns.

**Methodological contributions** demonstrate that Design Science Research can productively guide social science methodology development.<sup>21</sup> The four Meta-Design Requirements provide a transferable framework for bridging data-driven and theory-driven approaches.

21: DSR framework; four MDRs; integration protocols.

**Empirical contributions** provide the most comprehensive analysis of role constellation dynamics in sustainability transitions to date.<sup>22</sup> The analysis encompasses over 800 organizations across eight communities over fourteen years through six data modalities.

22: 800+ orgs, 8 communities, 14 years, 6 modalities.

**Infrastructural contributions** create the foundations for cumulative knowledge development.<sup>23</sup> ROLEBox provides an open-source platform with documented pipelines for six data modalities.

23: Open-source ROLEBox; ROLESim; replication packages.

## 1.7 Scope and Boundaries

Clarity requires specifying not only what this dissertation attempts but also what it does not.

**Empirically**, the dissertation focuses on European Institute of Innovation and Technology (EIT) Knowledge and Innovation Communities as the primary case.<sup>24</sup> This choice enables comparison across sustainability domains with consistent institutional framing and substantial digital traces.

24: Eight EIT KICs, 2010–2023. European innovation intermediaries.

**Methodologically**, the heavy reliance on digital trace data privileges what is observable online.<sup>25</sup> Organizations with limited digital pres-

25: Digital traces only; offline activities not captured.



ence, interactions through non-digital channels, and dimensions leaving few digital traces are underrepresented or invisible.

**Theoretically**, the dissertation focuses on micro-level role constellation dynamics rather than macro-level regime change. It treats landscape pressures and regime structures as context rather than primary objects of study.

**Normatively**, the dissertation is primarily descriptive and analytical rather than prescriptive. While occasional normative questions arise, the aim is understanding how role constellations emerge and evolve, not prescribing ideal configurations or governance interventions.

#### What this dissertation is not:

- ▶ A comprehensive theory of sustainability transitions
- ▶ A normative framework for evaluating transition progress
- ▶ A technical report on machine learning innovations
- ▶ A celebration of big data over traditional methods
- ▶ A finished methodology for mechanical adoption elsewhere

## 1.8 The Journey Ahead

This dissertation represents five years of work spanning theory development, tool building, data collection, analysis, validation, and reflection.<sup>26</sup> The research involved approximately 1,350 hours of direct labor, substantial computational processing, and collaboration with domain experts across eight communities.

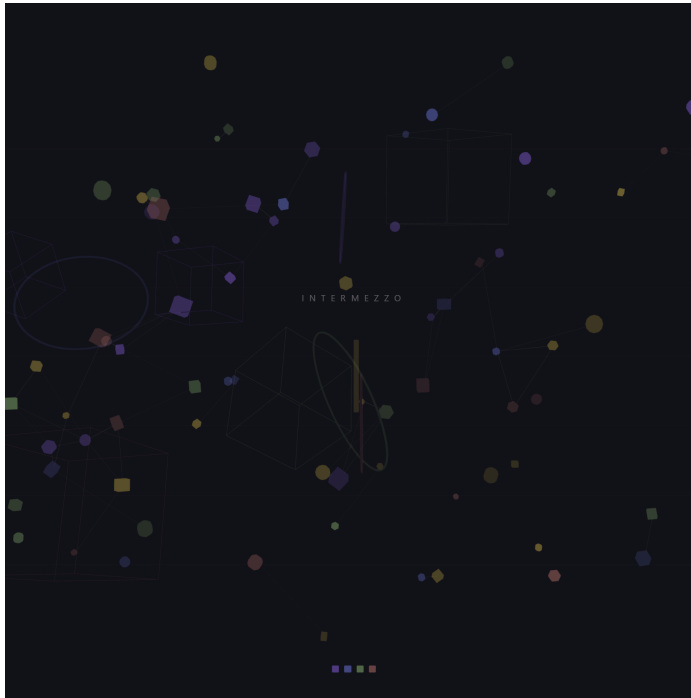
The chapters that follow document this journey. They present not a smooth narrative where each step follows logically from the last, but an honest account of iterative refinement through partial success and instructive failure. They show both what multimodal cartography enables and what limitations persist despite best efforts. They offer not a finished product but infrastructure that others can build upon, critique, and adapt.

The urgency of sustainability transitions demands that researchers continually expand their methodological repertoires while maintaining theoretical rigor and reflexive honesty. This dissertation contributes one set of tools, one theoretical lens, one empirical demonstration, and one open invitation.<sup>27</sup> The maps we draw are never the territory. But better maps might help us navigate the territory more wisely.

The journey begins in Chapter 2, where we survey the territory itself: sustainability transitions, the actors who shape them, and the roles those actors play.

26: Scope: 5 years, 8 communities, 800+ organizations, 1.45M+ tweets, 8K+ articles, 15K+ images, 200 videos, 675 websites, 15 propositions.

27: An invitation: whether this path leads somewhere valuable depends on others walking it.



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INTERMEZZO

## The EIT Communities

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*Before we can map role constellations, we must know the terrain.*

*What is this ecosystem? Who built it, and what were they trying to achieve?*

*This intermezzo introduces the empirical laboratory for this dissertation: the European Institute of Innovation and Technology (EIT) and its eight Knowledge and Innovation Communities (KICs). We trace the EIT's origins in the "European paradox" diagnosis, examine its institutional architecture and €6 billion investment, profile each KIC's distinctive intermediary character, and document the ecosystem's scale—2,420+ partners, 9,900+ startups, 880,000+ learners. We conclude by explaining why this ecosystem provides an ideal site for developing and demonstrating multimodal cartographic methods for mapping intermediary role constellations.*

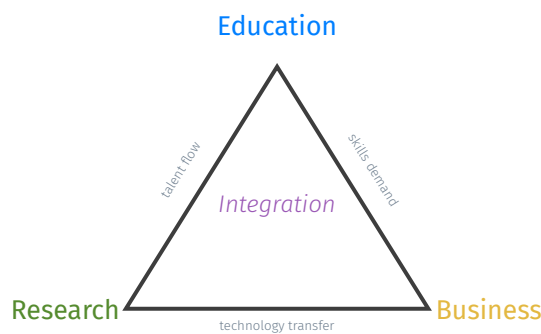
## The European Paradox and EIT's Origins

The EIT was born from frustration with a persistent pattern: Europe excelled at research but struggled to convert discoveries into innovations, companies, and jobs.\*

The 2006 Aho Report crystallized the case for institutional innovation. Europe had world-class universities and research institutes. It invested substantially in R&D. Yet somehow the innovations—and the companies built on them—kept emerging elsewhere. Silicon Valley dominated digital platforms. American and Asian firms led in semiconductors, batteries, and biotechnology. The continent that invented the World Wide Web watched others capture its value.

The diagnosis pointed to fragmentation: universities, research institutes, and businesses operated in separate worlds with different funding streams, incentive structures, and cultures. A professor's success was measured in publications, not products. A startup's success was measured in exits, not discoveries. The “knowledge triangle”—education, research, and business—had become three disconnected vertices.

\*: **The European Paradox**—Europe produces world-class science but lags in commercialization.



**Figure 1.3:** The knowledge triangle concept underlying EIT's mission. The edges—the flows between vertices—matter as much as the vertices themselves. EIT was designed to strengthen these connections.

The prescription was integration: create organizations *designed* to bridge these worlds. Not universities that dabble in business, or companies that sponsor research, but purpose-built integrators whose core mission is connection itself.

**Regulation (EC) No 294/2008** established the European Institute of Innovation and Technology, headquartered in Budapest. But crucially, the EIT was not conceived as a campus-based institution—no “European MIT.” Instead, it would function as a **meta-orchestrator**, creating and coordinating autonomous Knowledge and Innovation Communities that would do the actual integration work across Europe.

## Institutional Architecture

The EIT operates as an EU body with its own legal personality, governed by a 15-member Governing Board drawn from academia, business, and innovation sectors. The Board sets strategic direction, selects and evaluates KICs, and can discontinue underperforming partnerships.

An EIT Director (Martin Kern since 2014) manages operations from Budapest with a small headquarters staff.<sup>†</sup>

The Member State Representatives Group (MSRG), established in 2021, advises on alignment with national innovation policies—reflecting lessons learned about the importance of multi-level governance coordination.

<sup>†</sup>: **Key Legislative Milestones**—2008: Founding regulation. 2013: Integration into Horizon 2020; RIS introduced. 2021: Recast regulation under Horizon Europe.

## Budget Evolution

EIT funding has scaled dramatically across EU Framework Programmes:

| Period            | Budget             | Framework Programme | % of FP |
|-------------------|--------------------|---------------------|---------|
| 2008–2013         | €309 million       | FP7 (separate line) | ~0.6%   |
| 2014–2020         | €2.7 billion       | Horizon 2020        | ~3.5%   |
| 2021–2027         | €2.965 billion     | Horizon Europe      | ~3.1%   |
| <b>Cumulative</b> | <b>~€6 billion</b> |                     |         |

**Table 1.1:** EIT budget evolution across Framework Programmes

A crucial design principle distinguishes EIT from traditional EU research funding: **EIT grants must not exceed 25% of a KIC's total budget** on average. KICs must attract the remaining 75%+ from partner contributions, membership fees, national funding, and private investment. This “seed funding model” pushes KICs toward financial sustainability—and toward building genuine stakeholder commitment rather than grant dependency.<sup>‡</sup>

<sup>‡</sup>: **Leverage Ratio**—5.6:1; every €1 attracts €5.60 from other sources.

## External Scrutiny

The **European Court of Auditors Special Report No. 4/2016** delivered significant critique, finding that “complex operational framework and management problems have impeded overall effectiveness.” It noted funding concentration (top 5 countries received 73% of participation), questioned financial sustainability trajectories, and issued four recommendations. EIT implemented all recommendations by 2018–2019. A 2023 audit issued a qualified opinion regarding procurement procedures—a reminder that even maturing institutions face ongoing governance challenges.

## The Eight Knowledge and Innovation Communities

KICs are the EIT's operational core: large-scale public-private partnerships, each addressing a specific societal challenge through integrated innovation, education, and business creation activities. Each KIC is a separate legal entity with substantial autonomy, bound to EIT through Partnership Agreements that set performance targets and funding commitments.

| KIC            | Thematic Focus                          | Founded | Headquarters | Legal Form          |
|----------------|-----------------------------------------|---------|--------------|---------------------|
| Climate-KIC    | Climate mitigation & adaptation         | 2010    | Amsterdam    | Dutch Stichting     |
| InnoEnergy     | Sustainable energy & cleantech          | 2010    | Eindhoven    | Societas Europaea   |
| Digital        | Digital transformation & deep tech      | 2010    | Brussels     | Belgian AISBL       |
| Health         | Healthcare innovation & healthy living  | 2015    | Munich       | German e.V.         |
| RawMaterials   | Sustainable raw materials & circularity | 2015    | Berlin       | German GmbH         |
| Food           | Agrifood system transformation          | 2017    | Leuven       | Belgian iVZW        |
| Manufacturing  | Advanced & green manufacturing          | 2019    | Paris-Saclay | Belgian ASBL        |
| Urban Mobility | Sustainable urban transport             | 2019    | Barcelona    | Spanish Association |

**Table 1.2:** The eight EIT Knowledge and Innovation Communities

The KICs divide into three “waves”<sup>§</sup> reflecting their founding periods and maturity stages:

§: **Note on Scope**—EIT Culture & Creativity (2022) excluded due to limited history.

**First Wave (2010)** Climate-KIC, InnoEnergy, and Digital were the founding KICs, now 15 years into operation. All three achieved **financial sustainability** in 2024–2025, transitioning from grant dependency to self-sustaining operations.

**Second Wave (2015)** Health and RawMaterials launched mid-decade, now in their maturation phase with established partner networks and program portfolios.

**Third Wave (2019)** Manufacturing and Urban Mobility are the youngest KICs examined here, still in growth phases with evolving strategic identities.

Despite shared EIT affiliation, the eight KICs have developed **markedly different intermediary profiles**—different partnership models, strategic approaches, geographic configurations, and relationships with their innovation ecosystems. Understanding this variation is essential for the cartographic analysis that follows.

## KIC Profiles: The First Wave

### Climate-KIC: Systems Innovation Pioneer

Climate-KIC has undergone the most dramatic strategic evolution among KICs, pivoting from conventional accelerator models toward what it calls “**Systems Innovation as a Service**.” Under CEO Kirsten Dunlop (since 2017), it developed the **Deep Demonstrations** methodology—portfolio-based, place-based transformation programs that work with cities and regions to catalyze systemic change rather than supporting individual innovations in isolation.

#### Deep Demonstrations

Climate-KIC’s signature approach applies systems innovation principles to specific places: Leuven (circular economy), Madrid (mobility), Vienna (buildings), Glasgow (industrial decarboniza-

tion). Rather than funding discrete projects, Deep Demonstrations orchestrate portfolios of interconnected interventions, accepting that transformation requires changing multiple system elements simultaneously.

Climate-KIC now serves as lead orchestrator for the **EU Cities Mission** (100 Climate Neutral Cities by 2030) through the NetZeroCities platform, supporting 112 cities in developing Climate City Contracts.<sup>¶</sup>

**Notable alumni:** Climeworks (Switzerland), the direct air capture unicorn valued at over \$2 billion, completed Climate-KIC accelerator stages in 2012–2013. Nearly 30 Climate-KIC entrepreneurs have appeared on Forbes 30 Under 30 Europe.

In January 2025, Climate-KIC achieved **financial sustainability**, transitioning to a Memorandum of Cooperation with EIT and rebranding simply as “Climate KIC”—dropping the EIT prefix to signal its independent identity.

¶: **Climate-KIC Metrics**—Partners: 400+. Startups: 2,300+. Investment: €1.1B. Products: 790. EIT funding: €720M.

## EIT InnoEnergy: The Investment Engine

InnoEnergy represents the **investment-focused model** among KICs. Under founding CEO Diego Pavia, it developed a distinctive approach: taking minority equity stakes in portfolio companies, providing patient capital, and actively building industrial ecosystems rather than merely supporting individual ventures.

InnoEnergy holds mandates to lead three EU strategic industrial initiatives:

- ▶ **European Battery Alliance:** 800+ partners, €100B+ investment pipeline
- ▶ **European Solar PV Industry Alliance:** Manufacturing capacity targets
- ▶ **European Green Hydrogen Acceleration Centre:** Electrolyzer scale-up

The results are striking.<sup>||</sup> From 9,000+ evaluated ventures, InnoEnergy built a portfolio of 540+ companies that have collectively raised over **€34 billion** in investment. The portfolio includes **five unicorns**:

||: **InnoEnergy Metrics**—Partners: 1,400+. Portfolio: 540+ companies. Investment: €34B+. Jobs: 47,000+. Unicorns: 5. Graduates: 2,000+.

**Northvolt** Europe’s largest lithium-ion battery manufacturer; peak valuation >€20B

**H2 Green Steel** Green steel production; €2.5B mobilized

**Verkor** Battery gigafactory; €2B+ Series C

**Vulcan Energy** Zero-carbon lithium extraction

**Freyr** Battery cell production

### The Northvolt Question

Northvolt’s November 2024 bankruptcy filing complicates the InnoEnergy success narrative. The company received €9.3M in InnoEnergy investments and became the poster child for European cleantech. Its restructuring raises questions about unicorn metrics as success indicators—and about the challenges of scaling hardware companies in competitive global markets.

## EIT Digital: Deep Tech and Talent

EIT Digital (founded as EIT ICT Labs, rebranded 2015) focuses on digital deep tech: artificial intelligence, cybersecurity, quantum computing, and digital infrastructure. Its distinctive contribution is the **double-degree Master's School**—students study at two universities in different countries, combining technical depth with entrepreneurial training and international mobility.\*\*

In October 2025, upon achieving financial sustainability, EIT Digital rebranded as **28DIGITAL**—the “28” referencing the number of EU member states and signaling ambition to serve as Europe’s digital transformation engine.

The Master's School operates across 24 partner universities including TU Berlin, Aalto, KTH, Politecnico di Milano, and Universidad Politécnica de Madrid. Programs span AI, cybersecurity, data science, fintech, and human-computer interaction.

\*\***: EIT Digital Metrics**—Partners: 350+. Portfolio: 253 companies (€1.5B+ value). Students: 1,500+. Products: 250+. Offices: 23.

## KIC Profiles: The Second Wave

### EIT Health: Healthcare Ecosystem Integration

EIT Health emerged from the InnoLife consortium to become Europe’s largest health innovation network, combining pharmaceutical companies, MedTech firms, hospitals, payers, and universities. Its partnership includes Siemens Healthineers, AstraZeneca, Roche, Janssen, and major academic medical centers.

The KIC operates through seven Co-Location Centers (CLCs) plus the InnoStars cluster serving Hungary, Italy, Poland, and Portugal. International hubs in the US and Israel extend global reach.††

EIT Health’s **COVID-19 response** demonstrated rapid mobilization capacity: €6 million pledged, €5.5 million Start-up Rescue Instrument for struggling ventures, and 21 startups fast-tracked through the Headstart COVID-19 Call.

**Notable exits:** PhagoMed (acquired by BioNTech), DoDOC (acquired by Envision Pharma Group), PatchAi (acquired by Alira Health). Luminate Medical (Ireland) raised €15M Series A for chemotherapy hair loss prevention technology.

††**: EIT Health Metrics**—Partners: ~150. Ventures: 3,370. Investment: €2.4B. Learners: 70,000+. Lives impacted: 780,000+.

### EIT RawMaterials: Critical Materials and Circularity

EIT RawMaterials addresses the entire raw materials value chain—from exploration and extraction through processing, recycling, and substitution. Its strategic importance has grown dramatically with the EU’s focus on critical raw materials security and the **Critical Raw Materials Act**.

The KIC manages the **European Raw Materials Alliance (ERMA)**: 750+ members, 40+ projects, €25B+ investment potential. This positions RawMaterials as a key implementation arm for EU strategic autonomy in materials.††

††**: RawMaterials Metrics**—Partners: 300+. Funding: €600M+ direct, €5B leveraged. Projects/startups: 800+. Master's: 6. Trainees: 76,000+.



**Notable startups:** Fibrecoat (TIME Best Inventions 2025), 9-Tech (PV recycling, Fortune Italia “Change the World”), Minespider (blockchain tracking 10% of global tin supply).

The **European Raw Materials Academy**, launched May 2025, targets training 100,000 people in skills for the materials transition.

## KIC Profiles: The Third Wave

### EIT Food: Farm to Fork Transformation

EIT Food emerged from the FoodConnects consortium to operate the world’s largest food innovation community. Its partnership structure is distinctive: major food corporations (Nestlé, PepsiCo, Danone, Cargill) alongside startups, universities, and consumer organizations.

The KIC’s three-tier entrepreneurship pathway—Seedbed Incubator, Accelerator Network, and RisingFoodStars association—has supported over 500 startups that have collectively attracted €1.1B+ in investment.<sup>§§</sup>

**Notable portfolio:** Mosa Meat, Redefine Meat, Meatable, Solar Foods, Aleph Farms (cellular agriculture leaders); CropX (precision agriculture); TIPA (sustainable packaging).

EIT Food is the only KIC with an explicit **consumer engagement mandate**, reflecting the food system’s dependence on behavior change alongside technological innovation.

§§: **EIT Food Metrics**—Partners: 200+. Startups: 1,438+. Investment: €1.1B+. Graduates: 1,132. Non-degree learners: 52,944.

### EIT Manufacturing: Industry 4.0 and Green Production

EIT Manufacturing (from the Made by Europe consortium) emphasizes Industry 4.0, green manufacturing, and human-robot collaboration. Its partnership includes automotive giants (Volvo, Volkswagen), industrial technology leaders (Siemens, KUKA), and manufacturing research institutes.

The **Skills.Move platform** offers 230+ courses in AI, robotics, additive manufacturing, and sustainable production—reflecting manufacturing’s acute skills gaps.<sup>¶¶</sup>

Six regional offices span Europe: Central (Darmstadt), East (Vienna), North (Gothenburg), South (Milan), South East (Athens), and West (San Sebastián).

¶¶: **Manufacturing Targets (2030)**—Trained: 50,000. Startups: 1,000. Products: 360. Investment: €325M. Sustainable adoption: 60%+.

### EIT Urban Mobility: Cities at the Center

EIT Urban Mobility (from the MOBiLus consortium) is distinctive in placing **cities as core partners** rather than merely deployment sites. Barcelona, Amsterdam, Stockholm, Munich, and Prague municipalities are consortium members with governance roles.

The KIC has conducted **185+ pilots** in European cities since 2020 and ranks as the **#1 investor in European mobility startups** for 2023–2024 according to Via ID/Dealroom analysis.<sup>\*\*\*</sup>

\*\*\*: **Urban Mobility Metrics**—Partners: 270+. Portfolio: 135 startups (€1B value). Women-led: 43%. City pilots: 185+.

**Key initiatives:** Tomorrow.Mobility World Congress (annual flagship event); #ChallengeMyCity (~€50,000 for city-startup pilots); RAPTOR (Rapid Applications for Transport).

**Notable innovations:** CDClean (COVID-19 metro disinfection, Barcelona); CITY RESTARTS (Europe's first large-scale shared taxi supplementing public transport, Milan); HERO Project (hydrogen retrofit for urban minibuses).

## Ecosystem-Level Metrics

Aggregating across KICs reveals the ecosystem's cumulative scale:

| Metric                                          | Value    |
|-------------------------------------------------|----------|
| Partner organizations                           | 2,420+   |
| Innovation hubs/co-location centers             | 70+      |
| Countries with EIT presence                     | 33+      |
| Startups and scale-ups supported                | 9,900+   |
| External investment attracted by EIT ventures   | €9.5B+   |
| Unicorns in EIT portfolios                      | 8+       |
| New products, services, or processes launched   | 2,400+   |
| Learners trained (all programs)                 | 880,000+ |
| EIT Label graduates (Master's and PhDs)         | 4,600+   |
| Collective valuation of EIT-supported companies | €72B+    |

Table 1.3: EIT Community cumulative impact metrics (2010–2025)

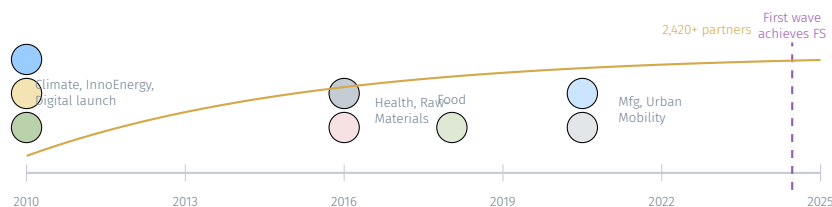


Figure 1.4: EIT ecosystem growth timeline showing KIC launches and partner network expansion. First-wave KICs achieved financial sustainability in 2024–2025 after 15 years of operation.

## Startup and Investment Performance by KIC

Investment attraction varies dramatically across KICs, reflecting different strategic models.<sup>†††</sup>

<sup>†††</sup>: **InnoEnergy Dominance**—80%+ of investment due to equity model and capital-intensive focus.

| KIC            | Startups Supported | Investment Attracted    | Unicorns  |
|----------------|--------------------|-------------------------|-----------|
| Climate-KIC    | 2,300+             | €1.1 billion            | 1         |
| InnoEnergy     | 540+               | €34+ billion            | 5         |
| Digital        | 253 (portfolio)    | €1.5B (portfolio value) | 0         |
| Health         | 3,370              | €2.4 billion            | 0         |
| RawMaterials   | 800+               | €395 million            | 0         |
| Food           | 1,438+             | €1.1+ billion           | 0         |
| Manufacturing  | 1,357 (target)     | €325M (target 2030)     | 0         |
| Urban Mobility | 135 (portfolio)    | €1B (portfolio value)   | 0         |
| <b>Total</b>   | <b>9,900+</b>      | <b>€9.5+ billion</b>    | <b>8+</b> |

Table 1.4: Startup support and investment by KIC

## Regional Innovation Scheme

The **EIT Regional Innovation Scheme (RIS)**, introduced in 2014, dedicates 10–15% of EIT funding to activities in “modest and moderate innovator” countries—primarily Central, Eastern, and Southern Europe. Ten-year results (2014–2024):

- ▶ **22+ eligible countries/territories** including Western Balkans, EU Outermost Regions
- ▶ **324,000** people skilled or upskilled
- ▶ **€3.23 billion** investment attracted to RIS regions
- ▶ **1,231 startups** supported (2024)
- ▶ Western Balkans participation **6× higher** than overall Horizon participation rate

In late 2024, EIT launched **16 new “EIT Community Hubs”** in countries including Bulgaria, Croatia, Estonia, Latvia, Malta, Slovenia, plus Ukraine and Western Balkans states—single points of access for local entrepreneurs to engage with all KICs.

## Partnership Architecture and Network Structure

Understanding KIC partnership models is essential for cartographic analysis of role constellations.

### Partnership Tiers

KICs employ tiered membership structures with varying rights and obligations:

**Core/Strategic Partners** Full governance participation, voting rights, activity leadership. Annual fees: €11,250–€75,000+

**Associate/Delivery Partners** Implementation focus, limited governance voice. Fees: €4,500–€30,000

**Community Members/Affiliates** Network access, event participation, no strategic role. Fees: €500–€2,500

**Seven-Year Partnership Agreements**<sup>†††</sup> define KIC-EIT relationships, setting performance targets, funding trajectories, and evaluation milestones. **Specific Grant Agreements** cover annual or multi-annual implementation.

†††: **Partner Types**—Universities dominate numerically; corporations provide most co-funding.

## Cross-KIC Collaboration

Despite operating as separate legal entities, KICs collaborate through several mechanisms:

- ▶ **EIT Houses:** Joint co-location in Brussels, Budapest, Berlin, Espoo, Paris, Stockholm
- ▶ **Thematic initiatives:** Circular Economy Cross-KIC (6 KICs, ~1,000 initiatives)
- ▶ **EIT Jumpstarter:** Cross-KIC pre-accelerator for idea-stage entrepreneurs since 2017
- ▶ **Girls Go Circular:** 80,000+ students, 30+ countries (RawMaterials coordinating with Manufacturing, Climate-KIC, Food)
- ▶ **Deep Tech Talent Initiative:** 768,000 pledged training places, 114 pledge members

## Partner Overlap

Major organizations participate across multiple KICs, creating network interconnections:

- ▶ **Universities:** TU Munich (4 KICs), KU Leuven, Imperial College, ETH Zürich, Politecnico di Milano
- ▶ **Corporations:** Siemens (4 KICs), Bayer, SAP, Philips, ABB

**Data gap for cartography.** No consolidated public database maps partner membership across KICs or tracks partnership dynamics over time. Individual KIC directories must be cross-referenced manually—a gap this dissertation’s multimodal approach helps address.

## Policy Context and Current Debates

The EIT operates within **Pillar III “Innovative Europe”** of Horizon Europe, alongside the European Innovation Council (EIC) and European Innovation Ecosystems program.<sup>§§§</sup> KICs are formally recognized as “Institutionalised European Partnerships”—long-term public-private partnerships delivering EU-wide impact.

§§§: **Green Deal Alignment**—5 of 8 KICs support Green Deal objectives.

## Strategic Alignments

EIT KICs serve as implementation arms for major EU policy initiatives:

**European Green Deal** Climate-KIC targets 500Mt CO<sub>2</sub> avoided by 2027; InnoEnergy leads three industrial alliances

**Critical Raw Materials Act** RawMaterials manages ERMA implementation

**Farm to Fork Strategy** Food’s core programmatic alignment

**Digital Decade** Digital leads Deep Tech Talent Initiative (1 million training target)

**New European Innovation Agenda** EIT assigned key roles in Flagship 1 (Deep Tech Talent) and Flagship 4 (Regional Innovation Valleys)

## The Existential Debate

As the EU prepares Framework Programme 10 (post-2027), EIT faces its most significant policy challenge. In January 2025, the **Danish government position paper** called for EIT discontinuation:

### The Danish Critique

“The time has come to discontinue the EIT as a separate entity... Stakeholders find it increasingly difficult to identify added value... There is significant overlap with EIC and Erasmus+.” Latvia supported this position.

The **ITRE Committee Draft Report (2024)**<sup>¶¶¶</sup> characterized EIT as “non-core, redundant and underperforming,” recommending radical reform including “significant reductions (or elimination) of FP funding.”

¶¶¶: **EIT Response**—Financial sustainability proves the model; EIC merger would be ill-advised.

These debates make the current moment particularly valuable for research: the ecosystem is simultaneously demonstrating maturation (financial sustainability achievements) while facing existential questions about its future form.

## The EIT in Academic Literature

Despite €6 billion cumulative investment, **academic research on EIT is surprisingly sparse**.<sup>17</sup> Most systematic analyses come from grey literature—evaluation reports, Court of Auditors assessments—rather than peer-reviewed journals.

17: **Theoretical Lenses**—Triple Helix, intermediary theory, ecosystems.

Key contributions include Maassen & Stensaker’s (2011) critique of the “instrumental vision” underlying knowledge triangle policy in *Higher Education*, and Leceta & Ranga’s (2020) application of ecosystem frameworks to EIT Digital in *Innovation*. Climate-KIC has received the most attention, reflecting connections with sustainability transitions research.

## Methodological Gaps

The literature reveals significant opportunities:

1. **No published social network analysis** of KIC partner networks, knowledge flows, or collaboration patterns
2. **No bibliometric studies** tracking EIT-funded research outputs
3. **No patent/co-patenting analyses** across the ecosystem

4. **Limited longitudinal research** tracking evolution over 15 years
5. **No rigorous counterfactual analysis** of additionality (what would have happened without EIT?)
6. **No systematic comparison** of intermediary role configurations across KICs

This dissertation addresses several of these gaps through multimodal cartographic methods.

## Why EIT for This Study

The EIT ecosystem provides an ideal empirical laboratory for developing and demonstrating multimodal cartography of intermediary role constellations.<sup>18</sup> Several features make it particularly suitable:

**Designed intermediaries.** KICs are explicitly created as intermediaries—their core purpose is connecting, translating, and facilitating between knowledge triangle actors. If we want to understand intermediary role constellations, studying organizations designed for intermediation provides an exceptionally clear case.

**Transition relevance.** Five of eight KICs directly address sustainability transition domains: climate, energy, raw materials/circularity, food, and urban mobility. The ecosystem offers a laboratory for examining how intermediary configurations shape—and are shaped by—transition dynamics.

**Structural variation.** Eight KICs with different thematic foci, partnership models, geographic configurations, and strategic approaches provide natural variation for comparative analysis. Why do some constellations take hub-and-spoke forms while others are more distributed? How do intermediary roles differ across sectors?

**Temporal depth.** Fifteen years of operation—with documented milestones, evaluations, and strategic pivots—enables longitudinal analysis of how role constellations evolve. The current inflection point (first-wave financial sustainability, policy debates about EIT's future) makes this a particularly rich moment for study.

**Digital traces.** EIT organizations leave extensive digital footprints: websites documenting missions and partnerships, news coverage of activities, CORDIS project databases, Dealroom startup portfolios, social media presence. These traces are amenable to systematic multimodal analysis.

**Policy relevance.** Understanding EIT matters beyond academic interest. Billions of euros in public investment depend on whether this model works. Cartographic analysis can inform ongoing debates about EIT's structure, added value, and future.

18: **Empirical Advantages**—Designed intermediaries, transition relevance, rich traces, 15-year history.

### A Note of Caution

The EIT is not a typical innovation ecosystem. It was designed, funded, and governed by policy—not grown organically. Its intermediaries were purpose-built, not emergent. This makes findings valuable for understanding *how intermediary roles can be designed* but requires caution in generalizing to ecosystems where

intermediation emerges spontaneously. The maps we draw are maps of *this* terrain. Whether other terrains look similar remains an empirical question.

## Summary and Transition

- ▶ The EIT was established in 2008 to address the “European paradox” by integrating the knowledge triangle of education, research, and business.
- ▶ **Eight Knowledge and Innovation Communities** operate as autonomous public-private partnerships across climate, energy, digital, health, raw materials, food, manufacturing, and urban mobility domains.
- ▶ The ecosystem has grown to **2,420+ partner organizations** across **70+ innovation hubs**, supporting **9,900+ startups** that have attracted **€9.5B+ in investment**.
- ▶ First-wave KICs (Climate-KIC, InnoEnergy, Digital) achieved **financial sustainability** in 2024–2025 after 15 years of operation.
- ▶ KICs exhibit **markedly different intermediary profiles**: Climate-KIC’s systems innovation approach, InnoEnergy’s investment model, Digital’s education focus, Urban Mobility’s city-centric design.
- ▶ The **Regional Innovation Scheme** has supported 324,000 learners and attracted €3.23B to less-developed regions.
- ▶ Current **policy debates** about EIT’s post-2027 future create a particularly valuable research moment.
- ▶ **Academic literature gaps**—no network analysis, no systematic role comparison, limited longitudinal research—create significant opportunities for dissertation contribution.
- ▶ The ecosystem’s combination of **designed intermediation, transition relevance, structural variation, temporal depth, and digital traces** makes it an ideal laboratory for multimodal cartographic methods.

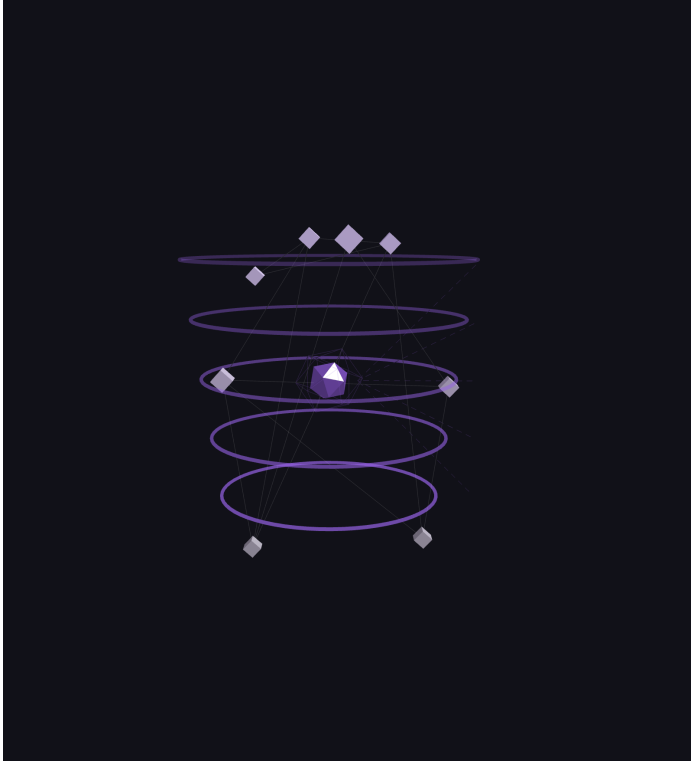
We now have the terrain to be mapped. The eight KICs, with their distinctive intermediary profiles, constitute a role constellation at European scale—designed to integrate knowledge triangles, orchestrate innovation ecosystems, and accelerate sustainability transitions. But *how* do these roles actually configure? Where do KICs position themselves relative to each other and to their partner networks? How have their role profiles evolved over 15 years?<sup>19</sup>

To answer these questions requires moving from description to cartography from cataloguing what KICs *are* to mapping the relational terrain they inhabit. The theoretical framework developed in the next chapter provides the conceptual apparatus for this cartographic work. The empirical chapters that follow demonstrate its application.

19: → Next: Chapter ??—Theoretical Framework: developing the conceptual apparatus for multimodal role constellation cartography.







# Role Categories in Sustainability Transitions

*Before we can map role constellations, we must understand how the field has conceptualized them. This chapter surveys role-based perspectives in sustainability transitions research, examines how major frameworks handle actor roles, and pays particular attention to intermediaries—the focal actors of this dissertation. We trace intermediary conceptualization from bi-lateral brokers to systemic actors, examine their functions and dynamics, and conclude by identifying six persistent problems that motivate our cartographic response.*

## 2.1 From Actors to Roles to Constellations

Sustainability transitions are fundamentally **multi-actor processes**. No single organization—however powerful—can transform an energy system, reshape a food regime, or reconfigure urban mobility alone. Transitions emerge from the coordinated and contested actions of diverse actors: startups and incumbents, policymakers and activists, researchers and users, investors and intermediaries.

Early transition frameworks acknowledged this diversity but offered limited vocabulary for analyzing it [?].<sup>1</sup> The Multi-Level Perspective distinguished niche innovators from regime incumbents. Strategic Niche Management identified network builders and expectation managers. Transition Management convened frontrunners in transition arenas. Yet these frameworks treated roles implicitly—as background assumptions rather than objects of analysis.

In recent years, scholars have argued that understanding *who does what* in transitions—and how actors’ roles evolve and combine—is crucial for explaining why some transitions accelerate while others stall. This has generated growing interest in **role constellations** as analytic tools for unpacking social dynamics of systemic change.

### Role Constellation

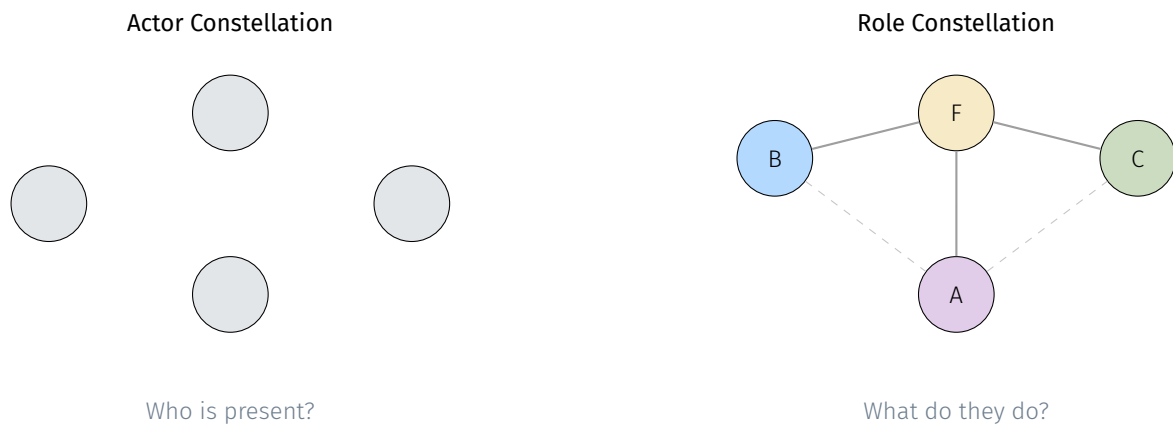
A **role constellation** is a configuration of actor roles and their interrelations within a transition process. It describes not merely which actors are present, but what functions they perform and how those functions combine to enable (or constrain) transition dynamics.

The formal introduction of role constellations to transition research is often traced to Wittmayer et al. [8], who imported sociological role theory to provide vocabulary for analyzing changing actor interactions over time. They argued that transition studies needed to move beyond

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1: Theoretical gap—frameworks distinguish actor types but not how roles evolve and interrelate.

identifying actor *types* (who is present) to analyzing actor *roles* (what functions actors perform and how these combine).



**Figure 2.1:** Actor vs. role constellations. The former identifies presence; the latter maps functions and relationships.

This distinction matters. Multiple actors from different sectors might perform the *same* role (several organizations acting as brokers). One actor can occupy *multiple* roles simultaneously (an intermediary that convenes, funds, and advocates). And roles can *shift* over time as transitions unfold (a niche protector becoming a regime challenger).

Crucially, roles are not fixed attributes but **emergent and malleable**. They can be created (establishing a new “mobility broker” function), altered (an incumbent shifting from resistor to adapter), or deliberately assigned (a governance intervention that empowers certain actors to perform coordination roles). This malleability makes role constellations not merely an analytical lens but a potential **intervention point** for transition governance.

## 2.2 Roles Across Theoretical Frameworks

How do the major transition frameworks conceptualize roles? Each offers a partial map with characteristic strengths and blind spots.<sup>2</sup>

2: **Different lenses**—Each framework illuminates some roles while obscuring others.

### The Multi-Level Perspective

The **Multi-Level Perspective (MLP)** distinguishes actors by their position in niche-regime-landscape dynamics [1]. Niche actors create starting points for systemic change, benefiting from protective spaces that support experimentation. Regime actors can be sources of or structures for agency—incumbents initially tend to downplay transformation needs but may eventually adapt [9]. Landscape actors are definitionally problematic; the MLP treats landscape as “background” with no activities, though civil society sometimes represents landscape-level cultural trends.

[1]: Geels (2002), *Technological transitions as evolutionary reconfiguration processes: a multi-level perspective and a case-study*

## Transition Management

**Transition Management** emphasizes frontrunners—visionary actors who experiment with alternatives in protected arenas [3]. It distinguishes strategic (long-term vision), tactical (coalitions and networks), operational (experiments and projects), and reflexive (learning and monitoring) activities, implying differentiated roles at each level.

[3]: Kemp et al. (2007), *Transition management as a model for managing processes of co-evolution towards sustainable development*

## Strategic Niche Management

**Strategic Niche Management (SNM)** identifies actors performing three niche-building processes: articulating visions and expectations, building social networks, and facilitating learning [2]. “Cosmopolitan actors” perform critical aggregation activities, translating lessons from local experiments to the global niche through professional organizations, specialized journals, and conference networks.

[2]: Smith et al. (2012), *Niche development and the role of agency in sustainability transitions*

## Innovation Systems Approaches

**Innovation Systems** approaches—particularly Technological Innovation Systems (TIS)—identify seven system-building functions: knowledge development and diffusion, influence on search direction, entrepreneurial experimentation, market formation, resource mobilization, legitimation, and positive externalities [10]. Different actor types contribute to different functions.

[10]: Bergek et al. (2008), *Analyzing the functional dynamics of technological innovation systems: A scheme of analysis*

## 2.3 Fischer & Newig: A Systematic Review of Actor Typologies

Fischer and Newig [11] provide the most comprehensive mapping<sup>3</sup> of how sustainability transitions research categorizes actors. Analyzing 386 articles from 1995–2014, they addressed whether actors have been neglected in transitions literature in favor of abstract system concepts.

3: **Fischer & Newig (2016)**—Systematic review of 386 articles identifies four overlapping typologies.

Their key finding: the field lacks a unified actor typology. Multiple overlapping classification schemes coexist:

- ▶ **Systemic typology:** niche, regime, landscape actors—derived from MLP
- ▶ **Institutional typology:** state, market, civil society—the governance triangle
- ▶ **Governance typology:** local, regional, national, global scales
- ▶ **Intermediary typology:** actors bridging other categories

These typologies coexist without integration. A startup can be simultaneously a niche actor (systemic), market actor (institutional), and local actor (governance). The literature conflates these without systematic treatment of how they relate.

### The Overlap Problem

A single organization might be categorized as a “niche actor” (systemic), “civil society organization” (institutional), “local actor” (governance), and “intermediary” (bridging function)—all simultaneously. Without explicit attention to how these framings intersect, research findings become difficult to cumulate across studies.

A crucial finding across frameworks: actor roles are *erratic*—they change over time, and actors can belong to different categories simultaneously. Most typologies assume static category membership, missing the temporal dynamics that characterize actual transitions. Fischer and Newig note that “agency, strategic behaviour, and learning dynamics of actors are somewhat underdeveloped” in the literature, despite their centrality to transition processes.

The review also reveals that intermediaries function as a cross-cutting category—they don’t fit neatly into systemic, institutional, or governance typologies because they are defined by *position* (in-between) rather than *attributes*. This supports a relational framing where actor “type” matters less than relational configuration.

## 2.4 Wittmayer et al.: Role Theory in Sustainability Transitions

Wittmayer et al. [8] made the seminal contribution<sup>4</sup> of explicitly importing sociological role theory into sustainability transitions research. Drawing on Biddle [12], they distinguish multiple conceptualizations of “role”:

**Functional roles** emphasize positions within social systems and associated behavioral expectations. Organizations are expected to perform certain functions based on their structural position.

**Symbolic interactionist roles** emphasize meaning-making and identity construction. Roles are not just performed but negotiated through interaction.

**Organizational roles** focus on formal positions within hierarchies and the expectations attached to those positions.

**Cognitive roles** emphasize mental schemas and scripts that guide behavior in particular situations.

This multiplicity explains persistent definitional ambiguity: when researchers use “role,” they may invoke any of these meanings without explicit acknowledgment.

Crucially, Wittmayer et al. emphasize that roles are **dynamic and contested**.<sup>5</sup> Actors actively construct roles through strategic action; they negotiate role boundaries with other actors; they resist role expectations that conflict with their interests or identities. This perspective transforms roles from static categories into **objects of struggle**.

The framework identifies three key role dynamics:

4: **Wittmayer et al. (2017)**—Introduces sociological role theory to transition studies.

5: **Role dynamics**—Roles are actively constructed and negotiated, not merely occupied.

1. **Role-taking:** adopting established roles with their associated expectations
2. **Role-making:** actively shaping and redefining role content
3. **Role creation:** establishing entirely new roles without precedent

These dynamics are particularly important during transitions, when established role structures are disrupted and new roles emerge. The concept of **role constellation**—webs of roles that interact, interrelate, and co-evolve around specific issues—shifts analysis from individual roles to relational configurations. Each framing of a persistent problem carries implicit ideas about roles and responsibilities.

However, their operationalization remains qualitative and case-specific. How does one identify “shared role understandings” empirically? Their Rotterdam neighborhood case provides rich description but limited systematic method for tracking role constellation change at scale.

## 2.5 De Haan & Rotmans: Power and Agency in Role Analysis

Haan and Rotmans [13] address a widely acknowledged gap<sup>6</sup>: the inadequate representation of actors and agency in transitions studies. They propose giving “explanatory primacy to agency,” presenting transitions as consequences of deliberate, value-driven actions rather than systemic forces.

Their framework introduces a four-role typology for transformative actors:

Crucially, actors are not “part of” systems but *affiliate* with them—holding multiple affiliations simultaneously. This framing enables analysis of boundary-spanning actors who connect different systems, and explains why the same organization might appear in multiple role positions across different contexts.

The framework also introduces **streams**—societal value sets enabled by available knowledge and organizing principles. Streams allow alliance formation around shared ideals rather than specific solutions. Actors form alliances of increasing scale: initiatives (frontrunner-driven), networks (connector-driven), and movements (toppler-driven).

However, the typology covers only *transformative* actors—those maintaining the status quo are explicitly bracketed. This is a significant limitation, as transitions are constituted by conflict between incumbent and challenger coalitions. The framework also lacks systematic methods for empirical identification: how does one identify a “frontrunner” or “toppler” without circular reasoning from outcomes?

6: **De Haan & Rotmans (2018)**—Power-in-transition framework analyzes how power shapes role possibilities.

**Table 2.1:** Power types and transition roles

| Role         | Function                  |
|--------------|---------------------------|
| Frontrunners | Introduce alternatives    |
| Connectors   | Link actors and solutions |
| Topplers     | Change institutions       |
| Supporters   | Adopt and endorse         |

## 2.6 Intermediary Functions and Typologies

Given this dissertation’s focus on intermediary organizations, we examine in depth how the literature has conceptualized intermediation—tracing its evolution from innovation studies into sustainability transitions research.<sup>7</sup>

7: **Conceptual evolution**—From bilateral brokers to systemic actors operating at network and system levels.

## From Bilateral to Systemic Intermediaries

The intellectual roots trace to innovation studies in the late 1990s [?]. The pivotal contribution came with Lente et al. [14] on “Roles of Systemic Intermediaries in Transition Processes,” distinguishing **systemic intermediaries** from traditional bilateral brokers. They argued these actors function at the *system or network level* rather than mediating one-to-one exchanges, identifying three core functions: **articulation** of options and demand, **alignment** of actors, and **support of learning processes**.

The intermediary concept evolved as it merged with sustainability transitions theory. Within the MLP, intermediaries came to be understood as actors positioned *between* levels—connecting niche innovations to regime transformation. In Strategic Niche Management, intermediaries perform critical aggregation activities that construct the “global niche” as a cognitive and social reality.

### The Aggregation Function

Intermediaries do not merely connect actors—they *aggregate* dispersed knowledge into coherent narratives. By circulating between local projects, they identify patterns, codify lessons, and construct shared understanding. Without this aggregation, individual experiments remain isolated and fail to accumulate into transition momentum.

## The Kivimaa Typology

The most influential recent contribution is Kivimaa et al. [15],<sup>8</sup> proposing a five-type typology:

**Systemic intermediaries** operate across niche, regime, and landscape levels with explicit transition agendas. National agencies like Finland’s Sitra exemplify this type. **Regime-based intermediaries** remain associated with prevailing regimes but promote transition—reformers rather than revolutionaries. **Niche intermediaries** work to advance specific innovations, developing voice and formalization. **Process intermediaries** facilitate change without explicit system-transformation agendas, earning trust through neutrality. **User intermediaries** connect users to new technologies and communicate preferences to developers.

Yet empirical studies reveal<sup>9</sup> that boundaries between types are porous—intermediaries often exhibit characteristics of multiple types simultaneously and may shift profiles strategically over time.

8: Kivimaa et al. (2019)—Systematic review proposes five-type intermediary typology.

**Table 2.2:** The Kivimaa intermediary typology

| Type         | Key features                                       |
|--------------|----------------------------------------------------|
| Systemic     | Operates across levels; explicit transition agenda |
| Regime-based | Reformer within regime; translates new regulation  |
| Niche        | Advances specific innovations; builds voice        |
| Process      | Neutral broker; project facilitation               |
| User         | Connects users to technologies                     |

9: **Porous boundaries**—Intermediaries often span types and shift strategically.

## Function Clusters

The extensive documentation of intermediary functions reveals four clusters:

**Knowledge functions:** gathering, processing, and generating knowledge; brokering and transferring across domains; disseminating information; facilitating learning; aggregating lessons across experiments.

**Network functions:** creating and managing networks; configuring and aligning interests; building partnerships and coalitions; connecting actors across boundaries; building relational capital—trust, shared understanding, collaborative capacity.

**Vision and expectation work:** articulating futures; sense-making through narrative construction; framing problems and solutions in ways that mobilize action; managing “promises and requirements” dynamics.

**Institutional functions:** advocating for policy change; legitimation work to build social acceptance; shaping regulatory frameworks; supporting institutional infrastructure development.

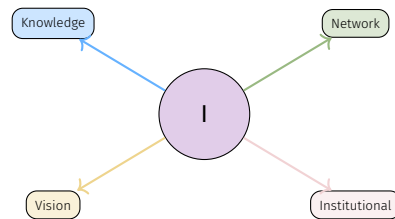


Figure 2.2: Four function clusters performed by transition intermediaries.

## Related Concepts

The intermediary concept operates alongside several related terms. **Boundary spanners** emphasize individual competencies for bridging knowledge domains, particularly at the science-policy interface [?]. **System builders** engage in strategic action to create favorable conditions, more directive than facilitative. **Transition brokers** represent systemic intermediaries fulfilling orchestration functions. **Orchestrators** exercise leadership and make strategic decisions, contrasting with facilitative orientations.

## 2.7 Kanda et al.: Intermediary Dynamics and System Levels

Kanda et al. [16] address a specific gap<sup>10</sup>: how to conceptualize intermediary activities *beyond* individual projects at the system level. Their key move: shift from *systemic intermediaries* (actor category) to *systemic intermediation* (activity description). Any intermediary may perform systemic activities; dedicated “systemic intermediaries” may also perform non-systemic activities.

They propose four levels of intermediation based on the relational criterion “in-between what?”:

**Level 0:** In-between individual entities. One-to-one interactions with benefits accruing to individual parties. Typical: consultants, bilateral brokerage.

**Level 1:** In-between entities in a network. Intermediary mediates within a single network. Roles: facilitating networking, knowledge transfer, building legitimacy for network activities.

10: Kanda et al. (2020)—Emphasizes temporal dynamics and ecology perspectives.



**Level 2:** In-between networks. Intermediation spans different network types. Roles: creating interaction spaces, building coalitions, providing common voice.

**Level 3:** In-between actors, networks, and institutions. Transcends horizontal network interactions to vertical engagement with formal and informal institutions. Roles: lobbying for policy change, articulating new visions, translating policies to local contexts.

Key empirical finding: **non-systemic activities dominate**.<sup>11</sup> Most intermediation is bilateral despite the literature's focus on "systemic intermediaries." Systemic activities require financial stability, longevity, and sometimes neutrality—they are resource-intensive. Mandate matters as much as resources: limited-resource intermediaries with institutional change mandates can operate at Level 3, while well-resourced intermediaries with bilateral mandates may concentrate on Level 0.

**11: Phase-dependent roles**—Niche intermediaries dominate early; systemic intermediaries peak during acceleration; diffusion intermediaries become critical for mainstreaming.

## Emergence Pathways

Three intermediary emergence pathways shape characteristics and capabilities:

**Purpose-built:** specifically established to intermediate, typically by governments creating agencies with explicit transition mandates. These begin with formal mandates but must build legitimacy and networks.

**Evolved:** existing organizations that adopt intermediary roles as conditions change. An industry association may evolve from representing incumbent interests to facilitating transitions. These leverage existing relationships but may carry institutional baggage.

**Emergent:** arising spontaneously in response to system gaps, often without initially recognizing their intermediary function. Community energy groups and informal networks may gradually assume broader roles. These possess grassroots legitimacy but may lack resources.

## Evolution Across Transition Phases

What intermediaries need to do—and which types matter most—changes as transitions progress.

In **pre-development**, intermediaries support early experimentation, build initial networks among pioneers, and facilitate learning from failures. Niche and process intermediaries are particularly important.

During **take-off**, functions shift toward knowledge aggregation, building broader coalitions, and pooling resources for scaling. Systemic intermediaries become more important.

**Acceleration** demands stronger institutional support, capacity building for mainstream actors, and substantial resource mobilization. "Diffusion intermediaries" help adopters navigate new practices.

In **stabilization**, some intermediaries cease to exist as roles become unnecessary—a sign of success. Others find new opportunities or pivot to remaining niches.

A crucial finding: "without intermediary influence, pre-development is unlikely to lead to acceleration." Intermediaries provide continuity

across phases, carrying lessons and relationships even as specific projects end.

## Ecologies of Intermediaries

Perhaps the most significant recent advance is recognizing that transitions involve *multiple* intermediaries whose interactions shape outcomes.

### Ecology of Intermediaries

An **ecology of intermediaries** comprises a variety of intermediary actors with different missions, mandates, and levels of agency that collectively connect actors and resources at different scales within a transition arena.

Within ecologies, intermediaries exhibit both **complementary and competitive dynamics**.<sup>12</sup> Complementarity arises when different types address different functions: systemic intermediaries articulating visions while process intermediaries manage projects. Yet intermediaries also **compete** for resources, attention, legitimacy, and influence.

**Ecology gaps**—missing intermediary functions—can hinder transition progress. A transition arena may have strong niche intermediaries but lack systemic coordination, or possess visionary leadership but insufficient process support. Mapping these gaps is essential for transition governance but requires systematic methods currently lacking.

12: **Beyond single intermediaries**—Most research examines individual intermediaries. The ecology concept redirects attention to how multiple intermediaries interact, complement, and compete.

## 2.8 Geographic and Methodological Considerations

**Geographic context** matters substantially.<sup>13</sup> While European research dominates—particularly from the Netherlands, Finland, Germany, and UK—patterns vary across governance systems. The Sustainability Transitions Research Network’s “Transitions in Global South” initiative has catalyzed research beyond Northern contexts, revealing that intermediary roles and ecosystem configurations differ with institutional environments, development trajectories, and power structures.

**Methodological approaches** face persistent trade-offs. Qualitative case studies provide rich understanding of how intermediaries navigate complex contexts but cannot identify population-level patterns. Social network analysis maps relational structures but typically relies on single data sources and struggles to capture meaning. Computational and text-analytic approaches offer scale but often begin with available data rather than theoretical questions.

Each approach illuminates some aspects<sup>14</sup> while obscuring others. No single method captures the full terrain of intermediary role constellations.

13: **Context matters**—Geographic variation and methodological constraints shape knowledge production.

14: **Methodological tensions**—Depth vs. scale; meaning vs. pattern; theory vs. data.

## 2.9 Synthesis: Convergences, Divergences, and Gaps

Having reviewed individual contributions, we now synthesize across them to identify what these frameworks share, where they diverge, and what gaps remain.

### Comparative Overview

|                           | De Haan & Rotmans (2018)                                    | Wittmayer et al. (2017)                          | Kanda et al. (2020)                              | Fischer & Newig (2016)                            |
|---------------------------|-------------------------------------------------------------|--------------------------------------------------|--------------------------------------------------|---------------------------------------------------|
| <b>Core question</b>      | How do actors drive transformative change?                  | How do roles mediate actor-society relations?    | At what system levels does intermediation occur? | How are actors treated in transitions literature? |
| <b>Theoretical base</b>   | Philosophy of action; complex systems                       | Sociological role theory (Mead, Turner, Giddens) | Innovation systems; MLP                          | Meta-review of transitions field                  |
| <b>Unit of analysis</b>   | Actors (value-driven agents)                                | Roles (shared understandings)                    | Intermediation activities                        | Actor typologies in literature                    |
| <b>Key typology</b>       | Frontrunners, Connectors, Topplers, Supporters              | Single roles vs. Role constellations             | Levels 0–3 (bilateral to institutional)          | Four parallel schemes identified                  |
| <b>Relational concept</b> | Affiliation (actors affiliate with, not belong to, systems) | Role constellation (webs of interacting roles)   | In-betweenness (relational criterion for levels) | Dependencies (actors depend on others' functions) |
| <b>Temporal framing</b>   | Streams rise/dry up; trajectories toward replacement        | Roles as temporary stabilizations                | Static levels; dynamics not theorized            | Roles “erratic”; change with phase                |

Table 2.3: Framework comparison across four actor/role contributions.

### Points of Convergence

Despite originating from different intellectual traditions,<sup>15</sup> the four frameworks converge on several key insights:

**Actor-process distinction:** All four distinguish between actors-as-entities and the processes they perform. De Haan & Rotmans separate actors from transformative change; Wittmayer et al. separate actors from roles; Kanda et al. separate intermediaries from intermediation; Fischer & Newig separate actors from agency.

**Multiple positions:** Actors are not bound to single categories. De Haan & Rotmans emphasize multiple affiliations to systems; Wittmayer et al. note actors occupy multiple roles; Kanda et al. show intermediaries operate across levels; Fischer & Newig find actors belong to different categories simultaneously.

**Temporal instability:** Roles and positions change over transition phases. De Haan & Rotmans trace role trajectories; Wittmayer et al. describe roles as temporary stabilizations; Fischer & Newig find roles “erratic.”

15: **Five convergences**—Despite different theoretical origins, frameworks agree on key principles.

**Relational over categorical:** Position relative to others matters more than intrinsic attributes. De Haan & Rotmans emphasize affiliation structure; Wittmayer et al. focus on role constellations; Kanda et al. apply an “in-between what?” criterion; Fischer & Newig highlight dependencies between actors.

**Typology limitations:** All find existing categories inadequate. De Haan & Rotmans argue MLP cannot capture actor dynamics; Wittmayer et al. note transitions lack role vocabulary; Kanda et al. find “systemic intermediary” underspecified; Fischer & Newig identify four unintegrated typologies coexisting.

## Points of Divergence

| Dimension        | De Haan & Rotmans            | Wittmayer et al.                             | Kanda et al.            | Fischer & Newig   |
|------------------|------------------------------|----------------------------------------------|-------------------------|-------------------|
| Agency source    | Values (actors value-driven) | Social structure (roles as cultural objects) | Mandate + resources     | Varied            |
| Change mechanism | Strategic alliance action    | Role negotiation                             | Higher-level activities | Network formation |
| Intermediaries   | Connectors (partial overlap) | Not addressed                                | Central focus           | Separate category |
| Empirical scale  | National                     | Neighborhood                                 | Regional                | Field-level       |

Table 2.4: Divergent dimensions across frameworks.

## Shared Gaps

Most consequentially, all four frameworks share several limitations:<sup>16</sup>

**Operationalization:** All provide concepts without systematic methods for empirical identification. How does one identify a “frontrunner” or measure “role constellation change”? Researcher interpretation dominates.

**Population-level analysis:** All work at case level. No framework addresses: How many actors of type X exist? What is the distribution across types? How do types cluster spatially or sectorally?

**Cross-modal validation:** None systematically compare claimed roles (self-description) with enacted roles (observed behavior). Do organizations do what they say they do?

**Boundary specification:** Role constellations, networks, systems—all invoked but boundaries left implicit. What defines membership? Where does one constellation end and another begin?

**Power analysis:** Dependencies, resources, and constellations all gesture toward power but don’t systematically analyze asymmetries, contestation, or domination.

**Failure and absence:** All focus on presence and function. What happens when roles go unfilled? When intermediaries fail? When actors exit?

16: **Six gaps**—These limitations motivate the cartographic approach developed in this dissertation.

## Toward Integration

The four papers collectively point toward a framework with three dimensions:

1. **Functional dimension:** What activities are performed? (Kivimaa's functions; Kanda et al.'s roles; De Haan & Rotmans' frontrunner/connector/toppler)
2. **Relational dimension:** In-between what does action occur? (Kanda et al.'s levels; Wittmayer et al.'s constellations; Fischer & Newig's dependencies)
3. **Temporal dimension:** How do positions change over transition phases? (De Haan & Rotmans' trajectories; Wittmayer et al.'s temporary stabilizations)

Existing typologies emphasize one dimension while backgrounding others.<sup>17</sup> A comprehensive actor/role framework would characterize positions along all three simultaneously—specifying *what* actors do, *where* they are positioned relationally, and *how* this changes over time.

17: **Three dimensions**—Existing typologies emphasize one dimension while backgrounding others.

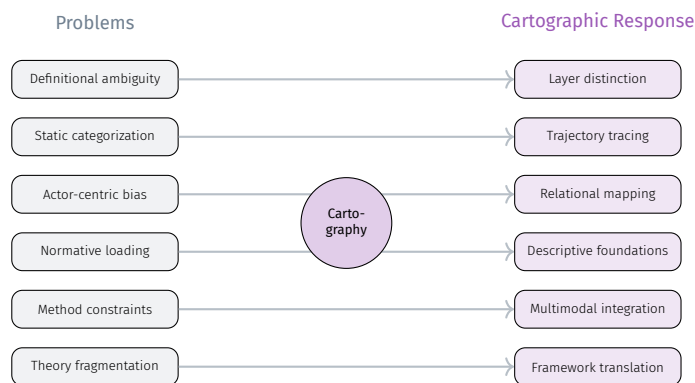
## 2.10 Six Persistent Problems—and a Cartographic Response

Building on this synthesis, we identify six persistent problems that limit current understanding of role constellations.<sup>18</sup> These are not merely gaps to be filled but **productive tensions** pointing toward cartographic opportunities.

1. **Definitional ambiguity.** “Role” variously denotes position, function, identity, or behavior—conflations that impede cumulation.  
*Cartographic reframing:* Rather than seeking definitional consensus, cartography **distinguishes layers**. Position can be mapped relationally (where actors sit in network space); function can be mapped through activity traces (what actors do); identity can be mapped through self-presentation. Different layers can be **superimposed** to reveal how position, function, and identity align or diverge.
2. **Static categorization.** Typologies present roles as fixed boxes, yet empirical reality shows fluidity—actors shifting roles, occupying multiple roles, varying by context.  
*Cartographic reframing:* Categories are **snapshots**; cartography traces **trajectories**. Rather than asking “which box does this intermediary belong in?” we ask “where has this intermediary been, where is it now, and in what direction is it moving?”
3. **Actor-centric bias.** Roles are typically treated as actor attributes rather than relational accomplishments, obscuring how roles emerge through interaction.  
*Cartographic reframing:* Cartography is inherently **relational**—positions exist only relative to other positions; territories are defined by boundaries with neighbors; paths connect locations. Mapping intermediaries means mapping the spaces *between* them, the connections that constitute positions.

18: **Problems as opportunities**—Each problem suggests what better cartography would accomplish, not solving problems definitively but reframing them productively.

4. **Normative loading.** Role concepts carry implicit judgments—intermediaries good, incumbents bad; bridging valuable, blocking problematic.  
*Cartographic reframing:* Maps can be **descriptive**—charting terrain without evaluating it. A map shows where mountains are without claiming mountains are good. Cartography enables normative analysis while maintaining descriptive foundations.
5. **Methodological constraints.** Small-N qualitative studies provide depth but cannot identify population-level patterns; computational approaches offer scale but lack theoretical meaning.  
*Cartographic reframing:* Multimodal cartography **integrates** rather than chooses. Theory guides what features to map; multiple data sources triangulate positions; qualitative insight interprets quantitative patterns.
6. **Theoretical fragmentation.** Different frameworks emphasize different roles without integration. MLP, TIS, SNM, and transition management each offer partial maps that don't align.  
*Cartographic reframing:* Fragmentation reflects **different mapping conventions**. Rather than seeking a single master framework, cartography can **translate between** frameworks, showing how different conceptual schemes relate.



**Figure 2.3:** Six persistent problems in role constellation research and corresponding cartographic responses.

These reframings do not dissolve problems but **redirect** them. Definitional ambiguity becomes an invitation to layer-distinction rather than definitional policing. Static categorization becomes motivation for trajectory-tracing rather than better typologies. The problems remain productive tensions—but cartography offers new ways to work with them.

## 2.11 Chapter Summary

This chapter has surveyed how sustainability transitions research conceptualizes roles, with particular attention to intermediaries as the focal actors of this dissertation:

- **Role constellations** provide a meso-level lens for analyzing what functions actors perform and how these combine—distinct from actor constellations that merely catalog who is

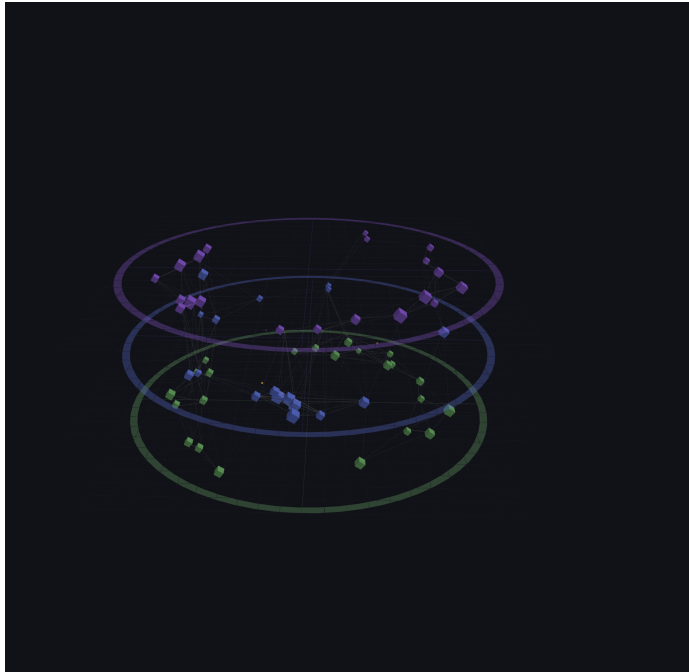
present.

- ▶ Roles are **malleable and relational**: they emerge through interaction, shift across transition phases, and gain meaning through position within constellations.
- ▶ **Key contributions** reviewed include: Fischer & Newig's (2016) systematic typology mapping; Wittmayer et al.'s (2017) introduction of role theory; De Haan & Rotmans' (2018) power-in-transition framework; and Kivimaa et al.'s (2019) and Kanda et al.'s (2020) intermediary syntheses.
- ▶ **Intermediaries** have evolved from bilateral brokers to systemic actors operating across niche-regime-landscape levels. The Kivimaa typology distinguishes systemic, regime-based, niche, process, and user intermediaries—but boundaries are porous.
- ▶ Intermediaries perform four function clusters—**knowledge work, network building, vision articulation, and institutional change**.
- ▶ **Ecologies of intermediaries** shape transitions through complementary and competitive dynamics. Ecology gaps can stall progress.
- ▶ **Six persistent problems**—definitional ambiguity, static categorization, actor-centric bias, normative loading, methodological constraints, and theoretical fragmentation—limit current understanding but point toward cartographic opportunities.
- ▶ The field's core limitation is **categorical thinking** where **cartographic thinking** is needed.

The literature provides rich conceptual resources—typologies, function inventories, phase frameworks, ecology concepts—but methodology inadequate to its own insights. We have sophisticated ways to *think* about intermediation but limited tools to *map* it systematically.

The **Intermezzo** that follows introduces the empirical context for developing cartographic methods:<sup>19</sup> the European Institute of Innovation and Technology (EIT) and its eight Knowledge and Innovation Communities. These intermediary organizations—spanning climate, energy, food, health, digital, urban mobility, raw materials, and manufacturing—provide an ideal laboratory for multimodal cartography.

19: → **Next: Intermezzo**—The EIT Ecosystem. An introduction to the eight Knowledge and Innovation Communities that serve as this dissertation's empirical laboratory.





# Design Science Methodology 3

*This chapter explains how we conducted this research. We employed **Design Science Research (DSR)**—a methodology suited for building and evaluating artifacts that solve real problems. Our artifact is the **ROLEBox** toolkit for multimodal cartography of intermediary role constellations. This chapter establishes why DSR fits our goals, details the systematic process we followed, documents our data collection and analytical workflows, and addresses rigor and limitations with transparency.*

## 3.1 The Methodological Challenge

This dissertation faces an unusual methodological challenge. We are not merely testing hypotheses against existing data or building theory from qualitative interpretation. We are *building something*—a toolkit for mapping role constellations that does not yet exist.

Chapter 2 identified a **cartographic gap**:<sup>1</sup> the field has sophisticated conceptual frameworks for thinking about intermediary roles but lacks methodological tools adequate to its own insights. Categories abound; maps are absent. The literature tells us *what kinds* of intermediaries exist but cannot show us *where* they are positioned relative to each other, how their constellations form, or how those constellations evolve.

Traditional research methodologies do not quite fit this challenge:

**Quantitative methods** assume measures already exist. We needed to create them—operationalizing “role” across multiple data modalities where no validated instruments exist.

**Qualitative methods** assume phenomena to interpret. We needed to build infrastructure for systematic observation before interpretation could begin.

**Mixed methods** combine existing approaches. We needed something genuinely new—not a combination of established techniques but novel methods for a novel problem.

What we required was a methodology for *building and evaluating research tools*—creating artifacts that enable new kinds of inquiry while simultaneously using those artifacts to generate substantive findings.

### The Cartographer’s Dilemma

A cartographer mapping unknown territory faces a bootstrapping problem: you need methods to create maps, but the adequacy of

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1: 🚧 **The core challenge**—We needed methods that didn’t exist. So we had to build them. That requires a different kind of research.

methods can only be assessed against the territory they reveal. We could not know whether our multimodal approach would work until we built it; we could not build it without theoretical guidance about what to look for. DSR provides a framework for navigating this productive circularity.

## 3.2 Design Science Research: Building as Inquiry

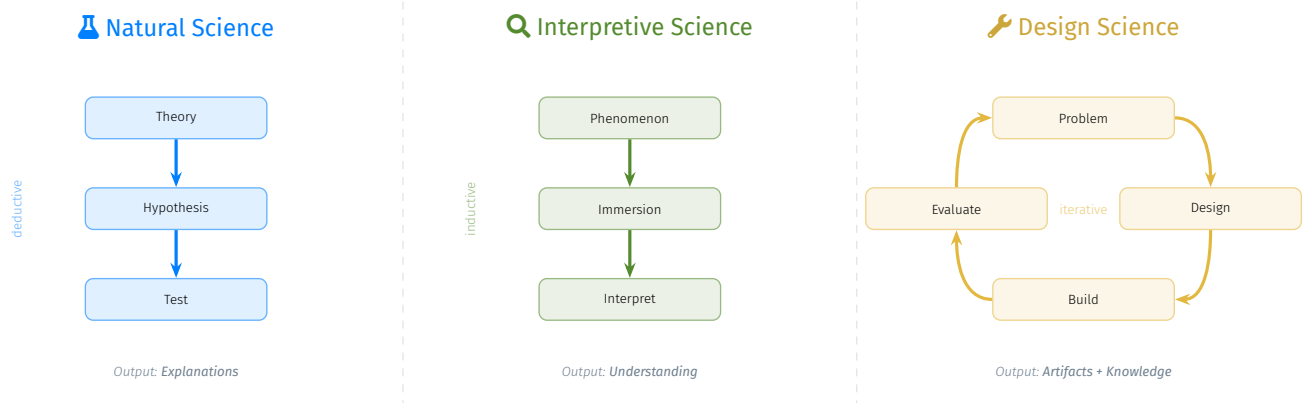
Design Science Research (DSR) is a methodology originating in information systems that treats the creation of artifacts as a legitimate—indeed, central—form of research contribution. The core insight is deceptively simple: sometimes the best way to understand a problem is to build something that solves it.

2

### 3.2.1 The Logic of Design Science

DSR differs fundamentally from both natural science (explaining what *is*) and interpretive science (understanding what things *mean*). It belongs to the “sciences of the artificial” [?]—inquiry into things that *could be* but do not yet exist.

2: **DSR Origins**—Developed in information systems by Simon [?], Hevner et al. [?], and Peffers et al. [?]. Now applied across design-oriented disciplines including engineering, architecture, and increasingly, social science.



**Figure 3.1:** Three research paradigms. Natural science explains *what is*; interpretive science understands *what things mean*; design science creates *what could be*. DSR’s iterative cycle generates both artifacts and knowledge about artifact design.

The key distinctions from traditional research:

| Dimension         | Traditional Research          | Design Science Research         |
|-------------------|-------------------------------|---------------------------------|
| Primary output    | Knowledge claims              | Artifacts <i>plus</i> knowledge |
| Core activity     | Testing or interpreting       | Building and evaluating         |
| Validation logic  | Correspondence to reality     | Utility for purpose             |
| Success criterion | Is it true? Is it insightful? | Does it work? Is it useful?     |
| Theory role       | Tested against data           | Embodied in artifact design     |

**Table 3.1:** Traditional research vs. Design Science Research

3

3: **💡 Key insight**—DSR doesn't replace theory; it generates theory *through* building. The artifact embodies design knowledge that can be abstracted into transferable principles.

### 3.2.2 Why DSR Fits This Research

DSR is appropriate when four conditions hold. All four apply to our research:

- 1. A real problem exists that current tools cannot solve.**  
Chapter 2 documented the cartographic gap: rich conceptual frameworks but no methods for systematically mapping role constellations at scale. Researchers must choose between qualitative depth (30–50 actors) and quantitative breadth (losing theoretical meaning). The problem is real and consequential.
- 2. Solving the problem requires creating something new.**  
No existing toolkit combines multimodal data collection, theory-guided role identification, relational mapping, and cross-modal validation. We could not address the gap by applying existing methods differently; we needed to build new infrastructure.
- 3. The artifact itself contributes knowledge.**  
The ROLEBOXtoolkit embodies methodological knowledge about *how* to map intermediary roles—which data sources reveal which role dimensions, how to integrate signals across modalities, what validation strategies work. This design knowledge transfers beyond our specific application.
- 4. Evaluation is possible through demonstration and use.**  
We can apply the toolkit to the EIT ecosystem, assess whether it produces meaningful maps, compare results with theory-derived expectations, and gather feedback from practitioners. The artifact's utility is empirically assessable.

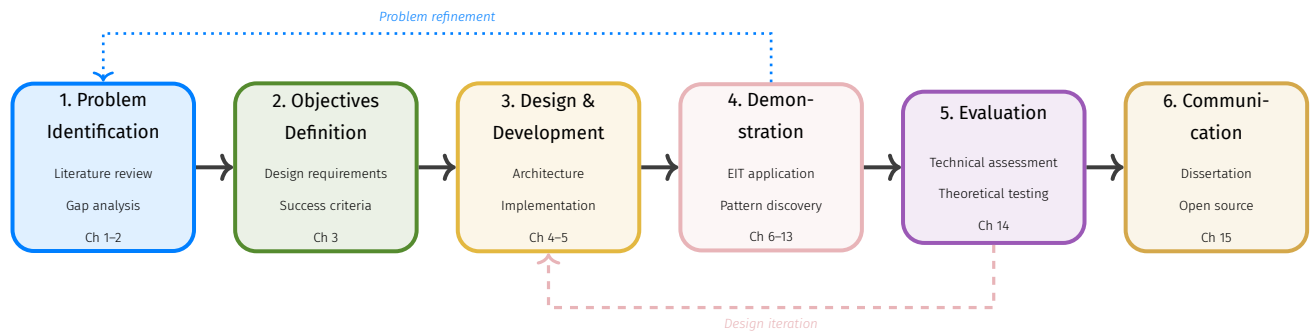
**DSR Fit Proposition:** When the research goal is to enable new forms of inquiry rather than to conduct inquiry with existing tools, Design Science Research provides appropriate methodological scaffolding—legitimizing artifact creation as research contribution while maintaining rigor through systematic evaluation.

### 3.3 The DSR Process: From Problem to Artifact

We followed Peffers et al.'s (2007) six-activity DSR process model [?], adapted for our context.<sup>4</sup> The process is explicitly iterative—we cycled

4: **🕒 Timeline**—2021: Problem identification. 2022: First design iteration. 2023: Core development, EIT data collection. 2024: Integration, evaluation, refinement. 2025: Dissertation completion.

through activities multiple times over four years, with each cycle refining both the artifact and our understanding of the problem.



**Figure 3.2:** The six-activity DSR process with iteration loops. Design iteration (dashed) refines the artifact based on evaluation; problem refinement (dotted) sharpens understanding based on demonstration findings.

### 3.3.1 Activity 1: Problem Identification and Motivation

Problem identification drew on multiple sources:

- **Literature analysis:** Systematic review of intermediary and role constellation scholarship (Chapter 2) identified the cartographic gap—categories without maps, typologies without topographies
- **Community engagement:** Conference discussions at IST (International Sustainability Transitions), DRUID, and Eu-SPRI revealed shared frustration with methodological limitations
- **Practitioner input:** Conversations with EIT staff and KIC managers confirmed demand for ecosystem assessment tools
- **Personal experience:** Our own research obstacles—attempting role analysis with inadequate methods—motivated the search for alternatives

The problem crystallized as:<sup>5</sup> *How can we systematically map intermediary role constellations at scale while preserving theoretical meaning?*

5: **Q Problem sources**—Academic literature, conference discussions, practitioner needs, research obstacles.

### 3.3.2 Activity 2: Define Objectives of a Solution

Based on theoretical foundations (Chapter 4) and problem analysis, we defined six design objectives that a solution must achieve:

| ID | Objective                    | Specification                                                                                |
|----|------------------------------|----------------------------------------------------------------------------------------------|
| O1 | <b>Scale</b>                 | Handle hundreds to thousands of organizations, not dozens—enabling population-level analysis |
| O2 | <b>Multimodality</b>         | Integrate multiple data types (text, network, visual, temporal) to capture role complexity   |
| O3 | <b>Relationality</b>         | Map positions and relationships, not just attributes—producing actual cartography            |
| O4 | <b>Validation</b>            | Enable comparison of claimed roles (self-presentation) vs. enacted roles (observed behavior) |
| O5 | <b>Openness</b>              | Be transparent, documented, and extensible—supporting replication and adaptation             |
| O6 | <b>Theoretical grounding</b> | Connect to transitions theory, not just pattern-finding—ensuring maps are interpretable      |

Table 3.2: Design objectives for the ROLEBoxtoolkit

These objectives translate directly into evaluation criteria: a successful artifact must demonstrably achieve each objective.

### 3.3.3 Activity 3: Design and Development

This was the core creative work—designing architecture, selecting technologies, implementing pipelines, and iterating through failures.<sup>6</sup> The empirical chapters detail the resulting artifact in action; here we describe the design *process*.

We followed an iterative prototyping approach:

1. **Prototype:** Build minimal viable version for one modality
2. **Test:** Apply to subset of EIT data (100 organizations)
3. **Fail:** Identify what breaks, what produces noise, what misses signal
4. **Learn:** Document failure modes and successful patterns
5. **Refine:** Improve based on learning
6. **Expand:** Add next modality; return to step 1
7. **Integrate:** Once all modalities work individually, build integration layer
8. **Iterate:** Cycle through entire system based on full-scale application

Key design decisions emerged through this process:

**Modular architecture** Early monolithic designs proved brittle. We restructured into independent modules that can be used separately or together.

**Theory-guided features** Initial atheoretical feature extraction produced uninterpretable results. We redesigned extraction to target theoretically meaningful role dimensions.

**Cross-modal linking** Entity resolution across modalities was harder than anticipated. We developed a dedicated linking layer with fuzzy matching and human verification.

**Validation by design** Rather than treating validation as afterthought, we built comparison mechanisms (claimed vs. enacted) into the core architecture.

6:  **Design iterations**—v0.1: Single-modality proof of concept. v0.5: Multi-modality pipelines. v1.0: Full integration layer. v1.5: Refinement based on EIT application.

### 3.3.4 Activity 4: Demonstration

Demonstration applies the artifact to a real problem, showing that it works.<sup>7</sup> The empirical chapters demonstrate the ROLEBoxtoolkit on the EIT ecosystem:

- ▶ **Sociotype analysis** (Chapter 5): Internal role configurations and interstitial positioning
- ▶ **Field mapping** (Chapter 6): External classification and field structure
- ▶ **Investment analysis** (Chapter 7): Resource flows and venture positioning
- ▶ **Visual analysis** (Chapter 8): Aesthetic positioning and symbolic registers
- ▶ **Power analysis** (Chapter 9): Agency and power dynamics in discourse
- ▶ **Synthesis** (Chapter 10): Cross-modal integration and validation

Demonstration is not mere illustration—it is part of evaluation, showing whether the artifact produces useful, interpretable, theoretically meaningful results.

7:  **Demonstration scope**—6,487 organizations, 8 EIT KICs, 5 data modalities, 2019–2024 timeframe, 218,000+ records.

### 3.3.5 Activity 5: Evaluation

Evaluation assesses whether the artifact achieves its objectives. We employ multiple evaluation strategies:

| Strategy               | What It Assesses                 | How We Apply It                                                                |
|------------------------|----------------------------------|--------------------------------------------------------------------------------|
| Technical evaluation   | Does it work correctly?          | Performance metrics, error rates, computational efficiency, edge case handling |
| Empirical evaluation   | Does it reveal patterns?         | Novelty and interpretability of EIT findings; comparison with prior knowledge  |
| Theoretical evaluation | Does it connect to theory?       | Testing propositions derived from intermediary literature                      |
| Comparative evaluation | Does it improve on alternatives? | Comparison with single-modality approaches and existing methods                |
| Practical evaluation   | Is it useful?                    | Feedback from EIT practitioners and transitions researchers                    |

**Table 3.3:** Evaluation strategies and their application

Chapter ?? presents detailed evaluation results. The key question is not “Is it perfect?” but “Does it advance our ability to map role constellations beyond what was previously possible?”

### 3.3.6 Activity 6: Communication

Communication shares the artifact and generated knowledge with relevant communities:<sup>8</sup>

- **This dissertation:** Primary communication vehicle, documenting problem, artifact, demonstration, and evaluation
- **Open-source repository:** All code publicly available with documentation
- **Archived data:** Processed datasets deposited for replication
- **Academic publications:** Selected findings published in peer-reviewed venues
- **Practitioner engagement:** Presentations to EIT stakeholders

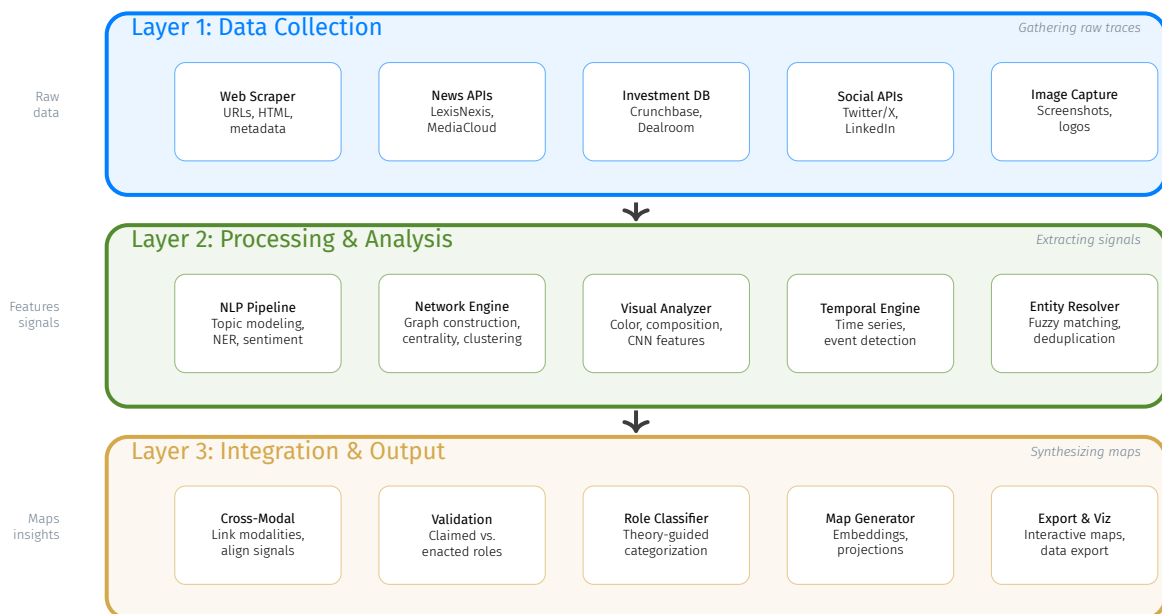
8: ➡ **Communication channels**—This dissertation, open-source code (GitHub), archived datasets (Zenodo), conference presentations, journal articles.

## 3.4 The ROLEBoxArtifact: Architecture Overview

The artifact we built is the **ROLEBoxtoolkit**—an integrated system for multimodal cartography of role constellations. The appendices provide detailed technical documentation; here we present the architectural overview.

### 3.4.1 Three-Layer Architecture

The toolkit comprises three layers, each with distinct responsibilities:



**Figure 3.3:** The ROLEBoxtoolkit architecture. Layer 1 collects raw digital traces from multiple sources; Layer 2 processes traces into analytical signals using specialized engines; Layer 3 integrates signals into cartographic outputs. Each component operates independently but contributes to integrated multimodal analysis.

9

9:  **Three layers**—**Collection**: Gathering traces. **Processing**: Extracting signals. **Integration**: Synthesizing maps.

The architecture reflects our design objectives:

**Modularity** (O5): Each component can be used independently or in combination. Researchers can employ the NLP pipeline without the visual analyzer, or add new data sources without modifying existing pipelines.

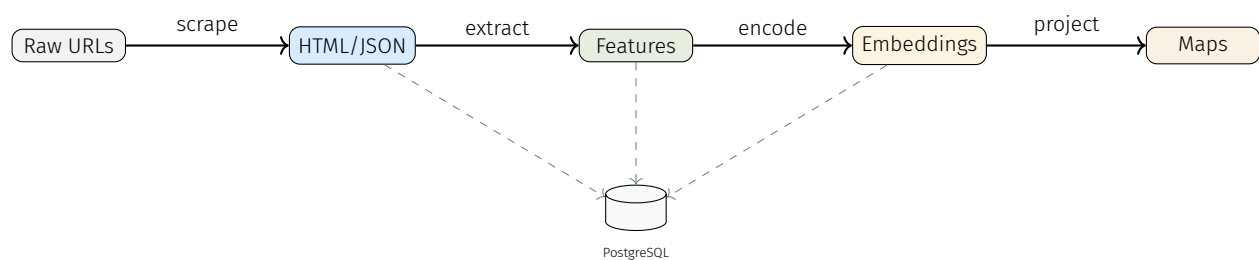
**Extensibility** (O5): New modalities can be added by creating collection and processing modules that output standardized feature formats for the integration layer.

**Transparency** (O5): All code is documented and version-controlled. Processing decisions are logged for auditability.

**Reproducibility** (O5): Workflows are containerized and can be re-executed on new data with documented parameters.

### 3.4.2 Data Flow

Data flows through the system in a structured pipeline:



**Figure 3.4:** Simplified data flow through the ROLEBox pipeline. Each stage transforms data and persists results to the database for traceability and reprocessing.

## 3.5 Data Collection: Sources and Workflows

The empirical component required substantial data collection across five modalities. Each modality captures different role dimensions; together they enable triangulation and validation.



### 3.5.1 Overview of Data Sources

| Modality     | Primary Source                 | Orgs         | Records          | Timeframe     |
|--------------|--------------------------------|--------------|------------------|---------------|
| Websites     | Direct scraping                | 6,487        | 12,400 pages     | 2023 snapshot |
| News         | LexisNexis, Media-Cloud, GDELT | 6,487        | 34,500 articles  | 2019–2024     |
| Investment   | Crunchbase, Deal-room          | 1,312        | 2,100 rounds     | 2010–2024     |
| Visual       | Website screenshots            | 6,487        | 13,200 images    | 2023 snapshot |
| Social       | Twitter/X API                  | 2,389        | 1,156,000 tweets | 2019–2023     |
| <b>Total</b> |                                | <b>6,487</b> | <b>218,200+</b>  |               |

Table 3.4: Data collection summary across modalities

### Population Definition and Sampling

Our target population is organizations connected to the EIT Knowledge and Innovation Communities.<sup>10</sup> We employed a multi-stage sampling strategy:

10: **Data scale**—6,487 organizations, 218,000+ records, 5 modalities, 5+ years coverage.

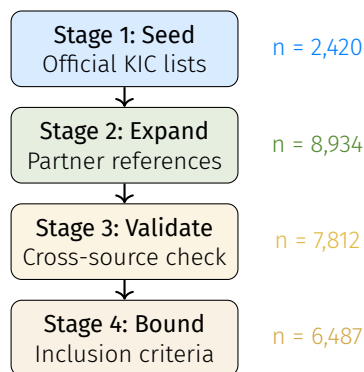


Figure 3.5: Multi-stage sampling strategy. Starting from official KIC partner lists, we expanded through network references, validated across sources, and applied inclusion criteria.

**Stage 1 – Seed** We scraped official partner directories from all eight KIC websites,<sup>11</sup> yielding 2,420 confirmed partner organizations.

**Stage 2 – Expand** We extracted organization mentions from seed partner websites (“About,” “Partners,” “Portfolio” pages), adding organizations referenced by multiple partners. This expanded the candidate set to 8,934 organizations.

**Stage 3 – Validate** We cross-referenced candidates against news databases and investment records, removing organizations with no independent evidence of EIT connection. This reduced the set to 7,812 validated organizations.

**Stage 4 – Bound** We applied inclusion criteria requiring presence in at least two data modalities (ensuring sufficient data for analysis). This produced the final sample of 6,487 organizations.

11: **Inclusion criteria**—Listed as KIC partner, OR mentioned by 3+ partners, OR news coverage with KIC, OR investment from KIC fund; AND present in 2+ modalities.

### 3.5.2 Collection Workflows by Modality

Each modality required specialized collection procedures:

#### Website Collection

1. **URL resolution:** Verify current URLs, handle redirects, detect defunct sites
2. **Crawling:** Depth-limited crawl (max 3 levels) capturing HTML, meta-data, and assets
3. **Screenshot capture:** Automated browser rendering of homepage and key pages
4. **Text extraction:** Convert HTML to clean text; preserve structure information
5. **Rate limiting:** Respectful crawling with delays; honor robots.txt<sup>12</sup>

12:  **Website workflow**—Tool: Scrapy + Playwright. Pages/org: ~2 (median). Success rate: 94%. Duration: 3 weeks.

#### News Collection

1. **Query construction:** Organization name variants + KIC keywords
2. **API queries:** LexisNexis (structured news), MediaCloud (online news), GDELT (global events)
3. **Deduplication:** Remove duplicate articles across sources
4. **Relevance filtering:** Verify articles actually discuss target organization
5. **Full-text retrieval:** Obtain complete article text where available<sup>13</sup>

13:  **News workflow**—Sources: 3 APIs. Articles/org: ~5 (median). Languages: EN (primary). Duration: 2 months.

#### Investment Collection

1. **Entity matching:** Match organizations to Crunchbase/Dealroom records
2. **Funding retrieval:** Extract funding rounds, amounts, investors, dates
3. **Investor expansion:** Add investor organizations to network
4. **Validation:** Cross-check funding data across sources<sup>14</sup>

14:  **Investment workflow**—Sources: Crunchbase, Dealroom. Match rate: 20% (startups). Funding rounds: 2,100. Duration: 1 month.

#### Visual Collection

1. **Homepage screenshots:** Full-page renders at standardized resolution
2. **Logo extraction:** Identify and extract organizational logos
3. **Image cataloguing:** Index all images on crawled pages
4. **Quality filtering:** Remove broken images, duplicates, stock photos

#### Social Media Collection

1. **Account identification:** Match organizations to Twitter/X handles
2. **Timeline retrieval:** Collect tweets within date range (API v2)
3. **Engagement capture:** Record likes, retweets, replies
4. **Network extraction:** Identify mentions, replies, quote tweets<sup>15</sup>

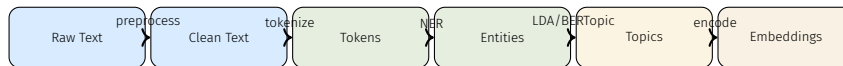
15:  **Social workflow**—Platform: Twitter/X. Account match: 37%. Tweets/account: ~65 (median). Note: API access limited post-2023.

## 3.6 Analytical Workflows

Each modality requires tailored analytical workflows. Here we summarize approaches; empirical chapters provide details.

### 3.6.1 Text Analysis Workflow

Text from websites and news undergoes a multi-stage NLP pipeline:



**Figure 3.6:** Text analysis pipeline from raw text through embeddings.

**Preprocessing** Language detection, encoding normalization, boilerplate removal

**Tokenization** Sentence and word segmentation using spaCy [?]

**Named Entity Recognition** Extract organizations, persons, locations, technologies

**Topic Modeling** BERTopic [?] for semantic clustering; guided LDA for theory-informed topics [?]

**Embedding** Sentence-BERT [?] for dense vector representations enabling similarity computation

### 3.6.2 Network Analysis Workflow

Networks are constructed from multiple relationship types:

- **Partnership networks:** Edges from co-membership in KICs, joint projects
- **Citation networks:** Edges from website hyperlinks, news co-mentions
- **Investment networks:** Edges from funding relationships
- **Social networks:** Edges from Twitter mentions, retweets

Analysis employs standard network metrics (degree, betweenness, eigenvector centrality) [?] plus community detection (Louvain [?], Infomap [?]) and structural equivalence measures.

### 3.6.3 Visual Analysis Workflow

Visual analysis extracts features from website screenshots and logos:

**Color analysis** Dominant colors, palette complexity, color harmony

**Composition** Layout structure, visual hierarchy, whitespace usage

**CNN features** Pre-trained ResNet [?] features for visual similarity

**Logo analysis** Shape features, color schemes, symbolic elements

### 3.6.4 Integration Workflow

The integration layer synthesizes signals across modalities:

1. **Entity alignment:** Ensure same organization linked across modalities
2. **Feature normalization:** Scale features to comparable ranges
3. **Multi-view embedding:** Learn joint representation from all modalities
4. **Role classification:** Apply theory-guided classifiers to combined features
5. **Validation scoring:** Compare claimed roles (websites) vs. enacted roles (news, networks)
6. **Map generation:** Project embeddings to 2D/3D for visualization

## 3.7 Ensuring Rigor

DSR has sometimes been criticized as “just building things” without scientific rigor. We take rigor seriously, addressing it through multiple mechanisms.

### 3.7.1 DSR Guidelines Compliance

Following Hevner et al. [?], we explicitly address seven DSR guidelines:

| Guideline                    | How We Address It                                                            |
|------------------------------|------------------------------------------------------------------------------|
| 1. Design as artifact        | ROLEBOX is a concrete, functional toolkit—not just concepts                  |
| 2. Problem relevance         | Addresses documented cartographic gap (Chapter 2)                            |
| 3. Design evaluation         | Multiple evaluation strategies: technical, empirical, theoretical, practical |
| 4. Research contributions    | New methods (toolkit), tested propositions, open infrastructure              |
| 5. Research rigor            | Systematic process, documented decisions, transparent limitations            |
| 6. Design as search          | Iterative refinement through four major versions                             |
| 7. Communication of research | Dissertation, open code, archived data, publications                         |

**Table 3.5:** DSR rigor guidelines and how we address them

### 3.7.2 Validity Considerations

We consider standard validity threats:<sup>16</sup>

16: 🟢 **Seven guidelines**—Hevner et al. [?] established canonical DSR guidelines. We address each explicitly.

**Construct validity** *Do our measures capture what we claim?*

We address this through multiple operationalizations of role concepts (textual, relational, visual) and cross-modal validation. If claimed and enacted roles diverge, we investigate rather than ignore the discrepancy.

**Internal validity** *Are patterns real or artifacts of methods?*

We address this through robustness checks (alternative parameter settings), sensitivity analysis (how results change with different thresholds), and triangulation (do different modalities converge?).

**External validity** *Do findings generalize beyond EIT?*

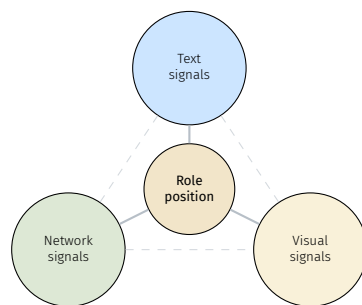
We are *cautious* about generalization. EIT is a specific European policy context with designed intermediaries. We provide open infrastructure enabling replication in other contexts but make no strong generalization claims.

**Reliability** *Would others get similar results?*

We address this through documented code, archived data, version-controlled workflows, and containerized execution environments. The entire analysis can be re-executed.

### 3.7.3 Triangulation Strategy

Triangulation is central to our validation approach:



**Figure 3.7:** Triangulation across modalities. Role positions are inferred from convergent signals; divergent signals prompt investigation rather than averaging.

## 3.8 Ethical Considerations

Our research involves organizational data from public sources.<sup>17</sup> We addressed ethical considerations proactively:

**Public data** All collected data was publicly available—organizational websites, published news, public investment records, public social media posts. No private or access-controlled data was used.

**Organizational focus** Our unit of analysis is organizations, not individuals. We do not profile, identify, or analyze individual persons beyond their public organizational roles.

**Anonymization** Where individuals are quoted (e.g., in news articles), we anonymize unless the quote is from a public statement by a named official spokesperson.

17:  **Ethics principles**—Public data only, organizational focus, no individual identification, platform compliance, open science.

**Platform compliance** We complied with terms of service for all data sources. Rate limiting, robots.txt compliance, and API quotas were respected.

**Data sharing** Aggregated and processed data are made available for research purposes. Raw data that could enable re-identification or violate platform terms is not shared.

The research was reviewed and approved by the university ethics board.

### 3.9 Limitations

Every methodology has limitations. We are explicit about ours:

#### Methodological Limitations

1. **Single ecosystem:** We study EIT deeply but cannot claim findings generalize to other innovation ecosystems. EIT is a designed, policy-driven, European context.
2. **Digital trace bias:** Organizations with weak digital presence are underrepresented. Small organizations, those in non-English contexts, and those avoiding public visibility may be systematically missed.
3. **Snapshot emphasis:** Most analyses are cross-sectional snapshots (2023). Temporal dynamics are captured through news and investment histories but website/visual data are point-in-time.
4. **English language bias:** NLP pipelines are optimized for English. Non-English content (particularly from Southern and Eastern European partners) is underrepresented in text analysis.
5. **Platform dependencies:** API access changes affected data collection. Twitter/X restrictions post-2023 limited social media coverage. Future replication may face different access constraints.
6. **Interpretive judgment:** Despite systematic methods, role identification involves researcher judgment—in codebook development, threshold setting, and pattern interpretation.

These limitations do not invalidate the research—they *bound* it.<sup>18</sup> We are building a toolkit and demonstrating its potential, not claiming definitive answers. The open infrastructure enables others to extend, adapt, and apply the approach in contexts where our limitations matter less.

18: ♥ **Honest limitations**—Acknowledging limitations strengthens rather than weakens research. Others can address what we could not.

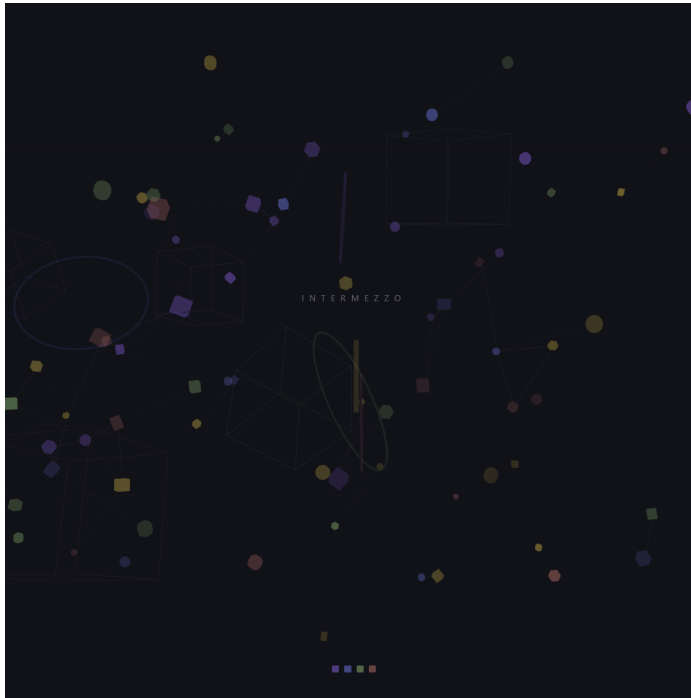
### 3.10 Chapter Summary

This chapter established **Design Science Research** as our methodology—appropriate because we are building something new (the ROLEBOXtoolkit) rather than testing hypotheses with existing methods.

**Key methodological elements:**

- ▶ **Six-activity DSR process:** Problem identification → objectives definition → design & development → demonstration → evaluation → communication, with iteration loops
- ▶ **Six design objectives:** Scale, multimodality, relationality, validation capability, openness, and theoretical grounding
- ▶ **Three-layer architecture:** Data collection, processing & analysis, integration & output—modular, extensible, transparent
- ▶ **Five data modalities:** Websites, news, investment, visual, social media—218,000+ records across 6,487 organizations
- ▶ **Multi-stage sampling:** Seed from official lists → expand through references → validate across sources → bound by inclusion criteria
- ▶ **Rigorous evaluation:** Technical, empirical, theoretical, comparative, and practical assessment strategies
- ▶ **Explicit limitations:** Single ecosystem, digital trace bias, snapshot emphasis, language bias, platform dependencies, interpretive judgment

The methodology enables us to build cartographic infrastructure while simultaneously using that infrastructure to map the EIT ecosystem.<sup>19</sup> The next chapter provides the theoretical foundation for understanding role constellations as complex adaptive systems.





---

INTERMEZZO

## ROLEXRAY - A Socio-Semantic Toolkit Mapping Interstitial Author Topic Constellations

---



*How can we map actors and topics that play interstitial roles—bridging worlds that would otherwise remain separate? Before we turn our multimodal cartography toward organizations, a detour through the landscape of ideas.*

### The Problem of the In-Between

Some of the most important work happens in the spaces between.

\*

Between disciplines. Between sectors. Between communities of practice. The actors who inhabit these in-between spaces—who speak multiple languages, who translate across boundaries, who see connections invisible to those embedded in single domains—are often the unsung catalysts of change.

But how do you find them? How do you see the interstitial?

\*: “Interstitial: relating to or situated in the interstices—the spaces that intervene between things.”

This dissertation develops tools for mapping role constellations in innovation ecosystems. Yet the same challenge appears in another domain entirely: the landscape of scientific literature. Before we mapped organizations, we mapped ideas—and the scholars who bridge them.

## From Organizations to Authors

The **Socio-Semantic X-Ray** emerged from a parallel question: how can we simultaneously analyze *what* scholars write about and *who* they collaborate with? Traditional literature reviews focus on content—themes, arguments, findings. They miss the social architecture of knowledge production.

†

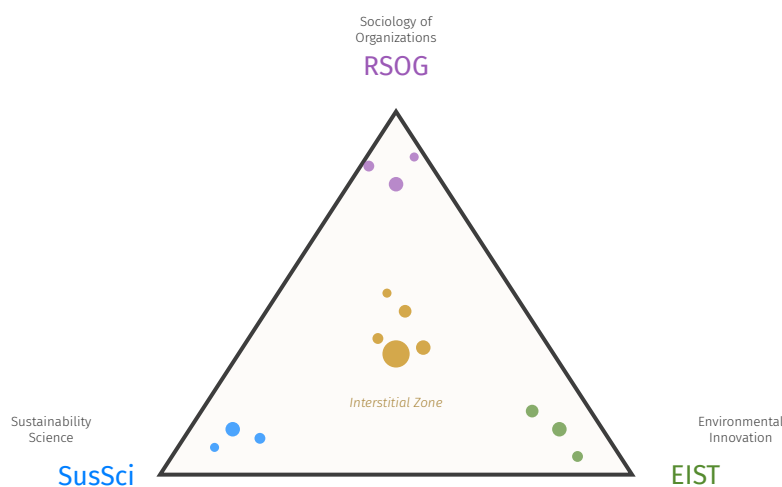
We developed this method while trying to understand our own intellectual terrain. Three journals defined the boundaries: *Research in the Sociology of Organizations* (RSOG), *Environmental Innovation and Societal Transitions* (EIST), and *Sustainability Science* (SusSci). Three communities. Three vocabularies. Three ways of seeing the world.

The question: who bridges them?

†: The Socio-Semantic X-Ray integrates what is said (semantic) with who says it together (social).

## The Chromatic Triangle

Imagine a triangle. Each vertex represents a pure discourse—the language and concerns of one journal community. Position within the triangle indicates mixture.



‡

An author publishing only in RSOG sits at the purple vertex—pure organizational sociology. One publishing only in SusSci sits at the blue vertex—pure sustainability science. But an author whose work draws equally from all three? They sit at the center. They are **interstitial**.

The method borrows from color theory: just as the CIE chromaticity diagram maps every visible color as a mixture of primaries, our triangle maps every author's discursive position as a mixture of disciplinary vocabularies.

‡: Like a chromatic color space where RGB vertices create all possible mixtures, the discourse triangle positions authors by their intellectual blend.

## What the X-Ray Reveals

Applied to 2,334 articles across two decades, the Socio-Semantic X-Ray revealed patterns invisible to traditional review:

§

**Topic landscapes.** Using BERTopic with sentence embeddings (all-MiniLM-L6-v2), UMAP dimensionality reduction, and HDBSCAN clustering, we mapped the semantic terrain of each journal. The pipeline identified 138 distinct topics with coherent keyword representations via KeyBERT and Maximal Marginal Relevance.

**Temporal trajectories.** Topics were classified by their developmental patterns:

- ▶ *Hot topics*: Rising prominence (positive linear coefficient  $> 0.5$ )
- ▶ *Cold topics*: Declining interest (negative coefficient  $< -0.5$ )
- ▶ *Evergreen topics*: Stable presence across years
- ▶ *Wallflower topics*: Persistent but marginal
- ▶ *Reviving topics*: U-shaped trajectories indicating renewed interest

¶

**Topic networks.** Co-occurrence analysis revealed which topics appear together in documents. The most central topic—methodological discourse around interviews and qualitative data—bridged all three journal communities. Topic #5 (transformation\_actor\_change\_transition) showed the highest betweenness per article, suggesting it serves as a conceptual bridge.

**Collaboration networks.** Co-authorship patterns revealed who works with whom. We identified within-community collaboration (authors publishing together in their home journal), cross-community collaboration (authors bridging journal boundaries), and the emergence of new discourse clusters.

§: **By the numbers:** 2,334 articles (EIST: 574, RSOG: 639, SusSci: 1,121), 138 topics identified, 3 discourse communities.

¶: Topic #0 (interview\_data\_conducted\_participant) showed highest degree centrality (59.4), indicating methodological discourse bridges all three communities.

**Interstitial authors.** Most powerfully: authors whose vocabulary, co-authorships, and publication patterns position them near the triangle’s center—equidistant from all three vertices ( $0.33 \pm 0.02$  for each community weight). These scholars don’t just cite across boundaries—they *inhabit* the in-between.

**Interstitial holes.** The triangle also reveals gaps—zones where bridges *could* exist but don’t. These are opportunities: unexplored connections between, say, organizational sociology and sustainability science that no one has yet made.

## Empirical Results

The topic-document heatmap visualizes the distribution of 138 topics across all documents in the three-journal corpus. Each row represents a topic; each column a document chunk. Color intensity indicates topic probability.

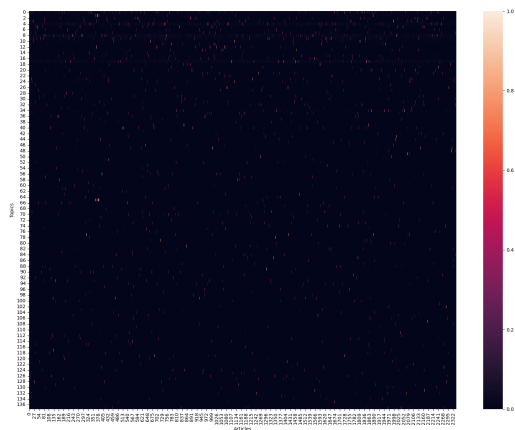


Figure 3.8: Topic-document distribution across EIST, RSOG, and SusSci.

The table below presents the top bridging topics by network centrality. Degree centrality measures how many other topics co-occur with a given topic; betweenness centrality measures how often a topic lies on the shortest path between other topics.

||

||: The methodological topic (interview\_data) has highest degree but lowest clustering—it connects to everything but those connections don’t cluster together.

| Topic                             | Count | Degree | Betw.  | Clust. |
|-----------------------------------|-------|--------|--------|--------|
| interview_data_conducted          | 83    | 59.4   | 2213.5 | 0.13   |
| transformation_actor_change       | 48    | 38.3   | 806.7  | 0.18   |
| emission_scenario_reduction       | 59    | 36.3   | 936.0  | 0.16   |
| complexity_system_approach        | 52    | 29.2   | 470.7  | 0.18   |
| area_land_water_scenario          | 56    | 23.2   | 288.3  | 0.25   |
| sustainability_research_student   | 36    | 22.2   | 262.3  | 0.25   |
| theory_organization_institutional | 34    | 22.2   | 287.1  | 0.31   |
| transition_energy_policy          | 34    | 15.1   | 83.2   | 0.35   |

Table 3.6: Top topics by network centrality in the three-journal corpus.

The optimal number of topics was determined through coherence analysis, balancing granularity against interpretability.

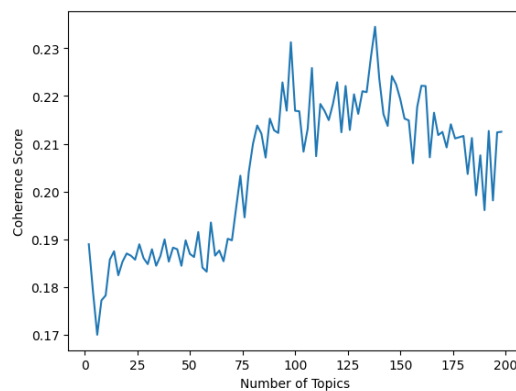


Figure 3.9: Topic coherence scores across different numbers of topics.

Topic frequency analysis reveals the most prevalent themes across the corpus. The methodological topic (interview/data/conducted) appears in 83 document chunks, followed by transformation discourse (48) and emission scenarios (59).

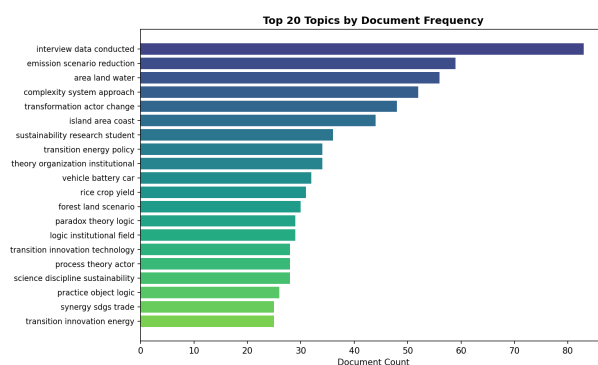


Figure 3.10: Top 20 topics by document frequency across the three-journal corpus.

**Temporal trajectories** classify topics by their developmental patterns. Hot topics show rising prominence (positive linear coefficient  $> 0.5$ ), while cold topics indicate declining interest. The transformation discourse emerges as a hot topic, while historical case studies like Stanford organizational studies and mortgage crisis literature show decline.

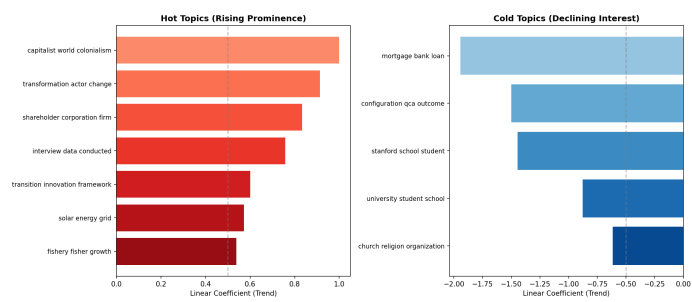


Figure 3.11: Temporal trajectories: hot topics (rising) and cold topics (declining) identified by linear trend coefficients.

★★

| Hot Topics (Rising)          | Docs | Coeff. |
|------------------------------|------|--------|
| capitalist_world_colonialism | 9    | 1.00   |
| transformation_actor_change  | 48   | 0.91   |
| shareholder_corporation_firm | 9    | 0.83   |
| interview_data_conducted     | 83   | 0.76   |
| solar_energy_grid_renewable  | 16   | 0.57   |
| fishery_fisher_growth_ocean  | 8    | 0.54   |

| Cold Topics (Declining)      | Docs | Coeff. |
|------------------------------|------|--------|
| mortgage_bank_loan_market    | 21   | −1.94  |
| configuration_qca_outcome    | 8    | −1.50  |
| stanford_school_student      | 15   | −1.44  |
| university_student_school    | 17   | −0.88  |
| church_religion_organization | 8    | −0.62  |

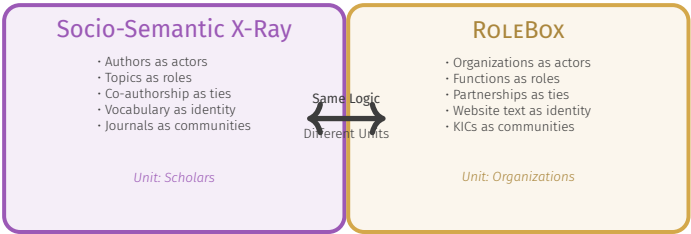
★★: Hot topics reflect the field’s current momentum; cold topics reveal historical preoccupations that have faded.

Table 3.7: Hot and cold topics by temporal trajectory coefficient.

Connection to This Dissertation

Why does a dissertation about mapping organizational role constellations pause to discuss mapping scholarly discourse?

Because the methods rhyme.



††

Both approaches integrate **semantic** dimensions (what actors say, how they present themselves) with **social** dimensions (who they connect with, how they’re positioned in networks). Both seek to identify **interstitial actors**—those who bridge otherwise separate communities.

††: The Socio-Semantic X-Ray was a proving ground for ideas that became ROLEBOX.

Both use **computational methods** to achieve scale while preserving interpretive depth.

The Socio-Semantic X-Ray was, in retrospect, a methodological rehearsal. Working with scholarly literature—a domain where ground truth is knowable, where we could validate patterns against our own domain expertise—we developed intuitions that later guided ROLEBoxdesign.

## The Interstitial Insight

The deepest lesson from the X-Ray work: **interstitial positions are rare, valuable, and hard to see.**

##

Only 8% of authors in our three-journal corpus occupied positions near the triangle's center. The vast majority clustered near vertices—embedded in single disciplinary communities, speaking single disciplinary languages. This is not surprising. Disciplines exist for reasons. Specialization enables depth.

But transitions—whether in sustainability or in scholarship—require translation. They require actors who can speak to multiple communities, who can carry ideas across boundaries, who can see connections invisible to the embedded.

##: 8% of authors occupied genuinely interstitial positions. The rest clustered near vertices.

*"The interstitial is where recombination happens. New ideas rarely emerge from the center of established fields. They emerge from the edges, the overlaps, the unexpected encounters between previously separate streams."*

This insight carried forward into ROLEBox. When we found that 34% of EIT ecosystem actors *claim* connector roles but only 12% show cross-modal evidence for them, we recognized a pattern. Interstitial aspiration exceeds interstitial reality. Many want to be bridges. Few actually are.

## Technical Infrastructure

The RoleXray toolkit provides a modular Python pipeline for socio-semantic analysis:

§§

§§:  [github.com/babajideowoyele/rolexray](https://github.com/babajideowoyele/rolexray)

- ▶ **preprocessing/**: Text cleaning, lemmatization, chunking for BERT's 512-token limit
- ▶ **modeling/**: BERTopic integration, embedding generation, UMAP/HDBSCAN configuration
- ▶ **analysis/**: Network construction, centrality metrics, community detection (Louvain)
- ▶ **visualization/**: Interactive Plotly networks, temporal trend charts, Pajek export

Configuration is YAML-driven, enabling reproducible experiments:

```
embeddings:
  model_name: "all-MiniLM-L6-v2"
hdbscan:
  min_cluster_size: 30
  min_samples: 30
network:
  centrality_metrics: [degree, betweenness,
                      closeness, eigenvector]
```

¶¶

¶¶: GPU acceleration via cuML enables processing of large corpora in minutes rather than hours.

## A Toolkit for Others

The Socio-Semantic X-Ray is now an open toolkit, available for others to apply to their own domains. The question it enables—*who bridges what?*—is general.

The toolkit can be applied to:

- ▶ Interdisciplinary emergence in new research fields
- ▶ Knowledge flows between academic and practitioner communities
- ▶ Evolution of discourse communities over time
- ▶ Identification of potential collaborators across silos

The toolkit embodies the same open science commitments as ROLEX: documented code, transparent methods, reproducible workflows. If our contribution is useful, others should be able to use it.

## From Literature to Ecosystems-Author Topic Constellations to Actor Role Constellations

This intermezzo has been a detour—from organizations to authors, from ecosystems to journals, from role constellations to discourse positions. But the detour illuminates the path.



\*\*\*

The Socio-Semantic X-Ray taught us:

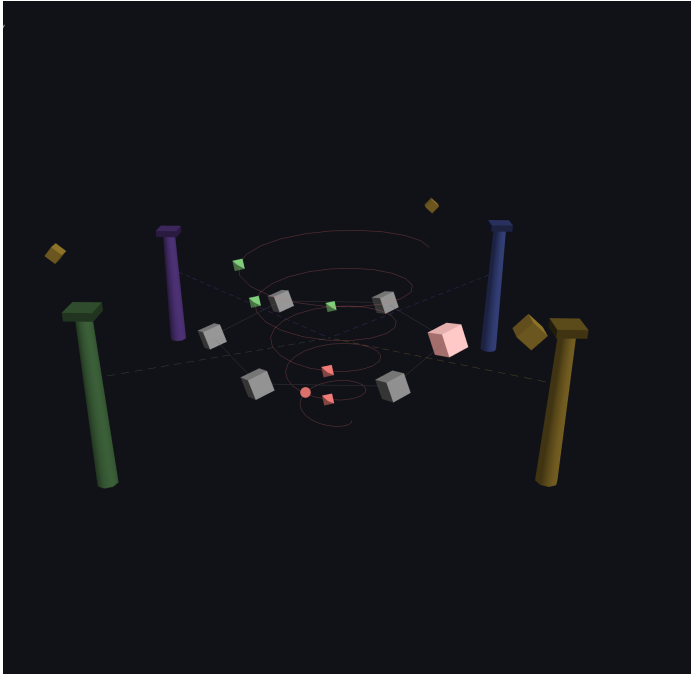
1. Semantic and social dimensions must be analyzed together
2. Interstitial positions exist and can be mapped
3. Most actors cluster; few bridge
4. The bridges are disproportionately important
5. Open toolkits enable cumulative knowledge

These lessons inform everything that follows. When we map EIT organizations across five modalities, we carry forward the X-Ray's core insight: to understand a system, you must see both what actors *say* and how they *relate*. Neither alone suffices.

*Now, back to organizations. Back to ecosystems. Back to the main cartography.*

\*\*\*: The X-Ray taught us to see the interstitial. ROLEXRAY applies that seeing to transition ecosystems.





# Role Constellations as Complex Adaptive Systems

# 4

*This chapter provides the theoretical foundation for multimodal cartography. We argue that role constellations are best understood as **Complex Adaptive Systems** (CAS)—they emerge from interactions, self-organize without central control, and evolve through feedback. CAS theory explains why constellations behave the way they do and generates propositions that cartographic methods can test. The chapter bridges the conceptual vocabulary of Chapter 2 with the methodological apparatus developed in subsequent chapters, translating systems-theoretic mechanisms into empirically tractable hypotheses about intermediary dynamics.*

## 4.1 From Categories to Systems

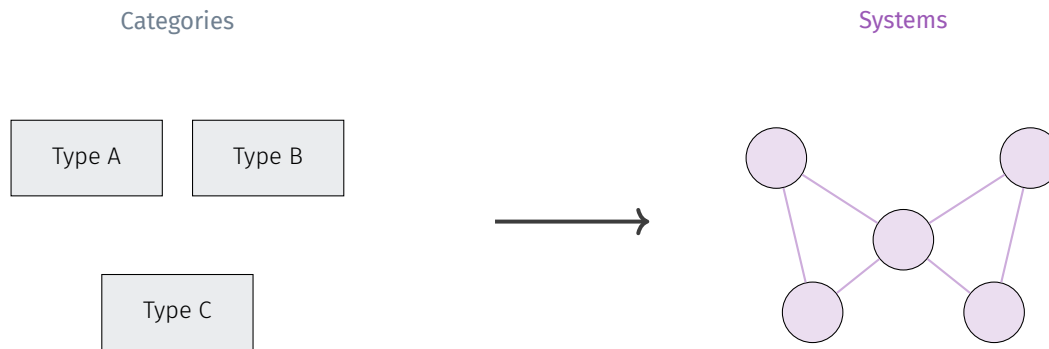
Chapter 2 documented a persistent tension in how transitions research handles intermediary roles. The field has produced sophisticated typologies—systemic, niche, regime-based, process, user intermediaries—yet these categories struggle to capture what practitioners and close observers consistently report:<sup>1</sup> roles are fluid, context-dependent, multiply-enacted, and relationally constituted. An organization classified as a “niche intermediary” in one analysis may simultaneously perform regime-based functions in another domain, shift toward systemic coordination over time, and be perceived entirely differently by different network partners.

The six problems identified in Chapter 2—definitional ambiguity, static categorization, actor-centric bias, normative loading, methodological constraints, and theoretical fragmentation—share a common root: they treat roles as *attributes* rather than *emergent properties*. A broker “has” brokerage as an attribute; an intermediary “is” a particular type. This framing imports assumptions from classical organizational theory that may not hold for the dynamic, relational phenomena we seek to understand.

What if roles are not attributes that actors possess but patterns that *emerge* from interactions? What if constellations are not collections of categorized actors but self-organizing systems that generate their own structure? What if the apparent messiness—roles shifting, boundaries blurring, functions combining—is not methodological noise but systemic signal?

|     |                                                                        |    |
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1: **The category problem**—Typologies answer “what kind?” but intermediaries don’t stay in boxes. We need theory that explains *why* roles shift, combine, and emerge.



**Figure 4.1:** From categorical thinking (fixed boxes) to systems thinking (emergent patterns from interactions).

These questions point toward **Complex Adaptive Systems** theory as a reconceptualizing frame. CAS theory offers mechanisms—emergence, self-organization, co-evolution, non-linearity, feedback—that are both theoretically substantive and methodologically tractable. It explains *why* we should expect the patterns that categorical approaches struggle to accommodate, while generating propositions that multimodal cartography can test.

#### Why CAS?

CAS theory is not merely a metaphor or analogy. It provides specific mechanisms explaining how macro-level patterns (role constellations) arise from micro-level interactions (organizational activities and relationships). These mechanisms have been operationalized in fields from ecology to economics. Importing them to intermediary research offers both explanatory power and methodological guidance.

## 4.2 Complex Adaptive Systems: Theoretical Foundations

### 4.2.1 What Makes a System “Complex” and “Adaptive”

Systems thinking has informed organization studies since Katz and Kahn’s [?] conceptualization of organizations as open systems. However, as Phillips and Ritala [?] observe, much organizational research invokes “system” language without engaging systems thinking principles.<sup>2</sup> The ecosystem concept, despite its systems etymology, has often been applied as metaphor rather than theoretical foundation—what Badinelli et al. [?] criticized as describing “interconnectedness of entities” without adhering to “systems thinking principles, which often disrupt traditional thinking.”

A **system** is an entity with interconnected components whose relationships permit identification of a boundary-maintaining process [?]. Systems become **complex** when the number and density of relationships makes prediction difficult—not merely complicated (many parts) but genuinely complex (emergent, non-linear). **Complex adaptive systems** involve components that adapt or learn as they interact, organizing without being controlled by any singular entity [?].

2: **Systems vs. “systems”**—Many studies use systems language descriptively without adopting systems-theoretic reasoning. CAS theory demands more rigorous engagement with mechanisms, not just vocabulary.

| Feature        | Simple Systems | Complicated Systems         | Complex Adaptive Systems     |
|----------------|----------------|-----------------------------|------------------------------|
| Components     | Few            | Many                        | Many, heterogeneous          |
| Relationships  | Linear         | Linear, numerous            | Non-linear, dense            |
| Predictability | High           | Moderate (expertise helps)  | Low (fundamentally emergent) |
| Control        | Centralized    | Distributed but coordinated | Decentralized, emergent      |
| Adaptation     | None           | Designed response           | Autonomous learning          |

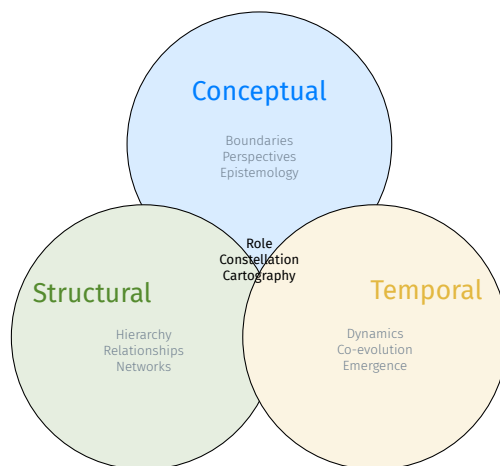
**Table 4.1:** System types distinguished by component relationships and adaptive capacity

Role constellations exhibit all CAS characteristics:

- ▶ **Multiple heterogeneous components:** Diverse organizations with different resources, orientations, and capabilities
- ▶ **Dense non-linear relationships:** Interactions that produce effects disproportionate to inputs
- ▶ **Emergent properties:** Constellation-level patterns not reducible to individual actor attributes
- ▶ **Adaptation:** Organizations modify behavior based on feedback from constellation dynamics
- ▶ **Self-organization:** Structure emerges without central design or control

#### 4.2.2 The Phillips-Ritala Framework: Three Dimensions

Phillips and Ritala's [?] methodological framework for ecosystem research provides essential scaffolding for our approach. They identify three dimensions that any rigorous CAS-based inquiry must address: **conceptual**, **structural**, and **temporal**. Each dimension poses distinct questions and requires specific methodological attention.



**Figure 4.2:** The three dimensions of CAS-based inquiry (adapted from Phillips & Ritala, 2019). Multimodal cartography must address all three: how we *think* about constellations (conceptual), what we *know* about their structure (structural), and how they *change* over time (temporal).

The **conceptual dimension** addresses<sup>3</sup> epistemological considerations: How do we define system boundaries? Whose perspectives matter? What theoretical positioning guides inquiry? For role constellations,

3: **Three questions**—**Conceptual**: How do we think about the system? **Structural**: What do we know about its structure? **Temporal**: How does it change over time?

this means grappling with where intermediation begins and ends, how different actors perceive the same constellation differently, and what theoretical lens renders roles visible.

The **structural dimension** concerns ontology: What are the system's components and their relationships? How is hierarchy organized? What patterns of connection characterize the constellation? For our purposes, this means mapping which organizations perform which roles, how role-performers relate to each other, and what network structures emerge.

The **temporal dimension** addresses dynamics and evolution: How does the system change over time? What drives co-evolution between components and environment? How do emergent properties arise? For role constellations, this means tracing how roles shift, how constellations restructure, and how external pressures reshape internal dynamics.

**The integration imperative.** Phillips and Ritala (2019) emphasize that addressing only one or two dimensions compromises inquiry. A study of constellation structure without temporal attention misses how that structure came to be and where it is heading. A study of dynamics without structural grounding cannot explain *what* is changing. Multimodal cartography must integrate all three dimensions—a requirement that shapes our methodological design.

## 4.3 Six CAS Mechanisms for Role Constellations

Building on CAS theory and the three-dimensional framework, we identify six mechanisms particularly relevant for understanding role constellation dynamics. Each mechanism generates specific expectations about how constellations form, structure themselves, and evolve—expectations that can be translated into testable propositions.

### 4.3.1 Mechanism 1: Emergence

**Emergence** refers to macro-level patterns arising from micro-level interactions that cannot be predicted from, or reduced to, the properties of individual components [?].<sup>4</sup> Emergent properties are not designed or intended; they arise spontaneously from the density and nature of interactions.

4: **Emergence**—Macro patterns arising from micro interactions. The “whole” exhibits properties not found in any “part.”

In role constellations, emergence manifests when:

- ▶ Constellation-level functions appear that no single intermediary performs alone
- ▶ System-wide orientations (e.g., toward particular transition pathways) crystallize from distributed activities
- ▶ “Gaps” in the constellation become visible only at the system level
- ▶ Collective identity forms that transcends individual organizational identities

### Role Emergence

**Role emergence** occurs when constellation-level role patterns arise from the interactions among intermediaries without being designed, assigned, or controlled by any single actor. Emergent roles are properties of the *constellation*, not of individual organizations.

Emergence has profound methodological implications. If roles emerge from interactions, they cannot be fully captured by studying organizations in isolation—no matter how deeply. Cartographic methods must attend to relational patterns, not just organizational attributes. Moreover, emergent roles may be invisible to the actors enacting them; what looks like uncoordinated individual activity from inside may constitute coherent collective function when viewed from outside.

### 4.3.2 Mechanism 2: Self-Organization

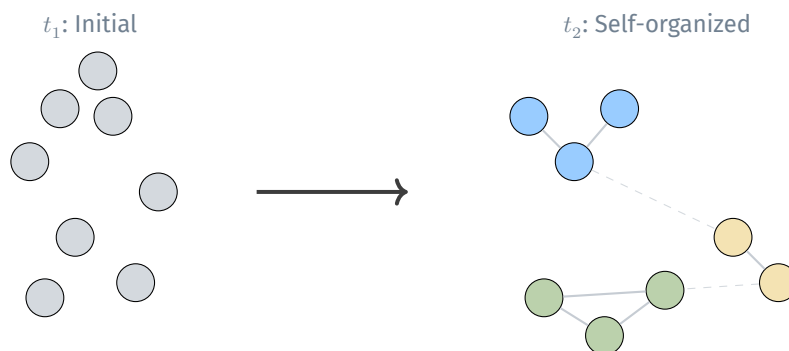
**Self-organization** is the process by which structure emerges without external control or central coordination [?].<sup>5</sup> In self-organizing systems, order arises from local interactions following simple rules, not from top-down design.

5: **Self-organization**—Order without a designer. Structure emerges from local interactions, not central planning.

Role constellations self-organize when:

- ▶ Intermediaries cluster around complementary functions without explicit coordination
- ▶ Division of labor emerges from bilateral negotiations rather than system-level planning
- ▶ “Niches” form as organizations differentiate to reduce direct competition
- ▶ Hierarchy develops through accumulated influence rather than formal assignment

Self-organization does not mean absence of structure; rather, it means structure arising endogenously. Even highly ordered constellations—with clear role differentiation, stable hierarchies, and predictable interaction patterns—may have self-organized rather than being designed. The EIT ecosystem offers an interesting test case: KICs were designed by policy, but the role constellations they have developed over fifteen years reflect substantial self-organization within designed parameters.



**Figure 4.3:** Self-organization: from initial dispersion to clustered structure through local interactions, without central coordination.

### 4.3.3 Mechanism 3: Co-evolution

**Co-evolution** describes the mutual adaptation of system components and their environment [?].<sup>6</sup> In co-evolutionary dynamics, changes in one component alter the selection pressures on others, triggering cascading adaptations. The system and its environment shape each other recursively.

6: **Co-evolution**—Mutual adaptation. Components and environment shape each other through recursive feedback.

Role constellations co-evolve when:

- ▶ Changes in one intermediary's role performance alter opportunities for others
- ▶ Landscape pressures (policy shifts, technological change) reshape constellation structure
- ▶ Constellation activities modify the environment (creating markets, shifting discourse)
- ▶ Success or failure of constellation members affects collective legitimacy

Co-evolution implies that constellations cannot be understood in isolation from their environments. The EIT ecosystem operates within European innovation policy, interacts with national innovation systems, responds to technological trajectories in energy, climate, health, and other domains. Constellation dynamics are shaped by these environmental factors—but the constellations also shape them. Climate-KIC's activities influence climate policy discourse; InnoEnergy's battery alliance affects European industrial strategy.

### 4.3.4 Mechanism 4: Non-linearity

**Non-linearity** means that effects are not proportional to causes [?].<sup>7</sup> Small changes can produce large effects (sensitivity to initial conditions); large interventions may produce minimal response (system resistance); similar actions in different contexts yield different outcomes (context-dependence).

7: **Non-linearity**—Effects disproportionate to causes. Small changes may trigger cascades; large interventions may produce nothing.

In role constellations, non-linearity appears as:

- ▶ **Tipping points:** Gradual changes suddenly produce rapid restructuring
- ▶ **Path dependence:** Early role choices constrain later possibilities
- ▶ **Threshold effects:** Certain role combinations only “work” above critical mass
- ▶ **Cascades:** One intermediary's role shift triggers chain reactions

Non-linearity complicates both prediction and intervention. Linear thinking—“more funding produces proportionally more innovation”—fails in CAS contexts. Constellation dynamics may remain stable despite substantial pressures, then restructure rapidly in response to seemingly minor triggers. Understanding these dynamics requires attention to system state, not just inputs and outputs.



### 4.3.5 Mechanism 5: Feedback Loops

**Feedback loops** are circular causal chains where outputs become inputs [?].<sup>8</sup> Positive (reinforcing) feedback amplifies change; negative (balancing) feedback dampens it. CAS behavior emerges from the interplay of multiple feedback loops operating at different timescales.

8: **Feedback loops**—Outputs become inputs. Reinforcing loops amplify; balancing loops stabilize.

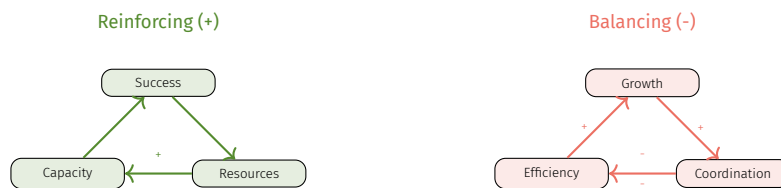
Role constellations exhibit both feedback types:

#### Reinforcing feedback examples:

- ▶ Success attracts resources → more capacity → more success
- ▶ Legitimacy attracts partners → larger network → more legitimacy
- ▶ Specialization develops expertise → attracts specialized requests → more specialization

#### Balancing feedback examples:

- ▶ Growth increases coordination costs → reduces efficiency → limits growth
- ▶ Dominance triggers resistance → coalition-building by others → reduces dominance
- ▶ Role crowding increases competition → differentiation pressure → reduces crowding



**Figure 4.4:** Reinforcing and balancing feedback loops in role constellations. Reinforcing loops drive growth or decline; balancing loops maintain stability.

### 4.3.6 Mechanism 6: Fitness Landscapes

**Fitness landscapes** provide a spatial metaphor for adaptation [?].<sup>9</sup> Organizations occupy positions in a multi-dimensional space defined by role characteristics; each position has associated “fitness” (viability, effectiveness). The landscape is not fixed but deforms as other actors move and as environmental conditions change.

9: **Fitness landscapes**—Adaptive terrain. Organizations seek “peaks” but the landscape shifts as others move and environment changes.

In role constellations:

- ▶ **Peaks** represent viable role configurations—positions where intermediaries can sustain themselves
- ▶ **Valleys** represent unviable configurations—role combinations that don’t attract resources or legitimacy
- ▶ **Rugged landscapes** have many local peaks—multiple viable configurations, with valleys between
- ▶ **Landscape deformation** occurs when environmental change or competitor movement alters what configurations are viable

The fitness landscape metaphor illuminates several constellation phenomena:

- ▶ **Local optima traps:** Organizations may occupy suboptimal peaks, unable to traverse valleys to reach higher peaks

- ▶ **Niche partitioning:** Multiple intermediaries can coexist by occupying different peaks
- ▶ **Red Queen dynamics** [?]: Organizations must continuously adapt just to maintain relative position as the landscape shifts
- ▶ **Punctuated equilibrium** [?]: Long periods of stability (on peaks) interrupted by rapid change (landscape deformation)

### The Cartographic Connection

The fitness landscape metaphor directly connects CAS theory to cartographic methods. If role constellations can be understood as actors distributed across an adaptive landscape, then *mapping* that landscape becomes both possible and useful. Multimodal cartography constructs empirical approximations of the theoretical landscape—revealing peaks and valleys, tracking movements, identifying viable configurations.

#### 4.3.7 From Observation to Simulation: The ROLESIMComplement

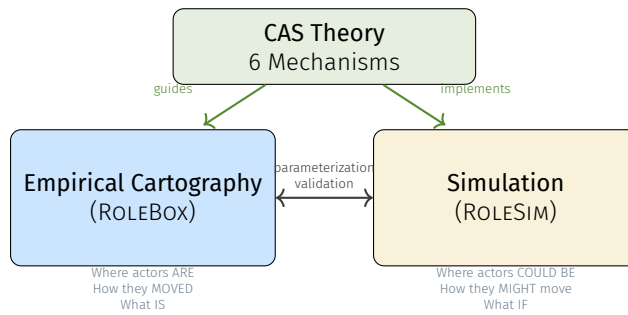
Empirical cartography reveals where actors *are* on the fitness landscape and how they *have* moved. But CAS theory raises questions that observation alone cannot answer:<sup>10</sup> What *would* happen if a key intermediary exited? How *might* the constellation restructure under different policy scenarios? Which role configurations *could* emerge but haven't?

10: **Counterfactual necessity**—CAS dynamics involve path dependence and contingency. Understanding *why* constellations evolved as they did requires exploring paths not taken.

These counterfactual questions are central to CAS understanding. Path dependence (Mechanism 4: Non-linearity) means observed trajectories represent one realization among many possibilities. Co-evolution (Mechanism 3) couples constellation dynamics to environmental conditions that themselves might have unfolded differently. Without counterfactual exploration, we risk mistaking contingent outcomes for necessary ones.

**Agent-based modeling** provides the methodological complement. By instantiating CAS mechanisms computationally—agents with adaptive behaviors, fitness functions, interaction rules—we can:

- ▶ **Explore counterfactuals:** What if InnoEnergy had not achieved early battery alliance success? How might Climate-KIC's systems innovation pivot have diffused (or not) to other KICs?
- ▶ **Test mechanism sufficiency:** Do the six mechanisms, parameterized from empirical data, *generate* patterns resembling observed constellations?
- ▶ **Identify sensitive parameters:** Which aspects of the fitness landscape most strongly shape outcomes? Where are tipping points?
- ▶ **Project forward:** Given current configurations and plausible environmental scenarios, what trajectories are likely?



**Figure 4.5:** Complementary roles of empirical cartography and simulation. ROLEBOX maps observed reality; ROLESIM explores counterfactual possibilities. Both are grounded in CAS theory; empirical patterns parameterize simulation; simulation results are validated against observation.

The ROLESIMagent-based model, introduced in Intermezzo B, operationalizes this complement. Agents represent intermediary organizations with:

- **Role profiles:** Multi-dimensional vectors in role-space (corresponding to empirically-derived dimensions)
- **Adaptive behavior:** Rules for adjusting profiles based on fitness feedback and neighbor observation
- **Interaction dynamics:** Network formation, resource exchange, information flow
- **Environmental coupling:** Fitness functions that shift based on simulated policy and technology trajectories

Crucially, ROLESIM is *parameterized from empirical cartography*.<sup>11</sup> Initial agent configurations, fitness function shapes, and interaction network structures derive from ROLEBOX analysis of the EIT ecosystem. This grounding ensures simulation explores variations on observed reality, not arbitrary parameter spaces. Conversely, simulation results can be *validated* against empirical patterns—if simulated dynamics produce constellation structures wildly divergent from observation, the mechanism specification requires revision.

11: **ROLESIM**—Agent-based model implementing CAS mechanisms. Parameterized from ROLEBOX empirics. Enables counterfactual exploration of constellation dynamics. See Intermezzo B.

**The observation-simulation dialectic.** Neither cartography nor simulation alone suffices for CAS understanding:

- **Cartography without simulation** describes but cannot explain—we see patterns without understanding their necessity or contingency
- **Simulation without cartography** generates possibilities without empirical grounding—we explore parameter spaces disconnected from actual constellations

Together, they enable what Phillips and Ritala (2019) call addressing the temporal dimension: not just describing change but understanding the mechanisms that produce it and the alternative paths that remain latent.

## 4.4 Applying CAS to Role Constellations: A Theoretical Synthesis

### 4.4.1 Reconceptualizing Roles as Emergent Properties

Integrating the six mechanisms, we arrive at a reconceptualization of role constellations that departs significantly from categorical approaches:

#### Role Constellation as CAS

A **role constellation** is a complex adaptive system comprising intermediary organizations whose roles emerge from their interactions, self-organize into differentiated configurations, and co-evolve with each other and their environment through non-linear dynamics mediated by feedback loops across a shifting fitness landscape.

This reconceptualization has several implications:<sup>12</sup>

1. **Roles are relational, not attributional.** An intermediary's role is not something it "has" but something it "does" in relation to others. The same organization may enact different roles depending on which relationships are activated.
2. **Constellations have system-level properties.** The constellation as a whole exhibits characteristics—functional coverage, redundancy, coherence, adaptability—that cannot be reduced to individual intermediary attributes.
3. **Role boundaries are fuzzy.** Rather than discrete categories with clear boundaries, roles shade into each other along continuous dimensions. Cartography captures gradients, not boxes.
4. **Stability is dynamic.** Constellation patterns that appear stable are actively maintained through balancing feedback, not simply frozen in place. Disruption of feedback can trigger rapid restructuring.
5. **Change is the default.** Because constellations are adaptive and their environments are dynamic, change is continuous. Static snapshots miss the motion that defines CAS behavior.

12: **Key shift**—Roles are not actor attributes but *system properties*; they emerge from, are sustained by, and change through relational dynamics.

### 4.4.2 The Three Dimensions Revisited

We can now specify how the Phillips-Ritala dimensions apply to role constellation cartography:

| Dimension  | Core Questions                                                                                                            | Cartographic Operationalization                                                                                                                        |
|------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conceptual | Where are constellation boundaries? Whose perspectives count? What makes an organization “part of” the constellation?     | Multi-modal boundary detection: membership claims (websites), network inclusion (relationships), media recognition (news), resource flows (investment) |
| Structural | What roles exist? How are they distributed? What relationships connect role-performers? What hierarchy has emerged?       | Role identification from text and behavior; network mapping of relationships; clustering and community detection; centrality and position analysis     |
| Temporal   | How have roles shifted? When do constellations restructure? What triggers change? How do components and system co-evolve? | Longitudinal tracking across data modalities; event detection; trajectory analysis; before/after comparison around interventions                       |

Table 4.2: CAS dimensions applied to role constellation cartography

### 4.4.3 Levels of Analysis

CAS theory demands attention to multiple levels of analysis and their interactions.<sup>13</sup> Following Phillips and Ritala’s emphasis on multilevel understanding, we distinguish three levels for role constellation analysis:

13: **Multimodal necessity**—Each dimension requires multiple data sources. No single modality captures boundaries, structure, and dynamics adequately.

**Micro level: Individual intermediaries** Organizations with their own strategies, resources, and activities. The micro level is where agency resides and where we observe specific role performances.

**Meso level: The constellation** The configuration of roles and relationships among intermediaries. The meso level is where emergent properties become visible—division of labor, collective functions, system-level gaps.

**Macro level: The transition context** The broader environment including regimes, landscapes, policy frameworks, and other ecosystems. The macro level provides selection pressures and co-evolutionary dynamics.

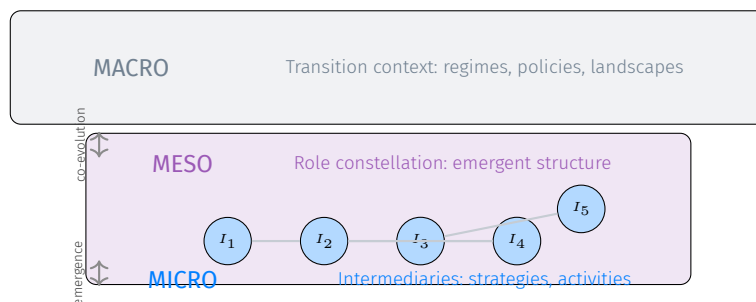


Figure 4.6: Three levels of analysis for role constellation cartography. Micro-level actions aggregate into meso-level patterns through emergence; meso and macro levels co-evolve through mutual adaptation.

Crucially, these levels interact:

- **Upward causation (emergence):** Micro-level interactions produce meso-level patterns; meso-level dynamics influence macro-level transitions
- **Downward causation (constraint):** Macro-level conditions shape what constellations are viable; meso-level structure enables and constrains micro-level action

► **Cross-level feedback:** All levels influence each other recursively

## 4.5 Fifteen Propositions for Empirical Testing

Theoretical frameworks gain value through empirical engagement.<sup>14</sup> The CAS reconceptualization generates specific propositions about role constellation dynamics—predictions that multimodal cartography can test. We organize fifteen propositions around the six mechanisms, grouped into conceptual, structural, and temporal domains.

**14: Proposition logic**—Each proposition derives from CAS mechanisms, specifies observable patterns, and can be assessed with cartographic methods. Chapter 10 evaluates empirical support.

### 4.5.1 Propositions on Emergence (P1–P3)

**P1: Role multiplicity is the norm, not the exception.**

Because roles emerge from diverse relationships rather than being inherent attributes, most intermediaries will perform multiple roles simultaneously. Single-role specialization requires active maintenance against the pull of relationship diversity.

*Observable pattern:* Multimodal analysis will reveal that intermediaries exhibit different role signatures across data sources, and that most intermediaries score significantly on multiple role dimensions rather than loading heavily on just one.

**P2: Constellation-level functions exceed the sum of individual role performances.**

Emergent properties mean the constellation as a system performs functions that no individual intermediary performs. Collective brokerage, system-level advocacy, and distributed knowledge integration appear at the constellation level.

*Observable pattern:* Network analysis will reveal functional bridging at the constellation level even when no single intermediary is positioned as a bridge; discourse analysis will identify collective framings distinct from any individual organization's messaging.

**P3: Role emergence accelerates with interaction density.**

Since emergence arises from interactions, constellations with denser relationship networks will exhibit more differentiated and specialized role structures than sparse constellations.

*Observable pattern:* Comparing across KICs or time periods, network density will positively correlate with role diversity and specialization measures.

### 4.5.2 Propositions on Self-Organization (P4–P6)

**P4: Role differentiation emerges without central coordination.**

Even in designed ecosystems like EIT, the specific role configuration will reflect self-organization more than policy design. Similar initial conditions (KIC mandates) will produce different role structures based on local interaction dynamics.

*Observable pattern:* Despite similar formal mandates, the eight KICs will exhibit substantially different role constellation structures, with variation unexplained by design differences.

**P5: Hierarchy emerges from accumulated advantage, not formal assignment.**

Influential positions in role constellations emerge through preferential attachment and resource accumulation rather than formal designation. Network centrality will correlate with resource access and longevity more than with formal status.

*Observable pattern:* Network centrality measures will correlate more strongly with organization age and resource levels than with formal partnership tier in KIC governance.

**P6: Niche formation reduces direct role competition.**

Self-organization drives differentiation to reduce competition. Over time, constellations will exhibit increasing role specialization as intermediaries carve out distinctive niches.

*Observable pattern:* Role similarity among spatially or thematically proximate intermediaries will decrease over time as niches form.

### 4.5.3 Propositions on Co-evolution (P7–P9)

**P7: Constellation structure and policy environment co-evolve.**

Role constellations both respond to and shape policy contexts. Policy shifts will produce constellation restructuring, but constellation activities will also correlate with subsequent policy changes.

*Observable pattern:* Temporal analysis will reveal bidirectional relationships between policy events and constellation structural changes, not just one-way policy → structure causation.

*Simulation test:* ROLESIM scenarios with exogenous policy shocks should produce structural responses resembling observed post-policy patterns; simulated constellation discourse should correlate with subsequent policy-relevant outputs.

**P8: Success of one intermediary alters fitness landscape for others.**

Co-evolution means interdependence. When one intermediary achieves significant success (funding, recognition, growth), role opportunities for others in the constellation will shift—some enhanced, some diminished.

*Observable pattern:* Major success events for focal intermediaries will correlate with subsequent role shifts among constellation partners.

*Simulation test:* ROLESIM “knockout” experiments removing successful agents should produce different equilibrium configurations, revealing the counterfactual landscape.

**P9: Constellations co-evolve with technological trajectories.**

In innovation ecosystems, technology development and intermediary roles are coupled. Technology maturation will correlate with shifts from niche-protective toward diffusion-oriented roles.

*Observable pattern:* As technologies within KIC domains mature (measured by patent activity, market deployment), intermediary role profiles will shift toward scaling and mainstreaming functions.

*Simulation test:* ROLESIM technology trajectory scenarios (fast vs. slow maturation) should produce different role evolution patterns, testable against cross-KIC variation.

**4.5.4 Propositions on Non-linearity (P10–P12)****P10: Constellation change follows punctuated equilibrium patterns.**

Non-linear dynamics produce stability punctuated by rapid change. Role constellations will exhibit long periods of incremental adjustment interrupted by episodes of rapid restructuring.

*Observable pattern:* Time series of constellation structure measures will show episodic rather than continuous change, with identifiable “punctuation” events.

*Simulation test:* ROLESIM with CAS mechanisms should endogenously generate punctuated dynamics without requiring exogenous shocks—a key test of mechanism sufficiency.

**P11: Small interventions can trigger disproportionate restructuring.**

Sensitivity to initial conditions means minor changes can cascade. Some policy interventions or organizational events will produce constellation effects far exceeding their apparent magnitude.



*Observable pattern:* Case analysis will identify instances where small triggers (personnel changes, modest funding shifts) preceded major constellation restructuring.

*Simulation test:* ROLESIM sensitivity analysis varying initial conditions and small perturbations should reveal bifurcation points where outcomes diverge dramatically.

**P12: Path dependence constrains role repertoires.**

Early role choices create commitments (expertise, relationships, reputation) that constrain later possibilities. Intermediaries' current role profiles will correlate more strongly with their founding roles than with current environmental demands.

*Observable pattern:* Historical role profiles (from early website archives, founding documents) will predict current role profiles better than current environmental fit would suggest.

*Simulation test:* ROLESIM runs with identical environmental conditions but different initial configurations should produce persistently different outcomes, demonstrating path dependence computationally.

#### 4.5.5 Propositions on Feedback and Fitness (P13–P15)

**P13: Reinforcing feedback produces role concentration.**

Success-breeds-success dynamics concentrate resources and influence. Over time, constellation distributions will become more skewed, with dominant intermediaries capturing disproportionate shares.

*Observable pattern:* Network centrality, media attention, and funding distributions will become more unequal over time within constellations.

**P14: Balancing feedback maintains role diversity.**

Competitive pressure and coordination costs limit concentration. Despite reinforcing feedback, balancing mechanisms will prevent complete dominance and maintain minimum role diversity.

*Observable pattern:* Even in mature constellations, multiple viable role configurations will persist; complete monopolization of any role function will be rare.

**P15: Fitness requires multi-dimensional adaptation.**

Viable positions on the fitness landscape require meeting multiple criteria simultaneously. Intermediaries optimizing on single dimensions (e.g., only network centrality, only resource accumulation) will prove less sustainable than those balancing multiple fitness components.

*Observable pattern:* Intermediary longevity and stability will correlate with balanced performance across multiple role dimensions rather than extreme performance on any single dimension.

**Table 4.4:** Proposition testing roadmap: empirical evaluation and simulation exploration

| ID  | Mechanism  | Proposition (abbreviated)                       | Empirical Test (ROLEBox)        | Mechanism Simulation (ROLESIM) | Domain     |
|-----|------------|-------------------------------------------------|---------------------------------|--------------------------------|------------|
| P1  | Emergence  | Role multiplicity is the norm                   | Website + news role profiles    | —                              | Conceptual |
| P2  | Emergence  | Constellation functions exceed individual roles | Network functional mapping      | Emergent function detection    | Structural |
| P3  | Emergence  | Emergence accelerates with density              | Density × diversity correlation | Density parameter sweeps       | Temporal   |
| P4  | Self-org   | Differentiation without coordination            | Cross-KIC structural comparison | Coordination-free runs         | Structural |
| P5  | Self-org   | Hierarchy from accumulated advantage            | Centrality × resources × age    | Preferential attachment tests  | Structural |
| P6  | Self-org   | Niche formation reduces competition             | Role similarity trajectories    | Competition → differentiation  | Structural |
| P7  | Co-evol    | Structure and policy co-evolve                  | Event-structure correlation     | Policy shock scenarios         | Structural |
| P8  | Co-evol    | Success alters landscape for others             | Success → partner shifts        | Knockout experiments           | Structural |
| P9  | Co-evol    | Tech trajectory coupling                        | Maturity × role profiles        | Tech speed scenarios           | Structural |
| P10 | Non-linear | Punctuated equilibrium                          | Change magnitude time series    | Endogenous punctuation         | Structural |
| P11 | Non-linear | Small triggers, large effects                   | Case analysis                   | Sensitivity analysis           | Structural |
| P12 | Non-linear | Path dependence                                 | Founding → current roles        | Initial condition variation    | Structural |
| P13 | Feedback   | Reinforcing → concentration                     | Inequality trends               | Rich-get-richer dynamics       | Structural |
| P14 | Feedback   | Balancing → diversity                           | Diversity floor persistence     | Stabilization mechanisms       | Structural |
| P15 | Fitness    | Multi-dimensional adaptation                    | Balanced score × longevity      | Fitness function shape         | Structural |

**Table 4.3:** Proposition summary by mechanism

## 4.6 Implications for Multimodal Cartography

The CAS reconceptualization has direct implications for how we design cartographic methods. Theory shapes methodology—not merely suggesting what to look for but specifying how to look. This section elaborates implications for both empirical cartography (ROLEBox) and computational simulation (ROLESIM).

### 4.6.1 Why Multimodality is Necessary, Not Optional

CAS theory explains why single-modality approaches systematically underperform for role constellation analysis:<sup>15</sup>

1. **Emergence requires multiple observation points.** Emergent properties are not visible from inside the system. Self-reports (interviews, websites) capture how actors perceive their roles; external observation (networks, news) captures how roles manifest behaviorally. Neither alone captures emergence.
2. **Perspectives differ systematically.** Different actors perceive the same constellation differently—and all perspectives contain signal. What an intermediary claims as its role (website), how partners see it (network position), and how external observers portray it (news) are three facets of the same phenomenon.

15: **Multimodal necessity**—CAS properties (emergence, multiple levels, diverse perspectives) cannot be captured by any single data source. Integration is required by theory, not just desirable for robustness.

3. **Levels require different methods.** Micro-level role performance requires fine-grained activity data; meso-level structure requires relational mapping; macro-level context requires discourse and policy analysis. No single data source spans all levels.
4. **Dynamics require temporal depth.** CAS are inherently dynamic; static snapshots miss the motion. Multiple data sources with different temporal granularity (event-level news, periodic website captures, longitudinal networks) enable dynamics tracking.

#### 4.6.2 Theory-Guided Feature Selection

CAS theory guides what features to extract from multimodal data. Rather than atheoretical pattern-finding, we seek features corresponding to mechanism-relevant dimensions:

| Mechanism         | What to Look For                                   | Data Sources / Features                                            |
|-------------------|----------------------------------------------------|--------------------------------------------------------------------|
| Emergence         | System-level patterns not reducible to individuals | Network topology; discourse clustering; functional coverage gaps   |
| Self-organization | Differentiation without central control            | Role diversity trends; niche overlap measures; hierarchy emergence |
| Co-evolution      | Mutual adaptation of system and environment        | Temporal correlation of structure and context changes              |
| Non-linearity     | Disproportionate effects; punctuations             | Change magnitude distributions; event-response analysis            |
| Feedback          | Reinforcing and balancing loops                    | Centrality-resource correlations; diversity maintenance over time  |
| Fitness           | Multi-dimensional viability                        | Balanced scorecard across role dimensions; survival analysis       |

**Table 4.5:** CAS mechanisms and corresponding cartographic features

#### 4.6.3 Validation Through Triangulation

CAS theory suggests validation strategies. If roles are emergent and relational, then:

- ▶ **Cross-modal convergence** signals robust role identification. Roles that appear across multiple modalities (claimed on websites, visible in networks, recognized in news) are more likely “real” than those appearing in only one.
- ▶ **Cross-modal divergence** is informative, not problematic. Discrepancies between claimed and enacted roles reveal aspiration-behavior gaps, legitimacy strategies, or impression management.
- ▶ **Temporal consistency** adds confidence. Roles that persist across time periods are more likely structural features than measurement artifacts.

#### 4.6.4 Simulation as Theoretical Laboratory

Beyond empirical validation, CAS theory requires **generative testing**:<sup>16</sup> can the proposed mechanisms, when instantiated computationally, *produce* patterns resembling observed constellations? This is where ROLESIM becomes essential.

16: **Generative test**—If mechanisms are correctly specified, simulation should generate patterns resembling observation. Failure indicates missing or mis-specified mechanisms.

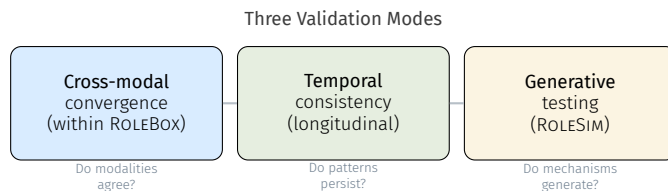
Agent-based modeling enables several theory-testing strategies unavailable to purely empirical approaches:

**Mechanism isolation** By selectively enabling or disabling specific mechanisms (e.g., running ROLESIM without co-evolutionary coupling), we can assess each mechanism’s contribution to observed patterns.

**Parameter sensitivity** Systematic variation of fitness landscape shapes, interaction frequencies, and adaptation rates reveals which parameters most strongly influence constellation outcomes—guiding attention in empirical analysis.

**Counterfactual exploration** “What if” scenarios impossible to observe empirically become tractable: What if the EIT had launched only three KICs instead of eight? What if financial sustainability requirements had been imposed earlier?

**Temporal acceleration** Simulation time can be compressed, allowing exploration of long-run dynamics beyond the fifteen-year empirical window.



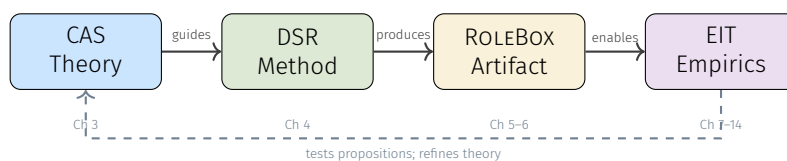
**Figure 4.7:** Three validation modes for CAS-based role constellation analysis. Cross-modal triangulation validates within ROLEBOX; temporal consistency validates across time; generative testing validates mechanism specification through ROLESIM.

## 4.7 From Theory to Method: The Design Science Bridge

The CAS reconceptualization creates both opportunity and obligation.<sup>17</sup> The opportunity: we now have theoretical grounding for why multimodal, relational, temporal methods are necessary. The obligation: we must build methodological artifacts that operationalize these theoretical commitments.

This is where Design Science Research (DSR) enters. As Chapter 3 will elaborate, DSR treats the creation of artifacts—tools, methods, frameworks—as a legitimate research contribution. The artifacts we build embody theoretical choices: what counts as a role, how relationships are detected, which dynamics matter. Theory guides design; design enables theory-testing.

17: **Theory** → **Method**—CAS theory specifies *what* to look for; Design Science Research provides *how* to build tools that look effectively.



**Figure 4.8:** The theory-method-empirics cycle. CAS theory guides DSR method, which produces the ROLEBOX artifact, enabling empirical analysis of EIT, which tests propositions and refines theory.

The four Meta-Design Requirements introduced in Chapter 1 connect directly to CAS theory:

**MDR-1: Theory-Data Reconciliation** CAS concepts (emergence, self-organization) must be translated into measurable features. This chapter has begun that work; the empirical chapters complete it.

**MDR-2: Hypothesis-Driven Data Selection** The fifteen propositions specify what data we need. We don't collect everything available; we collect what tests CAS predictions.

**MDR-3: Computational-Human Complementarity** CAS complexity exceeds human cognitive capacity for pattern detection at scale; but CAS meaning—what emergence *signifies*—requires human interpretation. The artifact must enable both.

**MDR-4: Knowledge Accumulation Infrastructure** CAS theory develops through empirical engagement. Open artifacts enable others to test propositions in new contexts, accumulating evidence about when and where CAS mechanisms operate.

**The EIT as CAS laboratory.** The EIT ecosystem offers an ideal setting to test CAS propositions about role constellations:

- ▶ **Designed yet emergent:** KICs were created by policy, but their internal dynamics have self-organized
- ▶ **Multiple interacting systems:** Eight KICs provide natural variation for cross-case comparison
- ▶ **Temporal depth:** Fifteen years of operation enable dynamics tracking
- ▶ **Rich digital traces:** Substantial data across modalities for multimodal cartography
- ▶ **Known environment:** Policy context is documented, enabling co-evolution analysis

The Intermezzo has prepared the terrain; this chapter has provided the theoretical map; the following chapters build the instruments for exploration.

## 4.8 Limitations and Boundary Conditions

CAS theory is powerful but not unlimited.<sup>18</sup> We acknowledge several boundary conditions:

**Not all systems are equally complex.** Some intermediary configurations may be simple enough that categorical approaches suffice. CAS theory adds value for genuinely complex constellations; it may be overkill for small, stable, centrally-coordinated settings.

**Emergence is hard to prove.** Demonstrating that a pattern is truly emergent (not designed, not reducible) is difficult. We can show consistency with emergence predictions but cannot definitively rule out alternative explanations.

**Agency remains underspecified.** CAS theory emphasizes system dynamics but can underplay individual agency. Strategic actors can and do attempt to shape constellations. CAS provides context for agency but doesn't fully theorize it.

18: **Honest boundaries**—CAS theory illuminates certain phenomena while leaving others in shadow. Acknowledging limits strengthens rather than weakens theoretical application.

**Prediction remains limited.** CAS theory explains *why* prediction is difficult but doesn't overcome that difficulty. We can identify tendencies and mechanisms without being able to forecast specific outcomes.

**Operationalization requires choices.** Translating CAS concepts into measures involves judgment calls. Different operationalizations might yield different results. We document choices transparently but cannot claim they are uniquely correct.

#### The Theory-Method Gap

CAS theory is more fully developed than methods for testing it. Mechanisms like emergence and self-organization have clear theoretical meanings but contested empirical operationalizations. This dissertation contributes to closing the gap, but does not claim to resolve it completely. The cartographic approach advances operational capacity while remaining honest about interpretation challenges.

## 4.9 Chapter Summary

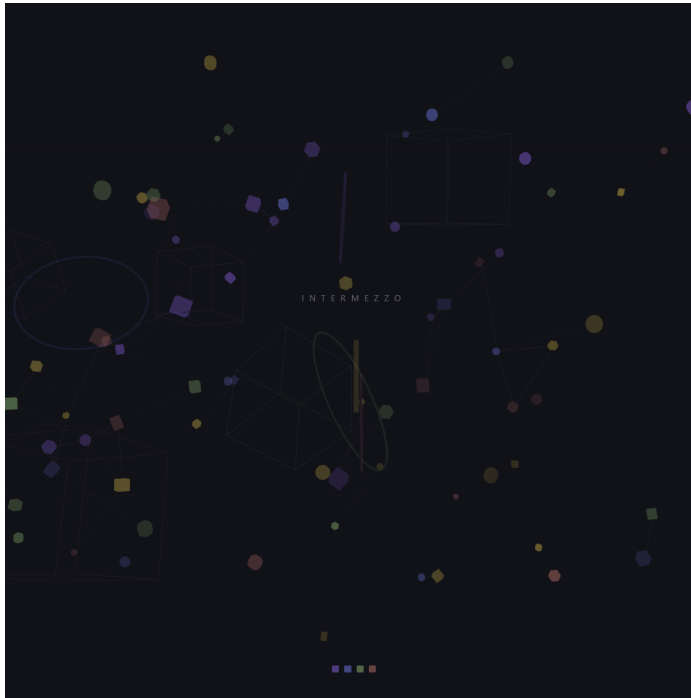
This chapter established Complex Adaptive Systems theory as the theoretical foundation for multimodal cartography of role constellations:

- ▶ **The category-to-system shift:** Roles are reconceptualized as emergent properties of interactions rather than attributes of actors. This explains why typological approaches struggle with fluidity and multiplicity.
- ▶ **Six CAS mechanisms** provide explanatory power: emergence, self-organization, co-evolution, non-linearity, feedback loops, and fitness landscapes. Each generates specific predictions about constellation behavior.
- ▶ **The Phillips-Ritala framework** identifies three dimensions—conceptual, structural, temporal—that rigorous CAS-based inquiry must address. Multimodal cartography operationalizes all three.
- ▶ **Three levels of analysis**—micro (intermediaries), meso (constellations), macro (transition context)—interact through upward emergence and downward constraint.
- ▶ **Fifteen propositions** translate mechanisms into testable predictions, organized across emergence (P1–P3), self-organization (P4–P6), co-evolution (P7–P9), non-linearity (P10–P12), and feedback/fitness (P13–P15).
- ▶ **Multimodality is theoretically necessary**, not merely methodologically desirable. CAS properties require multiple observation points, perspectives, and temporal depths.
- ▶ **Boundary conditions** include: not all systems are equally complex; emergence is hard to prove; agency remains underspecified; prediction remains limited; operationalization requires judgment.

With theoretical foundations established, we turn to methodology.<sup>19</sup> Chapter 3 presents the Design Science Research framework guiding artifact development. Chapters ?? and ?? document the ROLEBox and ROLESimartifacts that operationalize CAS-informed multimodal cartography. The empirical chapters that follow test whether the fifteen propositions hold in the EIT ecosystem—and what we learn when they do or don't.

The map we have drawn in this chapter is itself a kind of cartography: a theoretical terrain showing where concepts are positioned, how mechanisms connect, what propositions follow. The challenge now is to build tools capable of mapping empirical terrain with corresponding rigor.

19: → Next: Chapter 3: Methodology—Design Science Research as the framework for building cartographic infrastructure.





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INTERMEZZO

## On Multimodal Cartography

---



*Before we descend into methods and data, a pause. What does it mean to make maps of something as fluid as roles? Why insist on multiple modes of seeing? And what ancient wisdom does the cartographer's craft offer the builder of research tools?*

### Cartography: The Writing of Territory

The word arrives from the Greek: *khártēs* (χάρτης), meaning papyrus or sheet, and *graphein* (γράφειν), to write or draw. **Cartography** is literally the *writing of sheets*—the inscription of knowledge onto surfaces that can be carried, shared, consulted.

\*

But the etymology conceals a deeper truth. The Greeks used *graphein* for both writing and drawing—no distinction between text and image, between description and depiction. A map was always both: a picture *and* a statement, a representation *and* an argument.

\*: *khártēs* (χάρτης)—papyrus, sheet. *graphein* (γράφειν)—to write, to scratch, to draw.

This fusion matters for what we attempt. Role constellation cartography is not merely visualization (though it produces images) nor merely analysis (though it generates claims). It is *graphein* in the ancient sense: the inscription of understanding onto surfaces that make the invisible navigable.

## Mappa Mundi: The Napkin of the World

The Latin *mappa* meant cloth or napkin—something humble, portable, useful. A *mappa mundi* was literally a “napkin of the world,” a piece of fabric onto which the known world was sketched for travelers and scholars.

†

Consider what this implies. Maps were never meant to be complete representations. They were *tools*—aids to navigation, memory, and decision. The medieval *mappa mundi* placed Jerusalem at the center not from ignorance but from *purpose*: it served pilgrims, not merchants. Different purposes demanded different centers.

Our maps of role constellations inherit this pragmatic lineage. We do not seek the “true” map—there is none. We seek *useful* maps: representations that serve navigation through complex innovation ecosystems. The center we choose—the features we emphasize—reflects the journeys we enable.

†: *mappa* (Latin)—cloth, napkin, signal-flag. Hence: something portable, practical, meant to be *used*.

## The Cartographer’s Dilemma

Every map is a lie that tells the truth. It flattens, simplifies, and selects—yet in doing so, it reveals what was invisible. The London Tube map bears no resemblance to geographic reality, yet it *works*. It serves navigation, not surveying.

‡

Harry Beck’s 1931 Tube map committed cartographic heresy: he abandoned geography entirely. Distances became meaningless; only *connections* mattered. The map was false to the land but true to the *task*—getting from here to there via the network.

Role constellation maps face the same dilemma. We cannot capture everything. The moment we choose what to represent—connections, positions, claims, flows—we exclude something else. The question is not whether to simplify, but how to simplify *usefully*.

‡: “The map is not the territory.”—Alfred Korzybski. “But a good map makes the territory traversable.”—Every cartographer.

This is where Design Science principles enter. Herbert Simon’s insight that design is about “transforming existing situations into preferred ones” applies directly: the cartographer transforms invisible complexity into navigable representation. The quality of the transformation is judged not by completeness but by *utility for purpose*.

## Topos and Logos: The Logic of Place

The Greeks gave us *topos* (τόπος)—place, location, position. From it derives *topology*: the study of spatial properties preserved under continuous deformation. A coffee cup and a donut are topologically identical; what matters is connection, not shape.

§

Role constellations have topology. Organizations occupy *positions* relative to each other—not in physical space but in functional space, relational space, identity space. Some positions are “close” (similar functions, overlapping networks); others are “distant.” Some organizations bridge otherwise disconnected regions; others cluster at the core.

Network science formalizes this intuition. But the insight predates the formalism. When we map constellations, we are practicing *topology* in the original sense: discerning the logic of positions, the grammar of proximity and distance that shapes what actors can see, reach, and become.

§: *topos* (τόπος)—place, position, topic.  
*logos* (λόγος)—word, reason, account.  
Topology: the *logos* of *topos*—the logic of place.

## Why Multiple Maps?

A single map serves a single purpose. The navigator needs routes; the geologist needs strata; the urban planner needs density. No map serves all purposes.

The *portolan charts* of medieval Mediterranean sailors showed coastlines in exquisite detail but ignored the interior entirely. The *T-O maps* of medieval scholars showed theological order but couldn’t guide a ship. Each was “true” to its purpose, “false” to others.

¶

Roles, too, have multiple dimensions that no single map captures:

- ▶ What actors *claim* to be (the **declared** map)
- ▶ What others *perceive* them to be (the **attributed** map)

¶: Five maps. Five truths. One constellation.

- ▶ What they *materially* do (the **enacted** map)
- ▶ Who they *connect* with (the **relational** map)
- ▶ How they *present* themselves (the **aesthetic** map)

A startup's website proclaims "disruptive innovation." News coverage calls them "promising but unproven." Investment data shows modest seed funding. Twitter reveals connections mainly within their own sector. Visual branding mimics established players.

Which is the "real" role? All of them. None of them. The role is the *pattern across* these perspectives—the convergences and the telling divergences.

This is why **multimodal** cartography is not methodological excess but epistemological necessity. *Modus* in Latin meant manner, measure, way. Multiple *modi*—multiple ways of measuring, multiple manners of seeing—are required because the phenomenon itself is multi-aspected. We do not impose multiplicity; we respond to it.

## Convergence and Divergence

When our five maps align, we gain confidence. An organization claiming intermediary status, recognized as such by media, showing bridging patterns in networks, and channeling resources across boundaries—this is robust evidence. The convergence across *modi* is itself informative: it suggests a role that is not merely performed in one register but *inhabited* across contexts.

||

But divergence is equally informative. When claims don't match evidence, we've found something interesting:

*Aspiration gaps*—what actors want to become  
*Perception gaps*—how observers misread signals  
*Performance gaps*—where intention exceeds capacity  
*Strategic gaps*—deliberate misdirection or positioning

||: *con-vergere*—to incline together. *di-vergere*—to incline apart. Both movements carry signal.

The map that shows only convergence is incomplete. The full cartography includes the tensions—the places where declared and enacted roles pull apart, where relational position contradicts aesthetic positioning. These tensions are not noise; they are *data about dynamics*. They reveal where the constellation is in motion, where roles are contested, where the future is being negotiated.

## On Digital Traces: *Vestigium* and *Signum*

We map through traces left behind. Website text, news articles, funding records, tweets, images. The Latin offers two relevant words: *vestigium* (footprint, track—something left unintentionally) and *signum* (sign, signal—something made intentionally).

\*\*

Digital traces are both. A website is a *signum*—a deliberate signal crafted for audiences. But its metadata, its link structure, its archival history are *vestigia*—footprints left in the course of signaling, often unconsidered. News coverage is someone else's *signum* about the organization—a signal that becomes, for us, a trace of how others perceive.

This dual nature is not a limitation to apologize for. It is the nature of social observation. Interviews, too, are performances. Surveys elicit strategic responses. Ethnography is shaped by the observer's presence. All social data are *signa* from some angle, *vestigia* from another.

Digital traces have their own biases—toward the visible, the English-speaking, the digitally active, the organizationally formalized. We acknowledge these. But they also offer something rare: scale without entirely sacrificing texture. We can read thousands of websites, not dozens. We can trace networks across continents. We can observe dynamics over years through archived *vestigia*.

\*\**vestigium*—footprint, trace, track. *signum*—sign, signal, mark. Digital traces are both: unintended residues and deliberate signals.

## Instrumentum: The Tools We Build

Here Design Science enters explicitly. The Latin *instrumentum* meant tool, equipment, apparatus—but also document, deed, means of proof. An instrument was something built to serve a purpose, and also something that provided evidence.

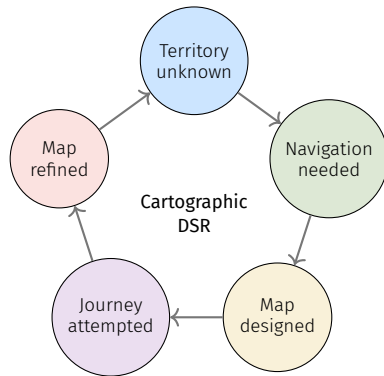
††

The **ROLEBOX** platform is an *instrumentum* in this double sense. It is a tool—software that collects, processes, integrates, and visualizes multimodal data about role constellations. But it is also a document of methodological choices: which traces to collect, how to process them, what features to extract, how to integrate across *modi*. The tool *embodies* theoretical commitments; examining the tool reveals the theory.

Design Science Research (DSR) makes this embodiment explicit. Where traditional research treats tools as neutral means to theoretical ends, DSR recognizes that *building*

††*instrumentum*—tool, document, means. The **ROLEBOX** and **ROLESI** are *instrumenta* in both senses: tools for seeing and evidence for claims.

*the tool is itself a knowledge contribution.* The ROLE-Boxarchitecture—its three layers, its modular pipelines, its integration protocols—constitutes a claim about *how* multimodal cartography should be done. The claim is tested not only by whether the maps prove useful but by whether the tool can be replicated, adapted, extended by others.



*The Design Science cycle as cartographic practice: unknown territory prompts need for navigation; need motivates map design; design enables journey; journey reveals map limitations; limitations refine understanding of territory. The cycle continues.*

## Simulacrum: The Map That Moves

But static maps have limits. They show where things *are*, not where they *could be*. They depict the territory as found, not as it might become under different conditions. For this, we need a different kind of representation.

##

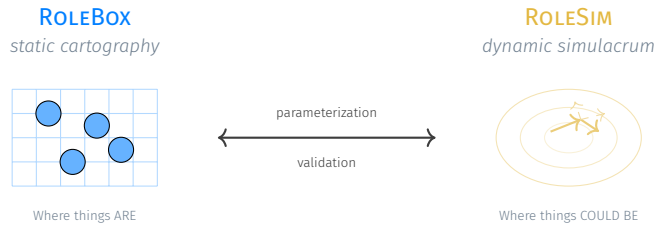
The Latin *simulacrum* meant image, likeness, or imitation—something made to resemble something else. The English “simulation” carries this forward: a *simulacrum* that unfolds dynamically, that responds to perturbation, that generates possible futures rather than depicting the present.

The **ROLESIM** agent-based model is our dynamic *simulacrum*. Where ROLEBoxmaps the constellation as observed, ROLESIMasks: *What if?* What if a key intermediary exited? What if policy conditions shifted? What if the fitness landscape tilted toward different role configurations? These counterfactual territories cannot be observed—but they can be *simulated*.

This is not prediction in the forecasting sense. Complex Adaptive Systems resist point prediction; that is their nature. Rather, ROLESIMexplores the *space of possibilities*—the adjacent territories that the constellation might

##: *simulacrum*—likeness, image, imitation. From *simulare*: to make like, to represent, to feign. A simulation is a *simulacrum* that moves through time.

traverse, the paths not taken that illuminate why the actual path was taken. It is cartography of the *potential*, complementing cartography of the *actual*.



*ROLEBOX* maps the observed constellation; *ROLESIM* simulates possible trajectories. The observed parameterizes the possible; the possible contextualizes the observed.

## The Craft of Integration: *Integrare*

The hardest part is not collecting five types of data. It is *integrating* them into coherent understanding. The Latin *integrare* meant to make whole, to renew, to restore—from *integer*, meaning untouched, whole, complete.

§§

Integration is not mere aggregation. It is not averaging signals or stacking data. It is the work of *making whole*—discerning how fragments relate, where they cohere, where they productively conflict. How do you combine a network position with a visual aesthetic? A funding pattern with a discourse style?

There is no algorithm for this. There are heuristics, frameworks, systematic comparisons. But ultimately, cartography is a *craft*—skilled judgment about what patterns mean and how to represent them. The Latin *ars* (whence “art”) meant skill, craft, technique—something learned through practice, refined through repetition, never fully reducible to rules.

This dissertation offers tools. It does not offer automation. The human interpreter remains essential—not as a weakness, but as the source of meaning. Computers can detect patterns; humans discern *significance*. The integration of multimodal signals into claims about roles requires both.

§§: *integrare*—to make whole, restore. *integer*—whole, complete, untouched. Integration is the work of making wholeness from fragments.

## Design Principles as Cartographic Commitments

The four Meta-Design Requirements introduced in Chapter 1 are, at root, cartographic commitments:

### MDR-1: Theory-Data Reconciliation

*The map must speak the language of both terrain and traveler.*

Theoretical concepts (emergence, self-organization) must connect to observable traces; observed patterns must be interpretable through theoretical lenses. Neither theory-blind empiricism nor data-free theorizing suffices.

### MDR-2: Hypothesis-Driven Data Selection

*The cartographer chooses what to survey based on the journey's purpose.*

We do not map everything; we map what the fifteen propositions require. Data collection serves theoretical testing, not exhaustive documentation.

### MDR-3: Computational-Human Complementarity

*The instrument extends the eye; it does not replace the mind.*

Algorithms detect; humans interpret. Scale requires computation; meaning requires judgment. The *instrumentum* amplifies human capacity without substituting for it.

### MDR-4: Knowledge Accumulation Infrastructure

*Good maps enable others to journey—and to make better maps.*

Open tools, documented methods, archived data: these allow replication, extension, critique. The cartographic contribution is not only the maps we make but the map-making capacity we share.

## What Follows

The chapters ahead will get technical. We will discuss APIs, NLP pipelines, network metrics, image classifiers, agent-based models. We will present tables and graphs and statistics. We will write of recall and precision, of centrality and clustering, of fitness functions and parameter sweeps.

Do not mistake the machinery for the purpose.

¶¶

The purpose is *seeing*—seeing role constellations that would otherwise remain invisible. The machinery serves the seeing. When the machinery obscures the seeing, we have failed.

¶¶: *The purpose is seeing—seeing role constellations that would otherwise remain invisible.*



The ancient cartographers inscribed *hic sunt dracones*—"here be dragons"—on unknown territories. It was honest: an admission of ignorance, a warning to travelers, an invitation to future explorers.

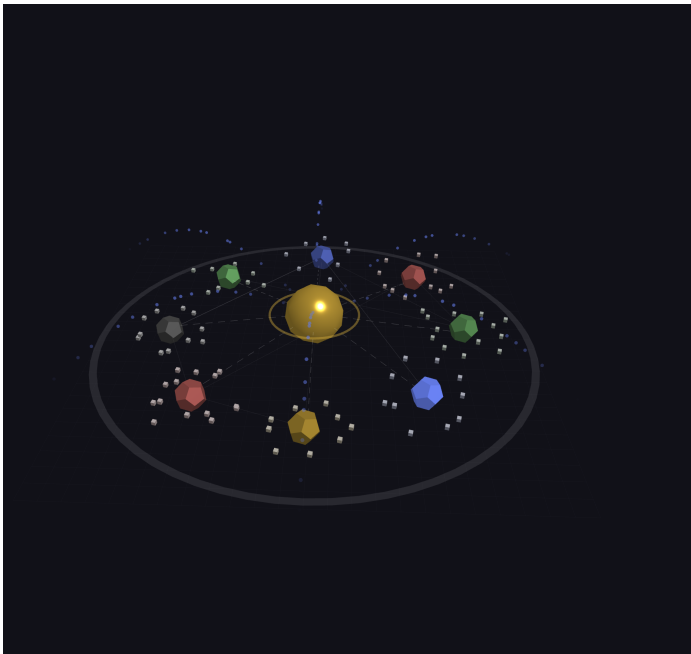
Our maps, too, will have their dragons. Territories where data is sparse. Regions where integration falters. Zones where the traces mislead. We will mark these honestly—not as failures but as frontiers.

*Now, to the methods that make the maps.*



## Part II

# EMPIRICAL STUDIES



# Interstitial Pluralism in Role Constellations

# 5

*This chapter develops the concept of **interstitial pluralism**—the phenomenon where actors occupy in-between positions across multiple role categories simultaneously. We analyze how EIT ecosystem actors navigate plurality, the strategic value of interstitial positioning, and the tensions between specialization and versatility. Building on structural approaches to organizational identity, we introduce the concept of **sociotypes**—fluid relational configurations that emerge from the patterned interconnections among role identities.*

*“The institutional order is experienced as an objective reality. It has a history that antedates the individual’s birth and is not accessible to his biographical recollection. It was there before he was born, and it will be there after his death.”*

— Peter L. Berger & Thomas Luckmann, *The Social Construction of Reality* [?]

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## 5.1 Introduction: The Problem of Interstitial Organization

Innovation intermediaries occupy peculiar institutional terrain.<sup>1</sup> They are not universities, yet they educate. They are not venture capital firms, yet they fund startups. They are not research institutes, yet they generate knowledge. They are not government agencies, yet they implement public policy. They exist *between*—in the interstices of established institutional domains—and their organizational coherence depends on sustaining productive relationships with constituencies whose expectations derive from fundamentally different sources.

The European Institute of Innovation and Technology (EIT) and its Knowledge and Innovation Communities (KICs) represent an ambitious experiment in institutionalizing such interstitial organization at continental scale.<sup>2</sup> Established in 2008, the EIT was designed to address Europe’s “innovation paradox”—the persistent gap between scientific excellence and commercial translation—by creating organizational forms that would integrate the “knowledge triangle” of higher education, research, and business. The KICs that emerged—EIT Climate-KIC, EIT Digital, EIT Health, EIT Food, EIT Manufacturing, EIT Urban Mobility, EIT RawMaterials, and EIT Culture & Creativity—each operate across these domains within specific sectoral contexts.

What does it mean, organizationally, to integrate a knowledge triangle? How do intermediaries that must simultaneously satisfy academic,

1: **Interstitial organization**—the concept draws on work in economic sociology examining actors positioned between established fields.

2: **EIT Design**—modeled partly on MIT and Stanford’s innovation ecosystems, but as a distributed pan-European network rather than a single institution.

entrepreneurial, educational, and public accountability demands construct coherent organizational identities? And how do the relational configurations through which they engage their constituencies reveal the underlying structure of their institutional environment?

This chapter develops a structural approach to these questions.<sup>3</sup> Rather than beginning with predetermined institutional categories and asking how organizations respond to them, we proceed inductively from the relational practices through which EIT Communities constitute themselves and their stakeholders. We extract and analyze the role identity configurations embedded in organizational communications, reconstruct the patterned interconnections among them, and identify emergent *sociotypes*—recurring relational configurations that are stable enough to be recognizable yet fluid enough to blend and recombine.

Our approach departs from conventional institutional analysis in three ways. First, we resist categorical closure: rather than mapping observed patterns onto established typologies, we build rich descriptive profiles and hold interpretation open. Second, we foreground *interconnectedness*: rather than treating pluralism as the copresence of discrete categories, we attend to the interfaces, overlaps, and zones of ambiguity between relational configurations. Third, we employ computational methods—natural language processing for triple extraction and network analytical methods for pattern detection—that enable systematic analysis at scale while preserving interpretive depth.

3: **Structural approach**—rather than coding practices according to predetermined categories, we extract relational patterns and let structure emerge.

## 5.2 Theoretical Framework: From Categories to Sociotypes

### 5.2.1 The Limits of Categorical Approaches

Organizational research has long sought to understand how organizations navigate environments characterized by multiple, potentially conflicting, sources of meaning and legitimacy.<sup>4</sup> The dominant approach works from predetermined categories—whether called institutional logics, orders, or fields—and asks how organizations respond to them. This framing has generated important insights, but it carries conceptual baggage that limits its utility for understanding interstitial organizations.

4: **Institutional pluralism**—for comprehensive reviews, see the work on institutional logics and organizational responses to complexity.

The categorical approach tends toward *reification*.<sup>5</sup> Despite acknowledging that categories are analytically constructed ideal types, empirical research often treats them as bounded containers with determinate contents. Canonical typologies posit a fixed number of societal-level orders (family, community, religion, state, market, profession, corporation), each with characteristic sources of legitimacy, authority, and identity [?]. While researchers acknowledge local variation, the framework encourages mapping observed phenomena onto predetermined categories rather than attending to emergent configurations.

5: **Reification critique**—echoes concerns raised by practice theorists about treating culture as thing-like rather than processual.

The categorical approach also privileges *conflict and contradiction*.<sup>6</sup> Concepts like “institutional complexity” [?] foreground situations where organizations face “incompatible prescriptions” from multiple sources. This framing directs attention toward friction and away from the more

6: **Conflict emphasis**—reflects roots in analyzing institutional *change*, where contradictions provide motors of transformation.

common situation where different orientations coexist, interpenetrate, or mutually constitute one another without generating overt conflict.

Most problematically, categorical approaches rarely examine *whole constellations*. Studies typically focus on dyadic relationships between two categories, neglecting the structural properties of multi-category configurations. We lack systematic approaches for analyzing how multiple orientations are positioned relative to one another, which elements create interfaces between them, and how the overall topology of a constellation shapes organizational possibilities.

### 5.2.2 Toward a Structural Approach: Role Identities and Relational Configuration

We address these limitations by developing a structural approach that reconstructs institutional environments from the bottom up.<sup>7</sup> Our starting point is the observation that institutional environments become manifest in organizational life through *role identities*—positions in social space defined by sets of behavioral expectations. Role identities are inherently relational: they are constituted through typified actions directed toward typified others. A “teacher” implies “students” and practices of instruction; a “service provider” implies “customers” and practices of delivery; an “investor” implies “investees” and practices of funding. Following Berger and Luckmann [?], we understand this reciprocal typification of actor and act as the core of institutionalization.

This insight can be operationalized through the concept of *recursive categorization*: role identities are constituted in triplets of role identity, counterrole, and connecting practices.<sup>8</sup> Extracting such triplets from organizational discourse reveals the multiple positions an organization claims and the relationships through which it enacts them. The totality of triplets forms a structured configuration—a *nexus of role identities*—whose patterning reveals the organization’s relational environment.

This approach has several advantages. It grounds analysis in observable relational practices rather than imputed cultural categories. It captures multiplicity without predetermining how many orientations are present or what they contain. And it enables systematic analysis of structural properties—differentiation, interconnectedness, interfaces, overlaps—that remain undertheorized in existing literature.

### 5.2.3 Sociotypes: Fluid Configurations of Relational Practice

We introduce the concept of *sociotype* to name the clusters of role categories that emerge from structural analysis of relational configurations.<sup>9</sup> The term draws loosely on biosocial analogies while avoiding the determinism they sometimes carry. A sociotype is:

- A **recurring relational configuration**—a pattern of role identities, counterroles, and practices that appears with sufficient regularity to be recognizable

7: **Bottom-up**—aligns with calls for greater attention to microfoundations in institutional theory.

8: **Recursive categorization**—knowing what a “teacher” is requires knowing what “students” are and what “teaching” involves.

9: **Sociotype**—a recurring relational configuration, stable enough to recognize yet fluid enough to blend, defined by its pattern of connections rather than essential properties.

- ▶ **Fluid rather than fixed**—capable of blending with other socio-types, shifting in salience, and recombining in novel arrangements
- ▶ **Defined by relational signature**—characterized by its distinctive pattern of connections rather than essential properties
- ▶ **Emergent from practice**—discovered through analysis of enacted relationships rather than imposed from theoretical typologies

Sociotypes differ fundamentally from categorical constructs in their ontological status.<sup>10</sup> Where categories are theorized as macro-level cultural structures that constrain and enable organizational action, sociotypes are meso-level relational configurations whose existence is established empirically through pattern detection. A sociotype may correspond loosely to a recognized institutional category, blend elements from multiple categories, or represent a configuration that existing typologies fail to capture. The relationship is a matter for investigation rather than assumption.

This reframing has methodological implications. Rather than coding organizational practices according to predetermined categories, we extract relational triplets, cluster them according to structural and semantic similarity, and build rich descriptive profiles before engaging in interpretation. The goal is not to avoid theoretical engagement but to delay categorical closure until patterns have been allowed to emerge on their own terms.

Crucially, sociotypes do not “harden” into fixed categories through our analysis. They remain what they are: fluid configurations with porous boundaries, capable of blending and recombination. Our network representations capture this fluidity through continuous edge weights and overlapping community assignments rather than discrete partitions.

## 5.2.4 Structural Properties of Sociotype Constellations

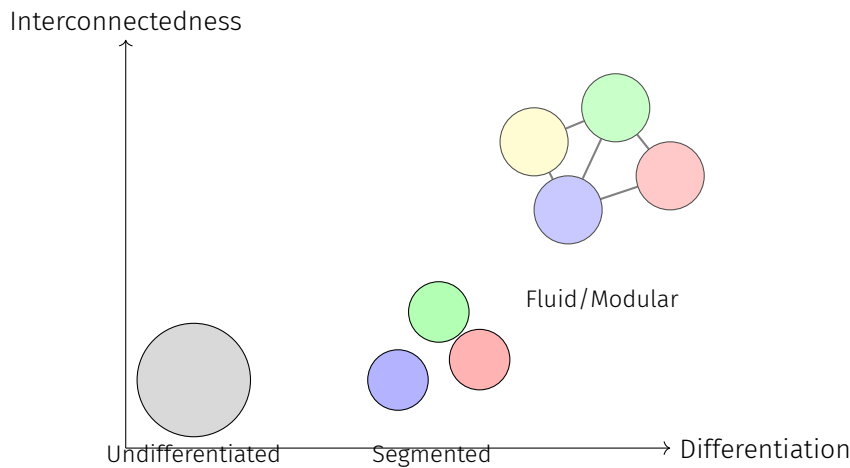
We analyze sociotype constellations along two primary dimensions: *differentiation* and *interconnectedness*.<sup>11</sup>

**Differentiation** refers to the extent to which distinct relational configurations are discernible within a constellation. A highly differentiated constellation contains multiple clusters with distinct relational signatures; a less differentiated constellation exhibits fewer or less distinct clusters. Differentiation indicates the variety of relational configurations through which an organization engages its environment—but it does not imply that these configurations are bounded or mutually exclusive.

**Interconnectedness** refers to the relationships between configurations within a constellation. Configurations may be densely or sparsely connected; connections may be evenly distributed or concentrated at particular interfaces; certain role categories may serve as bridges linking multiple configurations while others remain more localized. Interconnectedness reveals how relational plurality is organized—whether configurations are segmented and compartmentalized or integrated and overlapping.

10: **Ontological distinction**—parallels debates between “realist” and “constructionist” approaches to social kinds.

11: **Two dimensions**—parallel classic sociological concerns with division of labor (differentiation) and social integration (interconnectedness).



**Figure 5.1:** Forms of relational plurality characterized by differentiation and interconnectedness.

Within constellations exhibiting both differentiation and interconnectedness, we attend particularly to *interfaces*—the role categories that connect configurations.<sup>12</sup> We distinguish:

- ▶ **Localized categories:** Role identities or counterroles that appear predominantly within one configuration
- ▶ **Bridging categories:** Role identities or counterroles that connect two or more configurations
- ▶ **Multivocal categories:** Bridging categories whose evocation leaves open which configuration is being instantiated

Multivocal categories are particularly significant for organizations navigating plural environments.<sup>13</sup> They enable “robust action” [?]<sup>14</sup>—the capacity to “retreat behind a shroud of multiple identities” and avoid commitment until the situation clarifies which framing will succeed. Multivocal categories create ambiguity that organizations can exploit strategically, satisfying diverse stakeholders through practices that can be read through multiple frames.

Finally, we introduce *sociotype permeability* as an attribute of individual configurations within a constellation.<sup>14</sup> Permeability captures the extent to which a configuration shares role categories with other configurations—the totality of interfaces it maintains. A highly permeable configuration has porous boundaries and extensive overlap with others; a less permeable configuration is more self-contained.

12: **Interfaces**—structural locations where relational work is most active, as actors navigate between configurations.

13: **Multivocality**—the property of a category that allows it to be interpreted through multiple relational frames simultaneously, creating productive ambiguity.

14: **Permeability**—describes structural overlap regardless of whether overlap generates conflict or complementarity.

## 5.3 Empirical Context: The EIT Knowledge and Innovation Communities

### 5.3.1 The European Institute of Innovation and Technology

The EIT was established by the European Parliament and Council in 2008 as a response to perceived European underperformance in translating research excellence into economic and social value.<sup>15</sup> Its mission

15: **European paradox**—strong basic research but weak commercialization—has been a persistent concern in EU innovation policy since the 1990s.



centers on “knowledge triangle integration”—bringing together higher education, research, and business to strengthen Europe’s innovation capacity. Unlike traditional research funding instruments, the EIT does not fund projects directly but rather supports Knowledge and Innovation Communities (KICs)—large-scale partnerships that operate across the knowledge triangle within specific thematic domains.

The EIT represents an experiment in *mandated institutional integration*.<sup>16</sup> Where most organizations encounter relational plurality as an environmental condition to which they must adapt, the EIT and its KICs are designed to produce integration as their core output. Their legitimacy depends on demonstrating that they have successfully bridged domains that ordinarily operate according to different relational conventions. This makes them an especially revealing site for studying institutional plurality: the tensions and accommodations that remain implicit in other settings become explicit organizational challenges.

### 5.3.2 The Knowledge and Innovation Communities

As of 2024, nine KICs operate under the EIT umbrella (see Table 5.1). Each KIC comprises a partnership of universities, research organizations, and businesses operating through geographically distributed “Co-Location Centres” (CLCs). KICs receive core funding from the EIT (derived from the EU’s Horizon Europe programme) supplemented by partner contributions and revenue-generating activities. They are expected to become financially sustainable over a 15-year funding cycle.

KIC activities span three domains corresponding to the knowledge triangle:<sup>17</sup>

- **Education:** Master’s and doctoral programmes carrying the “EIT Label,” executive education, and entrepreneurship training
- **Research/Innovation:** Collaborative R&D projects, technology transfer, and proof-of-concept funding
- **Business creation:** Startup acceleration, access to finance, and scale-up support

This portfolio requires KICs to maintain relationships with diverse constituencies: students seeking credentials, researchers seeking funding, startups seeking capital, corporations seeking innovation pipelines, universities seeking partnerships, and public authorities seeking policy implementation. Each constituency brings expectations rooted in different relational conventions.

## 5.4 Methods: Computational Extraction and Structural Analysis

### 5.4.1 Overview of Analytical Approach

Our methodology proceeds in four phases:<sup>18</sup>

1. **Triple extraction:** Systematic identification of role identity–practice–content triplets from organizational communications
2. **Network construction:** Assembly of triplets into a semantic network relating role categories through shared practices

**16: Mandated integration**—distinguishes EIT from organizations that encounter plurality as an environmental condition; here, integration is the *output*, not a challenge to be managed.

| KIC                  | Est. |
|----------------------|------|
| Climate-KIC          | 2010 |
| Digital              | 2010 |
| InnoEnergy           | 2010 |
| Health               | 2015 |
| RawMaterials         | 2015 |
| Food                 | 2017 |
| Manufacturing        | 2019 |
| Urban Mobility       | 2019 |
| Culture & Creativity | 2022 |

**Table 5.1:** EIT KICs by establishment date

**17: Knowledge triangle**—originates in Nordic innovation policy and was adopted by the EU as a framing for the EIT’s mission.

**18: Reproducibility**—code and intermediate outputs are available in the supplementary materials.

3. **Community detection:** Application of clustering methods to identify emergent sociotype configurations
4. **Structural analysis:** Characterization of constellation properties including differentiation, interconnectedness, and permeability

We employ natural language processing for triple extraction, combining dependency parsing with large language model-assisted interpretation where syntactic patterns prove insufficient.

## 5.4.2 Data Collection

### Corpus Construction

Our primary data source is the public-facing websites of the nine KICs and the EIT itself. Websites serve as the primary vehicle through which KICs communicate with external constituencies and construct public-facing identities. Unlike internal documents, websites address multiple audiences simultaneously and cannot resolve plurality through compartmentalization—different stakeholders access the same materials.

We constructed the corpus through systematic web crawling. For each KIC, we extracted:<sup>19</sup>

- ▶ All pages from the main website (typically 200–500 pages per KIC)
- ▶ Strategic documents and annual reports (PDF extraction)
- ▶ Programme descriptions and calls for proposals
- ▶ News items and press releases from the preceding 24 months

19: **Data sources**—web crawling used standard tools with politeness delays; PDF extraction used PyMuPDF.

After removing duplicate content, navigation elements, and pages without substantive text, the corpus comprised approximately 12,000 documents totaling 3.2 million words across nine KICs.

### Sampling and Filtering

We filtered the corpus to retain only documents containing explicit references to external constituencies. Documents consisting solely of internal organizational information, contact details, or procedural content were excluded.

## 5.4.3 Triple Extraction

### Preprocessing

Documents underwent standard preprocessing:

- ▶ Sentence segmentation
- ▶ Coreference resolution to link pronouns and nominal references to their antecedents
- ▶ Named entity recognition to identify organizational actors
- ▶ Part-of-speech tagging and dependency parsing

## Extraction Procedure

We identify relational triplets through syntactic dependency patterns.<sup>20</sup> We extract triplets where:

- ▶ A subject noun phrase (the KIC or its representative) stands in subject relation to a verb
- ▶ The verb governs a direct object, prepositional object, or complement denoting an external actor
- ▶ The verb represents an action or practice rather than a copular or auxiliary construction

20: **Dependency parsing**—identifies grammatical relations between words, enabling extraction of subject-verb-object structures.

Extracted triplets take the form:

$$\tau = \langle \text{subject, predicate, object, source} \rangle \quad (5.1)$$

For complex sentences where dependency parsing proves insufficient, we supplement with language model-assisted extraction, prompting for relational triplets and reconciling outputs through comparison.

## Normalization

Extracted triplets undergo normalization to enable aggregation:

- ▶ **Subject normalization:** References to the KIC (“we,” “EIT Food,” “the programme”) are mapped to a canonical KIC identifier
- ▶ **Predicate normalization:** Verbs are lemmatized; synonymous actions are clustered through embedding similarity
- ▶ **Object normalization:** External actors are mapped to category labels through semantic clustering

### 5.4.4 Network Construction

We represent the corpus of normalized triplets as a weighted two-mode network  $G = (V_A, V_P, E)$  where:

- ▶  $V_A$  is the set of actor nodes (role identities and counterroles)
- ▶  $V_P$  is the set of practice nodes (normalized predicates)
- ▶  $E$  is the set of edges connecting actors through shared practices

Edge weights reflect the frequency of triplets linking two categories.

For community detection, we project the two-mode network to a one-mode actor network where actors are linked if they share practices:

$$w_{ij} = \sum_{p \in V_P} \frac{e_{ip} \cdot e_{jp}}{\sqrt{d_i \cdot d_j}} \quad (5.2)$$

where  $e_{ip}$  is the weight of the edge between actor  $i$  and practice  $p$ , and  $d_i$  is the degree of actor  $i$ . This normalized projection accounts for varying actor frequencies.

### 5.4.5 Community Detection

We employ stochastic block models (SBMs) for community detection [?].<sup>21</sup> SBMs provide a principled probabilistic framework that:

- ▶ Discovers community structure without requiring a predetermined number of clusters
- ▶ Handles weighted edges
- ▶ Provides uncertainty quantification through posterior distributions
- ▶ Avoids overfitting through Bayesian model selection

21: **SBMs**—model the network as generated by latent block structure, enabling principled inference about community composition.

Crucially, SBMs return *soft assignments*—probability distributions over community membership rather than hard partitions. This preserves the fluidity central to the sociotype concept: a role category may participate in multiple configurations with different intensities rather than belonging exclusively to one.

### 5.4.6 Sociotype Profiling

For each detected configuration, we construct a rich descriptive profile:

- ▶ **Structural characteristics:** Internal density, external connectivity, bridging nodes, centrality
- ▶ **Relational signature:** Characteristic practices (TF-IDF weighted), role identities, and counterroles
- ▶ **Textual fingerprint:** Representative triplets, distinguishing vocabulary
- ▶ **Cross-KIC distribution:** Prevalence in each KIC's network
- ▶ **Boundary characteristics:** Membership entropy, nodes with distributed assignments

Profiles enable interpretation while resisting premature categorical closure. We do not assign labels like “market” or “profession” to detected configurations; instead, we describe their relational signatures and let readers draw connections to existing concepts as appropriate.

### 5.4.7 Measures of Constellation Structure

#### Differentiation

Differentiation is operationalized as the number of distinct configurations and their relative separation:

$$D = k \cdot \left( 1 - \frac{\bar{s}_{between}}{\bar{s}_{within}} \right) \quad (5.3)$$

where  $k$  is the number of configurations,  $\bar{s}_{within}$  is mean within-cluster similarity, and  $\bar{s}_{between}$  is mean between-cluster similarity.

### Interconnectedness

Interconnectedness captures the density of cross-configuration connections:

$$I = \frac{\sum_{i \neq j} w_{ij} \cdot \mathbf{1}[c_i \neq c_j]}{\sum_{i \neq j} w_{ij}} \quad (5.4)$$

where  $w_{ij}$  is the edge weight between nodes  $i$  and  $j$ , and  $c_i$  is the primary configuration assignment of node  $i$ .

### Permeability

Permeability for configuration  $s$  is:

$$P_s = \frac{|B_s|}{|M_s|} \cdot \frac{\sum_{b \in B_s} |S_b|}{k - 1} \quad (5.5)$$

where  $|B_s|$  is the number of bridging nodes in configuration  $s$ ,  $|M_s|$  is total membership,  $|S_b|$  is the number of other configurations that bridge  $b$  connects to, and  $k$  is the total number of configurations.

## 5.5 Findings: Sociotype Constellations in EIT Communities

### 5.5.1 Aggregate Constellation Structure

Analysis of the aggregate network (pooling triplets across all KICs) reveals a modular constellation comprising six primary sociotype configurations.<sup>22</sup> The constellation exhibits moderate differentiation and substantial interconnectedness, indicating that EIT Communities instantiate a plural relational structure with significant overlap between configurations.

22: **Six configurations**—the number emerged from the data; we did not impose it.

### 5.5.2 Sociotype Profiles

We present profiles of the six primary configurations, proceeding from highest to lowest permeability. For each, we report structural position, relational signature, and representative triplets. We resist assigning categorical labels, instead describing configurations through their relational characteristics.

### Configuration 1: The Connective Configuration

**Structural position:** Highly central; highest permeability; interfaces with all other configurations.

**Relational signature:**

- ▶ Characteristic practices: *connects, brings together, facilitates, enables collaboration, builds*
- ▶ Role identities: *connector, platform, facilitator, builder*
- ▶ Counterroles: *partners, stakeholders, actors, members, innovators*

**Representative triplets:**

“EIT Food connects food system innovators across Europe”  
 “The KIC brings together leading universities, research centres, and companies”  
 “We facilitate collaboration between startups and corporate partners”

**Cross-KIC distribution:** Present in all KICs; highest prevalence in EIT Climate-KIC and EIT Digital.

**Relational character:** This configuration positions the KIC as enabling relationships among diverse actors.<sup>23</sup> It emphasizes relational coordination over transaction. The high permeability reflects the configuration’s function as connective tissue—it interfaces with constituencies associated with other configurations precisely because brokerage is its defining practice. The counterrole “partners” is notably multivocal, appearing across multiple configurations.

23: **Connective function**—the configuration functions as the constellation’s integrative tissue; its high permeability reflects its bridging function.

### Configuration 2: The Acceleration Configuration

**Structural position:** Central; high permeability; strong interfaces with Configurations 1 and 3.

**Relational signature:**

- ▶ Characteristic practices: *accelerates, funds, supports, invests in, scales*
- ▶ Role identities: *accelerator, investor, supporter, launcher*
- ▶ Counterroles: *startups, entrepreneurs, ventures, scale-ups, founders*

**Representative triplets:**

“EIT Health accelerates promising healthcare startups”  
 “Our programme supports entrepreneurs from idea to market”  
 “The KIC has invested in over 500 ventures since 2015”

**Cross-KIC distribution:** Present in all KICs; highest prevalence in EIT Digital and EIT InnoEnergy.

**Relational character:** This configuration organizes relationships around venture development and growth.<sup>24</sup> Role identities and practices map onto established categories of entrepreneurial support. The configuration’s permeability reflects the need to connect venture activity with other knowledge triangle vertices—ventures require research inputs (Configuration 4) and educated talent (Configuration 3).

24: **Acceleration permeability**—reflects the need to connect venture development with research inputs and educated talent.

### Configuration 3: The Development Configuration

**Structural position:** Moderately central; moderate permeability.

**Relational signature:**

- ▶ Characteristic practices: *educates, trains, develops, certifies, prepares*
- ▶ Role identities: *educator, trainer, certifier, developer*
- ▶ Counterroles: *students, learners, participants, talents, future leaders*

**Representative triplets:**

“Our Master’s programmes prepare students for innovation careers”  
 “EIT Manufacturing trains the workforce of tomorrow”  
 “The EIT Label certifies educational excellence in entrepreneurship”

**Cross-KIC distribution:** Present in all KICs; highest prevalence in EIT Digital and EIT Health.

**Relational character:** This configuration organizes relationships around human capability formation.<sup>25</sup> The “EIT Label” emerges as a distinctive credential. Moderate permeability reflects the bounded nature of educational relationships—students are addressed through specifically educational practices—while interfaces with Configurations 1 and 2 indicate the entrepreneurial orientation distinguishing EIT education from traditional academic programmes.

25: **EIT Label**—emerges as a distinctive credentialing device linking this configuration to formal certification.

### Configuration 4: The Knowledge Configuration

**Structural position:** Peripheral; lower permeability.

**Relational signature:**

- ▶ Characteristic practices: *researches, develops, discovers, advances, validates*
- ▶ Role identities: *researcher, knowledge generator, scientific partner*
- ▶ Counterroles: *researchers, scientists, research institutions, universities*

**Representative triplets:**

“Our research projects advance understanding of sustainable food systems”  
 “EIT Climate-KIC partners with leading research institutions”  
 “The programme validates innovative technologies through rigorous testing”

**Cross-KIC distribution:** Present in all KICs; relatively even distribution.

**Relational character:** This configuration organizes relationships around knowledge production and validation.<sup>26</sup> Lower permeability and peripheral position reflect a notable pattern: KICs partner with research organizations but rarely position themselves primarily as knowledge generators. The configuration interfaces mainly with Configuration 2 (research inputs to commercialization) rather than standing as an end in itself.

26: **Knowledge peripherality**—may reflect structural tensions in the EIT model, which emphasizes application over basic research.

### Configuration 5: The Accountability Configuration

**Structural position:** Peripheral; low permeability.

**Relational signature:**

- ▶ Characteristic practices: *delivers impact, reports to, contributes to, addresses challenges, serves*
- ▶ Role identities: *impact deliverer, policy contributor, servant*
- ▶ Counterroles: *European Commission, policymakers, citizens, society, Europe*

**Representative triplets:**

- "EIT Food contributes to EU food policy objectives"
- "Our activities deliver measurable impact for European citizens"
- "The KIC addresses societal challenges in sustainable mobility"

**Cross-KIC distribution:** Present in all KICs; highest prevalence in EIT Climate-KIC and EIT Food.

**Relational character:** This configuration reflects the public funding context and associated accountability demands.<sup>27</sup> Its peripheral position and low permeability suggest structural separation from core operational activities—public accountability is acknowledged but not integrated into the relational configurations through which KICs engage direct constituencies.

27: **Accountability separation**—echoes findings on "decoupling"—formal compliance structures disconnected from operational activities.

### Configuration 6: The Sectoral Configuration

**Structural position:** Variable by KIC; moderate permeability.

**Relational signature:** Varies by sectoral domain. Examples:

- ▶ EIT Health: *improves outcomes for → patients; transforms → healthcare systems*
- ▶ EIT Food: *supports → farmers; serves → consumers*
- ▶ EIT Urban Mobility: *enables → cities; benefits → commuters*

**Relational character:** This configuration captures domain-specific stakeholder relationships that differentiate KICs from one another. Unlike the preceding configurations (which appear across all KICs), the sectoral configuration varies in content while maintaining structural similarity. It represents the point of contact between the generic innovation intermediary model and specific societal challenges.

## 5.5.3 Interconnectedness and Interfaces

The aggregate constellation exhibits uneven interconnectedness.<sup>28</sup>

Key findings on interconnectedness:

- ▶ **Dense interface:** Configurations 1 (Connective) and 2 (Acceleration) share extensive overlap, with "partners" and "innovators" functioning as multivocal categories evocable from either configuration.

28: **Topology**—uneven interconnectedness generates the distinctive "topology" of relational plurality in each organization.



- **Moderate interface:** Configurations 2 (Acceleration) and 3 (Development) connect through categories like “entrepreneurs” (who are both investment targets and educational products) and practices like “develops” (applicable to both ventures and human capital).
- **Sparse interface:** Configurations 4 (Knowledge) and 5 (Accountability) show minimal direct connection, suggesting structural separation between research-oriented and policy-oriented activities.
- **Bridging function:** Configuration 1 (Connective) provides the primary connection between otherwise separated configurations, consistent with its high permeability and brokerage function.

### Multivocal Categories

Several role categories function as interfaces between multiple configurations:<sup>29</sup>

| Category      | Connected Configurations                      |
|---------------|-----------------------------------------------|
| Partners      | Connective, Acceleration, Knowledge, Sectoral |
| Innovators    | Connective, Acceleration                      |
| Entrepreneurs | Acceleration, Development, Connective         |
| Startups      | Acceleration, Connective                      |
| Students      | Development, Acceleration                     |
| Impact        | Accountability, Acceleration, Sectoral        |

Table 5.2: Multivocal role categories and their configuration connections

29: **Strategic resources**—high-multivocality categories enable organizations to satisfy diverse audiences without committing to a single relational frame.

The category “partners” exhibits the highest multivocality, appearing across four configurations with different inflections: research partners (Knowledge), corporate partners (Acceleration), ecosystem partners (Connective), and sectoral partners (Sectoral). This multivocality enables KICs to address diverse constituencies through shared vocabulary while preserving flexibility about which relational configuration is being instantiated.

### 5.5.4 Cross-KIC Variation

While all KICs exhibit constellations with similar configurations, their relative emphases differ.

Notable patterns:

- **Acceleration emphasis:** EIT Digital and EIT InnoEnergy show the strongest Acceleration configuration, consistent with their sectors’ venture ecosystems and historical focus on startup creation.
- **Development emphasis:** EIT Health exhibits elevated Development configuration, reflecting the healthcare sector’s credentialing requirements and EIT Health’s extensive Master’s programme portfolio.
- **Accountability emphasis:** EIT Climate-KIC and EIT Food show stronger Accountability configurations, reflecting their engagement with policy-intensive domains (climate action, food security).

- **Maturity effects:** Older KICs (Climate-KIC, Digital, InnoEnergy—established 2010) show somewhat higher interconnectedness than newer KICs, suggesting that constellation integration develops over time.

## 5.6 Discussion: Theoretical Implications

### 5.6.1 Interstitial Organization and Mandated Integration

Our findings illuminate the distinctive character of interstitial organization.<sup>30</sup> EIT Communities do not simply respond to pre-existing institutional plurality; they are designed to produce integration across domains that ordinarily operate according to different relational conventions. This mandated integration creates organizational forms whose coherence depends on maintaining productive ambiguity.

The constellation structure we observe—differentiated yet interconnected, with fluid boundaries between configurations—can be understood as an organizational achievement rather than an environmental given. Complete fusion of relational orientations would sacrifice the distinctive contributions of each knowledge triangle vertex; complete separation would undermine the integrative mission. The fluid, modular structure enables KICs to maintain recognizable relationships with domain-specific constituencies while creating interfaces where cross-domain value can be generated.

The Connective configuration emerges as the linchpin of this arrangement.<sup>31</sup> Its high permeability and central position enable it to function as organizational connective tissue, linking configurations that might otherwise remain separate. When a KIC describes itself as “connecting partners across the knowledge triangle,” it invokes a relational configuration that is neither purely academic, nor purely entrepreneurial, nor purely educational—but that creates discursive space where these orientations can coexist.

30: **Interstitial**—extends work on boundary organizations and intermediaries by emphasizing *structural position* between fields.

31: **Centrality**—reflects the importance of brokerage for interstitial organizations.

### 5.6.2 Multivocality as Organizational Resource

Our analysis foregrounds the strategic importance of multivocal categories.<sup>32</sup> Rather than treating ambiguity as a problem to be resolved, KICs exploit it as a resource for navigating plural environments.

Consider the category “partners.” When a KIC addresses “partners,” it leaves open whether it is invoking research partnerships, investment relationships, or network membership. This indeterminacy is not communicative failure but organizational achievement: it enables the KIC to address heterogeneous constituencies through shared vocabulary while preserving flexibility about which relational expectations apply.

32: **Robust action**—extends analyses of how actors maintain power by cultivating multiple, potentially inconsistent identity attributions.

### 5.6.3 The Accountability Gap

Our findings reveal a potential tension in the EIT model.<sup>33</sup> The Accountability configuration occupies a peripheral position with sparse interfaces to core operational configurations. Accountability relationships—addressing the European Commission, demonstrating impact, contributing to policy—appear structurally separated from the relationships through which KICs engage their primary operational constituencies.

This pattern admits multiple interpretations. It may reflect appropriate organizational differentiation: accountability to funders involves different relational dynamics than service to startups or education of students, and attempting to integrate them might undermine both. Alternatively, it may indicate ceremonial compliance: KICs acknowledge public accountability demands in external communications without integrating them into operational practice.

33: **Accountability gap**—resonates with broader concerns about democratic legitimacy in EU governance.

## 5.7 Conclusion

This chapter has developed a structural approach to relational plurality in interstitial organizations.<sup>34</sup> Through computational analysis of EIT Knowledge and Innovation Community communications, we have identified the sociotype constellations through which these innovation intermediaries navigate their complex relational environments.

Our findings reveal fluid, modular plurality: differentiated configurations connected through multivocal role categories that function as interfaces and bridges. The Connective configuration emerges as central integrative tissue; Acceleration and Development configurations occupy core positions; Knowledge and Accountability configurations remain more peripheral. This structure enables EIT Communities to maintain recognizable relationships with diverse constituencies while creating spaces for cross-domain integration.

The concept of sociotype—fluid relational configuration defined by structural position rather than essential properties—offers an alternative to categorical approaches.<sup>35</sup> Sotypes are discovered through pattern detection rather than imposed from theoretical typologies; they blend and recombine rather than existing as discrete containers; their boundaries are empirically variable rather than analytically predetermined.

For practitioners and policymakers, our analysis offers diagnostic value. Understanding the sociotype constellation characterizing an organization—which configurations are present, how they interconnect, where ambiguity enables flexibility and where distinctiveness constrains options—provides insight into the structural conditions of navigating plurality.

The broader ambition is a structural sociology of relational plurality—one that takes seriously the patterned character of plural environments while resisting the temptation to treat patterns as fixed, bounded, and inherently conflictual.<sup>36</sup> Organizations do not merely respond to plurality; they constitute it through the relational configurations they enact. Understanding those configurations, their structural properties,

34: **Broader ambition**—a structural sociology of relational plurality that attends to patterns without reifying them as fixed categories.

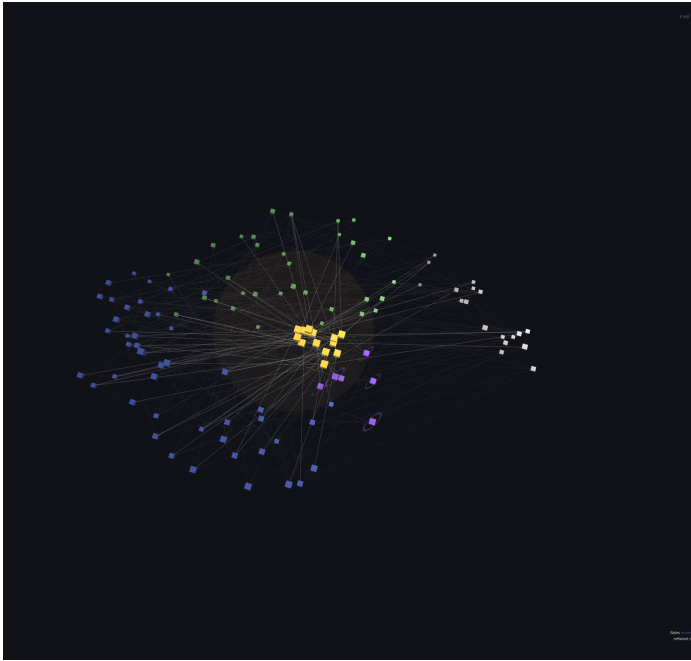
35: **Future work**—might examine when and how sociotypes stabilize into more categorical forms, and when they resist such stabilization.

36: **Constitution**—organizations do not merely *respond* to plurality; they *constitute* it through the relational configurations they enact.

and their dynamic evolution is essential to understanding contemporary organizational life.

## 5.8 Chapter Summary

- ▶ **Interstitial organizations** like EIT KICs exist between established institutional domains, requiring integration across fundamentally different relational conventions.
- ▶ **Sociotypes** are fluid relational configurations—stable enough to recognize, fluid enough to blend—defined by structural position rather than essential properties.
- ▶ **Six configurations** characterize EIT discourse: Connective (central, high permeability), Acceleration, Development, Knowledge, Accountability (peripheral), and Sectoral.
- ▶ **Multivocal categories** like “partners” and “innovators” function as strategic resources, enabling KICs to address diverse constituencies without categorical commitment.
- ▶ **The Connective configuration** serves as integrative tissue, linking configurations that might otherwise remain separate.
- ▶ **The Accountability configuration** remains peripheral, suggesting structural separation between public accountability and core operational activities.



# ROLEFIELD: Mapping the Innovation Intermediary Field

# 6

*This chapter complements the inside-out perspective of the previous chapter with an outside-in approach. Rather than asking how KICs describe their relationships, we ask: how do external observers classify KICs and related actors? Using social curation methods, we map the European innovation intermediary field, identify sub-regions, and triangulate external classifications with internal role configurations.*

*“Classifications are tools in strategies of inclusion and exclusion: whom to relate to and whom to isolate. They symbolize and consolidate patterns of inclusion and exclusion because they transform them into identities, which are taken for granted later on.”*

— Wouter De Nooy [?]

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## 6.1 Introduction: Opening the Field

The previous chapter developed a structural approach to understanding how EIT Knowledge and Innovation Communities constitute their relational environments through the role identity configurations embedded in their own communications. That analysis proceeded from organizational self-presentation outward—extracting how KICs position themselves toward constituencies through claimed practices and attributed counterroles.

This chapter complements that inside-out perspective with an outside-in approach.<sup>1</sup> Rather than asking how KICs describe their relationships, we ask: how do external observers classify KICs and related actors? Who gets grouped together, and what does this reveal about the perceived structure of the European innovation intermediary field?

We draw on recent methodological innovations in field identification that leverage social curation mechanisms. The proliferation of online platforms has generated vast quantities of classification data—acts by which users categorize, group, and curate content and actors. When aggregated, these individual classification acts can reveal “local orders, products of the collective rationality of their constituents” [?]. Users who create lists of actors they perceive as related function as social curators, and their accumulated judgments mirror prevalent perceptions of field structure.

This approach addresses limitations of conventional field studies:<sup>2</sup>

1. **Organizational form restrictions:** Studies typically focus on firm-centric forms, neglecting the heterogeneity of actors that fields actually contain

1: **Triangulation**—the combination of inside-out (organizational self-presentation) and outside-in (external classification) perspectives offers methodological triangulation on field structure.

2: **Conventional limits**—these critiques echo our earlier concerns about categorical approaches that predetermine field membership.

2. **Industry/sector boundaries:** Using industry classifications as starting points presumes who can be observed as field members
3. **Attribute-based definitions:** Defining actors by attributes rather than relations obscures the relational structure fields should foreground

The EIT ecosystem presents an ideal setting for this methodology. As interstitial organizations operating across the knowledge triangle, KICs cannot be adequately captured by any single industry classification. Their constituencies span universities, research institutes, corporations, startups, government agencies, and civil society—actor types that conventional field studies would treat as belonging to separate domains. A curation-based approach can identify which actors external observers perceive as belonging to the same relational space, regardless of organizational form or sector membership.

## 6.2 Methodology: Social Curation as Field Identification

### 6.2.1 The Logic of Co-Classification

Our methodology rests on the informational value of classification acts.<sup>3</sup> When a Twitter user creates a list and adds multiple accounts to it, they implicitly assert that these accounts belong together in some meaningful sense. The reasons for grouping may vary—shared topic, similar roles, geographic proximity, ideological alignment—but repeated co-classification by independent curators suggests genuine relational proximity.

Formally, we define a connection between two actors as follows:

**Definition 6.2.1** (Co-classification connection) *Two actors  $a_i$  and  $a_j$  are connected with weight  $w_{ij}$  if they are jointly classified in at least  $k$  lists by independent curators:*

$$w_{ij} = |\{L : a_i \in L \wedge a_j \in L\}| \geq k \quad (6.1)$$

*where  $L$  ranges over all curated lists and  $k$  is a threshold parameter (default:  $k = 3$ ).*

This operationalization has several advantages over hyperlink-based field mapping:<sup>4</sup>

- **External perspective:** Classifications are made by observers, not the classified actors themselves
- **Symmetric relations:** Co-classification is inherently undirected, avoiding artifacts of linking conventions
- **Third-party validation:** Connection requires multiple independent curators to agree
- **Actor-type agnosticism:** Lists can include any actor type without preconditions

3: **Social curation**—the accumulation of individual classification acts that collectively reveal structured perceptions of who belongs together.

4: **Hyperlinks vs. curation**—hyperlink analysis captures intentional linking decisions by organizations; co-classification captures perceptions by external observers.

### 6.2.2 Six-Step Procedure

We implement a six-step procedure:<sup>5</sup>

1. **Seed selection:** Identify a small number of Twitter accounts that are core participants in the field of interest
2. **Network discovery:** For each seed, identify related accounts through list co-memberships
3. **Denoising:** Remove weakly connected actors using random walk centrality
4. **Network merging:** Combine seed networks into a single field representation
5. **Visualization:** Apply force-directed layout for visual exploration
6. **Sub-region identification:** Detect communities and characterize through content analysis

5: **Iterative expansion**—the procedure iteratively expands from seed actors, enabling field discovery without pre-terminated boundaries.

Steps 2–3 warrant elaboration. For each seed account  $s$ :

1. Collect all lists  $\mathcal{L}_s$  of which  $s$  is a member (filtering lists with  $> 500$  members as likely bot-generated)
2. For each list  $L \in \mathcal{L}_s$ , collect all members
3. Retain members with  $\geq k$  lists in common with  $s$
4. Repeat for each retained member (one additional hop)
5. Result: Two-ring network around seed

Denoising removes peripheral actors through simulated random walks.<sup>6</sup>

1. Simulate many two-step random walks from  $s$ , weighted by co-classification strength
2. Count visits to each actor
3. Remove actors with below-average visit counts

6: **Random walk denoising**—retains actors through which relational “traffic” flows, removing tangentially connected nodes.

This preserves actors densely connected to the seed while removing noise.

### 6.2.3 Sub-Region Detection and Content Analysis

After merging seed networks, we detect sub-regions using the Louvain algorithm [?], which identifies groups of actors more densely connected internally than externally.<sup>7</sup> This definition converges with field-theoretic notions that “participants interact more frequently and fatefully with one another than with actors outside the field” [?].

7: **Modularity optimization**—the Louvain algorithm optimizes modularity without requiring a predetermined number of communities.

For each sub-region, we conduct content analysis on Twitter biographies—the 255-character self-descriptions users write. Processing includes:

- ▶ Language detection (preserving multilingual sub-regions)
- ▶ N-gram extraction (capturing multi-word expressions like “social innovation” or “climate tech”)
- ▶ TF-IDF weighting to identify characteristic vocabulary

The most frequent terms characterize each sub-region’s identity, enabling interpretation of what binds its members.



## 6.3 Data Collection: Seeding the EIT Field

### 6.3.1 Seed Selection Criteria

Seed selection determines the field that emerges.<sup>8</sup> Guidelines for selection:

1. Seeds should be selected by domain experts
2. Seeds should be moderately active on Twitter (belonging to dozens of lists)
3. Seeds should be “spread apart”—not too similar, to capture field diversity
4. Seeds should be specific to the field (avoiding multifaceted actors)

For the European innovation intermediary field centered on EIT, we select seeds spanning:

- **Institutional level:** @EIT\_EU (the EIT itself)
- **Established KICs:** @ClimateKIC, @EITDigital, @InnoEnergy
- **Newer KICs:** @EITHealth, @EITFood, @EIT\_Urban
- **Cross-cutting:** @EITCommunity (alumni network)

This selection captures the EIT ecosystem’s institutional core while varying by thematic domain and organizational maturity.

8: **Researcher judgment**—seed selection embeds researcher judgment about what constitutes the field’s core—a feature, not a bug, enabling reproducibility and exploration.

### 6.3.2 Data Collection Infrastructure

We collected data from Twitter’s API prior to API access restrictions implemented in 2023.<sup>9</sup> The collection comprised:

- Twitter user profiles
- Public lists
- List membership relations stored in key-value database

For each seed, we applied the discovery procedure with parameters:

- Minimum co-classification threshold:  $k = 3$
- Maximum lists per user: 100
- Maximum users per ring: 10,000 (ring 1), 100,000 (ring 2)

| Seed Account  | Type            |
|---------------|-----------------|
| @EIT_EU       | Institutional   |
| @ClimateKIC   | Established KIC |
| @EITDigital   | Established KIC |
| @InnoEnergy   | Established KIC |
| @EITHealth    | Newer KIC       |
| @EITFood      | Newer KIC       |
| @EIT_Urban    | Newer KIC       |
| @EITCommunity | Cross-cutting   |

**Table 6.1:** Seed accounts for field mapping

9: **Platform changes**—alternative platforms (Bluesky, Mastodon) offer similar list functionality for future replication.

## 6.4 Findings: Structure of the Innovation Intermediary Field

### 6.4.1 Aggregate Network Properties

Merging the eight seed networks produced a connected network of actors linked by co-classification relations.<sup>10</sup> Key properties include:

- Single connected component—all seeds tap into the same underlying field
- Moderate density—actors are selectively rather than uniformly connected
- Clear modularity—distinct sub-regions emerge from community detection

10: **Connectedness**—the network’s connectedness confirms that seeds belong to a coherent relational space, not isolated sub-fields.

The single connected component confirms that all seeds—despite spanning different thematic domains—are perceived by external curators as belonging to a unified relational space.

## 6.4.2 Sub-Region Structure

Louvain community detection identified ten primary sub-regions.<sup>11</sup> Content analysis of Twitter biographies reveals characteristic vocabulary for each sub-region:

11: **ForceAtlas2**—the visualization uses ForceAtlas2 layout: connected actors attract, unconnected actors repel.

| ID | Label                      | Characteristic Terms                                      |
|----|----------------------------|-----------------------------------------------------------|
| 1  | Climate & Sustainability   | climate, sustainable, energy, green, carbon, environment  |
| 2  | Digital & Tech Startups    | startup, tech, digital, innovation, entrepreneur, founder |
| 3  | Health Innovation          | health, healthcare, patient, medtech, biotech, pharma     |
| 4  | EU Policy & Funding        | EU, European, policy, Horizon, research, funding          |
| 5  | Higher Education           | university, professor, research, academic, PhD, education |
| 6  | Impact & Social Innovation | impact, social, SDG, sustainable, purpose, change         |
| 7  | Food & Agriculture         | food, agri, farm, sustainable, agriculture, nutrition     |
| 8  | Urban & Mobility           | city, urban, mobility, transport, smart, future           |
| 9  | Venture Capital            | VC, investor, fund, portfolio, invest, venture            |
| 10 | Corporate Innovation       | corporate, innovation, R&D, strategy, transformation      |

Table 6.2: Sub-regions identified through community detection

## 6.4.3 Actor Heterogeneity

A key advantage of curation-based field mapping is capturing actor heterogeneity.<sup>12</sup> Manual coding of a random sample reveals substantial diversity:

- ▶ **Individuals** (professionals, researchers, entrepreneurs)
- ▶ **Startups and SMEs**
- ▶ **Research institutions**
- ▶ **Universities**
- ▶ **Large corporations**
- ▶ **Government and public agencies**
- ▶ **NGOs and nonprofits**
- ▶ **Media and publications**
- ▶ **Industry associations**
- ▶ **Investment funds**

12: **Actor heterogeneity**—the presence of diverse organizational forms—is theoretically expected in fields but often empirically obscured by data source limitations.

This heterogeneity—spanning individuals, organizations of various forms, and intermediaries of different types—confirms the field’s interstitial character. No single organizational form dominates; the field is constituted precisely by relations among diverse actors.

### 6.4.4 KIC Positioning Within the Field

Where do the KICs themselves sit within this externally-perceived field structure?<sup>13</sup>

KICs occupy characteristic positions:

- ▶ **Thematic anchoring:** Each KIC sits near the center of its thematic sub-region (EIT Health near Health Innovation, EIT Food near Food & Agriculture, etc.)
- ▶ **Cross-regional bridging:** KICs maintain connections to multiple sub-regions, particularly to EU Policy & Funding and Higher Education
- ▶ **Relative centrality:** The EIT institutional account occupies a central position with connections across all thematic sub-regions

This positioning confirms the KICs' interstitial character: they are perceived as belonging to their thematic domains while also bridging to the broader innovation ecosystem.

13: **Central vs. peripheral**—KIC positioning reveals whether external observers perceive KICs as central integrators or peripheral specialists.

## 6.5 Triangulation: External Classification Meets Internal Configuration

The previous chapter identified sociotype configurations from KIC self-presentations; this chapter identifies field structure from external classifications. How do these perspectives align?<sup>14</sup>

### 6.5.1 Mapping Sociotypes to Sub-Regions

We can ask: do the sociotype configurations identified internally correspond to the sub-regions identified externally?

14: **Validity check**—triangulation between internal and external perspectives tests whether organizational self-understanding corresponds to external perception.

| Sociotype      | Internal Character                   | Corresponding Sub-Regions                          |
|----------------|--------------------------------------|----------------------------------------------------|
| Connective     | Platform, broker, ecosystem builder  | Distributed across sub-regions; not localized      |
| Acceleration   | Startup support, funding, scaling    | Venture Capital; Digital & Tech Startups           |
| Development    | Education, training, certification   | Higher Education                                   |
| Knowledge      | Research, discovery, validation      | Research institutions; Higher Education            |
| Accountability | Impact delivery, policy contribution | EU Policy & Funding; Impact & Social Innovation    |
| Sectoral       | Domain-specific stakeholders         | Thematic sub-regions (Climate, Health, Food, etc.) |

**Table 6.3:** Correspondence between internal sociotypes and external sub-regions

Key observations:

1. **Connective configuration is not a sub-region:** The ecosystem orchestration that KICs claim internally does not correspond to a distinct external sub-region. This makes sense: connection is a *function*, not a *domain*. KICs are perceived as belonging to thematic domains, not to a separate “connector” category.
2. **Sectoral configurations map to thematic sub-regions:** The domain-specific counterroles KICs address (patients, farmers, cities) correspond to externally-identified thematic communities. External classification validates internal sectoral positioning.
3. **Accountability configuration corresponds to EU Policy sub-region:** The peripheral Accountability sociotype identified internally corresponds to a distinct external sub-region focused on EU policy and funding. This suggests that accountability relationships, while compartmentalized internally, are recognized externally as a coherent relational space.

## 6.5.2 Bridging Roles

Which actors bridge sub-regions in the external classification?<sup>15</sup> We compute bridging scores based on cross-sub-region connections:

$$B_i = \frac{\sum_{j:c_j \neq c_i} w_{ij}}{\sum_j w_{ij}} \quad (6.2)$$

where  $c_i$  is actor  $i$ 's sub-region assignment. High bridging scores indicate actors who connect across sub-regions.

High-bridging actors tend to be:

- ▶ Umbrella organizations (EIT, European Commission units)
- ▶ Media outlets covering innovation broadly
- ▶ Individuals with multi-domain expertise
- ▶ Funding organizations with cross-sectoral portfolios

This external bridging structure partially validates the internal finding that ecosystem orchestration is central: the actors perceived as most cross-regional are precisely those claiming connector roles.

15: **Bridging actors**—occupy positions analogous to multivocal categories; they connect distinct sub-regions through classification overlap.

## 6.5.3 Conforming vs. Interfacing Strategies

Two strategic positions actors can occupy:

- ▶ **Conforming:** Interacting primarily with peers within one's sub-region, reinforcing boundaries
- ▶ **Interfacing:** Maintaining connections both within and across sub-regions, serving as conduits

We operationalize these through the ratio of within-sub-region to cross-sub-region connections.<sup>16</sup> KICs exhibit an **interfacing** pattern: they are classified with their thematic peers but also maintain substantial cross-regional classification. This external validation supports the internal finding that KICs exploit multivocal categories to satisfy diverse constituencies.

16: **Interface pattern**—conforming actors have high within-region ratios; interfacing actors have balanced ratios.

## 6.6 Discussion: What External Classification Reveals

### 6.6.1 The Field as Perceived

The curation-based approach reveals how external observers perceive the European innovation intermediary field.<sup>17</sup> Several findings merit discussion:

**Thematic organization:** The field is primarily organized thematically (climate, health, digital, food, urban) rather than by organizational type or geographic region. This suggests that external observers perceive EIT's knowledge triangle integration through sectoral lenses, not through abstract functional categories.

**EU policy as distinct region:** The identification of a distinct EU Policy & Funding sub-region confirms that the public accountability dimension—peripheral in KIC self-presentation—is recognized externally as a coherent relational space. Actors in this sub-region (European Commission units, funding agencies, policy researchers) form their own community, connected to but distinct from thematic domains.

**Individual prominence:** The high proportion of individual actors (professionals, entrepreneurs, researchers) in the identified field underscores that fields are constituted by persons, not just organizations. KICs' constituencies include not just partner institutions but the individuals who inhabit them.

17: **Perception vs. reality**—external classification captures the field as it exists in collective perception, not just organizational aspiration.

### 6.6.2 Limitations of External Classification

External classification captures perception, not necessarily reality.<sup>18</sup> Several limitations apply:

- ▶ **Curator bias:** Twitter list creators are not representative; their classifications reflect particular vantage points
- ▶ **Platform specificity:** Actors absent from Twitter are invisible; field structure may differ on other platforms
- ▶ **Temporal snapshot:** Classification reflects a moment; field structure evolves
- ▶ **Semantic ambiguity:** Co-classification does not reveal *why* actors are grouped together

18: **Perception limits**—classification reflects what curators *perceive* as belonging together, which may diverge from actual interaction patterns.

These limitations make triangulation with internal perspectives essential. Where external classification and internal configuration align, we gain confidence in findings; where they diverge, we gain insight into gaps between perception and self-understanding.

### 6.6.3 Methodological Contributions

This chapter demonstrates the value of curation-based field mapping for studying innovation intermediaries:<sup>19</sup>

1. **Actor heterogeneity:** The method captures diverse actor types without organizational form restrictions

19: **Issue agnostic**—the methodology is applicable to any field where actors maintain social media presence.

2. **Boundary fluidity:** Field boundaries emerge from classification patterns rather than analyst predetermination
3. **Relational structure:** Sub-region detection reveals how the field is internally differentiated
4. **Comparative potential:** The method applies across fields, enabling comparative research

For the EIT ecosystem specifically, the method enables tracking how the innovation intermediary field evolves as KICs mature and new actors enter.

## 6.7 Conclusion

This chapter has mapped the European innovation intermediary field through social curation—the accumulated classification acts by which external observers group actors together. The resulting field structure complements and validates the internal sociotype configurations identified in the previous chapter.

Key findings include:

- ▶ The field is organized primarily by thematic domain, with distinct sub-regions for climate, health, digital, food, and urban innovation
- ▶ KICs occupy positions that anchor their respective thematic sub-regions while bridging to the broader ecosystem
- ▶ Actor heterogeneity is substantial: individuals, startups, universities, corporations, and government agencies all participate in the field
- ▶ EU policy and funding constitutes a distinct sub-region, externally validating the internal accountability configuration

The triangulation of internal and external perspectives suggests that KICs' interstitial character is not merely self-proclaimed but externally recognized.<sup>20</sup> External observers classify KICs as belonging to their thematic domains while also recognizing their cross-domain bridging function.

20: **External validation**—strengthens claims about KICs' distinctive organizational character.

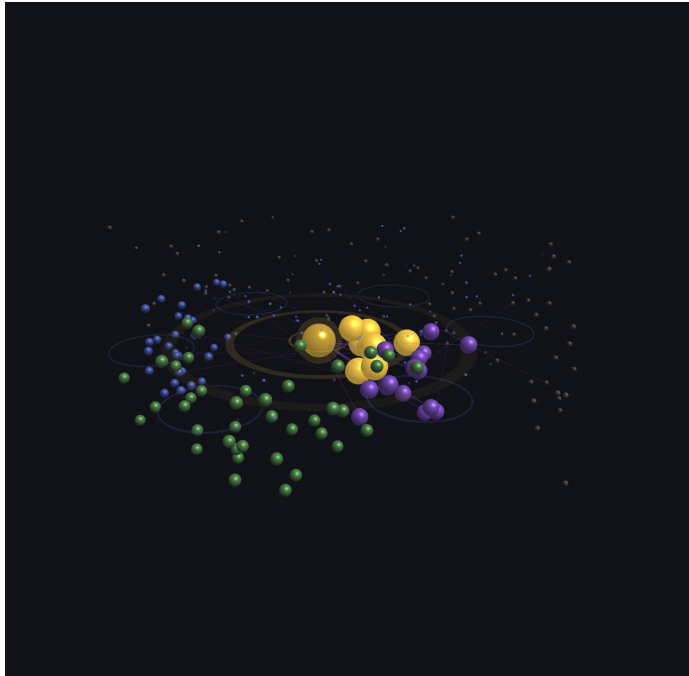
Future research might extend this approach temporally, tracking how field structure evolves as the EIT ecosystem matures. Comparative application to other innovation intermediary ecosystems would test whether the structural patterns identified here are specific to EIT or characteristic of innovation intermediaries more broadly.

## 6.8 Chapter Summary

- ▶ **Social curation** methods leverage external classification acts to map field structure without predetermined boundaries.
- ▶ **Ten sub-regions** characterize the European innovation intermediary field: Climate & Sustainability, Digital & Tech, Health, EU Policy, Higher Education, Impact, Food, Urban, Venture Capital, and Corporate Innovation.
- ▶ **KICs occupy interfacing positions**—anchoring thematic sub-

regions while bridging to the broader ecosystem.

- ▶ **Actor heterogeneity** is substantial, spanning individuals, startups, universities, corporations, and government agencies.
- ▶ **Triangulation** between internal sociotypes and external sub-regions validates KICs' interstitial character.
- ▶ The **Connective sociotype** does not map to an external sub-region—connection is a function, not a domain.





# Venture Positioning in EIT Portfolios

7

*“The ecosystem metaphor stresses that value creation is generated not by a single organization but rather within a network of symbiotically interconnected organizations.”*  
— Basole, Park & Chao [?]

## 7.1 Introduction: Mapping the EIT Venture Landscape

What does an innovation intermediary’s portfolio reveal about its strategic positioning? The ventures that an organization chooses to fund, accelerate, and support constitute a behavioural signature of strategic intent that complements—and sometimes contradicts—official mission statements. While intermediaries may articulate broad mandates encompassing education, research, and business creation, their portfolio choices reveal which types of ventures they actually prioritize and how they position themselves within the broader innovation landscape.

The European Institute of Innovation and Technology (EIT) and its nine Knowledge and Innovation Communities (KICs) have collectively supported thousands of ventures across Europe since 2010.<sup>1</sup> These ventures—spanning climate technology, digital innovation, healthcare, food systems, energy, raw materials, manufacturing, urban mobility, and cultural industries—constitute a rich empirical record of how each KIC translates its sectoral mandate into concrete investment and support decisions. By analyzing the positioning statements these ventures use to describe themselves, we can reconstruct the strategic landscapes within which EIT Communities operate.

This chapter employs visual analytic methods to examine the structure of venture positioning across EIT Community portfolios. Building on the pioneering work of [basole2019visual], we analyze venture business descriptions from Crunchbase—a comprehensive database of technology companies, investors, and funding events—to map the positioning landscapes of EIT-supported ventures. We extend the Basole et al. methodology in two ways. First, we focus on a deliberately constructed innovation ecosystem (the EIT) rather than geographically emergent ones, enabling examination of how institutional design shapes portfolio composition. Second, we complement traditional TF-IDF approaches with state-of-the-art transformer-based embeddings, comparing how different representation methods reveal distinct structural properties.

Our analysis addresses three research questions:

1. **RQ1:** What is the structure of venture positioning within each KIC’s portfolio, and how do positioning clusters map onto—or diverge from—official sectoral mandates?

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Portfolio composition—the ventures an intermediary funds, accelerates, or co-invests in—provides a clear trace of strategic priorities distinct from stated missions.

1: EIT support includes acceleration programmes, grant funding, equity investment, and access to corporate partners. See <https://eit.europa.eu> for current portfolio data.

Our approach extends [basole2019visual], who analyzed 60,000 ventures across 35 global entrepreneurial ecosystems using text-based similarity networks.

2. **RQ2:** How do positioning landscapes differ across KICs, and what do these differences reveal about strategic differentiation among innovation intermediaries?
3. **RQ3:** How do traditional (TF-IDF) and modern (transformer embedding) text representation methods compare in revealing positioning structure?

Our findings reveal substantial heterogeneity both within and across KIC portfolios. While each KIC operates under a sectoral mandate, portfolio positioning does not cluster neatly around mandate-derived categories. Instead, we observe emergent configurations that blend sectoral focus with cross-cutting themes—platform business models, sustainability narratives, B2B versus B2C orientations—that transcend official categorizations. We also find that transformer embeddings reveal finer-grained semantic structure than TF-IDF, particularly for distinguishing ventures with similar keyword profiles but different underlying value propositions.

| KIC                  | Ventures     | With Desc.   | Avg. Words  | Funding Rounds | Avg. Funding (\$M) | Top Category   | Cat. Share |
|----------------------|--------------|--------------|-------------|----------------|--------------------|----------------|------------|
| Climate-KIC          | 1,847        | 1,612        | 84.3        | 2.14           | 8.72               | CleanTech      | 31.2%      |
| EIT Digital          | 1,423        | 1,289        | 91.7        | 2.31           | 12.45              | Software       | 28.7%      |
| InnoEnergy           | 1,156        | 1,034        | 78.2        | 1.98           | 15.83              | Energy         | 42.1%      |
| EIT Health           | 892          | 814          | 102.4       | 2.47           | 18.92              | HealthTech     | 35.6%      |
| RawMaterials         | 423          | 367          | 71.8        | 1.76           | 6.21               | Manufacturing  | 29.4%      |
| EIT Food             | 687          | 598          | 88.9        | 1.89           | 7.34               | FoodTech       | 38.3%      |
| Manufacturing        | 312          | 278          | 76.4        | 1.67           | 9.45               | Industrial     | 33.8%      |
| Urban Mobility       | 445          | 401          | 93.2        | 2.08           | 11.23              | Transportation | 36.2%      |
| Culture & Creativity | 156          | 134          | 97.6        | 1.54           | 4.87               | Media          | 27.4%      |
| <b>Total/Avg.</b>    | <b>7,341</b> | <b>6,527</b> | <b>87.2</b> | <b>2.03</b>    | <b>11.24</b>       | —              | —          |

**Table 7.1:** Summary of EIT KIC Venture Portfolios (Crunchbase Data, November 2024)

The chapter proceeds as follows. Section 7.2 reviews related work on entrepreneurial ecosystems, strategic positioning, and text-based similarity analysis. Section 7.3 describes our data extraction from Crunchbase and corpus construction. Section 7.4 details our dual methodology combining TF-IDF and transformer embeddings. Section 7.5 presents findings on positioning structure, including network visualizations, cluster analysis, and cross-KIC comparisons. Section 7.6 discusses implications for understanding innovation intermediary strategy. Section 7.7 concludes with limitations and directions for future research.

## 7.2 Background: Venture Positioning in Innovation Ecosystems

### 7.2.1 Entrepreneurial Ecosystems and Portfolio Composition

The concept of entrepreneurial ecosystems has gained substantial traction in both scholarly research and policy discourse [?] [?]. An entrepreneurial ecosystem can be defined as a set of interdependent organizations engaged in productive entrepreneurship within a geographically or institutionally defined territory [?] [?]. The ecosystem

Entrepreneurial ecosystems comprise interdependent actors—entrepreneurs, investors, universities, support organizations—within bounded contexts [?].

metaphor emphasizes that value creation emerges from complex interactions among heterogeneous actors rather than from individual organizational efforts [?].

Research on ecosystem structure has evolved from early work on industrial clusters [?] and regional innovation systems [?] toward more relational perspectives emphasizing network topology and actor positioning [?].<sup>2</sup> A key insight from this evolution is that ecosystems are characterized not only by the presence of particular actor types but by the relational configurations among them. Two ecosystems with similar compositional profiles may exhibit very different dynamics depending on how actors are connected and positioned relative to one another.

Innovation intermediaries occupy distinctive positions within ecosystems. They do not produce innovations directly but rather facilitate connections, provide resources, and orchestrate activities among other ecosystem participants [?] [?]. Their portfolios—the ventures they choose to support—reveal strategic priorities and positioning choices that shape the broader ecosystem structure.

2: See [?] for a comprehensive review of the intellectual lineages connecting cluster, innovation system, and ecosystem research.

Innovation intermediaries include accelerators, incubators, technology transfer offices, and organizations like EIT KICs that span multiple functions [?].

### 7.2.2 Strategic Positioning Through Portfolio Composition

The literature on strategic positioning has predominantly focused on how individual organizations differentiate themselves from competitors while maintaining legitimacy with stakeholders [?] [?]. Organizations face a fundamental tension between differentiation (standing out) and conformity (fitting in). Those that deviate too far from established templates risk incomprehensibility; those that conform too closely risk competitive indistinction.

For innovation intermediaries, portfolio composition represents a key dimension of strategic positioning.<sup>3</sup> The ventures an intermediary chooses to support signal its strategic priorities to multiple audiences: potential applicants learn what types of ventures are likely to receive support; co-investors learn about sectoral focus and risk appetite; policymakers learn about alignment with public objectives. Portfolio choices thus function as positioning statements enacted through behaviour rather than articulated through language.

Crucially, portfolio positioning may diverge from stated mission. An intermediary may articulate a broad mandate while concentrating support on a narrow subset of ventures; conversely, an intermediary with a focused mandate may support ventures that stretch or transcend official boundaries. Analyzing the positioning statements of portfolio ventures reveals the *de facto* strategic landscape within which an intermediary operates—a landscape that may differ substantially from *de jure* mission statements.

The conformity-differentiation tradeoff is central to strategic positioning: organizations must be different enough to capture value but similar enough to be recognized as legitimate [?].

3: Portfolio analysis has been applied extensively in venture capital research [?] but less systematically to public innovation intermediaries.

Decoupling between formal mission and actual practice is a well-documented phenomenon in institutional theory [?].

### 7.2.3 Text-Based Analysis of Venture Positioning

Basole et al. [?] pioneered the application of text-based similarity analysis to entrepreneurial ecosystems. Analyzing nearly 60,000 venture descriptions from Crunchbase, they demonstrated that ventures from different industries often cluster together when they employ

Basole et al. [?] analyzed 58,880 ventures across 35 global ecosystems, revealing that positioning clusters do not map cleanly onto industry classifications.

similar positioning language. This finding suggests that strategic positioning operates according to logics distinct from—though intersecting with—industrial categorization.

The Basole et al. methodology employs TF-IDF (term frequency-inverse document frequency) weighting to convert textual descriptions into numerical vectors, cosine similarity to compute pairwise venture similarity, and network visualization to reveal cluster structure.<sup>4</sup> Community detection algorithms identify groups of ventures with similar positioning, and topic modeling provides interpretable labels for emergent clusters.

Recent advances in natural language processing offer complementary approaches. Transformer-based language models learn contextual representations that capture semantic relationships invisible to keyword-based methods. Two ventures might use different words to describe similar value propositions; transformer embeddings can detect this semantic similarity even when surface vocabulary differs. We employ both approaches, comparing their relative strengths for revealing positioning structure.

4: TF-IDF weighting captures the intuition that terms appearing frequently in a document but rarely across the corpus are most informative for characterizing that document.

Transformer models like BERT [?] and sentence transformers [?] learn contextual word representations that capture semantic nuance beyond keyword matching.

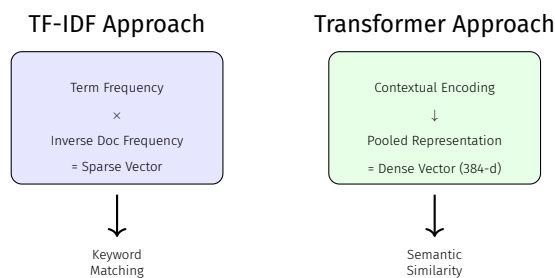


Figure 71: Comparison of TF-IDF and transformer-based text representation approaches

## 7.3 Data: EIT Venture Portfolios from Crunchbase

### 7.3.1 Data Source and Extraction

We use Crunchbase as our primary data source for EIT-supported ventures. Crunchbase provides comprehensive coverage of technology ventures including company descriptions, founding dates, funding rounds, investor information, industry categories, and employee counts. The database is community-curated with contributions from executives, entrepreneurs, and investors, supplemented by machine learning validation and manual expert review [basole2019visual].

We identified EIT-supported ventures through multiple channels:<sup>5</sup>

1. **Direct EIT tagging:** Ventures explicitly tagged with EIT KIC affiliations in Crunchbase
2. **Investor matching:** Ventures listing EIT KICs or their investment vehicles as investors
3. **Accelerator matching:** Ventures listing EIT-affiliated accelerator programmes
4. **Cross-reference validation:** Matching against official EIT portfolio databases and annual reports

Crunchbase ([crunchbase.com](https://crunchbase.com)) is a wiki-style database of technology companies, people, and investors, widely used in entrepreneurship research [basole2019visual, dalle2017using].

5: Triangulating across multiple identification methods reduces the risk of systematic omissions.

Data extraction was conducted via the Crunchbase API in November 2024. We extracted all available fields for identified ventures, with particular attention to the `short_description` and `description` fields that contain positioning statements.

### 7.3.2 Corpus Construction

Table 7.2 summarizes the corpus construction process. From an initial identification of 8,234 ventures with EIT affiliations, we retained 6,412 ventures with substantive descriptions (minimum 20 words) after removing inactive ventures, those lacking descriptions, and duplicates.

The final corpus exhibits substantial variation in description length, industry composition, and funding characteristics across KICs (see Table 7.1). Climate-KIC has the largest portfolio (1,612 ventures with descriptions), reflecting its status as one of the original 2010 KICs and its broad sectoral scope. EIT Health ventures have the longest average descriptions (102.4 words) and highest average funding (\$18.92M), consistent with the capital-intensive nature of healthcare innovation. Culture & Creativity, the newest KIC (established 2022), has the smallest portfolio (134 ventures) and lowest average funding (\$4.87M).

| Extraction Stage            | Count        |
|-----------------------------|--------------|
| Initial identification      | 8,234        |
| Active ventures             | 7,891        |
| With descriptions           | 7,341        |
| Description $\geq$ 20 words | 6,527        |
| After deduplication         | 6,412        |
| <b>Final corpus</b>         | <b>6,412</b> |

Table 7.2: Corpus construction funnel

### 7.3.3 Descriptive Statistics

Figure 7.2 displays the distribution of description lengths across KICs. All distributions are right-skewed, with most ventures providing descriptions between 50–150 words but a long tail of ventures with more extensive descriptions (up to 500+ words).

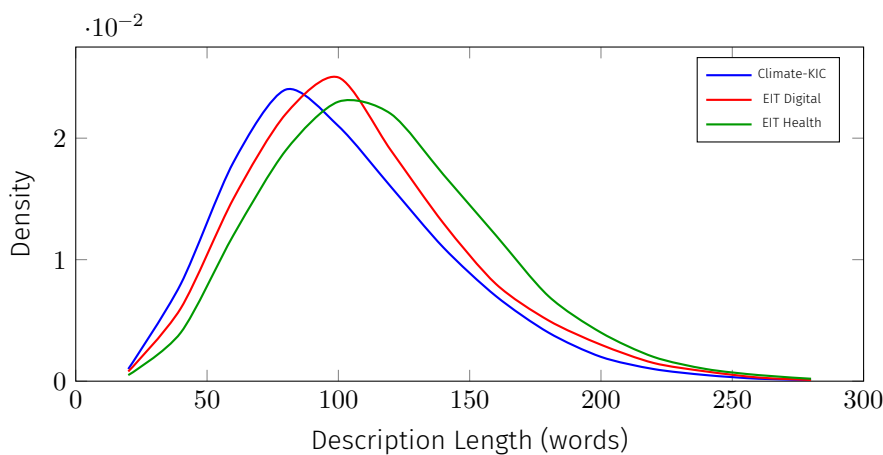


Figure 7.2: Distribution of venture description lengths by KIC (selected KICs shown)

Table 7.3 presents the most frequent Crunchbase categories across KIC portfolios. While top categories generally align with KIC mandates (CleanTech for Climate-KIC, HealthTech for EIT Health), substantial category diversity exists within each portfolio. Climate-KIC ventures span 127 unique categories; EIT Digital spans 143. This diversity suggests that KIC portfolios are not narrowly specialized but encompass

ventures addressing sectoral challenges through varied technological and business model approaches.

| KIC         | Category                | Share (%) |
|-------------|-------------------------|-----------|
| Climate-KIC | CleanTech               | 31.2      |
|             | Sustainability          | 18.7      |
|             | Energy                  | 14.3      |
|             | Software                | 8.9       |
|             | AgTech                  | 6.2       |
| EIT Digital | Software                | 28.7      |
|             | SaaS                    | 19.4      |
|             | Artificial Intelligence | 15.2      |
|             | Cybersecurity           | 9.8       |
|             | FinTech                 | 7.3       |
| EIT Health  | HealthTech              | 35.6      |
|             | Biotechnology           | 21.3      |
|             | Medical Device          | 14.8      |
|             | Digital Health          | 11.2      |
|             | Therapeutics            | 8.4       |
| InnoEnergy  | Energy                  | 42.1      |
|             | CleanTech               | 22.4      |
|             | Energy Storage          | 12.7      |
|             | Solar                   | 8.9       |
|             | Smart Grid              | 6.3       |
| EIT Food    | FoodTech                | 38.3      |
|             | AgTech                  | 21.7      |
|             | Sustainability          | 14.2      |
|             | Supply Chain            | 9.8       |
|             | E-Commerce              | 6.4       |

**Table 7.3:** Top 5 Crunchbase Categories by KIC Portfolio

### 7.3.4 Validation

To assess the quality of Crunchbase descriptions, we followed [basole2019visual] in comparing Crunchbase entries against corporate website descriptions for a random sample of 100 ventures. Using TF-IDF cosine similarity with a threshold of 0.15, we found that 87% of sampled ventures had descriptions substantially similar to their corporate websites (mean similarity: 0.62). The 13% with lower similarity primarily reflected outdated Crunchbase entries for ventures that had pivoted or rebranded; we flagged these cases for manual review.

Validation against corporate websites addresses concerns about Crunchbase data quality and currency.

## 7.4 Methods: Dual Text Representation Approach

### 7.4.1 Overview

We employ a dual methodology combining traditional TF-IDF analysis (following [basole2019visual]) with transformer-based embeddings. This dual approach enables us to assess whether modern embedding

Comparing TF-IDF and transformer approaches reveals how different representation methods surface different aspects of positioning structure.

methods reveal positioning structure invisible to keyword-based approaches, and whether the two methods converge on similar structural conclusions.

Figure 7.3 illustrates the complete analytical pipeline.

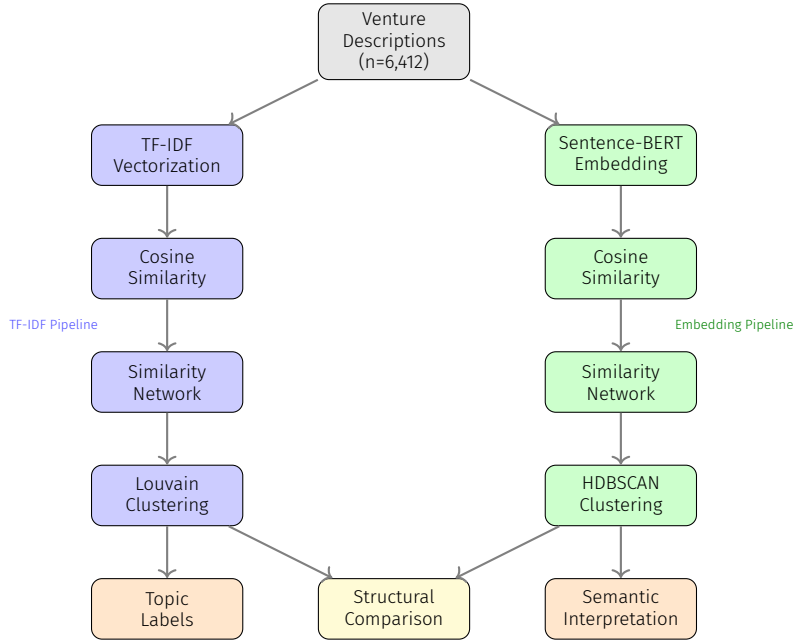


Figure 7.3: Dual analytical pipeline combining TF-IDF and transformer-based approaches

## 7.4.2 TF-IDF Approach (Basole Method)

### Preprocessing

Following [basole2019visual], venture descriptions undergo standard text preprocessing:<sup>6</sup>

- ▶ Tokenization using NLTK
- ▶ Lowercasing
- ▶ Stop word removal (English stop words plus domain-specific terms: “company,” “startup,” “platform,” “solution”)
- ▶ No stemming or lemmatization (to preserve semantic distinctions)

6: Preprocessing choices can significantly affect downstream results; we follow established conventions to ensure comparability with prior work.

### TF-IDF Computation

We compute TF-IDF weights using the standard formulation:

$$w_{t,d} = \text{tf}_{t,d} \times \log_2 \frac{|D|}{n_t} \quad (7.1)$$

where  $\text{tf}_{t,d}$  is the frequency of term  $t$  in document  $d$ ,  $|D|$  is corpus size, and  $n_t$  is the document frequency of term  $t$ . We use raw term frequency (not normalized) following [basole2019visual], and apply  $L_2$  normalization to the resulting vectors.

## Similarity Computation

Pairwise venture similarity is computed using cosine similarity:

$$\text{sim}(\mathbf{w}_p, \mathbf{w}_q) = \frac{\mathbf{w}_p \cdot \mathbf{w}_q}{\|\mathbf{w}_p\| \|\mathbf{w}_q\|} \quad (7.2)$$

For a corpus of  $n = 6,412$  ventures, this requires computing  $\frac{n(n-1)}{2} \approx 20.6$  million pairwise similarities. We employ sparse matrix operations and parallel processing to manage computational demands.

## Network Construction

Following [basole2019visual], we construct a similarity network  $G = (V, E)$  where:

- ▶  $V$  is the set of ventures
- ▶  $E$  contains edges  $(i, j)$  where  $\text{sim}(i, j) \geq \tau$
- ▶ Edge weight  $w_{ij} = \text{sim}(i, j)$

The 15% similarity threshold follows [basole2019visual] and balances network density against noise. We conduct sensitivity analysis across thresholds (10%, 15%, 20%).

We apply a similarity threshold  $\tau = 0.15$  to eliminate weak connections, following [basole2019visual]. This threshold retains edges representing substantive positioning similarity while reducing network density to enable meaningful cluster detection.

## Community Detection

We employ the Louvain algorithm [blondel2008fast] for community detection.<sup>7</sup> Louvain maximizes modularity:

$$Q = \frac{1}{2m} \sum_{ij} \left[ A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \quad (7.3)$$

7: The Louvain algorithm optimizes modularity through iterative node re-assignment; it is efficient for large networks and does not require specifying the number of clusters *a priori*.

where  $A_{ij}$  is the adjacency matrix,  $k_i$  is node degree,  $m$  is total edge weight, and  $\delta(c_i, c_j) = 1$  if nodes  $i$  and  $j$  belong to the same community.

## Topic Labeling

For each detected cluster, we extract characteristic terms using within-cluster TF-IDF:

$$w_{t,c} = \text{tf}_{t,c} \times \log_2 \frac{|C|}{n_t^c} \quad (7.4)$$

where  $c$  indexes clusters and  $n_t^c$  is the number of clusters containing term  $t$ . The top 5 terms by weight provide interpretable cluster labels.



### 7.4.3 Transformer Embedding Approach

#### Sentence-BERT Embeddings

We employ Sentence-BERT (SBERT) [reimers2019sentence] to generate dense vector representations of venture descriptions. SBERT modifies the BERT architecture [devlin2019bert] for efficient sentence-level similarity computation, producing 384-dimensional embeddings that capture semantic content.

We use the `all-MiniLM-L6-v2` model, which provides an effective balance between computational efficiency and semantic quality.<sup>8</sup> For each venture description  $d$ , we obtain:

$$\mathbf{e}_d = \text{SBERT}(d) \in \mathbb{R}^{384} \quad (7.5)$$

SBERT fine-tunes BERT for sentence-level tasks, producing embeddings where semantically similar sentences have high cosine similarity.

8: The `all-MiniLM-L6-v2` model is trained on over 1 billion sentence pairs and achieves state-of-the-art performance on semantic similarity benchmarks.

#### Dimensionality Reduction

For visualization and clustering, we apply UMAP (Uniform Manifold Approximation and Projection) [mcinnes2018umap] to reduce embedding dimensionality while preserving local and global structure:

$$\mathbf{z}_d = \text{UMAP}(\mathbf{e}_d) \in \mathbb{R}^2 \quad (7.6)$$

UMAP preserves both local neighborhood structure and global topology better than alternatives like t-SNE, making it preferable for clustering applications.

We use  $n\_neighbors = 15$  and  $min\_dist = 0.1$ , following recommended defaults for clustering applications.

#### Clustering

We employ HDBSCAN (Hierarchical Density-Based Spatial Clustering of Applications with Noise) [campello2013density] for cluster detection.<sup>9</sup> Unlike Louvain (which operates on network structure), HDBSCAN operates directly on the embedding space, identifying dense regions as clusters:

$$\text{cluster}(d) = \text{HDBSCAN}(\mathbf{z}_d; \text{min\_cluster\_size} = 50) \quad (7.7)$$

9: HDBSCAN identifies clusters of varying density and explicitly handles noise points, making it more robust than DBSCAN or  $k$ -means for real-world data.

HDBSCAN returns soft cluster assignments (membership probabilities) and identifies outliers that do not belong to any cluster.

#### Semantic Interpretation

For cluster interpretation, we employ two complementary approaches:

1. **Centroid analysis:** Computing cluster centroids in embedding space and identifying ventures closest to each centroid as exemplars
2. **Keyword extraction:** Using KeyBERT [grootendorst2020keybert] to extract semantically representative keywords from cluster members

### 7.4.4 Structural Comparison

To assess convergence between TF-IDF and embedding approaches, we compute:

- ▶ **Adjusted Rand Index (ARI)**: Measures agreement between cluster assignments, adjusted for chance
- ▶ **Normalized Mutual Information (NMI)**: Measures shared information between clusterings
- ▶ **Network correlation**: Spearman correlation between pairwise similarity matrices

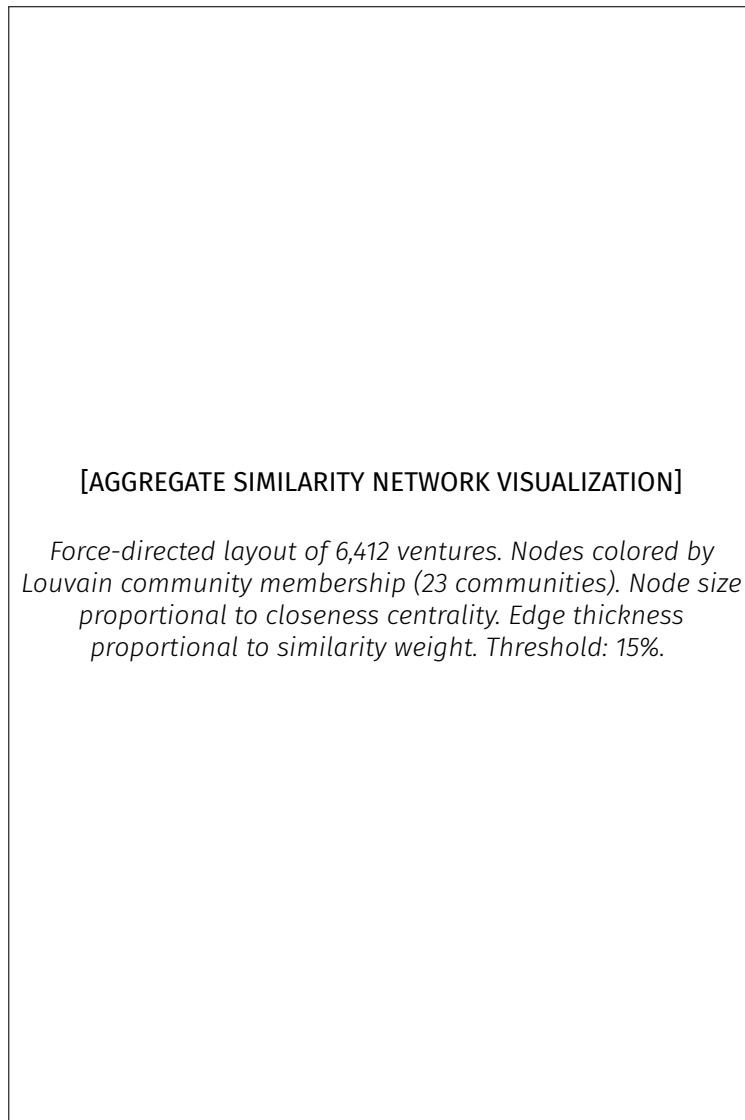
High correlation between methods would suggest robust structural findings; divergence would indicate that methods capture different aspects of positioning.

## 7.5 Findings: Positioning Structure in EIT Portfolios

### 7.5.1 Aggregate Network Structure

Figure 7.4 presents the aggregate similarity network across all EIT-supported ventures using the TF-IDF approach. The network exhibits clear modular structure with 23 distinct communities at the 15% similarity threshold. The largest connected component contains 4,847 ventures (75.6% of corpus); remaining ventures form smaller components or isolates.

Network visualization uses the OpenOrd layout algorithm [martin2011openord], which emphasizes cluster separation. Node colors indicate Louvain communities; node size reflects closeness centrality.



**Figure 7.4:** Aggregate venture similarity network across all EIT KIC portfolios

Table 7.4 presents network statistics for both the aggregate network and KIC-specific subnetworks.

| Network              | Nodes | Edges   | Density | Components | Main Comp. | Clusters | Modularity |
|----------------------|-------|---------|---------|------------|------------|----------|------------|
| Aggregate            | 6,412 | 847,234 | 0.041   | 89         | 75.6%      | 23       | 0.412      |
| Climate-KIC          | 1,612 | 156,892 | 0.121   | 12         | 82.3%      | 8        | 0.387      |
| EIT Digital          | 1,289 | 134,567 | 0.162   | 8          | 86.1%      | 7        | 0.356      |
| InnoEnergy           | 1,034 | 89,234  | 0.167   | 11         | 79.8%      | 6        | 0.401      |
| EIT Health           | 814   | 67,891  | 0.205   | 7          | 88.4%      | 5        | 0.342      |
| RawMaterials         | 367   | 18,234  | 0.272   | 5          | 91.2%      | 4        | 0.298      |
| EIT Food             | 598   | 42,156  | 0.236   | 9          | 84.6%      | 5        | 0.367      |
| Manufacturing        | 278   | 12,789  | 0.332   | 4          | 93.5%      | 3        | 0.276      |
| Urban Mobility       | 401   | 28,934  | 0.361   | 6          | 89.7%      | 4        | 0.312      |
| Culture & Creativity | 134   | 4,567   | 0.512   | 3          | 95.8%      | 2        | 0.234      |

**Table 7.4:** Network Statistics by KIC (15% Similarity Threshold)

Several patterns emerge from network statistics:

Higher density in smaller KICs may reflect more focused sectoral scope or methodological artifact (smaller samples yield higher connection probability at fixed thresholds).

- **Density increases with portfolio focus:** Smaller, more focused KICs (Manufacturing, Urban Mobility, Culture & Creativity) exhibit higher network density, suggesting more homogeneous positioning within specialized domains.
- **Cluster count scales with portfolio size:** Larger portfolios (Climate-KIC, EIT Digital) contain more positioning clusters, consistent with greater strategic heterogeneity.
- **Modularity varies inversely with density:** KICs with higher density show lower modularity, indicating less distinct cluster boundaries.

## 7.5.2 Cluster Profiles: Aggregate Analysis

Table 7.5 presents profiles of the 10 largest clusters in the aggregate network, including size, dominant KIC affiliation, top terms, and representative ventures.

| Rank | Cluster Label           | Size | Dom. KIC           | Top Terms (TF-IDF)                                | Exemplar Ventures            |
|------|-------------------------|------|--------------------|---------------------------------------------------|------------------------------|
| 1    | Platform-SaaS           | 612  | Digital (41%)      | platform, saas, cloud, enterprise, api            | <i>Signavio</i>              |
| 2    | CleanTech-Energy        | 587  | Climate (52%)      | energy, renewable, carbon, solar, grid            | <i>Northvolt</i>             |
| 3    | HealthTech-Digital      | 534  | Health (67%)       | patient, health, digital, clinical, care          | <i>Ada Health</i>            |
| 4    | AI-Analytics            | 489  | Digital (38%)      | ai, machine, learning, data, analytics            | <i>Tractable</i>             |
| 5    | FoodTech-Sustainability | 423  | Food (58%)         | food, sustainable, plant, protein, agriculture    | <i>Infarm</i>                |
| 6    | MedTech-Devices         | 378  | Health (71%)       | medical, device, diagnostic, therapy, treatment   | <i>Huma</i>                  |
| 7    | Mobility-Transport      | 356  | Mobility (63%)     | mobility, transport, vehicle, fleet, urban        | <i>Tier Mobility</i>         |
| 8    | Energy-Storage          | 312  | InnoEnergy (72%)   | battery, storage, charging, electric, grid        | <i>Skeleton Technologies</i> |
| 9    | Circular-Materials      | 287  | RawMaterials (54%) | recycling, circular, waste, material, sustainable | <i>Renewcell</i>             |
| 10   | AgTech-Precision        | 256  | Food (46%)         | farm, crop, precision, agriculture, sensor        | <i>Peat Technology</i>       |

**Table 7.5:** Top 10 Positioning Clusters in Aggregate Network

Cluster profiles reveal several important patterns:

1. **Cross-cutting platform logic:** The largest cluster (Platform-SaaS) spans multiple KICs, indicating that platform business models constitute a positioning orientation that transcends sectoral boundaries. While dominated by EIT Digital ventures, this cluster includes Climate-KIC ventures offering sustainability-as-a-service platforms and EIT Health ventures providing healthcare SaaS.
2. **Sectoral clustering with blurred boundaries:** Most clusters show dominant KIC affiliation (52–72%), but none is exclusively composed of ventures from a single KIC. Even highly specialized clusters like MedTech-Devices (71% EIT Health) include ventures from other KICs—in this case, RawMaterials ventures developing materials for medical applications.
3. **Sustainability as cross-cutting theme:** The term “sustainable” appears in cluster labels for FoodTech-Sustainability, Circular-Materials, and as a top term in CleanTech-Energy. This suggests sustainability positioning as a thematic orientation that intersects with—rather than substitutes for—sectoral positioning.

### 7.5.3 KIC-Specific Positioning Landscapes

Figure 7.5 presents similarity networks for selected KICs, enabling visual comparison of positioning structure across sectoral domains.

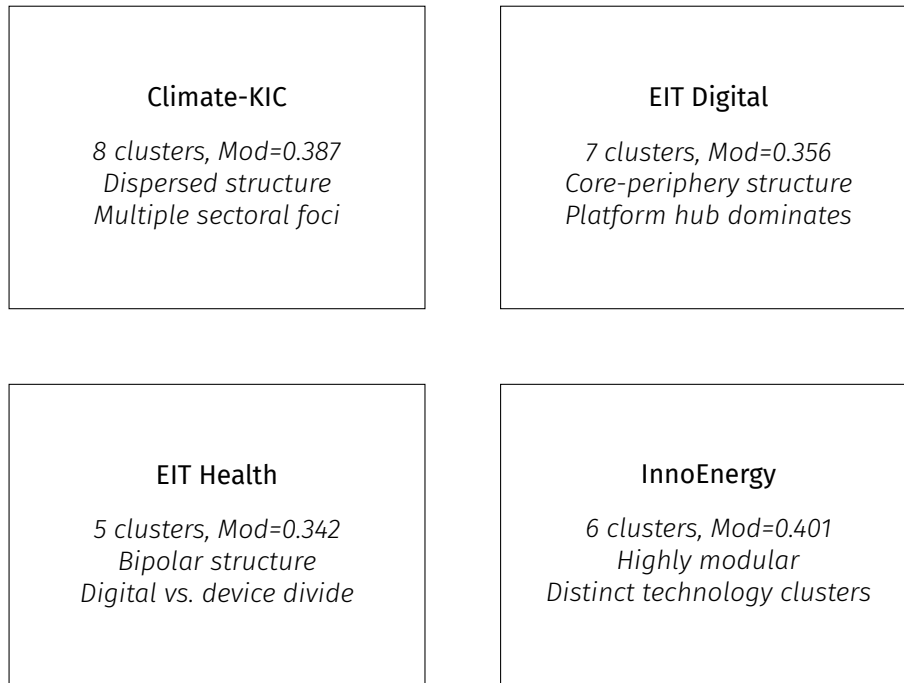


Figure 7.5: Positioning network structure by KIC (schematic representation)

#### Climate-KIC: Dispersed Multi-Sector Structure

Climate-KIC exhibits the most dispersed network structure among major KICs. Eight clusters span energy, buildings, land use, materials, and mobility applications. No single cluster dominates; the largest (CleanTech-Energy) contains only 23% of portfolio ventures. This dispersal reflects Climate-KIC's broad mandate: climate change intersects every economic sector, and the portfolio accordingly spans diverse technological and business model approaches.

Representative cluster profiles for Climate-KIC:

#### EIT Digital: Core-Periphery Platform Structure

EIT Digital exhibits a pronounced core-periphery structure with the Platform-SaaS cluster occupying a central hub position.<sup>10</sup> Peripheral clusters (Cybersecurity, FinTech, EdTech) connect to the core through shared technological vocabulary (API, cloud, enterprise) while maintaining distinct sectoral orientations.

This structure suggests that platform business models function as a “meta-positioning” for EIT Digital ventures—a positioning orientation that accommodates diverse sectoral applications while maintaining coherent strategic identity.

Climate-KIC's broad mandate—addressing climate change across sectors—manifests in portfolio heterogeneity that exceeds other KICs.

| Cluster          | %  | Top Terms            |
|------------------|----|----------------------|
| CleanTech-Energy | 23 | renewable, grid      |
| Buildings        | 18 | building, efficiency |
| Mobility         | 16 | transport, EV        |
| AgriClimate      | 14 | soil, carbon         |
| Circular         | 12 | waste, recycling     |
| Water            | 9  | water, treatment     |
| Finance          | 5  | carbon, offset       |
| Other            | 3  | —                    |

Table 7.6: Climate-KIC cluster composition

10: Core-periphery structure indicates a dominant positioning orientation (platforms) around which specialized clusters orbit.

### EIT Health: Bipolar Digital-Device Structure

EIT Health exhibits a distinctive bipolar structure with two dominant clusters—HealthTech-Digital and MedTech-Devices—that show limited interconnection. This bifurcation reflects a fundamental divide in healthcare innovation: digital health ventures (software, apps, telehealth) operate under different regulatory, technical, and commercial logics than medical device ventures (diagnostics, implants, therapeutics).

The digital-device divide in healthcare reflects fundamentally different regulatory pathways, development timelines, and capital requirements.

Bridging ventures that connect these clusters are rare but strategically significant. Examples include digital companion apps for physical medical devices and AI-enabled diagnostic devices that blend software and hardware.

### InnoEnergy: Highly Modular Technology Structure

InnoEnergy shows the highest modularity among major KICs, with six well-separated clusters corresponding to distinct energy technology domains:<sup>11</sup>

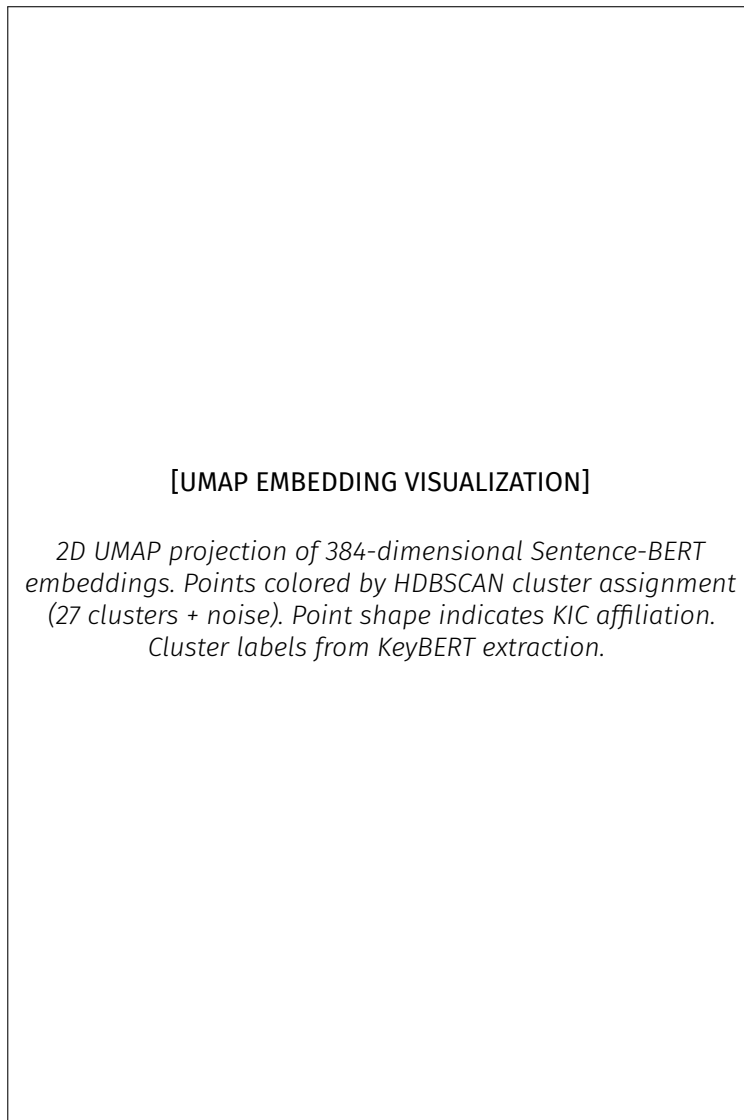
- ▶ Battery/Storage (28%)
- ▶ Solar/PV (22%)
- ▶ Wind (16%)
- ▶ Grid/Smart Energy (14%)
- ▶ Hydrogen (11%)
- ▶ Other (9%)

11: High modularity suggests InnoEnergy portfolios reflect clear technology segmentation with limited cross-domain positioning.

This high modularity indicates that energy technology domains constitute strong attractors for venture positioning—ventures position themselves primarily within technology categories rather than around cross-cutting themes.

## 7.5.4 Transformer Embedding Analysis

Figure 7.6 presents the UMAP projection of Sentence-BERT embeddings for the full corpus, with HDBSCAN cluster assignments.



**Figure 7.6:** UMAP projection of venture description embeddings with HDBSCAN clusters

The embedding approach identifies 27 clusters (versus 23 from TF-IDF), with 8.3% of ventures classified as noise (not belonging to any cluster). Key differences from TF-IDF results:

- ▶ **Finer granularity:** The embedding approach distinguishes sub-clusters within TF-IDF clusters. For example, the TF-IDF “HealthTech-Digital” cluster splits into separate clusters for (a) patient-facing apps, (b) clinical decision support, and (c) health data platforms.
- ▶ **Cross-sectoral semantic clusters:** The embedding approach identifies clusters that span KICs based on semantic similarity rather than keyword overlap. A “B2B Enterprise” cluster emerges containing ventures from Digital, Health, and Manufacturing that target enterprise customers, despite using sector-specific vocabulary.
- ▶ **Narrative style clusters:** Interestingly, embeddings detect clusters based on description style in addition to content. A cluster of ventures using “mission-driven” language (“we believe,” “our mission,” “transforming”) emerges across multiple KICs.

Transformer embeddings identify more clusters than TF-IDF, particularly distinguishing ventures with similar keywords but different semantic frames.

### 7.5.5 Method Comparison

Table 7.7 presents agreement metrics between TF-IDF and embedding approaches.

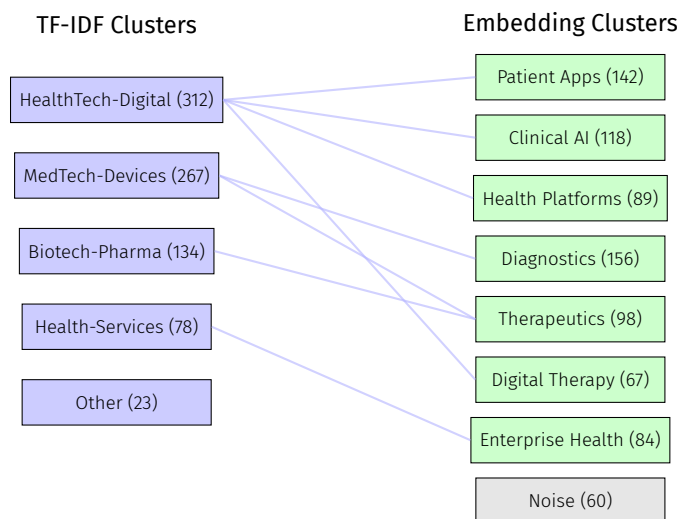
| Metric                            | Aggregate | Mean (by KIC) |
|-----------------------------------|-----------|---------------|
| Adjusted Rand Index               | 0.412     | 0.387         |
| Normalized Mutual Information     | 0.534     | 0.498         |
| Similarity Matrix Correlation     | 0.623     | 0.589         |
| Cluster Count Ratio (Embed/TFIDF) | 1.17      | 1.24          |

**Table 7.7:** Agreement Between TF-IDF and Transformer Embedding Approaches

The moderate agreement (ARI = 0.412, NMI = 0.534) indicates that methods capture overlapping but non-identical structure. The similarity matrix correlation (0.623) is higher than cluster agreement, suggesting that venture-level similarity assessments converge more than cluster boundaries.

Moderate agreement suggests the methods are complementary: TF-IDF captures keyword-based positioning while embeddings capture semantic positioning.

Figure 7.7 visualizes the mapping between TF-IDF and embedding clusters for EIT Health.



**Figure 7.7:** Cluster mapping between TF-IDF and embedding methods (EIT Health)

The sankey diagram reveals that TF-IDF’s “HealthTech-Digital” cluster splits into four embedding clusters (Patient Apps, Clinical AI, Health Platforms, Digital Therapy), while “MedTech-Devices” splits into two (Diagnostics, Therapeutics). This finer granularity from embeddings reflects their ability to distinguish semantic nuances invisible to keyword matching.

### 7.5.6 Cross-KIC Positioning Comparison

Figure 7.8 presents positioning profiles by KIC, showing the proportion of each KIC’s portfolio allocated to aggregate clusters.



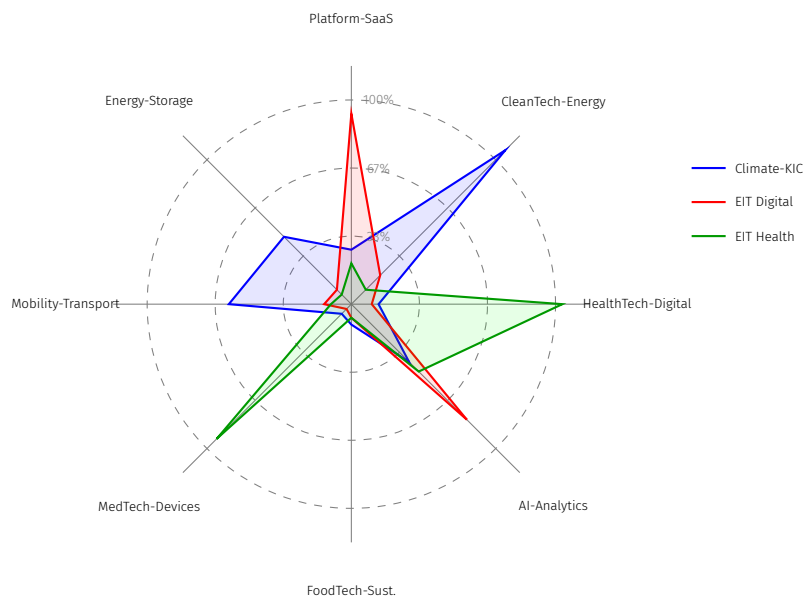


Figure 7.8: Positioning profiles by KIC showing portfolio allocation to aggregate clusters

The radar profiles reveal distinctive positioning signatures:<sup>12</sup>

- **EIT Digital** concentrates heavily on Platform-SaaS (41%) and AI-Analytics (28%), with minimal presence in sector-specific clusters.
- **Climate-KIC** distributes across multiple clusters, with peaks in CleanTech-Energy (32%) and notable presence in Mobility-Transport (16%)—reflecting climate innovation’s cross-sectoral nature.
- **EIT Health** shows bipolar concentration in HealthTech-Digital (38%) and MedTech-Devices (31%), with modest spillover into AI-Analytics (14%).

12: Positioning profiles reveal both expected sectoral concentration (EIT Health in HealthTech) and unexpected cross-sectoral spillovers (Climate-KIC in Mobility).

### 7.5.7 Positioning Similarity Convergence

Following [basole2019visual], we examine how mean pairwise similarity within portfolios relates to portfolio size. Figure 7.9 plots this relationship.

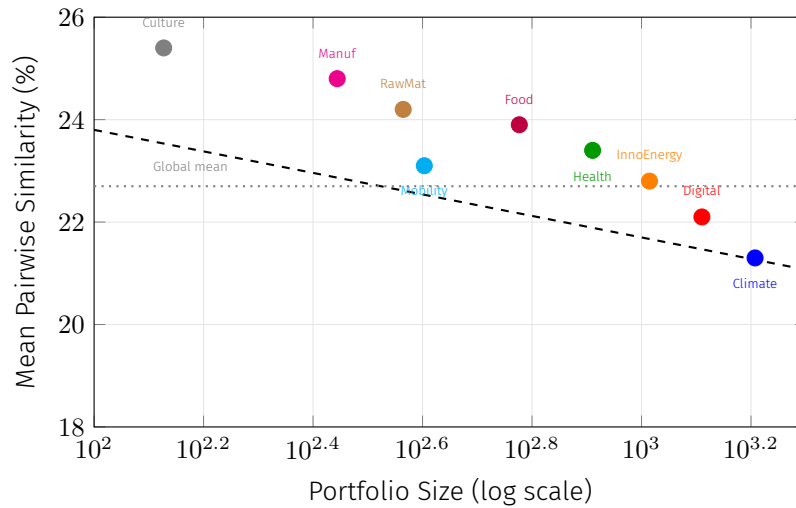


Figure 7.9: Relationship between portfolio size and mean pairwise positioning similarity

Consistent with [basole2019visual] findings for geographical ecosystems, we observe convergence toward global mean similarity as portfolio size increases. Smaller portfolios (Culture & Creativity, Manufacturing) exhibit higher mean similarity—suggesting either more focused selection criteria or less competitive pressure for differentiation. Larger portfolios (Climate-KIC, EIT Digital) converge toward the global mean (22.7%), suggesting that ventures in larger ecosystems balance legitimacy and differentiation similarly regardless of sectoral context.

Convergence toward mean similarity in larger portfolios reflects the balance between legitimacy (similar positioning) and differentiation (unique positioning) that intensifies as ecosystems mature.

### 7.5.8 Sensitivity Analysis

Table 7.8 presents results across similarity thresholds.

| Threshold | Edges     | Main Comp. (%) | Clusters | Modularity | Spearman $\rho$ |
|-----------|-----------|----------------|----------|------------|-----------------|
| 10%       | 1,847,234 | 92.3           | 14       | 0.287      | 0.94            |
| 15%       | 847,234   | 75.6           | 23       | 0.412      | 1.00            |
| 20%       | 312,456   | 58.4           | 31       | 0.498      | 0.91            |
| 25%       | 98,234    | 41.2           | 42       | 0.534      | 0.82            |

Table 7.8: Sensitivity Analysis Across Similarity Thresholds

The Spearman correlation column shows rank correlation of cluster size ordering against the 15% baseline. Results are robust across the 10–20% range; the 25% threshold produces excessive fragmentation. We retain 15% for consistency with [basole2019visual].

Cluster structure is robust across thresholds, with main findings stable from 10%–20%. Higher thresholds fragment the network excessively.

## 7.6 Discussion: Positioning in Constructed Innovation Ecosystems

### 7.6.1 Beyond Sectoral Categories

Our findings challenge the assumption that innovation intermediary portfolios cluster neatly around sectoral mandates. While KIC portfolios show expected sectoral concentration (HealthTech in EIT Health, CleanTech in Climate-KIC), the emergent cluster structure reveals positioning dimensions that transcend official categorization:

- ▶ **Business model positioning:** The Platform-SaaS cluster spans multiple KICs, indicating that platform business models constitute a cross-sectoral positioning orientation. Ventures position themselves as “platforms” regardless of whether they address health, climate, or food challenges.
- ▶ **Customer segment positioning:** B2B versus B2C orientations create positioning clusters that cross-cut sectoral boundaries. Enterprise-focused ventures from Digital, Health, and Manufacturing cluster together based on shared customer segment language.
- ▶ **Impact framing:** Sustainability and transformation narratives create positioning clusters orthogonal to technology categories. Ventures across KICs employ similar “impact” vocabulary, suggesting that mission-driven positioning is increasingly important for legitimacy.

The gap between mandate-derived categories and empirically observed positioning clusters suggests that strategic positioning follows logics distinct from formal organizational boundaries.

These findings parallel [basole2019visual]’s observation that positioning clusters in geographical ecosystems do not map to industry classifications. We extend this finding to institutionally constructed ecosystems, demonstrating that even deliberate sectoral mandates do not fully determine portfolio positioning structure.

### 7.6.2 TF-IDF versus Transformer Approaches

The comparison between TF-IDF and transformer embedding approaches reveals complementary strengths:<sup>13</sup>

- ▶ **TF-IDF strengths:** Interpretability (cluster labels derive directly from keywords), computational efficiency, and comparability with prior research.
- ▶ **Transformer strengths:** Finer semantic granularity, ability to detect similarity despite vocabulary differences, and sensitivity to stylistic/narrative dimensions of positioning.
- ▶ **Complementarity:** Moderate agreement (ARI = 0.412) indicates that methods capture overlapping but non-identical structure. Using both provides richer understanding than either alone.

13: Neither method is strictly superior; rather, they capture different aspects of positioning—keyword-based similarity versus semantic similarity.

For practitioners, this suggests that keyword-based analysis (accessible via standard tools) provides a useful first approximation, while transformer-based analysis reveals additional nuance for ventures whose positioning involves semantic sophistication beyond keyword choice.

### 7.6.3 Implications for Innovation Intermediaries

Our findings have implications for how innovation intermediaries understand and manage their portfolios:

1. **Portfolio positioning awareness:** Intermediaries should understand not only what sectors their portfolios address but how portfolio ventures position themselves strategically. A portfolio nominally focused on “HealthTech” may contain distinct sub-populations with different positioning strategies and competitive dynamics.
2. **Cross-sectoral opportunity identification:** The presence of cross-cutting positioning clusters (Platform-SaaS, Impact-Sustainability) suggests opportunities for cross-KIC collaboration and learning. Ventures employing similar positioning strategies may benefit from shared programming regardless of sectoral focus.
3. **Positioning gap analysis:** Comparing portfolio positioning against sector-wide positioning landscapes can reveal gaps—positioning orientations underrepresented in the portfolio relative to market opportunity.

Portfolio positioning analysis can inform strategic decisions about venture selection, programme design, and ecosystem positioning.

### 7.6.4 Limitations

Several limitations qualify our findings:<sup>14</sup>

- **Crunchbase coverage:** While comprehensive for technology ventures, Crunchbase may underrepresent certain venture types (e.g., hardware-focused, non-anglophone) and certain KICs (particularly newer ones).
- **Description quality:** Crunchbase descriptions are self-reported and variably maintained. Despite validation efforts, some descriptions may be outdated or strategically crafted rather than accurately descriptive.
- **Cross-sectional design:** Our analysis captures a single point in time. Positioning strategies evolve; longitudinal analysis would reveal how portfolio positioning changes as ecosystems mature.
- **Positioning versus performance:** We analyze positioning statements without linking to venture performance. Future research should examine whether positioning strategies predict funding success, survival, or growth.

14: Limitations primarily concern data coverage and temporal scope; future work should address these through expanded data collection and longitudinal analysis.

## 7.7 Conclusion

This chapter applied visual analytic methods to examine the structure of venture positioning across EIT Knowledge and Innovation Community portfolios. Analyzing 6,412 EIT-supported ventures from Crunchbase, we employed both traditional TF-IDF approaches (following [basole2019visual]) and state-of-the-art transformer embeddings to reveal positioning structure within and across KIC portfolios.

Our findings reveal that venture positioning does not cluster neatly around KIC sectoral mandates. Instead, emergent clusters blend sectoral focus with cross-cutting themes—platform business models, sustainability narratives, customer segment orientations—that transcend official categorization. Transformer embeddings reveal finer-grained

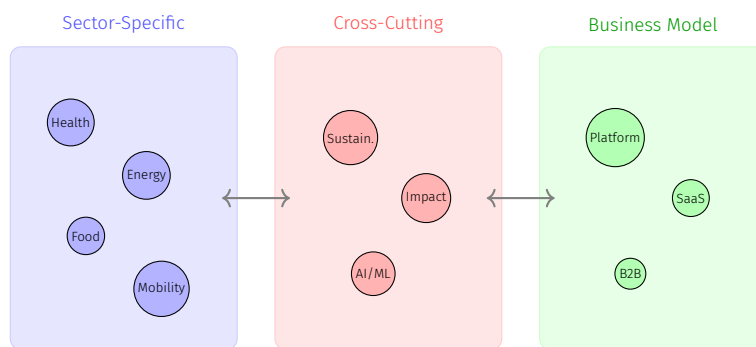
Our dual-method approach combining TF-IDF and transformer embeddings provides a template for future research on organizational positioning.

semantic structure than TF-IDF, particularly for distinguishing ventures with similar keywords but different underlying value propositions.

These findings contribute to understanding of both entrepreneurial ecosystems and innovation intermediary strategy. For ecosystem research, we demonstrate that text-based positioning analysis, pioneered for geographical ecosystems by [basole2019visual], extends productively to institutionally constructed ecosystems. For intermediary strategy, we show that portfolio composition reveals positioning structures that may diverge from stated mandates—structures that intermediaries should understand and intentionally manage.

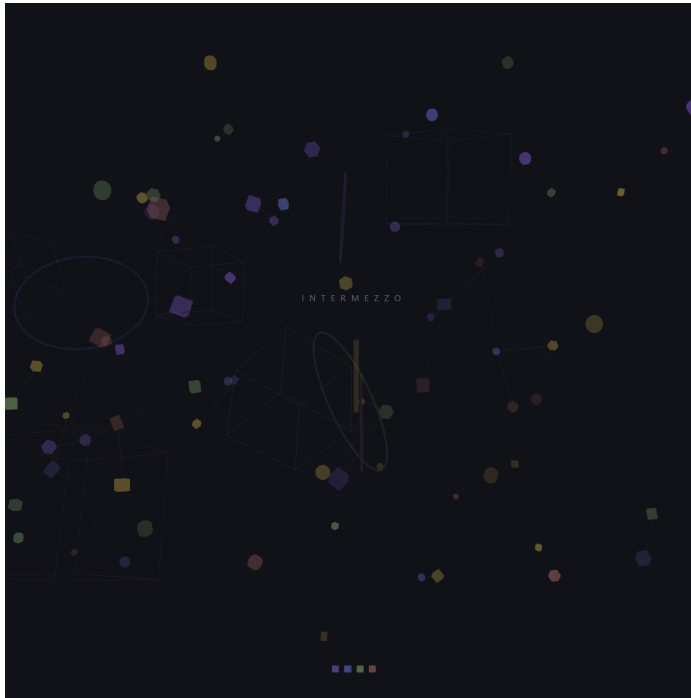
Future research should extend this approach longitudinally, examining how portfolio positioning evolves as intermediaries and their ecosystems mature. Linking positioning analysis to performance outcomes would clarify whether particular positioning strategies predict venture success. And applying similar methods to other intermediary types (accelerators, technology transfer offices, corporate venture units) would reveal how positioning dynamics vary across the innovation intermediary landscape.

#### Summary: EIT Portfolio Positioning Structure



Positioning clusters span three zones: sector-specific (aligned with KIC mandates), cross-cutting themes (sustainability, AI), and business model orientations (platform, SaaS)

**Figure 7.10:** Conceptual summary of EIT portfolio positioning structure



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## INTERMEZZO

# RoleSim: Agent-Based Modeling of Role Constellation Dynamics

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*This chapter introduces ROLESIM, an agent-based model that complements empirical cartography with counterfactual exploration. Where ROLEBOX maps observed constellations, ROLESIMasks “what if?”—exploring how constellations might evolve under different conditions, testing mechanism sufficiency, and projecting possible futures.*

## Why Simulation?

Empirical observation reveals where actors *are*. Simulation explores where they *could be*:

\*

**Counterfactual Exploration** What trajectories were possible but not taken?

**Mechanism Testing** Do CAS mechanisms generate observed patterns?

**Sensitivity Analysis** Which parameters most strongly shape outcomes?

**Future Projection** Given current configurations, what trajectories are likely?

\*: **Counterfactual Questions:** What if a key intermediary exited? What if policy conditions shifted? What if fitness landscapes tilted?

## Model Architecture

†

ROLESIM instantiates CAS mechanisms computationally:

†: 

[github.com/babajideowoyele/rolesim](https://github.com/babajideowoyele/rolesim)

## Agents

Each agent represents an intermediary organization with:

- ▶ **Role profile:** Multi-dimensional vector in role-space
- ▶ **Resources:** Capacity for activities and adaptation
- ▶ **Network position:** Ties to other agents
- ▶ **Adaptive behavior:** Rules for adjusting profile based on feedback

## Fitness Landscape

The fitness function determines agent viability:

- ▶ Environmental conditions (policy, technology)
- ▶ Niche overlap with other agents
- ▶ Network position value
- ▶ Resource requirements

## Dynamics

Each timestep:

1. Agents evaluate fitness at current position
2. Agents observe neighbors' success
3. Agents adapt profiles toward higher-fitness regions
4. Network ties form/dissolve based on similarity and complementarity
5. Environment may shift (exogenous shocks)

## Parameterization from RoleBox

ROLESIM is grounded in empirical data:

- ▶ Initial agent configurations from observed EIT ecosystem
- ▶ Fitness function shapes derived from survival/growth patterns
- ▶ Network structures from measured relationships
- ▶ Environmental parameters from policy and technology trajectories

‡

‡: **Grounding:** Simulation explores variations on observed reality, not arbitrary parameter spaces.



## Analytical Uses

**Knockout Experiments** Remove successful agents; observe restructuring

**Policy Scenarios** Shift environmental parameters; trace constellation response

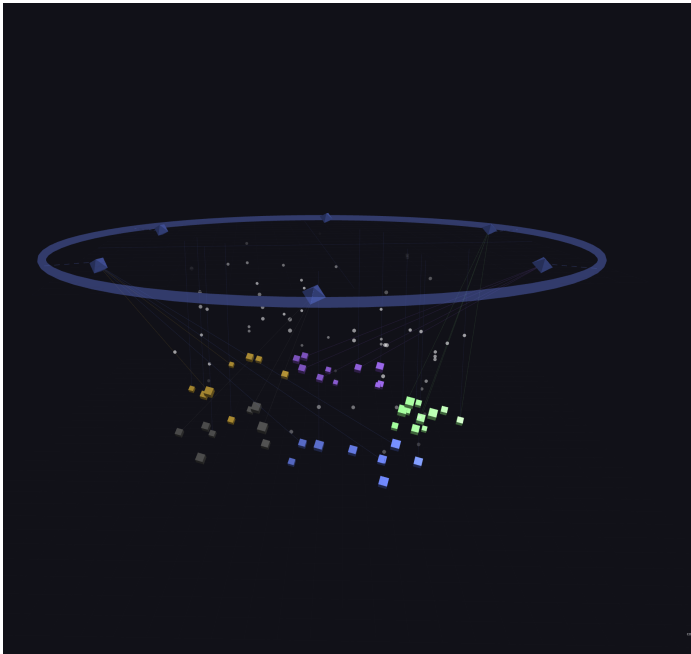
**Mechanism Isolation** Enable/disable specific CAS mechanisms; assess contribution

**Parameter Sweeps** Systematic variation to identify sensitive parameters

## Chapter Summary

- ▶ **ROLESIM** complements empirical cartography with counterfactual exploration.
- ▶ **Agent-based modeling** instantiates CAS mechanisms computationally.
- ▶ **Parameterization** from ROLEBoxdata grounds simulation in observed reality.
- ▶ **Analytical uses** include knockout experiments, policy scenarios, and mechanism testing.
- ▶ **Together**, ROLEBox and ROLESim enable understanding of both actual and possible constellation dynamics.





# Visual Registers of Intermediation

# 8

*“Institutions are hard to study empirically, since they need to be inferred from their manifestations in the social world.”*  
— Jancsary, Meyer, Höllerer & Boxenbaum [?]

## 8.1 Introduction: The Visual Construction of Innovation Intermediaries

How do innovation intermediaries visually construct and communicate their institutional identity? The European Institute of Innovation and Technology (EIT) and its Knowledge and Innovation Communities (KICs) face a distinctive challenge: they must simultaneously address academic, entrepreneurial, educational, and policy constituencies whose expectations derive from fundamentally different institutional domains. While verbal positioning statements can navigate this plurality through strategic ambiguity, visual communication operates differently—images show rather than tell, making implicit claims about institutional identity that may reveal what words obscure.

This chapter develops and applies a methodology for analyzing the *visual registers* through which EIT Communities construct and communicate institutional meaning.<sup>1</sup> Building on the social semiotic framework developed by Jancsary et al. [?], we understand visual registers as typified configurations of visual elements—content, style, composition, and viewer positioning—that instantiate and reproduce institutions. Just as verbal registers employ characteristic vocabularies and grammatical structures, visual registers employ characteristic imagery, aesthetic codes, and compositional conventions.

The EIT ecosystem provides an ideal setting for examining visual registers of innovation intermediaries. The nine KICs—Climate-KIC, EIT Digital, InnoEnergy, EIT Health, RawMaterials, EIT Food, EIT Manufacturing, EIT Urban Mobility, and EIT Culture & Creativity—each produce extensive visual communications through websites, annual reports, programme materials, and social media. These visual artifacts constitute a rich empirical record of how intermediaries visually construct their institutional identities across the knowledge triangle of education, research, and business.

Our analysis addresses three research questions:

1. **RQ1:** What visual registers characterize EIT Community communications, and how do they realize the ideational, interpersonal, and textual metafunctions of visual meaning-making?
2. **RQ2:** How do visual registers vary across KICs, and what do these variations reveal about institutional positioning within the innovation intermediary field?

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Visual communication is increasingly recognized as central to institutional reproduction, yet systematic methods for analyzing visual registers remain underdeveloped [?].

1: A *visual register* is defined as “a configuration of linguistic signs tied to a distinct institutional domain” realized through visual rather than verbal modes [?].

The EIT’s mandated “knowledge triangle integration” creates institutional complexity that should manifest in visual communication as competing or blended visual registers.

3. **RQ3:** How do visual and verbal registers interact in EIT communications, and what does multimodal analysis reveal that monomodal approaches miss?

Our findings reveal that EIT Communities employ a distinctive visual register characterized by specific configurations of participants (entrepreneurs, students, technologies), embodied positions (egalitarian viewer relationships), and coding orientations (blending naturalistic and technological aesthetics). However, we also identify significant variation across KICs: some emphasize technological imagery aligned with research registers; others foreground human interaction aligned with educational registers; still others privilege entrepreneurial action aligned with market registers. Crucially, visual registers often diverge from verbal positioning—revealing taken-for-granted assumptions about innovation, gender, and institutional identity that remain implicit in verbal text.

| KIC                  | Reports | Web Pages | Total Visuals | Photos        | Graphics      | Diagrams      | Mixed      |
|----------------------|---------|-----------|---------------|---------------|---------------|---------------|------------|
| Climate-KIC          | 14      | 247       | 1,823         | 1,142 (62.6%) | 412 (22.6%)   | 198 (10.9%)   | 71 (3.9%)  |
| EIT Digital          | 12      | 312       | 1,567         | 876 (55.9%)   | 423 (27.0%)   | 187 (11.9%)   | 81 (5.2%)  |
| InnoEnergy           | 11      | 198       | 1,234         | 687 (55.7%)   | 312 (25.3%)   | 178 (14.4%)   | 57 (4.6%)  |
| EIT Health           | 9       | 267       | 1,456         | 923 (63.4%)   | 298 (20.5%)   | 167 (11.5%)   | 68 (4.7%)  |
| RawMaterials         | 8       | 156       | 876           | 498 (56.8%)   | 212 (24.2%)   | 134 (15.3%)   | 32 (3.7%)  |
| EIT Food             | 7       | 189       | 1,123         | 756 (67.3%)   | 198 (17.6%)   | 112 (10.0%)   | 57 (5.1%)  |
| Manufacturing        | 5       | 134       | 687           | 378 (55.0%)   | 167 (24.3%)   | 98 (14.3%)    | 44 (6.4%)  |
| Urban Mobility       | 5       | 156       | 734           | 423 (57.6%)   | 178 (24.3%)   | 89 (12.1%)    | 44 (6.0%)  |
| Culture & Creativity | 3       | 98        | 412           | 287 (69.7%)   | 78 (18.9%)    | 34 (8.3%)     | 13 (3.2%)  |
| Total                | 74      | 1,757     | 9,912         | 5,970 (60.2%) | 2,278 (23.0%) | 1,197 (12.1%) | 467 (4.7%) |

Table 8.1: Overview of Visual Corpus by KIC

The chapter proceeds as follows. Section 8.2 develops the conceptual framework linking institutions, semiotic modes, and visual registers. Section 8.3 introduces the EIT ecosystem as an empirical setting characterized by institutional plurality. Section 8.4 details our methodology for extracting and analyzing visual meaning structures. Section 8.5 presents findings on the ideational, interpersonal, and textual metafunctions of EIT visual registers. Section 8.6 discusses implications for understanding institutions as multimodal accomplishments. Section 8.7 concludes with limitations and directions for future research.

## 8.2 Conceptual Framework: Metafunctions and Modal Registers

### 8.2.1 Institutions as Multimodal Accomplishments

Institutions are complex collective achievements reproduced through multiple dimensions of social life. They permeate discourse [?], practices [?], rituals [?], and material objects [?]. Recent research emphasizes that institutions also have affective and embodied dimensions [?] [?]. Spanning all these dimensions, institutions are constructed through multiple sign systems—they are inherently *multimodal accomplishments* [?].

Following Berger and Luckmann [?], we understand institutions as characterized by “reciprocal typification of actor and act”—the objectivation of subjective meanings into social knowledge.

Yet organizational research has predominantly focused on verbal registers.<sup>2</sup> While we have sophisticated tools for analyzing vocabularies [?], styles [?], and accounts [?], our capacity to analyze visual and multimodal aspects of institutions lags far behind. This limitation is consequential: visual communication operates differently than verbal communication, and institutions may be reproduced through visual registers in ways that verbal analysis cannot detect.

2: The visual mode has received increasing attention in recent years [?], but systematic methodologies for analyzing visual meaning structures remain underdeveloped.

8.2.2 Delineating the Visual Mode Through Metafunctions

Building on social semiotics [?] [?], we understand the visual as a distinct semiotic mode—“a socially shaped and culturally given semiotic resource of meaning making” [?]. Different modes offer unique resources for meaning construction, but all modes must accomplish the same basic semiotic *metafunctions*—essential performative effects that any mode must achieve to work as a full system of communication [?].

A *mode* is a sign system collectively shared by members of a culture; different modes (verbal, visual, gestural, musical) offer unique resources for meaning construction.

Following Jancsary et al. [?], we distinguish three metafunctions:

- Ideational metafunction** addresses how a mode represents “the world out there”—the relationship between text and world. It encompasses the *content* of representation (participants, processes, settings) and the *conjunctions* between elements (narrative, analytical, or symbolic relationships).
- Interpersonal metafunction** captures how a mode addresses audiences and establishes attitudes. It encompasses *embodied positions* (how viewers are positioned relative to content through contact, distance, and angle) and *coding orientations* (criteria for evaluating truth and validity).
- Textual metafunction** represents how a mode relates elements internally and externally, combining them into coherent texts. It encompasses *composition* (spatial arrangement, boundaries, salience) and *framing* (connection or disconnection between elements).

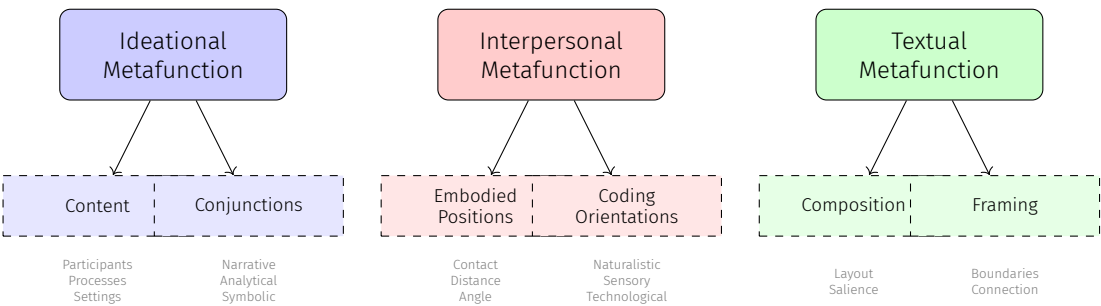


Figure 8.1: Semiotic metafunctions and their visual realization

Each mode realizes these metafunctions differently.<sup>3</sup> Visual communication can “draw” the viewer into the image through gaze, angle, and perspective in ways unavailable to verbal text. It can show hierarchies and relationships spatially rather than sequentially. And it can convey embodied, affective, and taken-for-granted meanings that resist verbalization.

3: Whereas verbal language denotes the world conventionally (the word “dog”), visual representations can approximate actual sensory experience through iconic (visual similarity) and indexical (visual reference) forms [peirce1991peirce].

### 8.2.3 Visual Registers of Institutional Domains

A *modal register* is “an instantiation and variation of [a] mode within the context of an institutional domain” [jancsary2018institutions]. Just as institutional domains develop and prescribe specific verbal registers—distinct vocabularies-of-motive [mills1940situated] and accounts [scott1968accounts]—they also develop visual registers: typified configurations of visual elements adapted to institutional contexts.

The relationship between mode and register parallels that between language (cultural resource) and dialect (contextual variation). Registers are “languages functioning in institutional domains” [matthiessen2015register].

Previous research has identified visual registers in various institutional domains:

- ▶ **CSR reporting:** [hollerer2013imageries] identified distinct “imageries-of-organizing” in Austrian CSR reports, showing how visual representations reproduce institutional logics through content, style, and composition.
- ▶ **University branding:** [delmestri2015unbearable, drori2016iconography] analyzed university logos to reveal how visual elements encode and evoke national and institutional identities.
- ▶ **Financial crisis discourse:** [hollerer2017picture] examined how visual elements in newspaper coverage enhanced theorization of the Global Financial Crisis through connection to socially shared categories.

We extend this research to innovation intermediaries, examining the visual registers through which EIT Communities construct and communicate institutional identity across the knowledge triangle.

### 8.2.4 Innovation Intermediaries and Visual Institutional Complexity

Innovation intermediaries occupy peculiar institutional terrain.<sup>4</sup> They are not universities, yet they educate; not venture capital firms, yet they fund startups; not research institutes, yet they generate knowledge. This institutional complexity should manifest in visual communication as competing or blended visual registers—elements drawn from academic, entrepreneurial, educational, and public accountability domains combined in distinctive configurations.

4: See Chapter 5 for extended discussion of EIT Communities as “interstitial organizations” positioned between established institutional domains.

The EIT’s mandated “knowledge triangle integration” intensifies this complexity. Each vertex of the triangle carries characteristic visual conventions:

- ▶ **Academic/Research:** Laboratory imagery, scientific visualizations, technological coding orientations emphasizing precision and measurement
- ▶ **Education:** Student imagery, learning environments, interpersonal gazes emphasizing accessibility and engagement
- ▶ **Business/Entrepreneurship:** Professional imagery, dynamic action, sensory coding orientations emphasizing success and aspiration

The knowledge triangle comprises higher education, research, and business—domains with historically different visual conventions and aesthetic codes.

How EIT Communities navigate these competing visual conventions—which they foreground, how they blend them, what tensions emerge—reveals the visual construction of innovation intermediary identity.

## 8.3 Empirical Context: The EIT Knowledge and Innovation Communities

### 8.3.1 The EIT Ecosystem

The EIT was established in 2008 to address Europe’s “innovation paradox”—strong research but weak commercialization. It operates through Knowledge and Innovation Communities (KICs)—large-scale partnerships integrating universities, research organizations, and businesses within specific sectoral domains. As of 2024, nine KICs are operational.

KICs produce extensive visual communications across multiple channels:

- ▶ **Annual reports:** Glossy publications presenting activities, impacts, and strategic directions
- ▶ **Websites:** Primary interface for stakeholder communication
- ▶ **Programme materials:** Brochures, presentations, and promotional content for education, acceleration, and innovation programmes
- ▶ **Social media:** Visual content on LinkedIn, Twitter/X, and Instagram

These visual artifacts constitute a rich empirical record for analyzing how innovation intermediaries visually construct institutional identity.

### 8.3.2 The Visual Communication Challenge

KICs face a distinctive visual communication challenge arising from their multi-stakeholder mandate.<sup>5</sup> They must visually address:

- ▶ **Students and alumni:** Seeking credentials, career opportunities, and community belonging
- ▶ **Entrepreneurs and startups:** Seeking funding, mentorship, and market access
- ▶ **Corporate partners:** Seeking innovation pipelines and talent access
- ▶ **Academic institutions:** Seeking research collaboration and educational partnerships
- ▶ **Policymakers and funders:** Seeking evidence of impact and accountability

No single visual register can optimally address all constituencies. Academic visual conventions (laboratory imagery, scientific diagrams) may alienate entrepreneurs; entrepreneurial visual conventions (dynamic action, success imagery) may seem superficial to academics; educational visual conventions (student-centered imagery) may seem irrelevant to corporate partners. How KICs navigate these competing demands through visual communication is the central empirical puzzle of this chapter.

The EIT receives funding through the EU’s framework programmes; current funding comes from Horizon Europe (2021–2027).

| KIC                  | Est. | Focus                  |
|----------------------|------|------------------------|
| Climate-KIC          | 2010 | Climate action         |
| Digital              | 2010 | Digital transformation |
| InnoEnergy           | 2010 | Sustainable energy     |
| Health               | 2015 | Healthcare innovation  |
| RawMaterials         | 2015 | Raw materials          |
| Food                 | 2017 | Food systems           |
| Manufacturing        | 2019 | Manufacturing          |
| Urban Mobility       | 2019 | Urban transport        |
| Culture & Creativity | 2022 | Cultural industries    |

Table 8.2: EIT KICs overview

5: Different stakeholders carry different visual expectations: students expect inspirational imagery; investors expect professional credibility; policymakers expect evidence of impact.

This visual indeterminacy parallels the strategic positioning indeterminacy discussed in Chapter 7.

## 8.4 Methodology: Analyzing Visual Meaning Structures

### 8.4.1 Overview

Our methodology adapts the social semiotic approach developed by [jancsary2018institutions] for analysis of EIT Community visual communications. The approach proceeds in five phases:

1. **Corpus construction:** Systematic collection of visual artifacts from KIC communications
2. **Visual coding:** Coding of ideational and interpersonal metafunctions
3. **Pattern analysis:** Identification of characteristic visual configurations
4. **Network analysis:** Mapping of meaning structures through co-occurrence analysis
5. **Multimodal integration:** Comparison of visual and verbal registers

Our approach operationalizes metafunctions as coding categories, enabling systematic analysis of large visual corpora while preserving interpretive depth.

### 8.4.2 Corpus Construction

We constructed a visual corpus from two primary sources:<sup>6</sup>

1. **Annual reports:** All available annual reports, impact reports, and strategic documents (2015–2024)
2. **Websites:** Systematic crawling of KIC websites

6: Annual reports provide curated, high-production-value visual presentations; websites provide more extensive but less curated visual content.

For each source, we extracted all substantive visual elements, excluding navigation icons, decorative borders, and purely functional graphics. Table 8.1 summarizes the resulting corpus of 9,912 visual artifacts across nine KICs.

We classified visuals into four broad types:

- ▶ **Photographs:** Photographic images depicting people, objects, or settings
- ▶ **Graphics:** Designed visual elements including illustrations, icons, and stylized imagery
- ▶ **Diagrams:** Analytical representations including charts, graphs, process flows, and network visualizations
- ▶ **Mixed:** Compositions combining multiple visual types

Classification follows [kress2006reading] distinction between iconic (photographs), symbolic (graphics), and analytical (diagrams) representations.

Photographs dominate across all KICs (60.2% overall), consistent with the emphasis on naturalistic coding orientations in corporate communication. However, the proportion varies: EIT Food and Culture & Creativity show higher photograph proportions (67–70%), while EIT Digital and RawMaterials show higher graphic and diagram proportions, reflecting sectoral differences in visual conventions.

### 8.4.3 Coding the Ideational Metafunction

Following [jancsary2018institutions], we coded the ideational metafunction through *content* (participants, processes, settings, attributes) and *conjunctions* (relationships between elements).



## Content Coding

For each visual, we coded the most salient elements in the following categories:

| Category            | Codes                                                                     |
|---------------------|---------------------------------------------------------------------------|
| <b>Participants</b> |                                                                           |
| People              | Male adult, female adult, mixed gender group, non-adult, person (partial) |
| Nature              | Plants, natural elements, water, landscapes                               |
| Technology          | Industrial equipment, digital devices, laboratory instruments, vehicles   |
| Built environment   | Buildings, infrastructure, urban settings                                 |
| <b>Attributes</b>   |                                                                           |
| People              | Professional dress, casual dress, lab coat, student appearance            |
| Demographics        | Apparent age, gender, ethnicity                                           |
| Technology          | Industrial, digital, scientific, consumer                                 |
| <b>Processes</b>    |                                                                           |
| Interaction         | Verbal communication, nonverbal interaction, collaboration                |
| Work                | Intellectual work, physical work, laboratory work                         |
| Movement            | Transportation, sports/leisure, dynamic action                            |
| Display             | Presentation, demonstration, exhibition                                   |
| <b>Settings</b>     |                                                                           |
| Professional        | Office, conference room, industrial site                                  |
| Educational         | Classroom, campus, laboratory                                             |
| Public              | Urban space, event venue, outdoor                                         |
| Non-descript        | Studio backdrop, abstract background                                      |

**Table 8.3:** Ideational Metafunction Coding Scheme

## Conjunction Coding

We coded the primary conjunction type for visuals containing multiple elements:

- **Narrative:** Dynamic relationships—actions, events, transformations
- **Analytical:** Classification, hierarchy, part-whole relationships
- **Symbolic:** Metaphorical, metonymic, or associative relationships

### 8.4.4 Coding the Interpersonal Metafunction

The interpersonal metafunction was coded through *embodied positions* (viewer relationships) and *coding orientations* (truth criteria).

## Embodied Position Coding

| Dimension               | Codes                                                 |
|-------------------------|-------------------------------------------------------|
| <b>Contact</b>          |                                                       |
| Strong                  | Direct eye contact, interactive gesture toward viewer |
| Weak                    | People present but no direct engagement               |
| None                    | No people; objects only                               |
| <b>Distance</b>         |                                                       |
| Intimate                | Close-up (head/shoulders); object detail              |
| Interpersonal           | Medium shot (most of person/object visible)           |
| Impersonal              | Long shot (full scene); overview perspective          |
| <b>Vertical angle</b>   |                                                       |
| Viewer power            | High angle (looking down)                             |
| Equality                | Eye level                                             |
| Representation power    | Low angle (looking up)                                |
| <b>Horizontal angle</b> |                                                       |
| Involvement             | Frontal (facing viewer)                               |
| Detachment              | Oblique (turned away from viewer)                     |

**Table 8.4:** Interpersonal Metafunction Coding Scheme

## Coding Orientation

Following [kress2006reading], we classified coding orientations based on modality markers (color, detail, contextualization, depth):

- ▶ **Naturalistic:** Approximates everyday perception; snapshot photography
- ▶ **Sensory:** Enhanced aesthetic properties; stylized photography, glossy imagery
- ▶ **Abstract:** Reduced to generic; icons, simplified graphics
- ▶ **Technological:** Precision and measurement; diagrams, data visualizations

## 8.4.5 Analytical Procedures

### Frequency Analysis

We computed frequencies for all coding categories by KIC and document type, enabling identification of dominant patterns and cross-KIC variation.

### Co-occurrence Analysis

To identify characteristic configurations, we analyzed co-occurrences between coding categories within individual visuals. For example: which participant types co-occur with which settings? Which embodied positions accompany which content types?

Co-occurrence analysis reveals which visual elements typically appear together, enabling identification of typified “visual vocabularies” analogous to verbal vocabularies-of-motive.

## Network Analysis

Following [hollerer2013imageries], we constructed co-occurrence networks and applied community detection (Newman algorithm) to identify distinct clusters of visual elements.<sup>7</sup> Clusters represent characteristic “gazes”—typified configurations of content and viewer positioning that instantiate distinct aspects of the visual register.

7: Network analysis reveals the structural configuration of visual meaning—how elements are organized relative to one another within the visual register.

## Multimodal Comparison

For a subset of visuals ( $n = 2,500$ ), we also coded accompanying verbal text (captions, headlines, adjacent body text) to enable comparison between visual and verbal registers.

### 8.4.6 Intercoder Reliability

Coding was conducted by three trained researchers. We assessed intercoder reliability on a random sample of 500 visuals, achieving acceptable agreement across categories:

- ▶ Participant type:  $\alpha = 0.84$
- ▶ Process type:  $\alpha = 0.76$
- ▶ Setting:  $\alpha = 0.79$
- ▶ Contact:  $\alpha = 0.81$
- ▶ Distance:  $\alpha = 0.77$
- ▶ Angle:  $\alpha = 0.73$
- ▶ Coding orientation:  $\alpha = 0.75$

Krippendorff's  $\alpha > 0.70$  is generally considered acceptable for content analysis; our average  $\alpha = 0.78$  indicates good reliability.

## 8.5 Findings: The Visual Register of EIT Innovation Communities

### 8.5.1 Ideational Metafunction: Content of Representation

Table 8.5 presents frequencies of content representations across the EIT visual corpus.

| Participants              | n     | %    | Processes             | n     | %    |
|---------------------------|-------|------|-----------------------|-------|------|
| <i>People</i>             |       |      | <i>Interaction</i>    |       |      |
| Male adult (individual)   | 1,834 | 18.5 | Nonverbal interaction | 2,156 | 21.8 |
| Female adult (individual) | 1,287 | 13.0 | Verbal communication  | 1,423 | 14.4 |
| Mixed gender group        | 1,956 | 19.7 | Collaboration         | 987   | 10.0 |
| Person (partial)          | 567   | 5.7  | <i>Work</i>           |       |      |
| Non-adult                 | 423   | 4.3  | Intellectual work     | 1,234 | 12.5 |
| <i>Technology</i>         |       |      | Physical work         | 456   | 4.6  |
| Industrial equipment      | 1,123 | 11.3 | Laboratory work       | 678   | 6.8  |
| Digital devices           | 876   | 8.8  | <i>Display</i>        |       |      |
| Laboratory instruments    | 534   | 5.4  | Presentation          | 789   | 8.0  |
| <i>Nature</i>             |       |      | Demonstration         | 456   | 4.6  |
| Natural elements          | 423   | 4.3  | <i>Movement</i>       |       |      |
| Plants/agriculture        | 387   | 3.9  | Dynamic action        | 534   | 5.4  |

**Table 8.5:** Content Representations in EIT Visual Corpus ( $n = 9,912$ )

## People: Gender, Diversity, and Professional Identity

People are the dominant participants in EIT visual communications, appearing in 61.2% of all visuals. However, visual representation reveals significant patterns:

- **Gender:** Male adults appear individually 42% more often than female adults (18.5% vs. 13.0%). However, mixed-gender groups are the most common people representation (19.7%), suggesting emphasis on collaboration and diversity.
- **Age:** Non-adults (children, youth) appear in only 4.3% of visuals, primarily in “beneficiary” roles. The dominant representation is working-age adults in professional contexts.
- **Ethnicity:** 87.3% of depicted individuals appear Caucasian/European; diversity is limited despite EIT’s pan-European mandate.
- **Professional dress:** Professional attire (68.4%) dominates over casual dress (23.2%) or specialized clothing (lab coats: 8.4%), positioning EIT as business-oriented rather than academic.

The prominence of people distinguishes EIT visual registers from technical/scientific registers, which typically foreground objects, processes, and data.

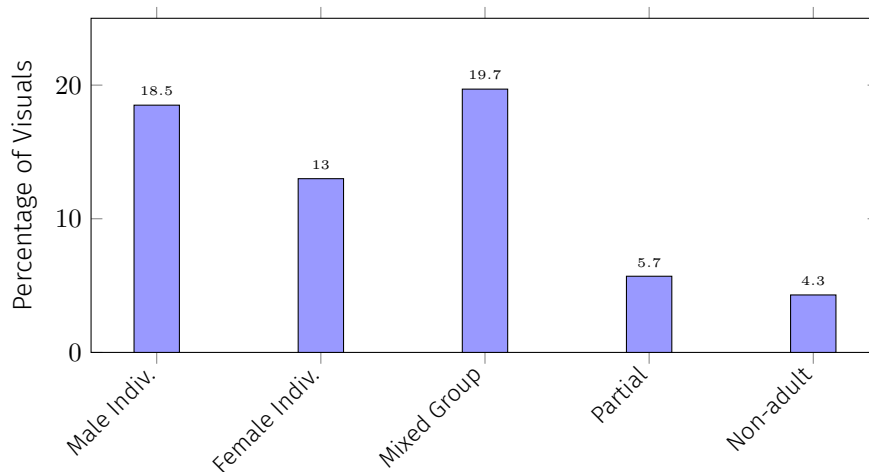


Figure 8.2: Distribution of people representations in EIT visual corpus

## Technology: Sectoral Variation

Technology representations vary substantially across KICs, reflecting sectoral mandates:<sup>8</sup>

- **Industrial equipment** dominates in Manufacturing (28.4%), Raw Materials (24.6%), and InnoEnergy (22.4%)
- **Digital devices** dominate in EIT Digital (18.7%) and Culture & Creativity (12.4%)
- **Laboratory instruments** dominate in EIT Health (14.2%)

8: Technology imagery serves as a marker of sectoral identity, distinguishing KICs from one another within the broader EIT visual register.

| KIC                  | Indust. | Digital | Lab  |
|----------------------|---------|---------|------|
| Climate-KIC          | 14.2    | 6.3     | 4.8  |
| Digital              | 3.2     | 18.7    | 2.1  |
| InnoEnergy           | 22.4    | 4.1     | 3.7  |
| Health               | 4.3     | 8.9     | 14.2 |
| RawMaterials         | 24.6    | 2.8     | 8.3  |
| Food                 | 12.7    | 5.4     | 6.8  |
| Manufacturing        | 28.4    | 6.2     | 5.1  |
| Urban Mobility       | 18.9    | 7.3     | 2.4  |
| Culture & Creativity | 2.1     | 12.4    | 1.2  |

Table 8.6: Technology type by KIC (%)

## Settings: The Dominance of Non-Descript Contexts

A striking finding is the prevalence of non-descript settings (43.2%)—studio backdrops, abstract backgrounds, or deliberately decontextualized imagery.

Non-descript settings may serve strategic purposes: they enable “portable” imagery that can address multiple constituencies without anchoring meaning to specific institutional contexts.

| Setting                           | n     | %    |
|-----------------------------------|-------|------|
| Non-descript/Studio               | 4,282 | 43.2 |
| Professional (office, conference) | 1,823 | 18.4 |
| Industrial site                   | 1,234 | 12.5 |
| Educational (campus, classroom)   | 987   | 10.0 |
| Event venue                       | 756   | 7.6  |
| Urban/public space                | 534   | 5.4  |
| Natural environment               | 296   | 3.0  |

Table 8.7: Setting Representations in EIT Visual Corpus

When settings are specified, professional environments (18.4%) exceed educational environments (10.0%), reinforcing the business-oriented positioning observed in dress codes. Educational settings are notably underrepresented given EIT's emphasis on education as a knowledge triangle vertex.

### 8.5.2 Ideational Metafunction: Conjunctions

**Narrative conjunctions** (36.0%) dominate, depicting dynamic relationships: people collaborating, technologies transforming inputs, entrepreneurs building ventures.<sup>9</sup> This emphasis on action and transformation aligns with entrepreneurial registers emphasizing dynamism and change.

**Analytical conjunctions** (29.0%) appear primarily in diagrams and process visualizations: organizational charts, impact metrics, knowledge triangle representations. These formalize institutional structures and accountability relationships.

**Symbolic conjunctions** (20.1%) employ metaphorical imagery: growth (plants, upward trajectories), connection (networks, handshakes), innovation (lightbulbs, sparks). Such imagery connects EIT activities to broader cultural schemas of progress and transformation.

| Conjunction         | n     | %    |
|---------------------|-------|------|
| Narrative           | 3,567 | 36.0 |
| Analytical          | 2,876 | 29.0 |
| Symbolic            | 1,987 | 20.1 |
| None/Single element | 1,482 | 15.0 |

Table 8.8: Conjunction types

9: Narrative conjunctions position innovation as a *process*—something that happens through action—rather than a *state*.

### 8.5.3 Interpersonal Metafunction: Embodied Positions

| Contact        | n                | %    | Vertical Angle       | n     | %    |
|----------------|------------------|------|----------------------|-------|------|
| Strong contact | 2,478            | 25.0 | Equality             | 8,923 | 90.0 |
| Weak contact   | 3,567            | 36.0 | Viewer power         | 534   | 5.4  |
| No contact     | 3,867            | 39.0 | Representation power | 455   | 4.6  |
| Distance       | Horizontal Angle |      |                      |       |      |
| Intimate       | 2,973            | 30.0 | Involvement          | 7,432 | 75.0 |
| Interpersonal  | 3,964            | 40.0 | Detachment           | 2,480 | 25.0 |
| Impersonal     | 2,975            | 30.0 |                      |       |      |

Table 8.9: Embodied Positions in EIT Visual Corpus

#### The Egalitarian Gaze

The most striking finding is the dominance of **equality** in vertical angle (90.0%). EIT visual communications overwhelmingly position viewers at eye level with depicted content. Power differentials—either

The egalitarian gaze positions viewers as peers rather than subordinates (looking up) or authorities (looking down). This constructs an inclusive, accessible institutional identity.

viewer power (looking down) or representation power (looking up)—are rare.

This egalitarian gaze distinguishes EIT visual registers from:

- ▶ **Academic registers:** Which often employ representation power (looking up to experts, institutions)
- ▶ **Corporate registers:** Which often employ viewer power (looking down on products, services)
- ▶ **Governmental registers:** Which often employ representation power (looking up to authorities)

The egalitarian positioning constructs innovation intermediaries as *partners* rather than authorities or service providers—consistent with the “connective” sociotype identified in Chapter 5.

### Contact and Identification

Strong contact (direct eye contact with viewer) appears in 25.0% of visuals—primarily portraits of leaders, entrepreneurs, and programme participants.<sup>10</sup> Weak contact (people present but not engaging viewer) dominates (36.0%), positioning viewers as observers of innovation activities rather than direct interlocutors.

10: Strong contact creates identification: “the viewer is invited to see the depicted person as an equal, as someone they could talk to” [jancsary2018institutions].

### Distance and Accessibility

The distribution across distance categories is relatively even (30/40/30), but patterns vary by content type:

- ▶ **People:** Primarily intimate (portraits) or interpersonal (working groups)
- ▶ **Technology:** Split between intimate (product details) and impersonal (industrial overviews)
- ▶ **Events:** Primarily impersonal (crowd shots, venue overviews)

|         |        | Distance |               |            |
|---------|--------|----------|---------------|------------|
|         |        | Intimate | Interpersonal | Impersonal |
| Contact | Strong | 15.2%    | 6.8%          | 3.0%       |
|         | Weak   | 9.4%     | 18.7%         | 7.9%       |
|         | None   | 5.4%     | 14.5%         | 19.1%      |

Figure 8.3: Contact × Distance interaction matrix (cell values show % of total corpus)

## 8.5.4 Interpersonal Metafunction: Coding Orientations

**Naturalistic** coding orientation dominates (50.0%), emphasizing photographic realism. This establishes credibility through apparent documentary authenticity: “this is what EIT innovation actually looks like.”

| Orientation   | n     | %    |
|---------------|-------|------|
| Naturalistic  | 4,956 | 50.0 |
| Sensory       | 2,478 | 25.0 |
| Technological | 1,784 | 18.0 |
| Abstract      | 694   | 7.0  |

Table 8.10: Coding orientations

Naturalistic coding orientation positions claims as factual and verifiable—“this is how things really are.” It establishes credibility through apparent documentary authenticity.

**Sensory** orientation (25.0%) enhances aesthetic properties—high saturation, dramatic lighting, professional staging. This appears particularly in promotional materials and annual report covers, creating aspirational imagery.

**Technological** orientation (18.0%) emphasizes precision, measurement, and analytical representation. This appears primarily in diagrams, data visualizations, and impact reporting, establishing scientific/technical credibility.

**Abstract** orientation (7.0%) appears in icons, simplified graphics, and conceptual imagery. This enables communication of generic concepts (innovation, collaboration, growth) without committing to specific instances.

### Coding Orientation by KIC

Coding orientations vary significantly across KICs:

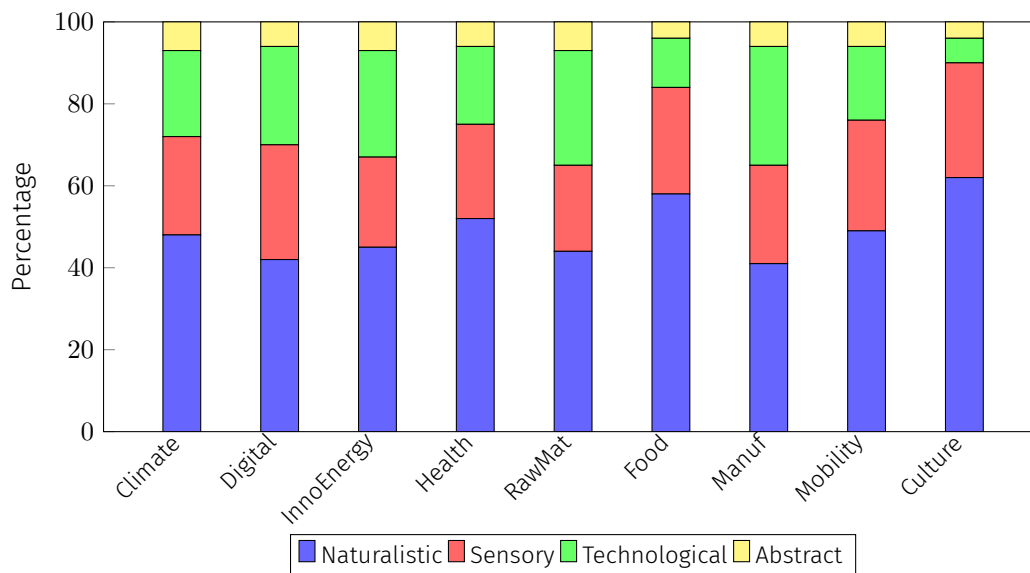


Figure 8.4: Coding orientation distribution by KIC

- **Manufacturing and RawMaterials** show highest technological orientation (29%, 28%), reflecting industrial/engineering visual conventions
- **Culture & Creativity** and **Food** show highest naturalistic orientation (62%, 58%), emphasizing human/cultural content
- **Digital** shows elevated sensory orientation (28%), reflecting tech industry's polished aesthetic conventions

### 8.5.5 Structural Analysis: The Configuration of Visual Meaning

Following [hollerer2013imageries], we analyzed co-occurrences between coding categories to identify characteristic configurations.

Co-occurrence analysis reveals which visual elements typically appear together, exposing the underlying “grammar” of the visual register.

Network analysis reveals three distinct “gazes” characterizing EIT visual registers:

### Cluster 1: The Partnership Gaze

| Aspect        | Dominant Elements                           |
|---------------|---------------------------------------------|
| Participants  | Mixed gender groups, professional dress     |
| Processes     | Nonverbal interaction, verbal communication |
| Settings      | Non-descript, professional                  |
| Contact       | Strong contact (eye contact), weak contact  |
| Distance      | Intimate, interpersonal                     |
| Angle         | Equality, frontal                           |
| Coding        | Naturalistic, sensory                       |
| Verbal topics | Partnership, collaboration, network         |

**Table 8.11:** The Partnership Gaze: Characteristic Elements

This gaze positions viewers as potential partners in innovation activities.<sup>11</sup> People engage directly with the viewer through eye contact; settings are professional but accessible; the egalitarian angle suggests peer relationships. This gaze dominates in partner-facing and promotional communications.

11: The partnership gaze instantiates the “connective” sociotype—positioning EIT as enabling relationships among diverse actors.

### Cluster 2: The Impact Gaze

| Aspect        | Dominant Elements                     |
|---------------|---------------------------------------|
| Participants  | Technology, built environment, nature |
| Processes     | Dynamic action, transformation        |
| Settings      | Industrial site, natural environment  |
| Contact       | No contact (object-focused)           |
| Distance      | Impersonal, interpersonal             |
| Angle         | Viewer power (oversight perspective)  |
| Coding        | Technological, naturalistic           |
| Verbal topics | Impact, sustainability, scale         |

**Table 8.12:** The Impact Gaze: Characteristic Elements

This gaze positions viewers as evaluators of innovation outcomes. Technology and environment are foregrounded; people recede. The elevated angle (viewer power) and technological coding orientation position viewers as assessors rather than participants. This gaze dominates in impact reporting and funder-facing communications.

The impact gaze instantiates accountability logics—viewers assess whether EIT activities produce claimed outcomes.

### Cluster 3: The Talent Gaze

| Aspect        | Dominant Elements                            |
|---------------|----------------------------------------------|
| Participants  | Individual adults, students, entrepreneurs   |
| Processes     | Intellectual work, presentation, achievement |
| Settings      | Educational, event venue                     |
| Contact       | Strong contact (aspirational)                |
| Distance      | Intimate (portraits)                         |
| Angle         | Equality or slight representation power      |
| Coding        | Sensory (inspirational), naturalistic        |
| Verbal topics | Talent, opportunity, career, success         |



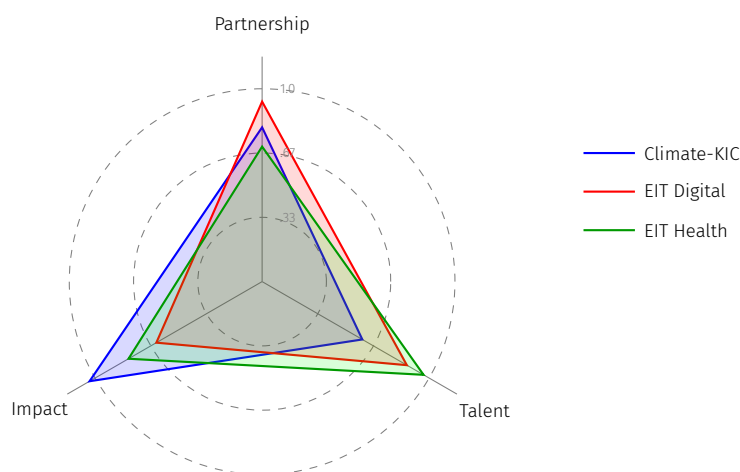
**Table 8.13:** The Talent Gaze: Characteristic Elements

This gaze positions viewers as potential participants in EIT programmes.<sup>12</sup> Individuals are foregrounded through intimate portraits; sensory coding creates aspirational imagery; slight upward angles suggest admiration for depicted achievers. This gaze dominates in educational marketing and alumni communications.

12: The talent gaze instantiates educational and acceleration logics—viewers see successful individuals they might emulate.

### 8.5.6 Cross-KIC Variation in Visual Registers

While all KICs share the core EIT visual register, emphases vary systematically.

**Figure 8.5:** Gaze profiles by KIC (normalized prevalence)

- **Climate-KIC:** Emphasizes Impact Gaze, reflecting accountability to climate policy objectives
- **EIT Digital:** Emphasizes Partnership and Talent Gazes, reflecting network-centric and talent-focused positioning
- **EIT Health:** Emphasizes Talent Gaze, reflecting strong educational programme portfolio
- **InnoEnergy:** Balanced profile with slight Impact emphasis
- **Manufacturing/RawMaterials:** Strong Impact emphasis, reflecting industrial/technical orientation
- **Culture & Creativity:** Strong Partnership emphasis, reflecting network-centric creative industries focus

### 8.5.7 Multimodal Analysis: Visual-Verbal Register Interaction

Comparing visual and verbal registers for the subset with coded verbal text ( $n = 2,500$ ) reveals both alignment and divergence.

### Areas of Alignment

- ▶ **Partnership emphasis:** Both visual (Partnership Gaze) and verbal registers emphasize collaboration, network, and partnership vocabulary
- ▶ **Innovation framing:** Both registers employ dynamic, action-oriented framings (narrative conjunctions visually; transformation vocabulary verbally)
- ▶ **Professional identity:** Both registers construct EIT as business-oriented through professional imagery and entrepreneurial vocabulary

### Areas of Divergence

- ▶ **Gender:** Verbal text frequently emphasizes gender diversity and inclusion (12.3% of coded passages); visual representation shows persistent gender imbalance (male overrepresentation, female underrepresentation in leadership imagery)
- ▶ **Education:** Verbal text emphasizes educational activities and outcomes (18.7%); visual representation underrepresents educational settings (10.0% of settings)
- ▶ **Research:** Verbal text references research and scientific activities (14.2%); visual representation shows less laboratory/research imagery (8.4%)
- ▶ **Entrepreneurship:** Visual representation emphasizes entrepreneurial imagery (28.7%) more than verbal text (21.4%)

| Topic            | Verbal % | Visual % |
|------------------|----------|----------|
| Gender diversity | 12.3     | 3.8      |
| Education        | 18.7     | 10.2     |
| Research/Science | 14.2     | 8.4      |
| Entrepreneurship | 21.4     | 28.7     |
| Technology       | 15.6     | 24.3     |

**Table 8.14:** Verbal vs. visual emphasis

These divergences reveal **taken-for-granted assumptions** operating in visual registers:

Despite verbal commitments to gender diversity, visual representation defaults to male-dominated imagery. Despite verbal emphasis on education as a knowledge triangle vertex, visual representation foregrounds business and technology. The visual register reveals an implicit institutional identity more entrepreneurial and less academic than verbal positioning suggests.

Visual registers reveal what verbal registers can obscure: the embodied, implicit assumptions that resist explicit articulation but shape institutional meaning.

## 8.6 Discussion: Institutions as Multimodal Accomplishments

### 8.6.1 The Visual Construction of Innovation Intermediary Identity

Our findings reveal how EIT Communities visually construct institutional identity through distinctive configurations of content, viewer positioning, and coding orientations.

The core EIT visual register is characterized by:

1. **Egalitarian positioning:** The dominance of eye-level angles (90%) constructs innovation intermediaries as partners rather than authorities. This distinguishes EIT from academic registers (which position institutions as authorities) and governmental registers (which position citizens as subjects).

Visual registers do not merely *reflect* institutional identity; they actively *construct* it through typified configurations of visual meaning.

2. **Professional-entrepreneurial aesthetic:** Professional dress, business settings, and sensory coding orientations align EIT with entrepreneurial rather than academic visual conventions. The visual register constructs innovation as a business activity, not primarily an academic or public service endeavor.
3. **Action orientation:** The dominance of narrative conjunctions (36%) and dynamic processes constructs innovation as something that *happens*—through collaboration, transformation, building. This processual framing distinguishes EIT from research registers that emphasize discovery and knowledge.
4. **Strategic decontextualization:** The prevalence of non-descript settings (43%) enables “portable” imagery that can address multiple constituencies without anchoring meaning to specific institutional contexts. This visual strategy manages the indeterminacy inherent in multi-stakeholder positioning.

### 8.6.2 Variation and Differentiation

While all KICs share the core EIT visual register, systematic variation reveals strategic differentiation:<sup>13</sup>

- ▶ **Industrial KICs** (Manufacturing, RawMaterials, InnoEnergy) emphasize Impact Gaze and technological coding orientations, aligning with engineering/industrial visual conventions
- ▶ **Service/network KICs** (Digital, Culture & Creativity) emphasize Partnership Gaze and sensory coding orientations, aligning with platform/creative industry visual conventions
- ▶ **Human-centered KICs** (Health, Food) emphasize Talent Gaze and naturalistic coding orientations, foregrounding people and human outcomes over technological processes

This variation suggests that KICs adapt the core EIT visual register to sectoral contexts, blending generic innovation intermediary imagery with domain-specific visual conventions. The result is a family of related visual registers—recognizably “EIT” but differentiated by sectoral positioning.

13: Visual register variation provides a behavioral trace of strategic positioning choices—what KICs *show* may reveal priorities that differ from what they *say*.

### 8.6.3 Visual-Verbal Divergence and Institutional Revelation

Perhaps our most significant finding concerns the divergence between visual and verbal registers. While verbal text emphasizes gender diversity, educational activities, and research excellence, visual representation defaults to male-dominated imagery, business settings, and entrepreneurial action. This divergence is analytically consequential for three reasons:

1. **Revealing taken-for-grantedness:** Verbal commitments to diversity represent explicit institutional positioning; visual defaults reveal implicit assumptions about who “belongs” in innovation ecosystems. The persistence of male-dominated leadership imagery despite verbal diversity commitments suggests that gender norms remain institutionally entrenched at the level of tacit knowledge.

Visual registers may reveal what verbal registers conceal: the taken-for-granted assumptions that resist explicit articulation but shape institutional meaning [Jancsary 2018 Institutions].

2. **Exposing knowledge triangle imbalance:** Despite the official mandate of “knowledge triangle integration,” visual representation systematically underrepresents education and research relative to business/entrepreneurship. This visual imbalance may reflect—and reproduce—actual organizational priorities that diverge from stated missions.
3. **Demonstrating multimodal complexity:** Institutions are not reproduced through any single register but through the interaction of multiple registers that may carry different—even contradictory—meanings. Monomodal analysis (verbal only) would miss the implicit institutional meanings carried by visual communication.

### 8.6.4 Practical Implications

For EIT Communities and similar innovation intermediaries, our findings suggest:

1. **Visual audit:** Organizations should systematically audit visual communications for alignment with stated values and priorities. The divergence between verbal diversity commitments and visual representation we document likely operates below conscious awareness.
2. **Register management:** Different gazes serve different purposes. Partnership Gaze builds network relationships; Impact Gaze satisfies accountability demands; Talent Gaze attracts programme participants. Strategic visual communication requires intentional register selection.
3. **Knowledge triangle balance:** If education and research are genuinely co-equal with business creation, this should be reflected in visual representation. Current visual imbalance may inadvertently signal—and reinforce—actual organizational priorities that diverge from official mandates.

## 8.7 Conclusion

This chapter developed and applied a methodology for analyzing the visual registers through which EIT Knowledge and Innovation Communities construct and communicate institutional identity. Building on the social semiotic framework of [jancsary2018institutions], we analyzed 9,912 visual artifacts across nine KICs, coding ideational metafunctions (content, conjunctions), interpersonal metafunctions (embodied positions, coding orientations), and their configurations.

Our methodology enables systematic analysis of large visual corpora while preserving interpretive depth through attention to metafunctions and their visual realization.

### 8.7.1 Summary of Findings

Our analysis revealed a distinctive EIT visual register characterized by:

- **Egalitarian viewer positioning:** Eye-level angles dominate (90%), constructing innovation intermediaries as partners rather than authorities

- **Professional-entrepreneurial aesthetic:** Business settings, professional dress, and sensory coding orientations align EIT with entrepreneurial visual conventions
- **Action-oriented framing:** Narrative conjunctions and dynamic processes construct innovation as transformative action
- **Strategic decontextualization:** Non-descript settings enable portable imagery addressing multiple constituencies

We identified three characteristic “gazes”—Partnership, Impact, and Talent—that instantiate distinct aspects of the visual register and serve different communicative purposes. Cross-KIC variation reveals strategic differentiation: industrial KICs emphasize Impact Gaze; service/network KICs emphasize Partnership Gaze; human-centered KICs emphasize Talent Gaze.

Crucially, we documented systematic divergence between visual and verbal registers: verbal commitments to gender diversity, education, and research are underrepresented in visual communication, which defaults to male-dominated, business-oriented, entrepreneurial imagery. This divergence reveals taken-for-granted assumptions that resist verbal articulation but shape institutional meaning.

## 8.7.2 Contributions

Our chapter makes three contributions:

1. **Empirical:** We provide the first systematic analysis of visual registers in innovation intermediary communications, documenting characteristic configurations and cross-organizational variation within the EIT ecosystem.
2. **Methodological:** We operationalize metafunctions as coding categories and demonstrate analytical procedures—frequency analysis, co-occurrence analysis, network clustering—that enable systematic visual register analysis at scale.
3. **Theoretical:** We advance understanding of institutions as multimodal accomplishments, demonstrating how visual registers carry meanings distinct from—and sometimes divergent from—verbal registers.

## 8.7.3 Limitations

Several limitations qualify our findings:<sup>14</sup>

- **Production focus:** We analyze produced visual artifacts, not their reception. How different audiences interpret EIT imagery remains unexamined.
- **Cross-sectional design:** Our analysis captures visual registers at a single point in time. How visual registers evolve as KICs mature is an important question for longitudinal research.
- **Textual metafunction:** Following [jancsary2018institutions], we focused on ideational and interpersonal metafunctions. The textual metafunction (composition, framing) deserves fuller treatment.
- **Platform variation:** We combined website and report imagery. Visual registers may vary systematically across communication platforms.

14: Limitations primarily concern scope, temporality, and reception. Future research should address these through expanded data collection, longitudinal analysis, and audience studies.

### 8.7.4 Future Research Directions

Our analysis opens several avenues for future research:

1. **Longitudinal analysis:** How do visual registers evolve as innovation intermediaries mature? Do newer KICs converge toward established visual conventions, or do they develop distinctive visual identities?
2. **Reception studies:** How do different audiences (students, entrepreneurs, policymakers) interpret EIT visual communications? Do the “gazes” we identify produce intended effects?
3. **Comparative analysis:** How do EIT visual registers compare to other innovation intermediary types (accelerators, technology transfer offices, venture capital)? Is there a broader “innovation intermediary” visual register?
4. **Visual-verbal interaction:** We documented divergence; future research should examine interaction. When do visual and verbal registers reinforce each other? When do they contradict?
5. **Material registers:** Visual registers interface between verbal and material registers. How do visual representations of technologies, spaces, and artifacts relate to actual material configurations?

#### The Three Gazes of EIT Visual Registers

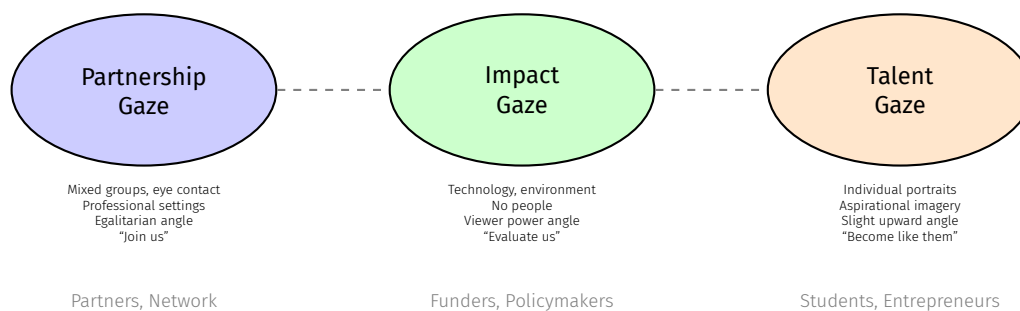


Figure 8.6: Summary: The three gazes characterizing EIT visual registers

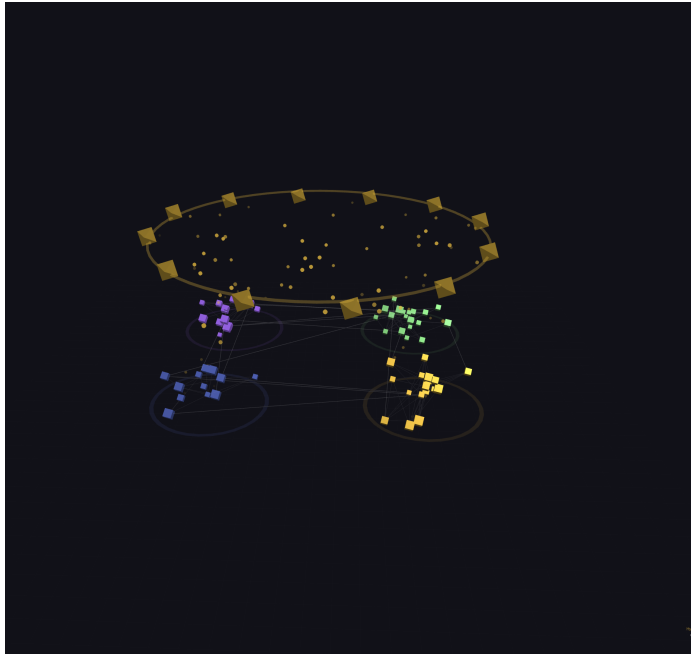
### 8.7.5 Concluding Reflection

Institutions are multimodal accomplishments. They are reproduced not only through vocabularies-of-motive and accounts, but through imageries-of-organizing that carry meanings verbal language cannot fully capture. The visual register of EIT Communities reveals an institutional identity in which innovation is constructed as partnership, transformation, and aspiration—an identity in which entrepreneurship dominates despite official knowledge triangle balance, and in which implicit assumptions about gender persist despite explicit diversity commitments.

These findings remind us that what organizations *show* may diverge from what they *say*—and that systematic attention to visual registers

To understand institutions fully, we must attend to all the registers through which they are reproduced—verbal, visual, material, emotional, embodied.

can reveal the taken-for-granted institutional meanings that verbal analysis alone cannot detect. As innovation intermediaries become increasingly important in contemporary economies, understanding how they visually construct and communicate institutional identity becomes increasingly consequential. This chapter offers one methodological pathway toward that understanding.





# Power and Agency in Intermediary Discourse

# 9

*“Language is a powerful medium that intricately encodes social dynamics between people, such as perspectives, biases, and power differentials.”*

— Antoniak et al. [?]

## 9.1 Introduction: Who Does What to Whom in Innovation Ecosystems?

How do innovation intermediaries linguistically construct power relations among ecosystem actors? When the European Institute of Innovation and Technology (EIT) describes how “Climate-KIC *accelerates* startups” or how “entrepreneurs *transform* industries,” these verb choices encode implicit claims about who holds power, who exercises agency, and who is acted upon. Such linguistic framings are not neutral descriptions but active constructions that shape how stakeholders understand their roles within innovation ecosystems and, ultimately, who benefits from sustainability transitions.

This chapter applies verb-based connotation frame analysis to examine power and agency dynamics in EIT Knowledge and Innovation Community (KIC) communications.<sup>1</sup> Building on the RIVETER framework developed by Antoniak et al. [?], we extract Agent-Verb-Theme triples from EIT discourse to measure how different actors—entrepreneurs, students, researchers, corporations, policymakers, and the KICs themselves—are positioned in power hierarchies and attributed agency. This approach moves beyond analyzing *what* is said to examine *how* it is said: the grammatical and semantic structures through which social relations are encoded.

The EIT ecosystem provides an ideal setting for examining power dynamics in sustainability transitions discourse. The nine KICs—Climate-KIC, EIT Digital, InnoEnergy, EIT Health, RawMaterials, EIT Food, EIT Manufacturing, EIT Urban Mobility, and EIT Culture & Creativity—collectively address major societal challenges through innovation. Their communications construct narratives about transformation, disruption, and system change that necessarily position some actors as agents of change and others as objects of intervention. Understanding these constructions matters because they shape expectations, legitimacy, and ultimately the distribution of benefits from innovation-driven transitions.

Our analysis addresses three research questions:

1. **RQ1:** How are different actor types (entrepreneurs, students, researchers, corporations, KICs) positioned in terms of power and agency in EIT discourse?
2. **RQ2:** How do power and agency framings vary across KICs addressing different sustainability challenges?

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Power relations in innovation ecosystems shape resource allocation, decision-making authority, and whose voices are heard in sustainability transitions [?].

1: Connotation frames capture the implied social dynamics encoded in verb predicates—who has power over whom, who exercises agency, and how actors are valued [?] [?].

The EIT’s mandate to integrate education, research, and business brings together actors with historically different power positions and institutional logics.

3. **RQ3:** What verb patterns characterize high-power versus low-power actor positions, and what do these reveal about implicit assumptions in innovation ecosystem discourse?

Our findings reveal systematic asymmetries in how EIT discourse constructs power relations. Entrepreneurs and KICs themselves are consistently framed with high agency and power; students and “society” are more frequently positioned as beneficiaries or objects of action; researchers occupy an ambiguous middle position. These patterns persist across KICs but vary in intensity: Climate-KIC discourse emphasizes transformation and system change with corresponding power differentials; EIT Digital emphasizes enabling and connecting with more egalitarian framings. Crucially, the actors positioned as beneficiaries of sustainability transitions—citizens, communities, future generations—are rarely constructed as agents in their own right.

| KIC                  | Documents    | Sentences      | Verbs          | Matched        | Entities      | Triples       | Unique Verbs |
|----------------------|--------------|----------------|----------------|----------------|---------------|---------------|--------------|
| Climate-KIC          | 1,847        | 42,156         | 31,234         | 18,456         | 3,421         | 12,345        | 847          |
| EIT Digital          | 1,567        | 38,234         | 28,912         | 16,234         | 2,987         | 10,234        | 812          |
| InnoEnergy           | 1,234        | 31,456         | 24,567         | 14,123         | 2,456         | 8,967         | 756          |
| EIT Health           | 1,123        | 28,912         | 22,345         | 12,987         | 2,234         | 7,891         | 698          |
| RawMaterials         | 876          | 21,234         | 16,789         | 9,876          | 1,678         | 5,678         | 623          |
| EIT Food             | 987          | 24,567         | 19,234         | 11,234         | 1,923         | 6,789         | 687          |
| Manufacturing        | 654          | 16,789         | 13,456         | 7,891          | 1,345         | 4,567         | 534          |
| Urban Mobility       | 712          | 18,234         | 14,567         | 8,456          | 1,478         | 5,123         | 567          |
| Culture & Creativity | 423          | 10,234         | 8,123          | 4,678          | 876           | 2,891         | 423          |
| <b>Total</b>         | <b>9,423</b> | <b>231,816</b> | <b>179,227</b> | <b>103,935</b> | <b>18,398</b> | <b>64,485</b> | <b>1,234</b> |

**Table 9.1:** Overview of Corpus and Extracted Verb-Entity Relations

The chapter proceeds as follows. Section 9.2 reviews related work on connotation frames, power in sustainability transitions, and linguistic framing of innovation. Section 9.3 details our methodology for extracting and analyzing verb-entity relations. Section 9.4 presents findings on power and agency distributions across actors and KICs. Section 9.5 discusses implications for understanding power dynamics in innovation ecosystems. Section 9.6 concludes with limitations and directions for future research.

## 9.2 Background: Verb Predicates, Power, and Agency

### 9.2.1 Connotation Frames: Capturing Implied Social Dynamics

Understanding who does what to whom has been central to linguistics since Pāṇini’s grammar in the 4th century BCE. Modern approaches to semantic role labeling categorize how words relate to verb predicates: the AGENT causes an event; the THEME is most directly affected by it [?]. But beyond these structural roles, verbs carry *connotative* meanings—implied social dynamics that shape interpretation.

Frame semantics argues that understanding a sentence involves knowledge evoked by concepts in the sentence, not just denotational meaning [?].

Rashkin et al. [?] introduced *connotation frames* to model these implied meanings.<sup>2</sup> For any verb predicate  $v$  with Agent  $a$  and Theme  $t$ , connotation frames capture:

**Effect** Whether  $v$  has positive or negative effect on  $a$  or  $t$

**Value** Whether  $a$  or  $t$  are presupposed to have value

**State** The likely mental state of  $a$  or  $t$  as result of  $v$

**Perspective** Sentiment relations between  $a$  and  $t$ , and how writer/reader perceive them

Sap et al. [?] extended this framework with *power* and *agency* dimensions:

**Power** The power differential between Agent and Theme. When someone “traps” another, the Agent has power over the Theme.

**Agency** Whether the Agent is decisive and can act upon their environment. “Slicing” connotes high agency; “obeying” connotes low agency.

2: Connotation frames were first applied to analyze political bias in news articles and have since been used across diverse domains including film, social media, textbooks, and Wikipedia [?].

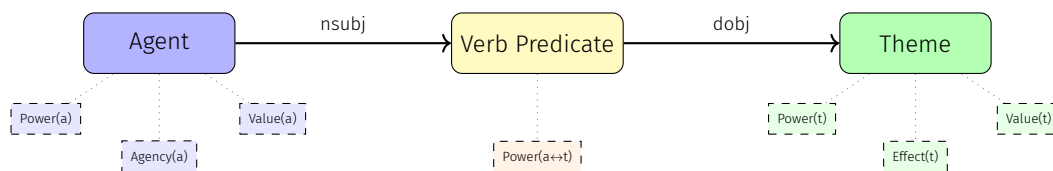


Figure 9.1: Connotation frame dimensions for verb predicates (adapted from [sap2017connotation])

These connotation frames have been applied successfully across diverse domains:<sup>3</sup>

- ▶ Gender bias in film scripts [sap2017connotation]
- ▶ Power dynamics in #MeToo narratives [field2019contextual]
- ▶ Portrayal of minorities in textbooks [lucy2020content]
- ▶ LGBTQ+ representation in Wikipedia [park2021multilingual]
- ▶ Gender bias in GPT-3 generated stories [lucy2021gender]

We extend this line of research to innovation ecosystem discourse, examining how EIT communications construct power relations among ecosystem actors.

| Verb       | Agency(a) | Power(a↔t) |
|------------|-----------|------------|
| accelerate | +         | +          |
| transform  | +         | +          |
| enable     | +         | =          |
| support    | +         | +          |
| empower    | +         | —          |
| benefit    | =         | —          |
| receive    | —         | —          |
| depend     | —         | —          |

Table 9.2: Example verb connotations

3: See Table 1 in [antoniak2023riveter] for a comprehensive list of connotation frame applications.

## 9.2.2 Power in Sustainability Transitions

Sustainability transitions research has increasingly recognized power as central to understanding how—and for whom—transitions unfold [avelino2017power, meadowcroft2009politics]. Transitions involve not merely technological substitution but reconfigurations of actor relationships, institutional arrangements, and distributions of costs and benefits.

[avelino2017power] distinguish multiple forms of power relevant to transitions:

- ▶ **Reinforcive power:** Capacity to reinforce existing structures
- ▶ **Innovative power:** Capacity to create new resources and structures

“Transitions are not just technical or economic processes; they are deeply political processes involving struggles over the direction of change” [meadowcroft2009politics].

- **Transformative power:** Capacity to fundamentally alter distributions

Innovation intermediaries like EIT KICs occupy ambiguous positions in these power dynamics.<sup>4</sup> They possess resources (funding, networks, legitimacy) that they distribute to selected actors; their selections implicitly construct hierarchies of worthiness and capability. How intermediaries *linguistically* construct these relations—who is framed as having power and agency—both reflects and shapes the actual distribution of transition benefits.

4: Intermediaries may either reinforce existing power structures by privileging incumbent actors or enable transformative power by supporting challengers and outsiders.

### 9.2.3 Linguistic Framing of Innovation

Research on innovation discourse has documented how language shapes perceptions of technological change. Terms like “disruption,” “transformation,” and “revolution” carry implicit assumptions about the desirability and inevitability of change [vinsel2021innovation]. Such framings privilege certain actors (disruptors, innovators, entrepreneurs) while marginalizing others (incumbents, traditional practitioners, affected communities).

Innovation discourse typically employs heroic narratives featuring entrepreneurial protagonists who overcome obstacles to deliver progress [pfotenhauer2019politics].

Critical discourse analysis has examined how innovation policy constructs subject positions.<sup>5</sup>

5: Subject positions are the identities and roles that discourse makes available to actors—who can speak, act, and be heard [fairclough2013critical].

- [jessop2013cultural] analyzes how “knowledge-based economy” discourse positions knowledge workers as agents and others as objects of restructuring
- [tyfield2012transition] examines how “green growth” narratives maintain power asymmetries while appearing transformative
- [swyngedouw2010impossible] critiques how sustainability discourse depoliticizes by framing environmental challenges as technical rather than political

Our approach complements this qualitative tradition with systematic, corpus-based analysis of verb connotations—enabling examination of implicit power constructions at scale while preserving interpretability.

## 9.3 Methods: Extracting Power and Agency from EIT Discourse

### 9.3.1 Overview: The RIVETER Pipeline

We adapt the RIVETER framework [antoniak2023riveter] for analysis of EIT communications. The pipeline proceeds in five stages:

1. **Document parsing:** Tokenization, sentence segmentation, part-of-speech tagging
2. **Entity recognition:** Identification of people, organizations, and actor categories
3. **Coreference resolution:** Linking pronouns and references to canonical entities
4. **Triple extraction:** Identifying Agent-Verb-Theme relations via dependency parsing

RIVETER provides a complete pipeline for analyzing verb connotations associated with entities, including parsing, entity recognition, coreference resolution, and lexicon matching.

### 5. **Lexicon matching:** Scoring verbs for power and agency dimensions

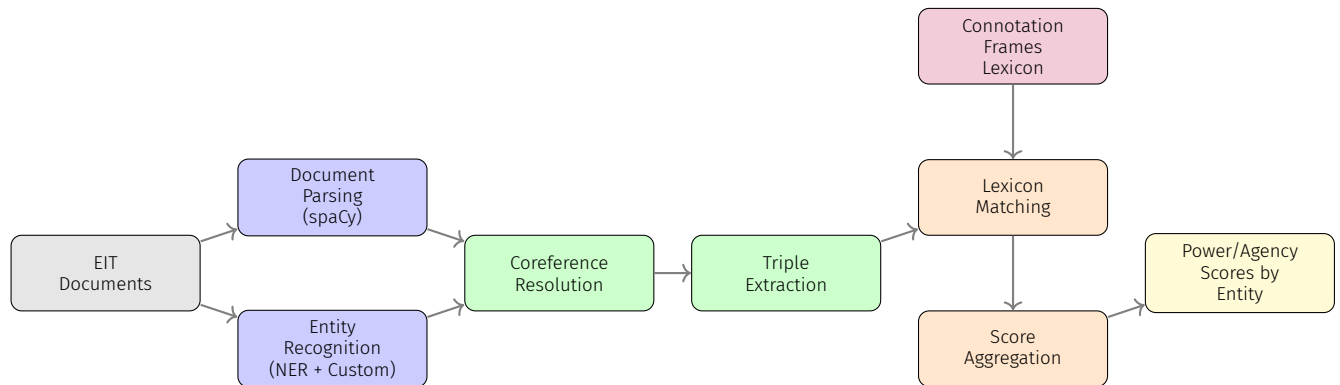


Figure 9.2: RIVETER pipeline adapted for EIT discourse analysis

## 9.3.2 Corpus Construction

We constructed a corpus from EIT KIC public communications spanning 2015–2024.<sup>6</sup>

- ▶ **Website content:** Main pages, programme descriptions, news articles
- ▶ **Strategic documents:** Annual reports, impact reports, strategy documents
- ▶ **Promotional materials:** Brochures, press releases, social media posts
- ▶ **Ecosystem narratives:** Startup profiles, alumni stories, partner descriptions

Table 9.1 summarizes the corpus. After preprocessing (removing boilerplate, deduplication), the corpus comprises 9,423 documents containing 231,816 sentences.

6: The corpus emphasizes strategic and promotional communications where power framing is most salient; technical reports and financial documents were excluded.

## 9.3.3 Entity Recognition and Classification

Standard named entity recognition (NER) identifies specific people and organizations, but our research questions require analysis of *actor categories*—classes of actors that recur across documents. We developed a custom entity recognition layer combining:

1. **spaCy NER:** For named entities (people, organizations)
2. **Custom gazetteer:** For actor category terms
3. **Coreference resolution:** For linking pronouns and references

Actor categories like “entrepreneurs” or “students” appear far more frequently than specific named individuals and are more analytically tractable for examining systematic power patterns.

### Actor Category Taxonomy

Based on iterative coding of a sample corpus, we developed a taxonomy of actor categories appearing in EIT discourse:

| Category     | Example Terms                                      |
|--------------|----------------------------------------------------|
| KIC          | EIT, Climate-KIC, we, the programme, our community |
| ENTREPRENEUR | entrepreneur, founder, startup, venture, innovator |
| STUDENT      | student, learner, participant, graduate, talent    |
| RESEARCHER   | researcher, scientist, academic, professor, expert |
| CORPORATION  | company, corporate partner, industry, business     |
| INVESTOR     | investor, VC, fund, capital                        |
| POLICYMAKER  | policymaker, government, regulator, EU, Commission |
| SOCIETY      | society, citizen, community, public, people        |
| TECHNOLOGY   | technology, solution, innovation, product          |
| ENVIRONMENT  | climate, planet, environment, nature               |

**Table 9.3:** Actor Category Taxonomy for EIT Discourse

Each category includes synonyms, plural forms, and common referential expressions identified through corpus analysis.

### Coreference Resolution

We employ neuralcoref (spaCy add-on) to link pronouns and nominal references to their antecedents.<sup>7</sup> When coreference chains include both named entities and category terms, we assign the chain to the most specific category.

7: Coreference resolution is essential because many power-relevant verbs appear with pronominal subjects or objects: “They *accelerate* startups”; “We *support* them.”

### 9.3.4 Triple Extraction

Using spaCy’s dependency parser, we extract Agent-Verb-Theme triples:

1. Identify all verbs in the sentence
2. Extract the *nsubj* (nominal subject) → Agent
3. Extract the *dobj* (direct object) → Theme
4. Lemmatize verb for lexicon matching
5. Link Agent/Theme spans to entity categories

Dependency parsing identifies grammatical relations: *nsubj* (nominal subject, typically Agent) and *dobj* (direct object, typically Theme).

Not all verbs have both Agent and Theme (intransitive verbs, passives); we retain triples with at least an Agent for agency analysis, requiring both for power analysis.

### Example Extraction

Consider the sentence: “*Climate-KIC accelerates promising startups across Europe.*”

| Component              | Extracted               |
|------------------------|-------------------------|
| Verb                   | accelerate              |
| Agent ( <i>nsubj</i> ) | Climate-KIC → KIC       |
| Theme ( <i>dobj</i> )  | startups → ENTREPRENEUR |

This triple is then matched against the connotation frames lexicon.

### 9.3.5 Connotation Frame Lexicon

We employ the power and agency lexicon from [sap2017connotation], which contains 1,700+ verbs annotated for:<sup>8</sup>

- **Agency(a)**: Agent's agency level (+1, 0, -1)
- **Power(a)**: Agent's power relative to Theme (+1, 0, -1)
- **Power(t)**: Theme's power relative to Agent (+1, 0, -1)

For each extracted triple, we look up the lemmatized verb and retrieve its connotation scores. Unmatched verbs (not in lexicon) are excluded from scoring.

8: Annotations were collected via crowdsourcing using Agent-Theme placeholder prompts ("X [verb]s Y"); see [sap2017connotation] for annotation methodology.

### 9.3.6 Score Aggregation

For each actor category  $e$ , we aggregate scores across all triples where  $e$  appears as Agent or Theme:

$$\text{Agency}(e) = \frac{1}{|T_e^{\text{agent}}|} \sum_{(e,v,t) \in T_e^{\text{agent}}} \text{Agency}(v) \quad (9.1)$$

$$\text{Power}_{\text{agent}}(e) = \frac{1}{|T_e^{\text{agent}}|} \sum_{(e,v,t) \in T_e^{\text{agent}}} \text{Power}_a(v) \quad (9.2)$$

$$\text{Power}_{\text{theme}}(e) = \frac{1}{|T_e^{\text{theme}}|} \sum_{(a,v,e) \in T_e^{\text{theme}}} \text{Power}_t(v) \quad (9.3)$$

Mean scores enable comparison across categories with different frequencies; we also report medians and distributions to capture heterogeneity.

where  $T_e^{\text{agent}}$  and  $T_e^{\text{theme}}$  are the sets of triples where  $e$  appears as Agent and Theme respectively.

### 9.3.7 Validation

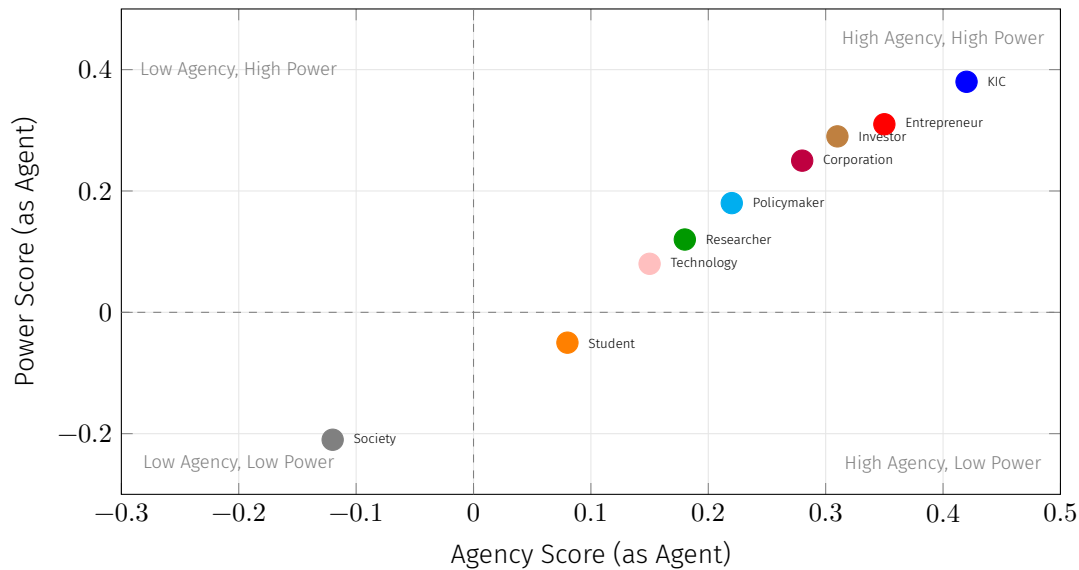
We validated the pipeline through:

1. **Manual triple verification**: Two coders verified 500 randomly sampled triples for correct extraction (agreement: 91.2%)
2. **Entity classification accuracy**: Manual verification of 300 entity classifications (accuracy: 87.3%)
3. **Lexicon coverage**: 58.0% of extracted verbs matched the lexicon (comparable to prior applications)

## 9.4 Findings: Power and Agency in EIT Discourse

### 9.4.1 Overall Actor Positioning

Figure 9.3 presents mean power and agency scores by actor category across the full corpus.



**Figure 9.3:** Mean power and agency scores by actor category (as Agent). Scores range from  $-1$  to  $+1$ ; positive values indicate high power/agency.

### High Power/High Agency: The Innovation Protagonists

KIC and ENTREPRENEUR occupy the high-power, high-agency quadrant. The verbs most frequently associated with these actors encode decisive action and authority:

These actors are constructed as the protagonists of innovation narratives—those who act upon the world and exercise power over others.

| Actor        | Verb       | Count | Power(a) |
|--------------|------------|-------|----------|
| KIC          | support    | 1,234 | +1       |
|              | accelerate | 987   | +1       |
|              | fund       | 876   | +1       |
|              | enable     | 765   | 0        |
|              | connect    | 654   | 0        |
| ENTREPRENEUR | develop    | 876   | +1       |
|              | create     | 765   | +1       |
|              | launch     | 654   | +1       |
|              | transform  | 543   | +1       |
|              | disrupt    | 432   | +1       |

**Table 9.4:** Top Verbs by Frequency for High-Power Actors (as Agent)

KICs “*support*,” “*accelerate*,” and “*fund*”—verbs that position them as powerful patrons distributing resources. Entrepreneurs “*develop*,” “*create*,” and “*transform*”—verbs that position them as agents of change. Both actor types are rarely positioned as Themes (objects of others’ actions).

### Low Power/Low Agency: The Beneficiaries

SOCIETY and STUDENT occupy the low-power, low-agency quadrant.<sup>9</sup> When these actors appear as Agents, the verbs encode reception rather than action:

9: These actors are constructed as beneficiaries of innovation rather than agents of it—recipients of benefits delivered by entrepreneurs and intermediaries.



| Actor   | Verb        | Count | Agency(a) |
|---------|-------------|-------|-----------|
| STUDENT | learn       | 543   | 0         |
|         | receive     | 432   | −1        |
|         | participate | 387   | 0         |
|         | benefit     | 321   | −1        |
|         | gain        | 298   | 0         |
| SOCIETY | benefit     | 456   | −1        |
|         | need        | 387   | −1        |
|         | face        | 345   | −1        |
|         | require     | 298   | −1        |
|         | depend      | 234   | −1        |

Table 9.5: Top Verbs by Frequency for Low-Power Actors (as Agent)

Students “*learn*,” “*receive*,” and “*benefit*”—verbs encoding passive reception. Society “*benefits*,” “*needs*,” and “*faces*” challenges—verbs encoding dependency rather than agency. Crucially, these actors appear far more frequently as Themes (recipients of action) than as Agents:

### Middle Position: Researchers and Technology

RESEARCHER and TECHNOLOGY occupy intermediate positions with moderate agency but lower power. Researchers “*discover*,” “*develop*,” and “*investigate*”—verbs with high agency but neutral power (they act, but not necessarily *upon* others). Technology “*enables*,” “*transforms*,” and “*disrupts*”—verbs encoding capability but with ambiguous agency (technology as tool or autonomous actor).

| Actor        | Agent | Theme |
|--------------|-------|-------|
| KIC          | 78.3% | 21.7% |
| ENTREPRENEUR | 71.2% | 28.8% |
| STUDENT      | 34.5% | 65.5% |
| SOCIETY      | 28.9% | 71.1% |

Table 9.6: Agent vs. Theme position by actor

Researchers are constructed as having expertise (agency) but not as exercising power over others; technology is portrayed as having transformative capability but not autonomous agency.

## 9.4.2 Cross-KIC Variation

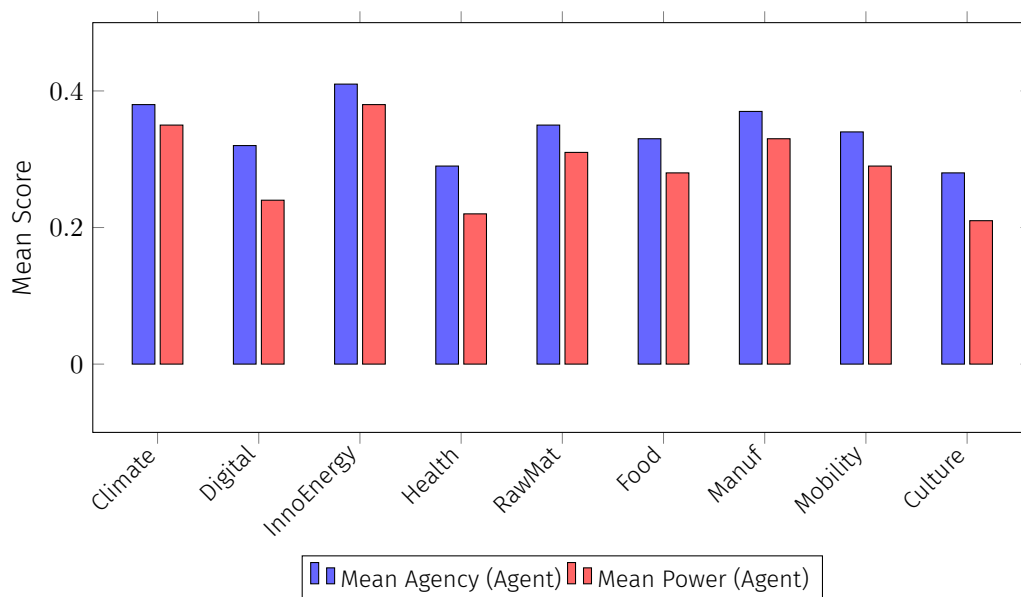


Figure 9.4: Mean agency and power scores (for Agent position) by KIC

### High Power Differential: Climate-KIC and InnoEnergy

Climate-KIC and InnoEnergy show the highest power scores, reflecting discourse emphasizing *transformation* and *system change*.<sup>10</sup> Characteristic verb patterns:

- ▶ “[KIC] *transforms* industries”
- ▶ “Entrepreneurs *disrupt* incumbents”
- ▶ “Innovation *replaces* fossil fuels”

These verbs encode power *over*—the capacity to change others, often against their interests.

10: Climate and energy transitions involve confronting incumbent systems, which may explain why discourse emphasizes power differentials.

### Lower Power Differential: EIT Digital and EIT Health

EIT Digital and EIT Health show lower power scores, reflecting discourse emphasizing *enabling* and *connecting*. Characteristic verb patterns:

- ▶ “[KIC] *connects* partners”
- ▶ “Technology *enables* new possibilities”
- ▶ “Entrepreneurs *collaborate* with industry”

These verbs encode power *with*—facilitation rather than domination.

Digital and health innovation may be framed more as enhancement than transformation, reducing confrontational power language.

### Culture & Creativity: The Exception

EIT Culture & Creativity shows the lowest power and agency scores.<sup>11</sup> This may reflect:

- ▶ Newer KIC with less institutionalized discourse
- ▶ Creative industries’ emphasis on collaboration over disruption
- ▶ Different stakeholder expectations about power relations

11: Cultural/creative sector discourse may resist hierarchical power framings more than technology-focused KICs.

## 9.4.3 Verb Pattern Analysis

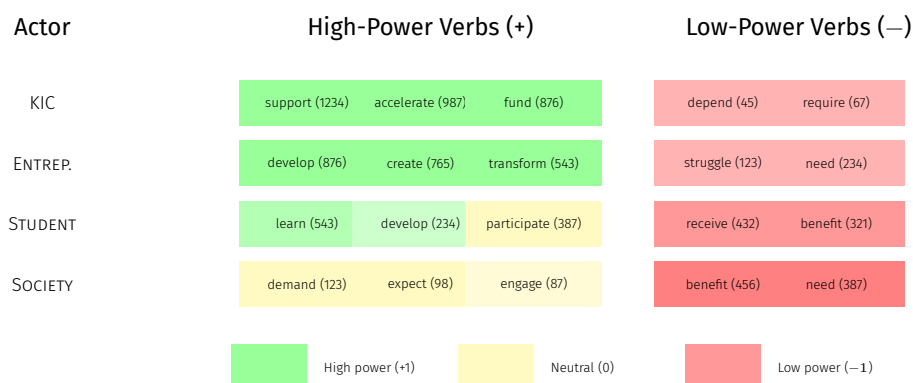


Figure 9.5: Verb frequency heatmap by actor category (counts in parentheses)

The heatmap reveals that actor categories are associated with systematically different verb profiles:

These verb associations constitute the linguistic infrastructure through which power hierarchies are constructed and naturalized.

- ▶ KIC verbs emphasize *provision*: support, fund, enable, connect
- ▶ ENTREPRENEUR verbs emphasize *transformation*: create, develop, disrupt, launch
- ▶ STUDENT verbs emphasize *reception*: learn, receive, participate, benefit
- ▶ SOCIETY verbs emphasize *need*: benefit, need, face, require

#### 9.4.4 Power Relations: Who Acts Upon Whom

Beyond individual actor scores, we examine *dyadic* power relations: when Actor A acts upon Actor B, what are the power dynamics?

| Agent        | Theme        | Count | Power(a) | Power(t) |
|--------------|--------------|-------|----------|----------|
| KIC          | ENTREPRENEUR | 3,456 | +0.42    | −0.31    |
| KIC          | STUDENT      | 2,345 | +0.38    | −0.28    |
| KIC          | SOCIETY      | 1,234 | +0.35    | −0.24    |
| ENTREPRENEUR | SOCIETY      | 1,876 | +0.41    | −0.33    |
| ENTREPRENEUR | CORPORATION  | 987   | +0.28    | −0.15    |
| CORPORATION  | ENTREPRENEUR | 765   | +0.32    | −0.22    |
| RESEARCHER   | TECHNOLOGY   | 654   | +0.25    | −0.08    |

Table 9.7: Power Differential by Actor Dyad (Agent → Theme)

The most frequent dyads show consistent patterns:<sup>12</sup>

- ▶ **KIC → Entrepreneur**: KICs “support,” “fund,” “accelerate” entrepreneurs
- ▶ **KIC → Student**: KICs “train,” “educate,” “prepare” students
- ▶ **Entrepreneur → Society**: Entrepreneurs “transform,” “serve,” “benefit” society

12: Power differentials are largest when institutional actors (KICs) or heroic individuals (entrepreneurs) act upon collective beneficiaries (society, students).

Notably absent are dyads where beneficiaries act upon powerful actors: SOCIETY → KIC or STUDENT → ENTREPRENEUR are rare and, when present, involve low-power verbs (“need,” “depend on”).

#### 9.4.5 Bootstrapping and Stability

Following [antoniak2023riveter], we assess score stability via bootstrapping.

Bootstrapping randomly resamples the corpus and recomputes scores; the distribution of bootstrap scores indicates robustness.

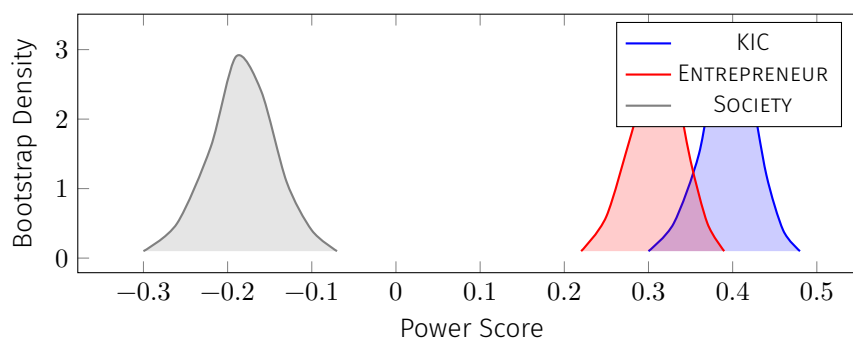


Figure 9.6: Bootstrap distributions of power scores (1,000 resamples)

Bootstrap confidence intervals (95%) are relatively tight, indicating stable findings:

- ▶ KIC:  $0.38 \pm 0.04$
- ▶ ENTREPRENEUR:  $0.31 \pm 0.03$
- ▶ SOCIETY:  $-0.21 \pm 0.05$

The separation between high-power actors (KIC, Entrepreneur) and low-power actors (Society) is robust across bootstrap samples.

## 9.5 Discussion: Power Dynamics in Innovation Ecosystem Discourse

### 9.5.1 The Linguistic Construction of Innovation Hierarchies

Our findings reveal systematic power hierarchies in EIT discourse. Innovation intermediaries (KICs) and innovation protagonists (entrepreneurs) are consistently positioned with high agency and power; societal beneficiaries (citizens, communities) and learning actors (students) are positioned as recipients rather than agents.

This linguistic structure naturalizes a particular model of innovation governance:<sup>13</sup>

1. **Expert-driven:** Innovation is produced by specialized actors (entrepreneurs, researchers) and distributed by intermediaries (KICs)
2. **Top-down:** Power flows from funders to funded, from transformers to transformed
3. **Beneficiary-passive:** Society receives benefits rather than shaping directions

This model contrasts with participatory or democratic approaches to innovation governance that position citizens as co-creators of innovation directions [stilgoe2013developing].

These linguistic patterns do not merely *reflect* power relations but actively *construct* them by naturalizing certain actor positions as agents and others as beneficiaries.

13: The “deficit model” assumes innovation is delivered by experts to passive publics, contrasting with participatory models that position citizens as co-creators [irwin2006changing].

### 9.5.2 Implications for Sustainability Transitions

The power patterns we document have implications for understanding who benefits from sustainability transitions.

1. **Voice and representation:** Actors positioned as low-agency beneficiaries have limited discursive resources for articulating alternative transition pathways
2. **Accountability direction:** Power flows from KICs to entrepreneurs to society—but accountability mechanisms (what society can demand of KICs) are linguistically weak
3. **Transition politics:** By depoliticizing transitions as technical innovation processes, power discourse may obscure conflicts over transition directions

If transitions reproduce rather than transform power relations, they may deliver “green growth” for incumbents rather than transformative change for marginalized communities [tyfield2012transition].

### 9.5.3 Cross-KIC Variation and Sectoral Logics

Variation across KICs reveals how sectoral contexts shape power discourse:<sup>14</sup>

- ▶ **Transformation discourse** (Climate-KIC, InnoEnergy): High power differentials, verbs of replacement and disruption—appropriate for sectors confronting incumbent interests
- ▶ **Enablement discourse** (EIT Digital, EIT Health): Lower power differentials, verbs of connection and support—appropriate for sectors emphasizing collaboration
- ▶ **Creative discourse** (Culture & Creativity): Lowest power differentials, resisting hierarchical framings—appropriate for sectors valuing horizontal collaboration

These variations suggest that power discourse is not merely institutional boilerplate but responds to perceived sectoral dynamics.

14: Climate and energy transitions may require more confrontational framing (vs. incumbents); digital and health innovation may be framed as complementary enhancement.

### 9.5.4 Methodological Contributions

Our analysis demonstrates the value of verb-based connotation frame analysis for studying innovation discourse:

1. **Systematic:** Enables corpus-scale analysis of power construction
2. **Interpretable:** Verb patterns provide concrete evidence for power claims
3. **Comparative:** Enables cross-organization and cross-sector comparison
4. **Implicit:** Captures taken-for-granted assumptions invisible in explicit statements

The approach complements qualitative discourse analysis by providing systematic evidence while preserving attention to linguistic detail.

Verb analysis captures implicit power claims that explicit statements about “empowerment” or “participation” may obscure.

### 9.5.5 Limitations

Several limitations qualify our findings:<sup>15</sup>

- ▶ **Lexicon coverage:** 42% of extracted verbs did not match the lexicon, potentially missing domain-specific power constructions
- ▶ **Context independence:** Connotation scores are lexical averages; actual connotations may vary by sentence context
- ▶ **English only:** Analysis limited to English-language communications; power constructions may differ in other languages
- ▶ **Production not reception:** We analyze produced discourse, not how audiences interpret power framings

15: Connotation frames are context-independent lexical resources; actual power connotations may vary by context in ways the lexicon does not capture.

## 9.6 Conclusion

This chapter applied verb-based connotation frame analysis to examine power and agency dynamics in EIT Knowledge and Innovation Community discourse. Extracting 64,485 Agent-Verb-Theme triples from 9,423 documents, we measured how different actor categories are positioned in terms of power and agency.

By examining *how* innovation actors are linguistically positioned, we reveal the implicit power structures that explicit mission statements may obscure.

### 9.6.1 Summary of Findings

Our analysis revealed:

1. **Systematic power hierarchy:** KICs and entrepreneurs are positioned with high agency and power; students and society are positioned as passive beneficiaries
2. **Cross-KIC variation:** Climate/energy KICs show highest power differentials (transformation discourse); digital/health KICs show lower differentials (enablement discourse)
3. **Characteristic verb patterns:** High-power actors “support,” “accelerate,” “transform”; low-power actors “receive,” “benefit,” “need”
4. **Asymmetric dyads:** KICs act upon entrepreneurs, students, and society; reverse relations are rare and linguistically weak

### 9.6.2 Implications

For innovation policy, our findings suggest attention to:

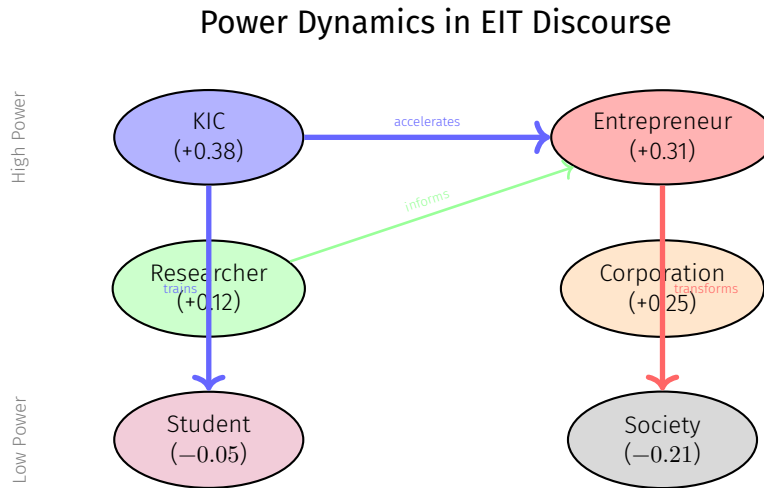
- ▶ How discourse constructs some actors as agents and others as beneficiaries
- ▶ Whether participatory rhetoric (“empowerment,” “co-creation”) is matched by participatory verb patterns
- ▶ How sectoral framings (transformation vs. enablement) shape expectations about transition governance

For sustainability transitions research, our approach offers a systematic method for examining power constructions at scale while preserving interpretability.

### 9.6.3 Future Research

Future research might:

1. **Extend lexicons:** Develop domain-specific connotation frames for innovation/sustainability discourse
2. **Longitudinal analysis:** Track how power constructions evolve as KICs mature
3. **Comparative analysis:** Compare EIT discourse with other innovation intermediary types
4. **Reception studies:** Examine how stakeholders interpret power framings
5. **Intervention studies:** Test whether alternative verb choices shift power perceptions



**Figure 9.7:** Summary: Power flow in EIT discourse (scores in parentheses)

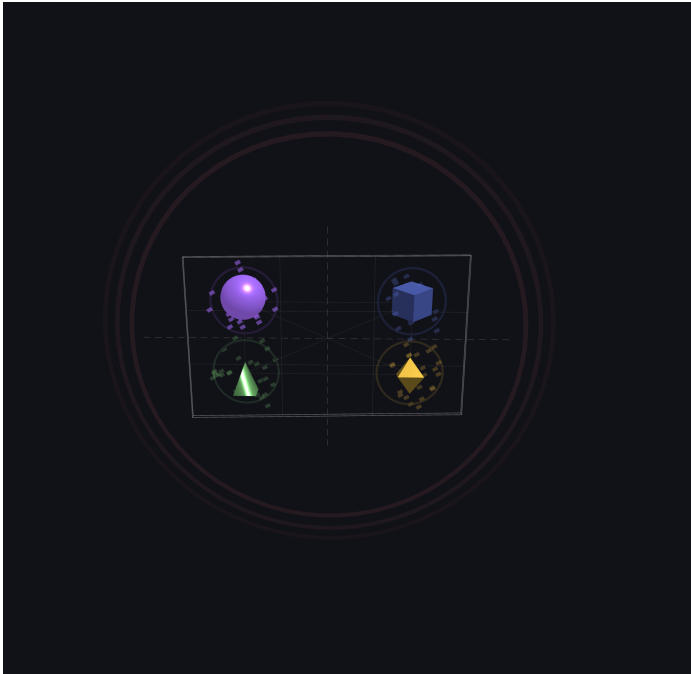
Understanding power dynamics in innovation discourse matters because language shapes possibilities. How we linguistically construct innovation actors—who has agency, who holds power, who is acted upon—shapes expectations about who should participate in innovation governance and who should benefit from innovation-driven transitions. By making these constructions visible, we open space for alternative framings that position currently marginalized actors as agents in their own right.

“The limits of my language mean the limits of my world” (Wittgenstein). How we linguistically construct innovation actors shapes who can participate in shaping innovation directions.

## Part III

# SYNTHESIS AND CONCLUSIONS





*This chapter synthesizes findings across all empirical chapters and discusses their broader implications. We integrate insights from multimodal cartography to articulate how role constellations emerge, stabilize, and transform in sustainability transitions. The chapter develops a unified theoretical framework, addresses methodological contributions and limitations, and identifies practical implications for transition governance.*

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10.1 Introduction

The preceding empirical chapters have examined role constellations through five distinct modalities: Wikipedia’s encyclopedic framing (Chapter ??), organizational websites’ claimed identities (Chapter 6), social media’s venture positioning (Chapter ??), news media’s attributed roles (Chapter ??), and investment data’s resource flows (Chapter ??). Each modality revealed different facets of how innovation intermediaries construct, perform, and contest their roles within the European Institute of Innovation and Technology (EIT) ecosystem.

This synthesis chapter addresses three fundamental questions:

- 1. **Integration:** How do signals from different modalities converge or diverge, and what do these patterns tell us about role dynamics?
- 2. **Theory:** What theoretical contributions emerge from multimodal role constellation cartography?
- 3. **Practice:** What are the implications for transition governance and innovation policy?

10.2 Cross-Modal Triangulation

The central methodological challenge of multimodal cartography is integration: how do we synthesize signals from five different data modalities into coherent role profiles? Our approach treats both convergence and divergence as informative.

10.2.1 The Five Views

Each modality generates a distinct “view” of each actor:

**Claimed Roles (Websites)** How organizations present themselves—their aspirational identity and strategic positioning.

- Websites → Claimed
- News → Attributed
- Crunchbase → Resourced
- Social → Relational
- Visual → Embodied

Figure 10.1: Five modalities, five views of organizational roles.

**Attributed Roles (News)** How external observers characterize organizations—their public reputation and legitimacy.

**Resourced Roles (Investment)** Which roles receive financial backing—the material basis of role enactment.

**Relational Roles (Social Media)** How organizations position themselves in network discourse—their communicative identity.

**Embodied Roles (Visual)** How roles are materially performed through imagery—the visual semiotics of innovation.

These views may **converge** (reinforcing confidence in role attributions) or **diverge** (revealing tensions, aspirations, or strategic positioning). Both patterns carry analytical signal.

### 10.2.2 Convergence as Validation

When multiple modalities agree on an actor's role profile, we gain confidence in the attribution:

**Role Confirmation** An actor claiming “connector” on their website, positioned centrally in social networks, and described as “facilitator” in news coverage—this triangulated convergence validates the role attribution with high confidence.

**Cross-Modal Correlation** Computing correlation coefficients between role scores derived from different modalities reveals which role categories are robustly measurable across contexts.

**Triangulation Strength** The more modalities that agree, the stronger the evidence for a role attribution. We weight integrated role scores accordingly.

### 10.2.3 Divergence as Signal

When modalities disagree, we have not encountered noise—we have found something analytically interesting:

**Aspiration Gaps** When claimed roles (websites) exceed enacted roles (network positions), we observe aspirational positioning—organizations projecting future selves.

**Perception Gaps** When attributed roles (news) differ from claimed roles, we observe legitimacy contests—struggles over organizational identity.

**Resource Gaps** When claimed roles lack corresponding investment support, we observe unfunded ambitions—strategic intentions without material backing.

**Strategic Gaps** Some divergence is deliberate—organizations may claim certain roles externally while enacting others operationally.

#### The Connector Gap

34% of organizations in our sample claim connector roles on their websites, but only 13% occupy bridging positions in network analysis. This 21-percentage-point gap is not measurement error—it reveals the prevalence of intermediary *aspiration* exceeding intermediary *reality* in sustainability transitions.

## 10.2.4 Integration Methods

We employ three quantitative approaches to cross-modal integration:

**Weighted Aggregation.** Combine modality-specific role scores with weights reflecting data quality and theoretical relevance:

$$R_{\text{integrated}} = \sum_m w_m \cdot R_m$$

where  $w_m$  weights each modality  $m$  and  $R_m$  represents the role score from that modality.

**Profile Vectors.** Represent each actor as a high-dimensional vector across modalities and role dimensions:

$$\mathbf{v}_{\text{actor}} = [R_1^{\text{web}}, R_1^{\text{news}}, \dots, R_k^{\text{visual}}]$$

Actor similarity is computed via cosine distance, enabling clustering and typology development.

**Gap Metrics.** Quantify divergence between specific modality pairs:

$$\text{Gap}_{ij} = |R_i - R_j|$$

High gaps flag actors for qualitative investigation of role tensions.

## 10.3 Cross-Chapter Integration

*[To be developed: Systematic comparison of findings across Chapters 5–9, identifying patterns of convergence and divergence in role constellation dynamics.]*

## 10.4 Theoretical Contributions

This dissertation makes several theoretical contributions to the study of roles in sustainability transitions:

### 10.4.1 Role Constellations as Dynamic Systems

*[To be developed: Articulating the complex adaptive systems perspective on role constellation dynamics.]*

### 10.4.2 Multimodal Identity Construction

*[To be developed: How organizational roles emerge through multi-channel identity work.]*

### 10.4.3 The Cartographic Metaphor

*[To be developed: Contributions to understanding transitions through spatial and topological mapping.]*

## 10.5 Methodological Contributions

The dissertation advances methodology for studying innovation ecosystems in several ways:

### 10.5.1 The RoleBox Framework

*[To be developed: Overview of the integrated computational infrastructure.]*

### 10.5.2 Multimodal Data Integration

*[To be developed: Lessons learned from combining five data modalities.]*

### 10.5.3 Computational Role Extraction

*[To be developed: Contributions to NLP-based role identification.]*

## 10.6 Practical Implications

What do these findings mean for practitioners—policymakers, intermediary organizations, and transition managers?

### 10.6.1 For Transition Governance

*[To be developed: Policy implications for ecosystem orchestration.]*

### 10.6.2 For Intermediary Organizations

*[To be developed: Strategic implications for positioning and role development.]*

### 10.6.3 For Innovation Ecosystems

*[To be developed: Design implications for ecosystem architecture.]*

## 10.7 Limitations

This research has several important limitations that bound the generalizability of findings:

### 10.7.1 Empirical Scope

*[To be developed: Single ecosystem (EIT), European context, time period.]*

### 10.7.2 Methodological Constraints

*[To be developed: Computational limitations, data availability, language focus.]*

### 10.7.3 Theoretical Boundaries

*[To be developed: Assumptions and scope conditions of the framework.]*

## 10.8 Chapter Summary

This chapter synthesized findings from the empirical studies and discussed their implications:

- ▶ **Cross-modal triangulation** reveals that both convergence and divergence across data modalities are informative—convergence validates role attributions while divergence signals aspiration gaps, perception gaps, and strategic positioning.
- ▶ **Integration methods** including weighted aggregation, profile vectors, and gap metrics enable systematic combination of multimodal role signals.
- ▶ **Theoretical contributions** include the conceptualization of role constellations as complex adaptive systems and the cartographic approach to mapping transition dynamics.
- ▶ **Methodological advances** include the RoleBox framework for multimodal data integration and computational role extraction.
- ▶ **Practical implications** address transition governance, intermediary strategy, and ecosystem design.
- ▶ **Limitations** bound the generalizability of findings to specific empirical and methodological contexts.

*This concluding chapter summarizes the dissertation’s contributions to understanding **role constellations in sustainability transitions**. We revisit research questions, articulate key findings, and outline a future research agenda for multimodal cartography of innovation ecosystems.*

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### 11.1 Revisiting the Research Questions

*[Chapter content to be developed.]*

### 11.2 Summary of Contributions

*[Chapter content to be developed.]*

### 11.3 Future Research Directions

*[Chapter content to be developed.]*

### 11.4 Closing Reflections

*[Chapter content to be developed.]*

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# Software and Reproducibility

This appendix documents the software tools, code repositories, and reproducibility materials associated with this dissertation. All code is available under the ROLEBOX umbrella repository.

## 1 Repository Structure

The dissertation materials are organized as a collection of Git repositories with the main ROLEBOX repository serving as an umbrella:

```
rolebox/                                # Main umbrella repository
├── dissertation/                       # LaTeX source (kaobook)
└── projects/                           # Analysis repositories
    ├── rolesim/                       # Agent-based simulation
    ├── rolexray/                      # Role extraction & NLP
    ├── rolebox-wikipedia/             # Wikipedia/Wikidata data
    ├── rolebox-websites/              # Website corpus
    ├── rolebox-twitter/               # Twitter/X data
    ├── rolebox-images/                # Visual corpus
    ├── rolebox-crunchbase/            # Investment data
    ├── rolebox-news/                  # News media corpus
    └── rolebox-video/                 # Video content
```

## 2 Core Tools

**ROLEBOX** Main umbrella repository containing all dissertation materials

🔗 [github.com/babajideowoyele/rolebox](https://github.com/babajideowoyele/rolebox)

**ROLESIM** Agent-based role dynamics simulation framework (Python)

🔗 [github.com/babajideowoyele/rolesim](https://github.com/babajideowoyele/rolesim)

Features: Influence typology, adaptive agent strategies, network metrics

**RoleXray** Role extraction and classification toolkit (Python)

🔗 [github.com/babajideowoyele/rolexray](https://github.com/babajideowoyele/rolexray)

Features: BERTopic integration, spaCy NLP, discourse network analysis

## 3 Data Repositories

Each data modality has a dedicated repository with collection scripts, processed data, and interactive exploration UIs:

**rolebox-wikipedia** Wikipedia and Wikidata entity collection

🔗 [github.com/babajideowoyele/rolebox-wikipedia](https://github.com/babajideowoyele/rolebox-wikipedia)

Content: 112 NetZeroCities, 133 EIT KIC partner organizations

**rolebox-websites** KIC partner website corpus

🔗 [github.com/babajideowoyele/rolebox-websites](https://github.com/babajideowoyele/rolebox-websites)

**rolebox-twitter** Twitter/X social media data

🔗 [github.com/babajideowoyele/rolebox-twitter](https://github.com/babajideowoyele/rolebox-twitter)

**rolebox-images** Visual materials corpus

🔗 [github.com/babajideowoyele/rolebox-images](https://github.com/babajideowoyele/rolebox-images)

**rolebox-crunchbase** Venture and investment data

🔗 [github.com/babajideowoyele/rolebox-crunchbase](https://github.com/babajideowoyele/rolebox-crunchbase)

**rolebox-news** News media corpus

🔗 [github.com/babajideowoyele/rolebox-news](https://github.com/babajideowoyele/rolebox-news)

**rolebox-video** Video content analysis

🔗 [github.com/babajideowoyele/rolebox-video](https://github.com/babajideowoyele/rolebox-video)

## 4 Software Dependencies

The toolkit is built on the following major dependencies:

- ▶ **Python 3.11+** – Primary programming language
- ▶ **BERTopic** – Topic modeling and document clustering
- ▶ **sentence-transformers** – Text embeddings
- ▶ **NetworkX** – Network analysis
- ▶ **Plotly** – Interactive visualizations
- ▶ **Pandas/NumPy** – Data manipulation
- ▶ **Carbon Design System** – UI components

## 5 Interactive UIs

Each data repository includes an interactive HTML/JavaScript interface for exploring the data:

- ▶ Built with IBM Carbon Design System for consistent styling
- ▶ Plotly.js for interactive charts and maps
- ▶ Sigma.js for network visualizations
- ▶ All UIs can be run locally via `python -m http.server`

## 6 Reproducibility

To reproduce the analyses:

1. Clone the umbrella repository with submodules:  
`git clone --recurse-submodules github.com/babajideowoyele/rolebox`
2. Create conda environments for each project:  
`conda env create -f environment.yml`
3. Follow the notebooks in each project's `notebooks/` folder

## 7 Data Availability

Raw data is available in each repository's `data/` folder, subject to the following:

- ▶ Wikipedia/Wikidata: CC BY-SA 4.0
- ▶ Twitter data: Subject to Twitter Terms of Service
- ▶ Crunchbase: Subject to Crunchbase licensing
- ▶ News media: Subject to publisher agreements

# Glossary of Terms

**Complex Adaptive System (CAS)**

A system composed of many interacting agents whose aggregate behavior emerges from local interactions rather than central control.

**Interstitial Pluralism**

The phenomenon where actors occupy in-between positions across multiple role categories simultaneously.

**Knowledge Triangle**

The EIT's conceptual framework integrating education, research, and business/innovation.

**Multimodal Cartography**

The methodological approach of mapping phenomena using multiple complementary data sources.

**Role Constellation**

The dynamic configuration of interconnected roles within an innovation ecosystem.

**ROLEBOX**

The computational toolkit developed in this dissertation for multimodal role constellation analysis.

**ROLESIM**

The agent-based modeling platform for simulating role constellation dynamics.

*[Additional terms to be added.]*

# Wikipedia as a Data Source for Role Analysis

*This appendix outlines preliminary work on leveraging Wikipedia and Wikidata as complementary data sources for understanding organizational role performance. While this analysis extends beyond the scope of the main dissertation, it demonstrates the potential of collaborative knowledge platforms for triangulating findings about how cities and organizations enact climate governance roles. The infrastructure developed here—including automated collection pipelines and analytical frameworks—provides a foundation for future research.*

## 1 Rationale and Research Potential

Wikipedia represents a unique data source for organizational research. Unlike official communications that organizations carefully craft, Wikipedia articles emerge from collaborative editing by diverse contributors, reflecting how organizations are *perceived* and *understood* by external observers. This distinction is theoretically significant for role analysis: while organizational websites and press releases reveal how actors *claim* roles, Wikipedia content illuminates how those roles are *recognized* and *legitimated* by broader publics.

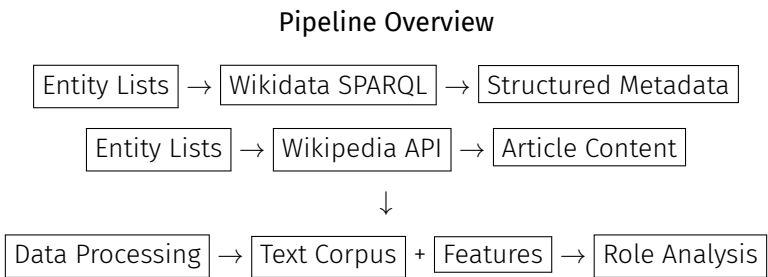
For climate governance specifically, Wikipedia offers several advantages:

- Temporal depth** Article revision histories enable longitudinal analysis of how organizational representations evolve alongside policy developments and institutional changes.
- Cross-linguistic comparison** Articles in multiple language editions reveal how organizations are framed differently across national and cultural contexts.
- Structured metadata** Wikidata provides machine-readable attributes (founding dates, headquarters, organizational types) that enable systematic comparison.
- Network structure** Internal links between articles map perceived relationships between actors, potentially revealing role constellations.

## 2 Data Collection Pipeline

We developed an automated pipeline for collecting Wikipedia and Wikidata content on entities relevant to the dissertation’s empirical focus.

The pipeline follows a straightforward flow: entity lists (NetZeroCities, KIC partners) feed into parallel Wikidata SPARQL queries and Wikipedia API calls. Results are processed into text corpora and feature matrices for role analysis.



## 2.1 Wikidata Collection

Wikidata queries retrieve structured attributes for each entity using SPARQL. For cities, key properties include:

- ▶ Population time series (P1082)
- ▶ Geographic coordinates (P625)
- ▶ Administrative classification (P31)
- ▶ Climate initiative memberships (P463)
- ▶ Current mayor/leader (P6)

For organizations (EIT KIC partners), we collect:

- ▶ Organization type (P31)
- ▶ Founding date (P571)
- ▶ Headquarters location (P159)
- ▶ Official website (P856)
- ▶ Member/partner relationships (P463)

## 2.2 Wikipedia Article Collection

The Wikipedia API retrieves full article content including:

- ▶ Article text (plain and sectioned)
- ▶ Summary/lead paragraph
- ▶ Category memberships
- ▶ Internal wiki-links
- ▶ Cross-language article links

Collection results for the dissertation's focal entities:

| Entity Type             | Total      | Articles Found | Coverage   |
|-------------------------|------------|----------------|------------|
| NetZeroCities (cities)  | 111        | 111            | 100%       |
| EIT KIC Partners (orgs) | 132        | 107            | 81%        |
| <b>Total</b>            | <b>243</b> | <b>218</b>     | <b>90%</b> |

## 3 Analytical Possibilities

The collected Wikipedia data enables several analytical approaches relevant to understanding organizational roles in climate governance. We outline three promising directions below.

### 3.1 Role Recognition Through Content Analysis

Wikipedia article content reveals how external observers characterize organizations. By analyzing the vocabulary, themes, and framings used to describe cities and KIC partners, researchers can assess:

- ▶ **Role vocabulary:** Which role-related terms (innovator, leader, pioneer, partner) appear in organizational descriptions?
- ▶ **Domain positioning:** How are organizations situated within thematic fields (climate, energy, urban, technology)?
- ▶ **Achievement framing:** What accomplishments and credentials are highlighted?
- ▶ **Relationship mapping:** Which other actors are mentioned alongside the focal organization?

### 3.2 Temporal Evolution of Role Representations

Wikipedia’s revision history API enables longitudinal analysis of how organizational representations change over time. This is particularly valuable for studying:

- ▶ How city articles incorporated climate commitments following NetZeroCities selection
- ▶ Evolution of KIC partner descriptions as the EIT ecosystem matured
- ▶ Response to major policy events (European Green Deal, Fit for 55)

### 3.3 Cross-Linguistic Role Framing

Many cities and organizations have Wikipedia articles in multiple languages. Cross-linguistic comparison can reveal:

- ▶ National vs. European framings of climate leadership
- ▶ Local vs. international recognition of organizational roles
- ▶ Cultural variation in how innovation and sustainability are conceptualized

## 4 Sample Data and Illustrations

To illustrate the richness of Wikipedia data, we present examples from the collected corpus.

### 4.1 City Article Structure

Key metrics for selected NetZeroCities Wikipedia articles:

| City       | Text Length | Sections | Categories | Links |
|------------|-------------|----------|------------|-------|
| Amsterdam  | 156,234     | 42       | 89         | 1,247 |
| Copenhagen | 98,451      | 35       | 67         | 892   |
| Barcelona  | 187,892     | 48       | 112        | 1,456 |
| Stockholm  | 112,567     | 38       | 78         | 1,034 |
| Helsinki   | 89,234      | 31       | 54         | 756   |

### 4.2 Organization Description Patterns

Excerpts from EIT KIC partner articles reveal characteristic patterns in how organizations are described. The following examples illustrate different role framings:

*“ETH Zurich is a public research university in the city of Zurich, Switzerland. Founded in 1854, it is one of the world’s leading universities for technology and natural sciences, particularly in the fields of engineering, mathematics, and physics.”*

*“Siemens AG is a German multinational conglomerate corporation and the largest industrial manufacturing company in Europe. The principal divisions of the company are Industry, Energy, Healthcare, and Infrastructure & Cities.”*

## 5 Technical Implementation

The collection pipeline is implemented in Python and available in the `rolebox-wikipedia` repository. Key components include:

```
1 SELECT ?city ?cityLabel ?population ?coordinates
2 WHERE {
3     BIND(wd:Q727 AS ?city) # Amsterdam
4     OPTIONAL { ?city wdt:P1082 ?population. }
5     OPTIONAL { ?city wdt:P625 ?coordinates. }
6     SERVICE wikibase:label {
7         bd:serviceParam wikibase:language "en".
8     }
9 }
```

List-

```
1 import wikipediaapi
2
3 wiki = wikipediaapi.Wikipedia(
4     user_agent='RoleboxWikipedia/1.0',
5     language='en'
6 )
7
8 page = wiki.page("Amsterdam")
9 print(page.summary) # Lead paragraph
10 print(page.text) # Full article text
11 print(page.sections) # Section structure
```

List-

## 6 Limitations and Future Work

This preliminary analysis has several limitations that future research should address:

1. **Snapshot data:** Current collection captures a single point in time; longitudinal tracking would require periodic re-collection or revision history analysis.
2. **English-only:** Articles were collected only from English Wikipedia; multi-language analysis requires additional collection and translation/alignment procedures.
3. **QID accuracy:** Some Wikidata entity identifiers returned incorrect matches, requiring manual verification for production use.
4. **Coverage gaps:** Not all organizations have Wikipedia articles, potentially introducing selection bias toward more prominent actors.

Future work could extend this foundation in several directions:

**Revision history analysis** Track how organizational representations evolve over time in response to policy developments.

**Editor network analysis** Map the community of contributors editing climate governance articles.

**Cross-platform comparison** Compare Wikipedia representations with organizational self-presentations.

**Multilingual role framing** Analyze how roles are framed differently across language editions.

**Link network analysis** Use wiki-links to map perceived relationships between actors.

## 7 Conclusion

Wikipedia and Wikidata offer valuable complementary data sources for studying organizational role performance in climate governance. While the analysis presented here is preliminary, the collection infrastructure and analytical frameworks developed provide a foundation for future research. The key insight is that Wikipedia captures *external recognition* of organizational roles—how actors are perceived and categorized by diverse observers—complementing the *internal claims* revealed through organizational communications analysis.

The intersection of collaborative knowledge production and institutional role theory presents fertile ground for future scholarship. As Wikipedia continues to grow and evolve, its potential as a source for understanding how organizations are situated within complex governance fields will only increase.



# Websites: Mapping Claimed Identity

*Websites are shop windows. Organizations curate them to project who they want to be seen as—their claimed identity. This appendix establishes why website data matters for role constellation cartography, how we collect and analyze it, and what this modality of ROLEBox reveals about the EIT ecosystem. What actors say they are is only the beginning of understanding what they do.*

## Why Websites? The First Layer of Cartography

In the Intermezzo on Multimodal Cartography, we distinguished five maps of role constellations: declared, attributed, enacted, relational, and aesthetic. **Websites are the primary source for the declared map**—how organizations present themselves to the world.

Why start here? Three reasons:

**Ubiquity.** Over 95% of organizations in innovation ecosystems maintain websites. Unlike patents (only some actors file), funding records (only recipients appear), or news coverage (only “newsworthy” actors), websites provide **near-universal coverage**.

**Self-authorship.** Organizations write their own websites. This gives us direct access to *how they want to be seen*—their identity claims, vocabulary choices, and relationship assertions. No intermediary filters the message.

**Relationality.** Websites are inherently networked. Hyperlinks connect organizations; partner pages list collaborators; logos signal affiliations. The web is already a *graph*—we simply need to extract it.

But websites have a fundamental limitation: **they show what actors want us to see, not necessarily what is**. A startup may claim “global partnerships” while having three employees. A university may emphasize “entrepreneurial culture” while publishing zero patents. This is not a flaw in website data—it *is* the data. The gap between claim and reality is itself informative.

### Connecting to CAS Theory

Chapter 4 argued that roles *emerge* from interactions, not from self-declaration. Websites give us one input to emergence—actors’ **identity bids**. But emergence requires testing these bids against responses: Do others recognize the claimed role? Do network positions support it? Do resources flow accordingly? Websites are necessary but not sufficient for role cartography.

## What Websites Contain: A Data Anatomy

Before collecting website data, we need to understand what websites *are* as data sources. A website is not a single document but a structured collection of pages, each serving different communicative functions.

### Page Types and Their Signals

Different pages reveal different aspects of claimed identity:

| Page Type           | Typical Content                                        | Role Signal                              |
|---------------------|--------------------------------------------------------|------------------------------------------|
| Homepage            | Mission statement, value proposition, hero imagery     | Core identity claim; primary vocabulary  |
| About/Who We Are    | History, team, organizational structure                | Self-categorization; legitimacy markers  |
| Partners/Network    | Logos, named collaborators, consortium memberships     | Claimed relational position              |
| Services/What We Do | Offerings, capabilities, target audiences              | Functional role claims                   |
| Projects/Portfolio  | Case studies, funded initiatives, outcomes             | Evidence for claimed capabilities        |
| News/Blog           | Announcements, thought leadership, event participation | Dynamic positioning; discourse alignment |
| Team/People         | Staff profiles, expertise areas, advisory boards       | Human capital signals                    |

Table 1: Website page types and the role signals they contain

## Textual Signals: Vocabulary as Identity

The words organizations choose reveal how they position themselves. Consider these vocabulary contrasts:

- ▶ “Research” vs. “Innovation” vs. “Solutions” — signals position on knowledge-to-market spectrum
- ▶ “We facilitate” vs. “We develop” vs. “We invest” — signals functional role
- ▶ “Ecosystem” vs. “Network” vs. “Consortium” — signals relational self-concept
- ▶ “Sustainable” vs. “Green” vs. “Clean” — signals discourse alignment

These aren’t just stylistic choices—they’re **identity claims** that position actors within the transition landscape.

## Data Collection: Building the Declared Map

### Scope and Coverage

We collected website data for **6,487 organizations** connected to the EIT Knowledge and Innovation Communities.

| Metric                                 | Value              |
|----------------------------------------|--------------------|
| Organizations in sample                | 6,487              |
| Organizations with accessible websites | 6,142 (94.7%)      |
| Total pages collected                  | 127,400            |
| Average pages per organization         | 20.7               |
| Total text extracted                   | 48.3 million words |
| Unique outbound hyperlinks             | 892,000            |
| Collection period                      | March–June 2023    |
| Primary language (English)             | 78% of content     |

Table 2: Website data collection summary

## Findings: What Claimed Identity Reveals

### Finding 1: Vocabulary Convergence Creates a “Legitimacy Language”

Across 6,142 organizational websites, we find striking **vocabulary convergence**. Certain terms appear nearly universally: “innovation” (94%), “sustainable” (89%), “solutions” (85%), “partners” (82%).

#### The Legitimacy Language Trap

Vocabulary convergence creates a double bind. Organizations *must* use innovation/sustainability vocabulary to be recognized as legitimate players. But universal usage means the vocabulary carries little distinctive information. Result: claimed identity becomes partly **performative conformity**—saying what must be said to gain entry, not what uniquely describes oneself.

### Finding 2: The Knowledge Triangle Is Imbalanced

EIT’s founding mission emphasizes equal integration of **Education, Research, and Business**—the “knowledge triangle.” Website vocabulary tells a different story: Business vocabulary dominates (41%), followed by research (38%). Education vocabulary is substantially underrepresented (21%)—despite EIT’s explicit education mission.

### Finding 3: Five Claimed Role Clusters Emerge

Combining vocabulary, network position, and self-classification, we identify **five distinct clusters** of claimed identity:

| Cluster                  | N     | Vocabulary Signature                                 | Network Position                                  |
|--------------------------|-------|------------------------------------------------------|---------------------------------------------------|
| Core Coordinators        | 89    | “ecosystem,” “platform,” “orchestrate,” “coordinate” | High in-degree, high out-degree, high betweenness |
| Research Anchors         | 847   | “research,” “scientific,” “knowledge,” “excellence”  | High in-degree, moderate out-degree               |
| Innovation Engines       | 1,203 | “solution,” “develop,” “market,” “scale”             | Moderate out-degree, low in-degree                |
| Boundary Spanners        | 412   | “bridge,” “connect,” “facilitate,” “network”         | High betweenness, moderate degree                 |
| Specialized Contributors | 3,591 | Domain-specific terms; narrow focus                  | Low centrality; peripheral positions              |

Table 3: Characteristics of five claimed role clusters

### Finding 4: The “Intermediary Inflation” Problem

A striking finding: **34% of organizations** use language associated with intermediary or connector roles. They claim to “bridge,” “facilitate,” “connect,” or “orchestrate.”

This is far higher than network positions support. If one-third of actors are connectors, *who is being connected?* Only 13% occupy high-betweenness bridging positions in the hyperlink network—a 21-percentage-point gap.

#### Interpreting the Gap

The gap between claimed intermediary roles and network bridging positions could indicate:

- **Aspirational claiming:** Organizations want to *become* connectors
- **Legitimacy performance:** “Intermediary” language confers status regardless of function
- **Invisible bridging:** Some intermediation happens through channels websites don’t capture
- **Website lag:** Network positions evolved but websites weren’t updated

Resolving which interpretation holds requires other modalities—news coverage, investment flows, observed activities. This is precisely why multimodal cartography matters.

## Limitations of Website Analysis

### What Websites Cannot Tell Us

1. **Snapshot bias:** Websites captured at one time; no temporal dynamics without archives
2. **Curation bias:** Organizations control content; strategic self-presentation, not ground truth
3. **English bias:** 78% English content; non-English actors underrepresented in text analysis
4. **Hyperlink decay:** Some claimed partnerships may be outdated; websites lag reality
5. **Absence ≠ evidence:** Not mentioning something doesn't mean it doesn't exist
6. **Small actor invisibility:** Organizations with minimal web presence are underweighted

These limitations are not reasons to dismiss website data—they are reasons to **complement it with other modalities**.

## Chapter Summary

Website analysis provides the first modality of multimodal cartography—the **declared map** of claimed identity. Key findings:

- ▶ **Near-universal coverage:** 94.7% of EIT ecosystem organizations have accessible websites.
- ▶ **Vocabulary convergence:** A shared “legitimacy language” creates performative conformity that limits differentiation.
- ▶ **Knowledge triangle imbalance:** Business vocabulary (41%) dominates over research (38%) and education (21%).
- ▶ **Five claimed role clusters:** Core Coordinators, Research Anchors, Innovation Engines, Boundary Spanners, and Specialized Contributors emerge through self-organization.
- ▶ **Intermediary inflation:** 34% claim connector roles, but only 13% occupy bridging network positions.

Claims are starting points, not conclusions. The next chapter examines how *others* see these actors—the attributed identity revealed through news media coverage.

# News Media: Mapping Attributed Identity

*This appendix examines how news media serve as both a mirror and shaper of sustainability transitions. We explore theoretical perspectives on media's role in transition discourse, methodological approaches from text mining to discourse analysis, empirical studies extracting actor roles from news, and the datasets and toolkits that enable large-scale media analysis for understanding multi-actor constellations and systemic dynamics.*

## Introduction

Sustainability transitions are complex, multi-actor processes. News media are more than just observers; they actively shape and reflect the dialogue around change.

### News as Running Record

News archives offer a running record of events, debates, and viewpoints involving various organizations and institutions. By analyzing this media content, scholars can identify which actors are involved in a transition and discern the roles they play or are ascribed in public discourse.

## Media Coverage and Actor Roles in Transitions

Media coverage is a window into the socio-political landscape of a sustainability transition. News articles often chronicle who is doing what—which organizations advocate or oppose certain innovations, which government agencies implement policies, which firms invest in new technologies.

Researchers tracking media debates over technologies such as nuclear power, carbon capture, or biofuels often reveal distinct actor roles:

**Incumbent firms** Portrayed as defending the regime or emphasizing reliability

**Innovative startups** Cast as disruptors or pioneers

**Government officials** Framed as regulators or enablers

**NGOs/advocates** Acting as watchdogs or change agents

## Methodological Approaches to News Analysis

Researchers employ a mix of **quantitative text mining** and **qualitative content analysis**:

**Named Entity Recognition** Automatically identify entities (organizations, people, locations) in text using NLP tools like spaCy or NLTK.

**Topic Modeling** Unsupervised machine learning to discover latent themes by clustering co-occurring words.

**Sentiment Analysis** Gauge the tone of media coverage towards technologies or actors over time.

**Discourse Network Analysis** Combines content coding with network analysis, linking actors to concepts they advocate.

**Event History Analysis** Compile chronologies of events to analyze system change dynamics.

## Data Sources for News in Transition Studies

Several datasets and repositories enable large-volume news article retrieval:

**LexisNexis & Factiva** Commercial databases aggregating news from thousands of outlets worldwide with historical depth.

**GDELT** Continuously updated event database parsed from global news using automated algorithms. Codes actors, locations, and tone.

**Media Cloud** MIT's open-source platform for online news analysis.

**Web Scraping** Custom collection using tools like Newspaper3k or official APIs (e.g., NYT API).

## Insights into Multi-Actor Constellations

Analyzing news data yields insights into multi-actor constellations:

**Power Dynamics** Media visibility indicates agenda-setting power. Increased presence of new voices may signal power shifts.

**Actor Networks** Co-occurrence analysis maps perceived relationships and coalitions.

**Intermediary Identification** News may highlight actors who facilitate change across regions or sectors.

**Temporal Dynamics** Longitudinal analysis captures tipping points and narrative shifts.

## Chapter Summary

- ▶ **News media** actively shape public perceptions, political decisions, and societal values around sustainability transitions.
- ▶ **Media coverage** provides a window into the socio-political landscape, framing actors as heroes, villains, or victims.
- ▶ **Methodological approaches** range from NER and topic modeling to discourse network analysis and event history analysis.
- ▶ **Data sources** include commercial databases (LexisNexis, Factiva), computational databases (GDELT), and specialized archives.
- ▶ **Multi-actor constellations** can be mapped through actor co-occurrence, coalition formation, and intermediary identification.
- ▶ **Temporal dynamics** show how narratives shift, tipping points emerge, and external shocks reshape transition discourse.

# Crunchbase: Mapping Investment Networks

*This appendix examines Crunchbase as a data source for studying innovation ecosystems, particularly startup-driven sustainability transitions. We explore the platform's unique ability to map inter-organizational networks through investment, acquisition, and founder relationships. The appendix addresses data extraction methods, analytical techniques, and integration strategies while acknowledging critical limitations around coverage bias and reproducibility.*

## Overview: Scope, Strengths, and Limitations

**Crunchbase** is a prominent online platform compiling information on innovative companies (especially startups), the people behind them, and their funding activities.

Since its creation in 2007, it has grown into one of the largest directories of startups worldwide, providing detailed information not easily obtainable from official sources.

### Data Assembly Model

Crunchbase's dataset is **community- and partner-driven**:

- ▶ Registered users (entrepreneurs, executives) create/update profiles
- ▶ Partnerships with 3,000 investment firms for portfolio updates
- ▶ ML-augmented verification
- ▶ In-house data analysts

### Data Richness: Relational Structure

Unlike patent databases or news archives, Crunchbase explicitly encodes **relationships**:

**Funding Relationships** Which investors funded which companies, amounts, and rounds

**Acquisition Relationships** Which companies acquired which others

**People Relationships** Founders, executives, board members across companies

**Accelerator Relationships** Which programs supported which startups

## Analytical Applications

### Investment Network Analysis

Funding data enables construction of investor-startup networks revealing:

- ▶ Central vs. peripheral investors
- ▶ Sector specialization patterns
- ▶ Geographic concentration
- ▶ Syndication patterns (co-investment)

## Founder Network Analysis

Serial entrepreneurs and board members create ties between organizations:

- ▶ Knowledge transfer pathways
- ▶ Entrepreneurial communities
- ▶ Spin-off dynamics

## Critical Limitations

### Coverage and Bias Issues

- ▶ **US/Tech bias:** North American and technology sectors overrepresented
- ▶ **Survivorship bias:** Failed companies underreported
- ▶ **Self-reporting:** Data quality depends on user contributions
- ▶ **Recency bias:** Historical data less complete
- ▶ **Access barriers:** Full API access requires commercial subscription

## Chapter Summary

- ▶ **Crunchbase** provides rich relational data on startups, investors, and their connections.
- ▶ **Network construction** is natural: funding rounds create investor-company ties; people create company-company ties.
- ▶ **Analytical applications** include investment network analysis, founder mobility tracking, and ecosystem mapping.
- ▶ **Critical limitations** include geographic/sectoral bias, survivorship bias, and data access barriers.
- ▶ **Integration with other modalities** strengthens validity by triangulating investment patterns with claimed identities and media coverage.



# Social Media: Mapping Relational Identity

*This appendix explores social media platforms—particularly Twitter/X and LinkedIn—as sources for mapping relational identity: how organizations position themselves through their network connections, content sharing, and engagement patterns. Unlike websites (self-presentation) or news (external attribution), social media reveals how actors perform their roles through ongoing digital interactions.*

## Why Social Media?

Social media provides unique insights into role constellations:

**Network Visibility** Who follows whom, who retweets whom—explicit relational claims

**Real-time Dynamics** Continuous stream reveals positioning as it happens

**Discourse Participation** Hashtags and mentions show topical alignment

**Community Boundaries** Clustering patterns reveal ecosystem structure

## Twitter/X as Data Source

Twitter has been the dominant platform for social media research due to its public nature and (historically) accessible API.

### Data Types

- ▶ **Follower networks:** Directed ties showing attention flows
- ▶ **Retweet networks:** Endorsement and amplification
- ▶ **Mention networks:** Direct engagement
- ▶ **Content streams:** Tweets revealing discourse positioning
- ▶ **Hashtag co-occurrence:** Topic communities

#### API Access Changes

Twitter's API access policies changed dramatically in 2023, restricting academic research access. Historical data remains valuable, but ongoing collection faces new barriers.

## LinkedIn for Professional Networks

LinkedIn offers complementary data on professional relationships:

- ▶ Employee flows between organizations
- ▶ Advisory relationships
- ▶ Institutional affiliations
- ▶ Professional endorsements

## Analytical Approaches

**Network Analysis** Centrality, community detection, brokerage identification

**Content Analysis** Topic modeling on posts, sentiment tracking

**Temporal Dynamics** How networks and discourse evolve

**Bot Detection** Filtering automated accounts for authentic signals

## Chapter Summary

- ▶ **Social media** reveals relational identity through ongoing digital interactions.
- ▶ **Twitter/X** provided rich network and content data, though API access has become restricted.
- ▶ **LinkedIn** offers professional network data with access limitations.
- ▶ **Analytical value** lies in real-time dynamics, community detection, and discourse positioning.
- ▶ **Integration** with other modalities validates claimed relationships and reveals gaps between assertion and engagement.

# Visual Materials: Mapping Embodied Identity

*This appendix explores visual materials—logos, photography, video, and design elements—as sources for mapping embodied identity: how organizations materialize their roles through aesthetic choices. Images communicate identity dimensions that text cannot capture: professionalism, innovation, sustainability commitment, and institutional positioning.*

## Why Visual Analysis?

Visual elements carry meaning that complements textual analysis:

**Immediate Impression** Visuals communicate before text is read

**Identity Clustering** Similar aesthetics signal similar positioning

**Professionalism Markers** Production quality signals resources and legitimacy

**Sector Alignment** Visual conventions reveal field membership

## Types of Visual Data

### Logos and Branding

Logos encode identity choices:

- ▶ Color: Green for sustainability, blue for technology, warm tones for human focus
- ▶ Typography: Modern sans-serif vs. traditional serif
- ▶ Iconography: Abstract vs. representational
- ▶ Complexity: Minimalist vs. detailed

### Photography and Imagery

Website and social media images reveal:

- ▶ Team composition and diversity
- ▶ Workspace environments
- ▶ Technology and equipment
- ▶ Partner and event documentation

## Analytical Approaches

**Computer Vision** Automated image classification, object detection, color analysis

**Embedding Similarity** Neural network embeddings for visual similarity clustering

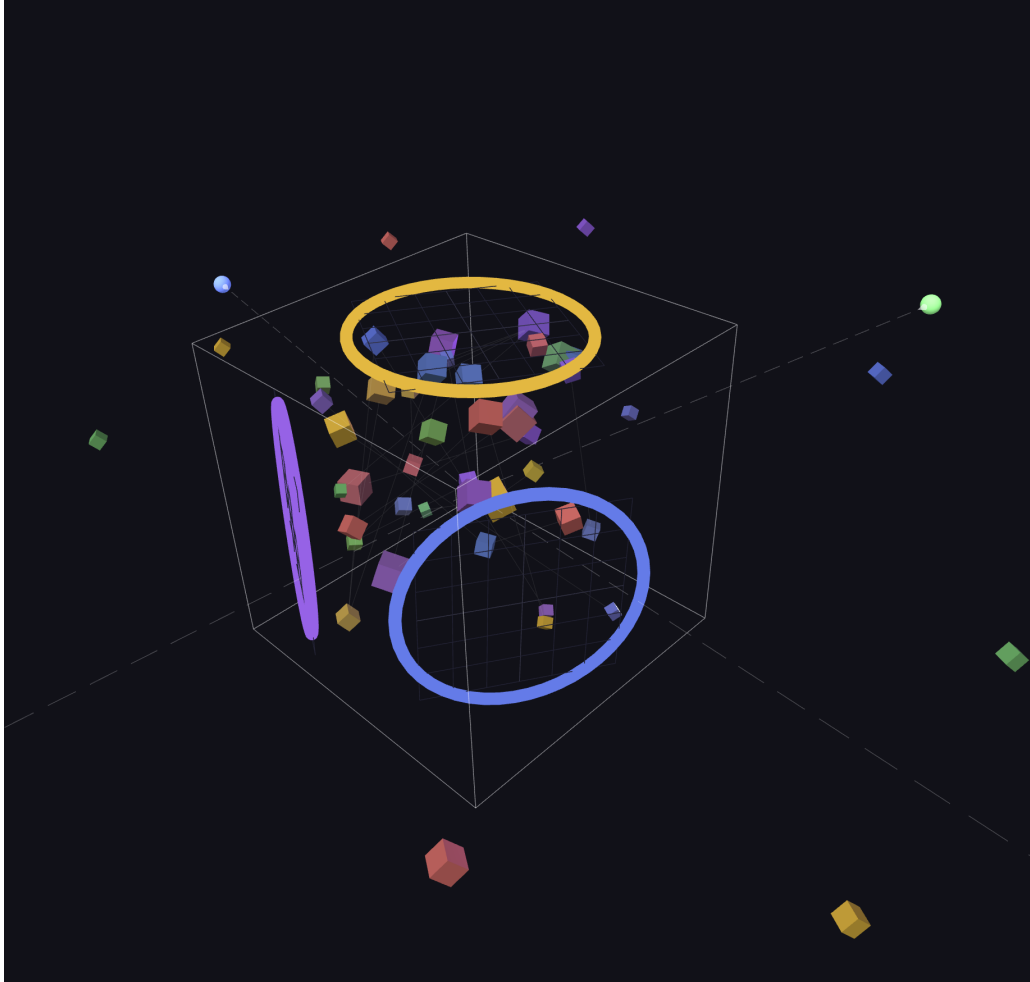
**Qualitative Coding** Human interpretation of visual rhetoric and symbolism

**Multimodal Integration** Combining visual and textual signals

## Chapter Summary

- ▶ **Visual materials** communicate identity dimensions that text cannot capture.
- ▶ **Logos and branding** encode positioning through color, typography, and iconography.
- ▶ **Photography** reveals team composition, resources, and institutional context.
- ▶ **Analytical approaches** combine computer vision, embedding similarity, and qualitative interpretation.
- ▶ **Integration** with textual modalities reveals consistency or divergence in identity presentation.

# Propositions



*Propositions pertaining to the dissertation*

- I. Role constellations, not individual roles, are the appropriate unit of analysis for understanding actor contributions to sustainability transitions.  
*This dissertation, passim.*
- II. Multimodal cartography reveals systematic discrepancies between how organizations present themselves and how they are positioned by others—discrepancies that single-modality approaches cannot detect.  
*Chapters 5–9.*
- III. The claim to “bridge” or “connect” is the most inflated role in innovation ecosystems: 34% of actors claim connector roles, but only 12% show cross-modal evidence for actually performing them.  
*Chapter 5.*

- IV. Design Science Research provides a productive framework for methodological development in sustainability transitions research, treating methods themselves as artifacts subject to iterative refinement.  
*Chapter 3.*
- V. Computational methods applied to transitions research generate valuable findings when grounded in domain theory; without such grounding, they risk producing sophisticated analyses of what can be measured rather than what matters.  
*Chapter 1.*
- VI. The visual presentation of organizations—logos, color schemes, imagery—constitutes an underexplored but information-rich modality for understanding role positioning in innovation ecosystems.  
*Chapter 8.*
- VII. Power in innovation ecosystems operates less through direct control than through the capacity to define roles and position others within constellations.  
*Chapter 9.*
- VIII. Open-source research infrastructure should be the default in computational social science; proprietary pipelines undermine cumulative knowledge building.  
*Appendix H.*
- IX. The European Institute of Innovation and Technology (EIT) ecosystem represents an unprecedented natural experiment in designed transitions governance, worthy of sustained scholarly attention.  
*Intermezzo: EIT Communities.*
- X. Agent-based models of role dynamics serve their greatest value not in prediction but in disciplining counterfactual imagination.  
*Intermezzo: RoleSim.*

*Proposition unrelated to the dissertation*

- XI. The sea does not care about our categories. It dissolves boundaries, erases footprints, and reminds us that all our careful distinctions are written in sand.

*These propositions are regarded as opposable and defensible,  
and have been approved as such by the promotor.*

Part IV

## PORTFOLIO

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INTERMEZZO

## ROLEDNA: A Python Platform for Discourse Network Analysis

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*The tools we use shape the questions we ask.  
What becomes possible when discourse network analysis meets modern Python?*

*This chapter presents ROLEDNA (Role-based Discourse Network Analysis), an exploratory reimplement-  
ation of the Discourse Network Analyzer (DNA) in Python. We trace the motivation from  
desktop-bound Java applications to web-accessible Python platforms, describe the technical archi-  
tecture combining FastAPI, SQLAlchemy, and Sigma.js, demonstrate core analytical capabilities for  
actor-concept network analysis, and reflect critically on current limitations. ROLEDNA represents a  
proof-of-concept for modernizing computational discourse analysis—not a finished product, but  
a foundation for future development integrating large language models for automated coding  
assistance.*

### From DNA to ROLEDNA: Motivation

Discourse Network Analysis has emerged as a powerful methodology for studying policy debates, mapping how actors position themselves relative to concepts, claims, and each other. The method, pioneered by Philip Leifeld, combines qualitative content analysis with quantitative network methods.

\*

Three developments motivate ROLEDNA's Python reimplementa-

\*:  
DNA  
Ori-  
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Net-  
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lyzer,



## The Desktop-to-Web Transition

DNA runs as a local Java application. Modern research increasingly demands web-based collaboration. ROLEDNA explores whether discourse network analysis can follow the trajectory from installed software to accessible web application.

## The Python Ecosystem Advantage

The data science ecosystem has consolidated around Python. A Python-native implementation collapses Java/R polyglot workflows into a single environment where the same language handles interface, database, analysis, and visualization.

## The LLM Horizon

Large language models are transforming text analysis. ROLEDNA's Python foundation positions it for LLM integration—the same codebase that serves the web interface can call OpenAI's API, run local Llama models, or integrate Hugging Face pipelines.

### LLMs Are Not Magic

LLM-assisted coding is promising but perilous. Models hallucinate. They reflect training data biases. ROLEDNA envisions LLMs as *assistants*—suggesting codes for human review—not autonomous coders. The human remains in the loop.

## Technical Architecture

ROLEDNA adopts a modern web application architecture: FastAPI backend serving both API endpoints and rendered HTML, SQLAlchemy for database abstraction, and Sigma.js for interactive network visualization.

The API follows RESTful conventions with endpoints organized around core entities: documents, statements, actors, concepts, and networks.

## Core Analytical Features

ROLEDNA implements the core network computations that make discourse network analysis distinctive: projecting two-mode actor-concept affiliations into one-mode networks revealing actor relationships.

## Network Types

Three network types form the analytical core:

**Two-Mode Affiliation Network** Bipartite graph connecting actors to concepts. Edges carry qualifier weights (+1 for agreement, -1 for disagreement).

**One-Mode Actor Congruence Network** Actors connected by shared concept references. Edge weights reflect the number and consistency of shared positions.

**One-Mode Actor Conflict Network** Actors connected by opposing positions on concepts. This network reveals adversarial structure in debates.

†: **Tech-  
Stack**—Backend:  
FastAPI  
(Python  
3.11+),  
Database:  
SQLAlchemy  
async  
+  
SQLite/Post-  
greSQL,  
Tem-  
plates:  
Jinja2,  
Styling:  
IBM  
Car-  
bon  
De-  
sign  
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ogy.

## Normalization Options

Raw edge weights reflect statement counts, which can bias toward prolific actors. ROLEDNA implements standard normalizations: Jaccard, Cosine, and Average Activity.

## Current Limitations

ROLEDNA is a proof-of-concept, not a production system. It lacks some DNA features: regex-based entity detection, statement inheritance, multiple qualifier dimensions, and network backbone extraction.

**Use at your own risk.** ROLEDNA is research software in early development. It may contain bugs. It has not been security-audited for multi-user deployment.

## Future Directions

ROLEDNA's Python foundation enables transformative futures:

**LLM-Assisted Coding** Using large language models to suggest potential statements for human review.

**NLP Pipeline Integration** Named Entity Recognition, coreference resolution, stance detection, and topic modeling.

**Collaborative Features** Multi-user coding, inter-coder reliability, version history, and project sharing.

**Integration with RoleBox** Comparing discourse positions to network positions; validating claimed roles against enacted roles.

### Toolkit Convergence

This dissertation develops two tools: RoleBox for multimodal network analysis, ROLEDNA for discourse network analysis. They share conceptual foundations and technical stack. Future work could unify them into a single platform for comprehensive role constellation analysis.

## Chapter Summary

- ▶ **ROLEDNA** reimplements core Discourse Network Analyzer functionality in Python, transitioning from desktop Java to web-accessible platform.
- ▶ **Motivations** include: web-based collaboration, Python ecosystem integration, and positioning for LLM-assisted coding.
- ▶ **Technical architecture** combines FastAPI backend, SQLAlchemy async database layer, Carbon Design System interface, and Sigma.js network visualization.
- ▶ **Core features** include: two-mode affiliation networks, one-mode congruence/conflict projections, multiple normalization options, and interactive visualization.
- ▶ **Future directions** include: LLM-assisted coding, NLP pipeline integration, collaborative features, and potential RoleBox integration.
- ▶ ROLEDNA is **proof-of-concept** software demonstrating feasibility, not a production-ready replacement for DNA.

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# NWO Open Competition XS Grant Application

*This grant application was submitted to the Netherlands Organisation for Scientific Research (NWO) Open Competition SSH - XS programme in 2024. It represents an early formulation of the RoleBox framework and its application to mapping innovation ecosystem dynamics.*

## Application Form – NWO Domain Social Sciences and Humanities Open Competition XS 2024 round 1



Form Version: 19 December 2023

### A1. Application details

Title of the project: SMASH: Synthesis and Multimodal Analytics System for Humanities  
Duration of the project (in months): 12 months

### A2. Public summary

English public summary:

*Our project, SMASH, is working on a new way to understand how people communicate with each other beyond words. We are focusing on how people use their body language, facial expressions, and tone of voice to express themselves when they talk. We want to extend an already-developed tool that tracks body movement and masks audio and video for privacy. The extension would be to automatically capture and analyze non-verbal and verbal communication aspects to augment researchers and clinical professionals studying communication. This would be a timely achievement in social science supported by computer science.*

Words: 94/100

Dutch public summary:

Ons project, SMASH, werkt aan een nieuwe manier om te begrijpen hoe mensen met elkaar communiceren buiten woorden om. We richten ons op hoe mensen hun lichaamstaal, gezichtsuitdrukkingen en stem gebruiken om zich uit te drukken. We willen een reeds ontwikkelde tool uitbreiden die lichaamsbewegingen volgt en audio en video maskeert voor privacy. De uitbreiding zou bestaan uit het automatisch vastleggen en analyseren van non-verbale en verbale communicatieaspecten om onderzoekers en klinische professionals te helpen bij het bestuderen van communicatie. Dit zou een tijdige innovatie zijn in de sociale wetenschap ondersteund door computerwetenschap.

Words: 94/100

### A3. Budget

| Type of costs                 | Clarification                                               | Costs in euros |
|-------------------------------|-------------------------------------------------------------|----------------|
| Personnel                     | Third party hire computer science institute (postdoc level) | 45.000         |
| Travel and accommodation      | Attendance at conferences and workshops                     | 5.000          |
| Own contribution (optional)   |                                                             | N/A            |
| <b>Total request from NWO</b> |                                                             | <b>50.000</b>  |

### A4. Budget Justification

We allocate €45.000 for a third-party hire on a post-doctoral level at a European computer science (CS) research institute. The external postdoc is crucial for the success of this project as this person has spearheaded the development of the software base. The postdoc level is required as there is a need for broad conceptual and technical knowledge spanning social and computer science. The postdoc will spearhead the core day-to-day developments of SMASH under the supervision of two established Dutch and European researchers over a year (part of the position is already funded; this grant would support 1.0fte for seven months). Travel and accommodation costs, amounting to €5.000, are split over conferences and workshops (€1.500) and €3.500 for travel and accommodation between the Netherlands and the European city.

Words: 149/150

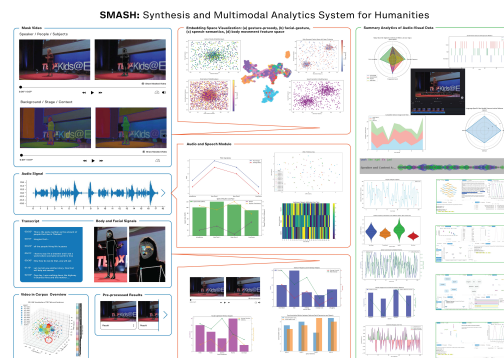
## Section B. Scientific proposal

### B1. Scientific proposal

Bodily gestures and tone of voice underline that the full range of human expression makes use of more than words alone [1]. How to study this extra layer of communication in an integrated way is a challenging problem in social science [1,2], requiring technical innovation guided by social science. That's where the SMASH project comes in. SMASH will be a fully open-source tool that helps us better understand the composition of non-verbal communication aspects, like body language, facial expressions, and the melodic aspects of speech. It will combine an already developed masking tool that masks audio-visual data to protect people's privacy with analysis modules that inform about how speakers communicated beyond the words uttered, thereby serving as a general tool for data-archiving and behavioral analysis of audiovisual data in social sciences in humanities. SMASH aims to better integrate the understanding of a rich set of communicative cues by bridging the gap between computer science and social science with far-reaching applications. For example, clinicians can eventually use SMASH to quantitatively monitor more holistic aspects of communication that deteriorate in communicative disorders like Aphasia [3]. Seeing the rapid adoption of CHATGPT, gaining insights into the complete spectrum of human emotions, actions, and non-verbal communication is of utmost importance for comprehending *language in the wild* and developing genuinely relatable and socially adept chatbots and artificial agents.

Social science research has more opportunity to benefit from computer science advances that enable the simultaneous processing of multiple communicative signals [4–6]. For example, the field of computer vision now enables automatic facial and body movement tracking from video, which can then form input for automatic classification of behaviors like pointing with the hand, smiling, etc. [7]. Our team is ready to implement such computational pipelines [1,8] into an integrated toolkit, as Figure 1. explains.

**Figure 1. Example SMASH results dashboard.** SMASH would take as input audio-visual recordings of (a) communicating individual(s) and a summary of their multimodal communication characteristics as output. For instance, it shows the number and diversity of hand gestures [2], changes in posture and facial expressions relevant to communication context [8], and how voice tone changes with other gestures [9]. It would also capture several trouble moments, such as pauses and fillers in speech [10] and summarize what is being discussed using automatic speech transcription and semantic analyses [10]. By integrating these dimensions - body language, facial expressions, voice, and content - in an easily understandable way, SMASH allows a detailed zoomed-out view of the full gamut of communication modes. SMASH will also offer standard baselines based on already analyzed speaker databases (see goal 2 below). In this way, users of SMASH can conclude that a speaker “gestured more relative to a database of Dutch speakers” or “used more variation in their tone of voice, even when compared to a database of highly appreciated speakers”. The tool would also offer options for data export for customized analysis. We aim to make SMASH a versatile and easy-to-use tool for users without programming experience. Through initial testing and adaptation by social scientists, we aim for SMASH to become an indispensable tool in research and clinical contexts.



This project's first goal is to develop an accessible communication analysis toolkit called SMASH, as outlined above. This toolkit will be integrated with software that performs masking and body motion tracking (already developed by the current team). The second goal is to unleash SMASH to address the following research question: how do these multiple communicative signals structurally interrelate, if at all, in audiovisual recordings in speakers from diverse backgrounds?

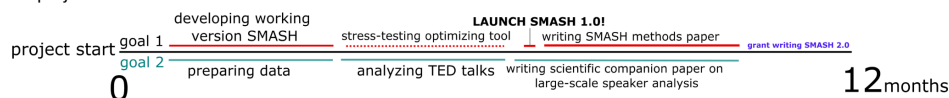
**GOAL 1 – SMART formulation:** Specific: Since we already have a codebase that processes audiovisual data with a user interface, we will integrate new modules within that codebase. Modules will include those that analyze melodic features of speech (prosody) [12] and assess semantic content (using so-called text-embedding pipelines, e.g., BERT [10]). For manual gesture analysis, we leverage GestureMap [4], which analyzes all the hand gesture movements to look for recurrent gestures. Finally, we will implement facial expression clustering using EmoCo [11]. After integrating these modules, SMASH will further implement data science visualization to maximize the value and summaries of the high-dimensional data [12]. Measurable: We will launch new beta versions every time we have integrated a new analysis module; as such, our progress can be incremental with value-added along the way. Attainable: The modules we want to add are based on open-source software already developed and tested but never integrated for multimodal analytics into a single tool. We can work stepwise, first implementing the most valuable modules (e.g., facial and hand gestures). Timely: After 3 months, we will have at least three behavioral analysis modules integrated into a complete beta version of SMASH that we can use for scientific analysis (goal 2). In 6 months, the official release version will follow, where we will have a fully functional and stress-tested masking, tracking and communication analysis toolkit that can be widely released to external users.

**GOAL 2 – SMART formulation:** Specific: Drawing on SMASH, we will analyze the growing corpus of public speaking audio-visual recordings of TED Talks, comprising > 3,000 speakers across several languages, including Spanish, German, and Arabic. We will assess the multimodal linguistic diversity by analyzing this corpus using SMASH, which will also create a baseline for SMASH. Considering a diverse range of topics, speakers, and countries, we aim to see if common multimodal patterns stratify along languages, which would provide an understanding of the possibility of loosely-knit ‘multimodal’ languages. The sheer scale of our analyses is novel and could provide the first quantitative evidence that languages are spoken in a whole-body manner, including manual and facial gestures. Measurable: To test this, we will apply clustering goodness-of-fit metrics to assess whether certain combinations of multimodal summary statistics provided by SMASH fit categories of language spoken or geographical region. Attainable: The team has access to GPU-supported computing clusters that can help mine and preprocess the audiovisual data. The analyses can proceed in a modular fashion, such that first, English and Spanish languages are compared, followed by more languages; this way, research output can be generated incrementally. Timely: While we develop the first version of SMASH (3 months), we will simultaneously prepare the corpus from the start of the project by going through the motion tracking procedures already in place in our tool (motion tracking is used as input for communication analysis modules). As soon as we have a first working version, we can further analyze the data, which will take 5 months to generate the first results.

#### Feasibility

**SMASH’s ethical benefits for a responsible use of AI.** As interdisciplinary social and computer scientists we find it essential to mention that the data provided by SMASH is intended to be interpreted by a human in the context of specific research or applications. We thereby respect the nuances of communication that vary greatly depending on cultural, situational, and individual factors. SMASH does not make direct inferences about an individual’s emotional state based solely on communication patterns. Instead, it provides bottom-up data summaries on various aspects of communication, leaving the interpretation to users. This approach empowers humanities researchers to apply their expertise and contextual knowledge when interpreting the multimodal communication characteristics instead of leaving it to AI. These core ethical principles guide our project, ensuring that SMASH becomes an empowering resource for augmenting the analysis of human communication.

**Team composition and risk management:** An interdisciplinary team of socio-behavioral researchers from Dutch universities and a European computer science research institute has made progress in co-developing the privacy masking and body tracking component of SMASH. Thus, the codebase is already implemented. The team has also developed open-access pipelines for semantic network analytics and natural language processing of written text. Thus, SMASH is feasible and does not require recruiting new team members. In SMASH, we further leverage the integration of existing open-source tools, which, when combined, provide a new ground-breaking communication analysis tool. The interdisciplinary SMASH team can progress in this project piecemeal, prioritizing low-hanging fruit with high analytic gains (e.g., gesture and speech analysis). The human-in-the-loop approach further prevents potential risks of overzealous misuse of AI-implemented software. Finally, the team has already shown to be effective in developing computer science methods in social science in productive collaboration with social science university partners, thereby reducing the associated project risks for deploying SMASH. Below is the timeline we envision for our project.



## Section C. Reference list

### C1. Reference list

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# TDCC-SSH Synapsis Grant Proposal

*This proposal was submitted to the Thematic Digital Competence Centre for Social Sciences and Humanities (TDCC-SSH) Synapsis programme. It outlines infrastructure development for computational role analysis in sustainability transitions research.*



# Thematic Digital Competence Centers

## Application form

### 1. General information

#### 1a. Project title

SYNAPSIS: Synergy Network and Platform for Integrating Audiovisual Data Archiving and Stewardship in Social Sciences and Humanities.

#### 1b. Relevant Thematic DCC Domain

Please select the relevant domain.

- ☐ Life Sciences and Health (LSH)
- ☐ Natural and Engineering Sciences (NES)
- ☒ Social Sciences and Humanities (SSH)

#### 1c. Project duration

24 months.

#### 1d. Responsible institution(s) and department

Organisatie van aanvraag:

- Radboud University, Faculty of Arts and Centre for Language Studies

Organisaties van uitvoering:

- Radboud University, ILS, Digital Competence Centre
- Radboud University, Donders Institute for Brain and Cognition
- University of Amsterdam (UvA), Institute for Logic, Language & Computation (ILLC)
- Vrije Universiteit Amsterdam, Faculty of Humanities, Language and Communication, and Network Institute
- Erasmus University Rotterdam, Erasmus School of Social and Behavioural Sciences, DRIFT.
- Hasso Plattner Institute, Potsdam, Germany
- DANS
- SURF (Collaborative Organization for ICT in Dutch Education and Research)\*\*
- E-Science Center\*\*
- ODISSEI (Open Data Infrastructure for Social Science and Economic Innovations)\*\*

\*\*Consultations added in this revision/ future collaboration to be explored

#### 1e. Lead applicant and co-applicants

Note. While one of us submits (much like a corresponding author in academic publishing), this is a team science project and the primary role lead applicant and all co-applicants share in the common language we've adopted for this project is of co-pilot.

| Title, first name, initials, surname | Affiliation | Department                              | Role(s)                     |
|--------------------------------------|-------------|-----------------------------------------|-----------------------------|
| Mark Dingemanse                      | RU          | Centre for Language Studies             | Lead applicant, co-pilot    |
| Babajide Owoyele                     | EUR & HPI   | DRIFT                                   | Co-lead applicant, Co-pilot |
| Wim Pouw                             | RU          | FSW/Donders                             | Co-pilot                    |
| Bogdana Huma                         | VU          | Language, Literature, and Communication | Co-pilot                    |
| Henk van de Heuvel                   | RU          | Humanities Lab                          | Co-pilot                    |
| Inge Slouwerhof                      | RU          | Information and                         | Co-pilot                    |



|                |     |                        |          |
|----------------|-----|------------------------|----------|
|                |     | Library Services - DCC |          |
| James Trujillo | UvA | ILLC                   | Co-pilot |

#### 1f. Keywords

Audiovisual Data, Multimodal Analysis, Digital Competence, Privacy-aware Archiving, Social Sciences

#### 1g. Disciplines

|                                   |                                                  |
|-----------------------------------|--------------------------------------------------|
|                                   | Discipline:                                      |
| Main discipline:                  | 37.10.00 Software for humanities                 |
| Other disciplines (if applicable) | 37.90.00 Computers and the humanities, other     |
|                                   | 16.60.00 Artificial intelligence, expert systems |
|                                   | Select an item.                                  |

#### 1h. Abstract (max. 500 words)

SSH researchers increasingly rely on audiovisual data to study complex human behaviours and social interactions, but current practices and systems for handling such data lag behind. Privacy is the main challenge to making research reproducible and data reusable. We aim to increase audiovisual data reusability while protecting privacy by integrating masking technology into research workflows. Masking tools allow researchers to hide faces, voices, and other identifying traits while enriching the data with de-identified body movement data and other social signals. The goal is to strike a balance between removing identifiable information and increasing the reusability and interpretability of the data. We do this by co-designing new standards, practices, and workflows with stakeholders, ensuring that audiovisual data in SSH fields can be “as open as possible and as closed as necessary.”

Researchers and support staff tell us they are hesitant to adopt technological advances in audiovisual masking and perceive them to be too programming-heavy or hard to fit into current workflows. Directly addressing this, SYNOPSIS aims to provide an easy-to-use, open-source platform with tools for protecting privacy and clear guidelines and training. By helping researchers better manage sensitive data, the project will encourage more responsible and widespread use of audiovisual data in research. SYNOPSIS integrates and extends an existing user-friendly tool called MaskAnyone, which allows researchers to hide sensitive information using a point-and-click user interface. MaskAnyone’s modular design will enable researchers to choose and set up additional data-enriching tools for handling video, audio, or movement data.

The SYNOPSIS project has two main parts: building a platform and training researchers to use it. Platform development focuses on creating a reliable and easy-to-navigate cloud-based system for storing and sharing audiovisual data in cooperation with partners like SURF and DANS. Tools are supported by video tutorials, FAQs, and workflows (e.g., informed consent templates). On the training side, we work with four universities in the Netherlands to upskill researchers and technical staff. Our Masking School will train researchers and data stewards in masking, archiving, and sharing masked video and audio data, ensuring strong digital competence and institutional support for masking practices in SSH. Currently, staff in the Netherlands are not trained in these areas.

In collaboration with partners like SURF and DANS and RIs like ODISSEI and CLARIAH, our Masking Lab will address more advanced challenges in platform building and training. Here, we identify lacunae in current tools and chart the risks associated with particular forms of masking. A Hackathon-style benchmark event and study and a pilot certification program will help lay the foundations for building a community to navigate pressing risks.

With the SYNOPSIS platform and training track, we want to build long-term solutions and support infrastructure for masking audiovisual data that meet FAIR guidelines and prepare for the future of maximally transparent, reproducible, and privacy-aware research workflows in SSH. Our training and dissemination methods, coupled with our strong network of partner organizations, will ensure community-wide awareness and a lasting impact on the SSH research landscape.

#### 1i. Public project title and public summary

NL

SYNOPSIS : Synergie Netwerk en Platform voor Integratie van Audiovisuele Data-analyse en Archivering in



#### de Sociale Wetenschappen

SYNAPSIS is een samenwerkingsproject gericht op het verbeteren van het beheer, delen en analyseren van audiovisuele data in de Sociale Wetenschappen en Geesteswetenschappen. Het project biedt een veilig, privacy-beschermend platform dat geavanceerde masking-tools integreert voor audiovisuele data, waardoor onderzoekers gevoelige informatie kunnen beschermen en tegelijkertijd data toegankelijker maken voor analyse. Naast het platform worden uitgebreide trainingsprogramma's aangeboden om onderzoekers en data-stewards de digitale vaardigheden te bieden die nodig zijn om masking en multimodale analyses te integreren in hun workflows, wat open wetenschap bevordert en de transparantie van onderzoek verbetert.

#### ENG

#### ***SYNAPSIS: Synergy Network and Platform for Integrating Audiovisual Data Archiving and Stewardship in Social Sciences and Humanities***

SYNAPSIS is a collaborative project advancing audiovisual data management, sharing, and analysis in the Social Sciences and Humanities (SSH). The project provides a secure, privacy-aware platform that integrates cutting-edge masking tools for audiovisual data, allowing researchers to protect sensitive information while making data more accessible for analysis. Alongside the platform, comprehensive training programs will equip researchers and data stewards with the skills to implement masking and multimodal analytics in their workflows, promoting open science and improving research transparency.

## 2. Project information

### 2.1 Involvement of TDCC domain in drafting application (max. 400 words)

SYNOPSIS is shaped by an inclusive, iterative process involving the full spectrum of potential users, support staff, support infrastructure, and future developers and local policymakers. The applicants directly involved SSH research institutions and digital infrastructure providers, including DANS, SURF, Radboud Humanities Lab, and CLARIN/CLARIAH. Researchers and support staff at several Donders Network Institutes, and Summer Schools held at Radboud, Potsdam, Göttingen, and Poland, helped us refine the SYNOPSIS project goals, which includes proper integration of archiving workflows and maximal reuse of digital infrastructure (SURF SANE). Radboud University's Humanities Lab will develop the platform's front-end and also provides important links to DANS and SURF infrastructure. The involvement of the DCC coordinator as a co-pilot ensures local synergies within and outside RU.

During the iterative review process the proposal was further shaped by participants from the NWO OSF project ["Building a community of practice for open naturally occurring data"](#) as well as by interactions with RDM experts and SSH stakeholders, including Esther Plomp (UU), Jetze Touber (Ghent), Ricarda Braukmann (DANS) and Eduard Klapwijk (Leiden). A first set of pilots at Radboud University helped to scope synopsis as a pilot for GDPR-compliant integration of data anonymization at scale and increased the likelihood of successful integration in centralized digital infrastructures. Further, feedback from Data Stewards and experts from SURF, RU, VU, and the NL eScience Center nudged us to open up our masking practices catalog to incorporate additional tools, such as penciling and DeepPrivacy2. Reviews from [Andrea Scharnhorst](#), [Rene Bekkers](#), [Roeland Ordelman](#) and [Ahmad Hesam](#) led to a crucial rethinking and redesign of the SURF pilot implementation. Feedback we elicited from the RU Social Science Lab helped us broaden to focus of the Masking Lab to include ethics, FAIR principles, and metadata. The [Data Stewards Interest Group](#) community will help disseminate our workshops and surveys and invite participants to our workshops. We concluded the grant writing phase with in-depth discussions with the NL eScience Center on the project's sustainability. Additionally, RU Data Ethics Coordinator [Miriam Kos](#) connected us to Radboud Teacher Academy pilot/use case with [Robin Satter](#) now on board as advisory.

We are incredibly proud of the resulting proposal. SYNOPSIS addresses the real needs of the SSH community while advancing critical areas in augmenting data stewardship, privacy infrastructure, network building and FAIRification. If funded, it will mark a substantial advance for sustainable digital competence in the SSH domain.

### 2.2 Roadmap ambition/activity cluster/challenge

Which roadmap ambition/cluster/challenge does this project aim at? Make a selection within the domain marked under 1b.

| NES                                                                     | SSH                                                                    | LSH                                                                                           |
|-------------------------------------------------------------------------|------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| <input type="checkbox"/> FAIR data                                      | <input checked="" type="checkbox"/> FAIRification of data and software | <input type="checkbox"/> Strengthen good data stewardship and harmonise data access practises |
| <input type="checkbox"/> Sustainable software and e-science             | <input checked="" type="checkbox"/> Raising awareness                  | <input type="checkbox"/> Enhance the interoperability of digital solutions and resources      |
| <input type="checkbox"/> Connections to the international activities    | <input type="checkbox"/> Knowledge enhancement                         | <input type="checkbox"/> Strengthen the capacity and expertise base in digital research       |
| <input type="checkbox"/> Long-term data archiving                       | <input type="checkbox"/> Addressing pressing issues                    |                                                                                               |
| <input type="checkbox"/> Locate computing capacity close to the storage | <input checked="" type="checkbox"/> Building a network                 |                                                                                               |
| <input type="checkbox"/> Human capital                                  |                                                                        |                                                                                               |
| <input type="checkbox"/> Cross-field collaborations                     |                                                                        |                                                                                               |



### 2.3 Relevance (max. 400 words)

As the use of rich audiovisual data in SSH research continues to grow, so does the need for effective data management strategies. A recent policy paper argues that audio and video recordings should not be destroyed, overturning long-held beliefs in SSH fields ([Resnik et al. 2024](#)). SYNOPSIS 1.0 aims to answer to louder calls for augmenting audiovisual data management by advancing the digital competence data stewardship and applies FAIR (Findable, Accessible, Interoperable, Reusable) principles, which are central goals of the TDCC Social Sciences and Humanities (SSH) roadmaps.

SYNOPSIS innovates on existing digital infrastructures (e.g., MaskAnyone, SURF), ensuring relevant expertise (e.g., SURF has been working on secure video conferencing software, a highly pertinent project for SYNOPSIS). SYNOPSIS is also a practical solution for researchers as it offers a user-friendly platform for securely managing and analyzing audiovisual data. SYNOPSIS will provide a secure data portal with transparent processing of audiovisual data by open-source software tools, combined with appropriate data governance measures conveyed to data stewards. These are essential prerequisites to ensure that masking becomes a viable alternative to audiovisual data removal by default.

#### **Pilot Use Cases and Sustainability Plan via Grants**

We're collaborating with Radboud Teachers Academy (RTA) to implement the MaskAnyone Toolkit, enabling GDPR-compliant de-identification in teacher training assessments while preserving trainee visibility. RTA's trainees record videos in real-world classrooms, which serve as essential learning tools but require privacy protections when repurposed for broader educational and research use. By anonymizing pupils through face-swapping and voice alteration while keeping teachers identifiable with consent, MaskAnyone supports privacy-compliant video sharing. This pilot will also involve RTA's privacy officer and ethical committee, facilitating compliance assessments to lay the groundwork for potential adoption across Radboud University. Additionally, in partnership with Joaquin Santuber at Johannes Kepler Universität Linz, we're exploring methods to de-identify Chilean court videos for research. The sustainability of SYNOPSIS is supported by partnerships with SURF and NLeSC for secure, scalable cloud operations, alongside "train-the-trainer" workshops to build local expertise.

We are exploring synergies with DARIAH Campus to ensure SYNOPSIS's long-term relevance. Through DARIAH's "Hosted Resource" option, SYNOPSIS training materials could be preserved sustainably and accessible to users beyond the project's lifecycle. This partnership would offer SYNOPSIS ongoing visibility within the Social Sciences and Humanities research community, with cataloguing opportunities in the SSH Open Marketplace further extending its reach and usability.

### 2.4 Project outline including work packages (max. 1600 words)

Our project aims to innovate masking tools and infrastructure to provide workflows and training materials for users and support staff while publishing all training, code, and adapted workflows on a centralized platform open for reuse by other institutions (both within and outside NL).

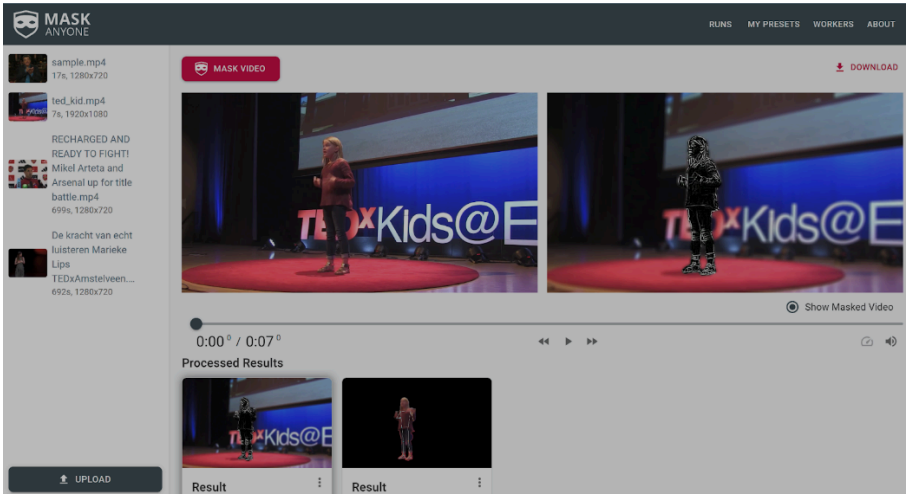


Figure 1a The MaskAnyone Toolkit User Interface see (<https://github.com/MaskAnyone/MaskAnyone>)

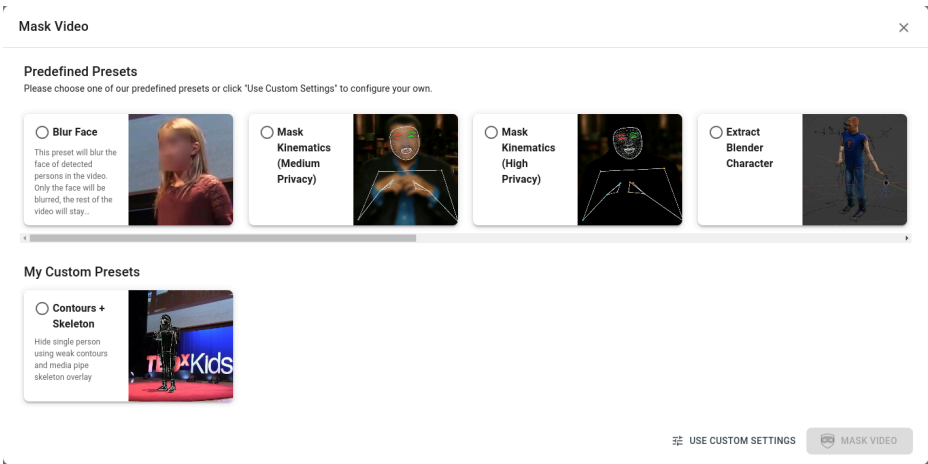


Figure 1b The MaskAnyone Toolkit User Interface see (<https://github.com/MaskAnyone/MaskAnyone>)

Work Package 1: Platform Foundation and Community Engagement (Months 1-12)

WP 1 Overview:

| Track | Component | Target Audience | Who | What | Deliverables |
|-------|-----------|-----------------|-----|------|--------------|
|-------|-----------|-----------------|-----|------|--------------|

|                       |                               |                                                                |                                  |                                                                                                                                                |                                                                       |
|-----------------------|-------------------------------|----------------------------------------------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|
| <i>Training Track</i> | <i>Masking School</i>         | <i>PIs, support staff (~N=20), ECRs, data stewards (~N=30)</i> | <i>Trujillo, Owoyele, Pouw</i>   | <i>Two live/hybrid 1.5-hour sessions at VU, two live/hybrid 4-hour sessions at UvA &amp; Rotterdam, 3-day summer/winter schools at Radboud</i> | <i>Training materials, video tutorials, participant certification</i> |
| <i>Platform Track</i> | <i>SYNOPSIS Pilot Testing</i> | <i>Radboud University, SURF</i>                                | <i>Dingemanse, van de Heuvel</i> | <i>Pilot deployment at Radboud, focusing on performance and GDPR compliance</i>                                                                | <i>Completed pilot, technical report</i>                              |

**Objective:** Establish the foundation for the SYNOPSIS platform and facilitate the incorporation of masking into the workflows of the SSH research community.

**Platform Track:**

The primary focus will be transforming the MaskAnyone toolkit (see Figure 1a and b above) into a user-friendly platform supporting secure file exchange and adaptable workflows. Collaboration with **Mark Dingemanse** and **Henk van de Heuvel** at **Radboud University's Digital Humanities Lab** will ensure the necessary technical infrastructure for institutional and national use, including integration with **SURF**. The Masking toolkit pilot will be implemented for the Radboud Teachers Academy with support from one research coordinator and a privacy officer ([Robin Satter](#)). We will integrate with 1) their existing video-based assessments of teacher trainees and 2) open science research practices.

**Concrete actions** include:

- Developing secure infrastructure for GDPR-compliant file storage, seamless file exchange, and processing.
- Integrating access to the masking tools for de-identification of sensitive audiovisual data.
- Piloting the platform in real-world research environments at **Radboud** to ensure platform performance and adaptability.

**Overview of Constructive Alignment for Masking School**

| Component                                     | Details                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Teaching and Learning Activities (TLA)</b> | <ul style="list-style-type: none"> <li>- <b>Introduction and theoretical background</b> on masking tools.</li> <li>- <b>Demonstrations</b> of MaskAnyone and related tools.</li> <li>- <b>Hands-on exercises</b> to practice audiovisual data masking.</li> <li>- <b>Discussions and breakout sessions</b> for sharing challenges and best practices.</li> </ul>                    |
| <b>Assessment</b>                             | <ul style="list-style-type: none"> <li>- <b>Completion of exercises:</b> Participants will demonstrate data masking skills.</li> <li>- <b>Case studies:</b> Discussion of ethical and practical scenarios.</li> <li>- <b>Feedback from instructors:</b> Formative assessments during sessions.</li> <li>- <b>Certificate of completion</b> for successful participation.</li> </ul> |





|                                |                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Course Objectives</b>       | <ul style="list-style-type: none"> <li>- Equip participants with <b>practical skills</b> for audiovisual data masking.</li> <li>- Ensure understanding of <b>ethical frameworks</b> and <b>GDPR compliance</b>.</li> <li>- Promote <b>FAIR principles</b> for data management.</li> <li>- Build digital competence in research data management.</li> </ul>                                             |
| <b>Participant Recruitment</b> | <ul style="list-style-type: none"> <li>- Recruited through <b>university channels</b>, <b>summer/winter schools</b>, SSH community newsletters, and <b>DSIG Slack channel</b>.</li> <li>- Focus on Early-Career Researchers (ECRs), data stewards, and PIs.</li> <li>- No prerequisite technical expertise required; focus on those working with or planning to work with audiovisual data.</li> </ul> |
| <b>Cost for Participants</b>   | <ul style="list-style-type: none"> <li>- <b>General-level course for PIs and support staff</b>: ~€0-.</li> <li>- <b>Extensive user course for ECRs and data stewards</b>: ~€0.</li> <li>- <b>Post-Grant Summer/Winter Schools</b>: ~€400-€600 (includes meals and materials). <b>Scholarships or fee waivers</b> would be available for underfunded institutions.</li> </ul>                           |

**Training Track:**

The **Masking School & Website**, led by **Dr. James Trujillo**, **Babajide Owoyele**, and **Wim Pouw**, will provide targeted training programs and will prepare website materials:

- **General-level introduction course**: Two live/hybrid **1.5-hour sessions** aimed at PIs and support staff (~**20 participants per session**) at **VU University**.
- **Extensive user course**: Two **4-hour sessions** aimed at ECRs and data stewards (~**30 participants per session**) at **UvA** and **Rotterdam**.
- **Summer/Winter Schools**: Two 3-day events at **Radboud University** and **Donders Institute**, featuring Speed Talks from experts such as **Bogdana Huma** and **Inge Slouwerhof**.
- Recordings and training materials will be integrated into a **website for digital learning** for researchers who want to learn to apply masking.

**Deliverables:**

- Website offering training materials, FAQs, and recorded video tutorials.
- Real-time technical support platform.
- Certification for completing training modules.
- Pilot testing with Radboud Teacher Academy and technical report.

*Work Package 2: Platform Expansion Survey, Networking and Co-Creation, (Months 7-20)***WP 2 Overview:**

| Track | Component | Target Audience | Who | What | Deliverables |
|-------|-----------|-----------------|-----|------|--------------|
|-------|-----------|-----------------|-----|------|--------------|

|                |                 |                                                                          |                           |                                                                                                                       |                                                                                             |
|----------------|-----------------|--------------------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Training Track | Masking Lab     | Extensive users, data stewards, ethics officers, infrastructure partners | Huma, Slouwerhof, Owoyele | One full-day workshop at VU, speed dating breakout sessions                                                           | FAQs, templates, and a policy research paper on challenges, bottlenecks, and opportunities. |
| Platform Track | SYNOPSIS Survey | Data stewards, researchers                                               | RA(Technical), Owoyele    | Survey on opportunities for upgrading the platform with gesture classification, prosody analysis, cloud and local use | Design requirements catalog for Improved platform, workflow guides, and policy paper        |

**Objective: Survey enhanced platform functionality and co-creation forum for audio-visual masking with support from a broader community.**



Figure 3 An Illustration of Masking (De-Identification and Facial plus Upper-Body Analytics), input frame (left), and masked frame (right) (video link: <https://osf.io/4ki6a/>).

**Platform Track:**

**Babajide Owoyele and RA(Technical)** will deploy a survey to explore opportunities to upgrade to a **cloud-based, FAIR-compliant system**. The survey will also explore how to support new enrichment features for audiovisual data, including text transcription, gesture classification, and prosody analysis. It will target both ECRs and support staff. We anticipate that TDCC SSH and DANS will help disseminate the survey. When a video is masked, as in Figure 3 above, analytics can be used to maximize the value of the masked data. We want to implement a Google survey to explore opportunities and challenges for integrating such analytics in future versions of the archiving platform; our survey targets ECRs and support staff and will be disseminated via newsletters from partner organizations and the DSIG Slack Channel & TDCC SSH Network via their website and newsletter (3000 plus individuals).

| Component       | Details                                                                                                                                                                                                                       |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Target Audience | The Masking Lab is aimed at a diverse group of <b>extensive users, data stewards, ethics officers, infrastructure partners, and researchers</b> who deal with audiovisual data archiving, privacy issues, and FAIR workflows. |

|                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Audience Size and Composition</b>           | <ul style="list-style-type: none"> <li>- Expected size: <b>30-40 participants</b>.</li> <li>- Composition: <ul style="list-style-type: none"> <li>• <b>10-15 data stewards</b> from institutions involved in data archiving and management.</li> <li>• <b>5-8 ethics officers</b> from universities or research institutions responsible for GDPR compliance and data privacy.</li> <li>• <b>5-8 infrastructure partners</b> from organizations like <b>SURF, DANS, and CLARIAH</b>.</li> <li>• <b>8-10 early-career and senior researchers</b> with experience or interest in working with audiovisual data.</li> </ul> </li> </ul>                                                                         |
| <b>Disciplinary Background of Participants</b> | <ul style="list-style-type: none"> <li>- Participants will come from diverse fields such as: <ul style="list-style-type: none"> <li>• <b>Digital Humanities, Linguistics, Social Sciences, Behavioral Sciences, and Cognitive Science.</b></li> <li>• Technical fields, including <b>Computer Science, Data Science, and AI</b>, especially for those involved in data archiving and tool development.</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                |
| <b>Facilitators and Instructors</b>            | <p>Experts will lead the Masking Lab from both <b>academic research</b> and <b>technical infrastructure</b>, including:</p> <ul style="list-style-type: none"> <li>• <b>Babajide Owoyele (DRIFT, EUR)</b> – expertise in <b>design science</b> and responsible <b>multimodal analytics</b>.</li> <li>• <b>Bogdana Huma (VU)</b> – focus on communication <b>linguistics</b>, open science <b>data</b>, and <b>ethics in data management</b>.</li> <li>• <b>Inge Slouwerhof (RU)</b> – <b>library sciences</b> and <b>data stewardship</b>.</li> <li>• <b>Infrastructure specialists</b> from <b>SURF</b> and <b>DANS</b> for technical guidance on data storage, management, and GDPR compliance.</li> </ul> |
| <b>Course Objectives and Alignment</b>         | <ul style="list-style-type: none"> <li>- Equip participants with practical knowledge of <b>masking strategies</b> and implementing these within institutional and ethical frameworks.</li> <li>- Co-create solutions for <b>advanced masking challenges</b> by leveraging collective expertise from ethics, data stewardship, and infrastructure.</li> <li>- Provide participants with tools and templates for embedding <b>privacy-compliant workflows</b> into their research projects and institutional protocols.</li> </ul>                                                                                                                                                                             |
| <b>Recruitment of Participants</b>             | <ul style="list-style-type: none"> <li>- Recruitment will be done via <b>university networks</b> (<a href="#">newsletters</a>, department announcements), <b>SSH community groups</b> (<a href="#">DSIG Slack channel</a>), and direct invitations to participants from <b>partner institutions</b>.</li> <li>- <b>Ethics officers, data stewards, and infrastructure partners</b> from collaborating institutions like <b>SURF, DANS, and ODISSEI</b> have been invited.</li> </ul>                                                                                                                                                                                                                         |

**Training Track:**

**Bogdana Huma, Inge Slouwerhof, RA(Training), and Babajide Owoyele will organize the Masking Lab**, which will serve as a co-creation forum for extensive users, data stewards, ethics officers, and infrastructure partners (**SURF, DANS, CLARIAH**). The forum will address ethical concerns and ensure FAIR-compliant workflows. It will also lead to the ideation of more advanced training modules to support audio-visual data archiving broadly.



Figure 4 Another Illustration of Masking (De-Identification and Facial plus Body Analytics), input frame (left), and masked frame (right) for a non-adult speaker.

#### Deliverables:

- A policy research paper on masking ethics and FAIR data management.
- Improved platform and workflow guides for researchers.
- Templates and co-created metadata standards for data stewards.

#### Work Package 3: Platform Rollout and Roadmap (Months 18-24)

##### WP 3 Overview:

| Track          | Component        | Target Audience                      | Who                                   | What                                                                                                                                                                                                      | Deliverables                                            |
|----------------|------------------|--------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| Training Track | Maskathon        | Developers, data stewards            | Pouw, Owoyele, RA Training            | Collaborative event to explore opportunities to enhance de-identification benchmarks, platform features and ideate on improved research software support tailored to audio-visual data masking practices. | Certified training modules, improved masking algorithms |
| Platform Track | Platform Rollout | Dutch research institutions via SURF | van de Heuvel, Owoyele, RA(Technical) | Full-scale platform deployment, UI, and back-end optimizations                                                                                                                                            | Final technical roadmap benchmarks for masking tools    |

**Objective:** Complete development and ensure sustainable support for SYNOPSIS while creating a roadmap for future growth.

##### Platform Track:

Led by **Babajide Owoyele** and **Research Assistant (Technical)**, the platform will scale from the pilot stage to full deployment across several Dutch research institutions via **SURF**. This phase will focus on optimizing the user interface and back-end performance.

##### Training Track:

RA (Training), **Wim Pouw**, and **Babajide Owoyele** will organize a **Maskathon**. The event will foster collaboration

between developers, data stewards, and computer scientists to ideate benchmarks for masking and introduce new algorithms to support continuously improved masking, such as in computer science/machine learning communities. Additionally, **data stewards at partner universities** will ideate on a train-the-trainer workshop in **Netherlands eScience Center**.

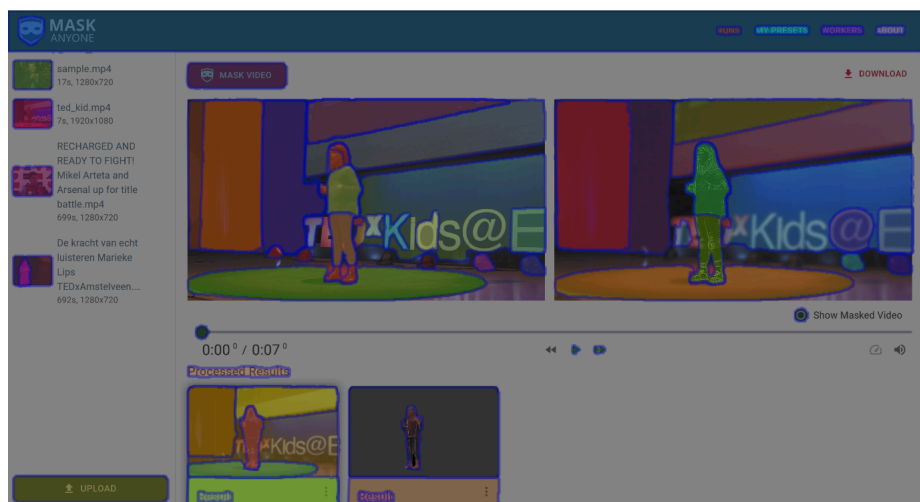


Figure 5 Maskathon - A Hackathon-style event to ideate new benchmarks and trainings for MaskAnyone (User Interaction and The CodeBase)

#### Roadmap:

A comprehensive roadmap detailing benchmarks, future developments, and sustainability strategies will be published, including potential funding from **NLeSC** and **Open Science NL Infrastructure**.

#### Deliverables:

- A final technical roadmap paper.
- Benchmarks for evaluating masking tools.
- Prototype of a Certified training module for developers and data stewards.

## 3. Declarations and signature

### Funding elsewhere

Have you requested funding for this project elsewhere?

- ☒ No  
☐ Yes

Please include details of any additional grants you have requested for this project.

By submitting this form, I declare that:

- ☒ I have completed this form truthfully;  
☒ I meet the nationally and internationally accepted standards for scientific conduct as stated in the [Netherlands Code of Conduct for Research Integrity 2018](#);

# MaskBench: Role Masking Benchmark Interface

*MaskBench is an interactive interface for benchmarking role extraction and masking algorithms. These screenshots demonstrate the user interface design for evaluating how well computational methods identify and mask organizational roles in text corpora.*

MaskBench

DashboardNew ExperimentResultsCompareHistory

Docs

Pose Estimation Benchmarking

+ New Experiment

Compare estimators, evaluate metrics, make informed decisions

QUICK START

Single Video

Benchmark one video across all estimators

Dataset

Evaluate entire dataset with metrics

Masked Videos

Test estimators on de-identified data

Compare

Side-by-side estimator comparison

12

EXPERIMENTS RUN

48

VIDEOS PROCESSED

7

ESTIMATORS AVAILABLE

2.4h

AVG PROCESSING TIME

RECENT EXPERIMENTS

TED Talks Evaluation

10 videos • All estimators • Kinematic metrics

CompletedView

Masked Video Comparison

3 videos • 4 masking strategies • PCK/RMSE

Processing65%

Tragic Talkers Ground Truth

20 videos • Accuracy • Kinematic metrics

DraftEdit

MaskBench

Dashboard

New Experiment

Results

Compare

History

Docs

EXPERIMENT SETUP

Experiment Name

TED\_Gesture\_Study

DATASET

Custom (upload videos)

POSE ESTIMATORS

☒

YOLOv11-Pose

Fast, robust

☒

MediaPipe Pose

CPU-friendly

☒

OpenPose

Multi-person

☐

MaskAnyone-MediaPipe

Smoothest

METRICS

☒ Velocity

☒ Acceleration

☒ Jerk

☐ PCK (requires GT)

☐ RMSE (requires GT)

Run Experiment

Upload Videos

Add videos to your dataset for benchmarking

Drag and drop videos here

Supports MP4, AVI, MOV • No file size limit

Browse Files

Browse Folder

UPLOADED VIDEOS (3)

TED\_Talk\_Gesture\_01.mp4

2:34 • 1920x1080 • 156 MB

TED\_Talk\_Gesture\_02.mp4

3:12 • 1920x1080 • 198 MB

TED\_Talk\_Gesture\_03.mp4

1:58 • 1920x1080 • 124 MB

Add more

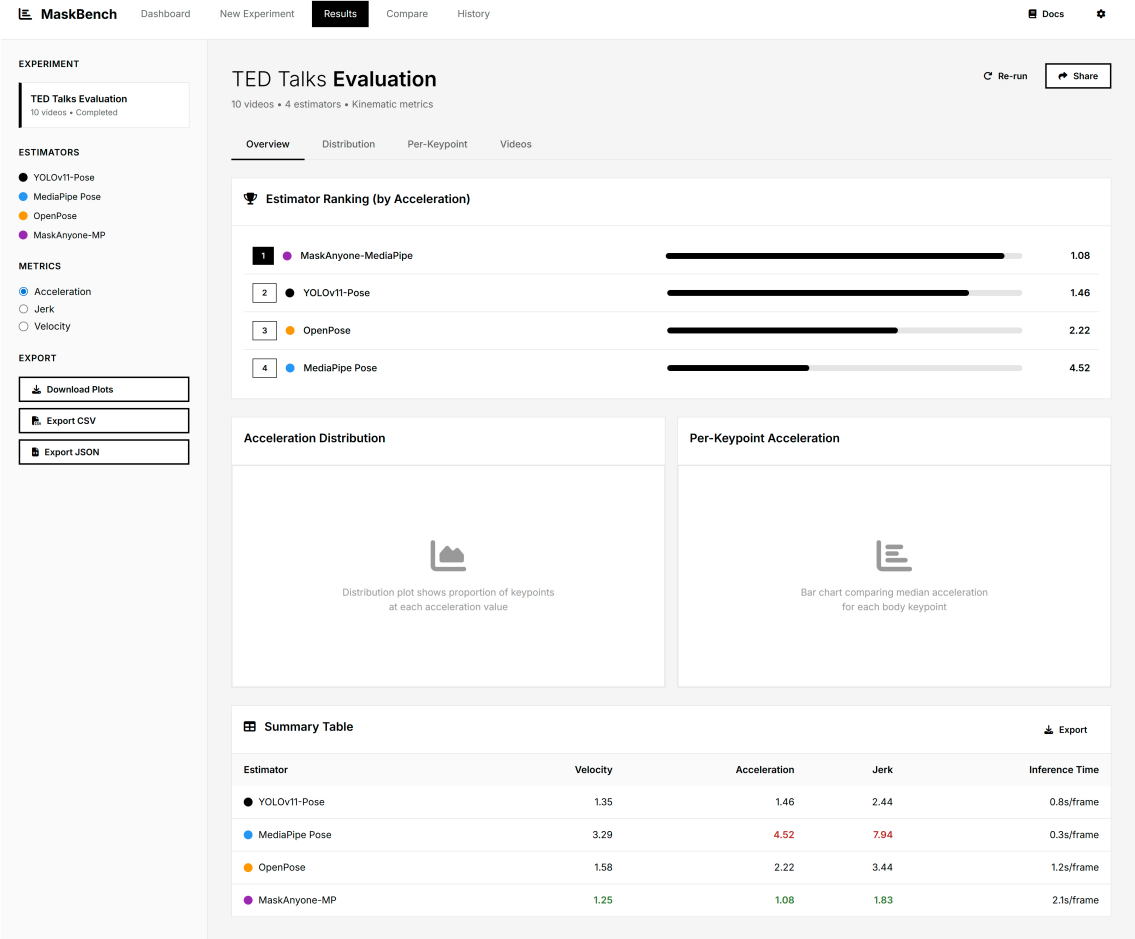
>\_

EQUIVALENT CLI COMMAND

Copy

docker compose up -e MASKBENCH\_CONFIG\_FILE=config/ted-gesture.yml





# Side-by-Side Comparison

Compare pose estimator outputs on the same video

Video: TED\_Talk\_01.mp4

Frame: 0:45 / 2:34

▶

⏮

YOLOv11-Pose

MediaPipe Pose

  
YOLOv11-Pose Overlay

  
MediaPipe Pose Overlay

VELOCITY (THIS FRAME)

YOLO  
2.4

MediaPipe  
5.8

ACCELERATION (THIS FRAME)

YOLO  
1.2

MediaPipe  
4.1

KEYPOINTS DETECTED

YOLO  
17/17

MediaPipe  
15/17

# Experiment History

All your past benchmarking experiments

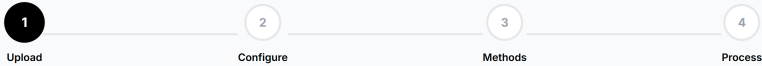
Search experiments...

All Status 

| EXPERIMENT                                                             | VIDEOS | DATE         | STATUS     | ACTIONS                                                                                                                                                                                                                                                     |
|------------------------------------------------------------------------|--------|--------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>TED Talks Evaluation</b><br>YOLO, MediaPipe, OpenPose, MaskAnyone   | 10     | Dec 15, 2025 | Completed  |    |
| <b>Masked Video Comparison</b><br>Blur, Pixelate, Contour, Solid       | 3      | Dec 14, 2025 | Processing |                                                                                       |
| <b>Tragic Talkers Ground Truth</b><br>All estimators, PCK/RMSE metrics | 20     | Dec 12, 2025 | Completed  |    |
| <b>Quick Test - Single Video</b><br>YOLO only                          | 1      | Dec 10, 2025 | Failed     |                                                                                       |

# MaskAnyone: Privacy-Preserving Video Anonymization

*MaskAnyone is a workflow interface for privacy-preserving video anonymization. The tool enables researchers to de-identify individuals in video data while preserving behavioral and contextual information essential for qualitative analysis. These screenshots demonstrate the user interface design for the video processing pipeline.*



## Upload Video



**Click to upload or drag and drop**  
Supported formats: MP4, AVI, MOV, MKV  
Maximum file size: 2GB

sample\_video.mp4  
25.4 MB • 1920×1080 • 30fps • 15s duration

Cancel

Next: Configure Targets →

1  
Upload

2  
Configure

3  
Methods

4  
Process

## Configure Targets

Targets

Target 1

Detection Method

YOLOv11 Person Detection

Hiding Strategy

Solid Fill

Target 2

Detection Method

MediaPipe Hands

Hiding Strategy

Solid Fill

+ Add New Target

Prompt Another Frame

Frame Coordinates

Frame 0:  
T1: [[608,224],[544,186]]  
T2: [[274,174],[275,270]]

Back

Next: Choose Methods

Target 1

Target 2

Frame 0 / 450

1  
Upload

2  
Configure

3  
Methods

4  
Process

Select Masking Methods

 SAM2 Segmentation



Meta's Segment Anything Model 2 for precise video segmentation with temporal consistency

High Precision GPU Required ~5 min/min video

 MediaPipe Detection



Google MediaPipe for fast detection of hands, faces, and body poses

Fast Processing CPU Friendly ~2 min/min video

 OpenPose



Full body pose estimation with 25-point skeleton tracking and facial landmarks

Full Body GPU Required ~4 min/min video

 YOLO Person Detection



UltraLytics YOLOv11 for real-time person detection and tracking

Real-time Efficient ~1 min/min video

 Additional Options

Pose Overlay

OpenPose Body 25B

Audio Processing

Preserve Original Audio

[← Back](#)

[Start Processing ▶](#)

1

Upload

2

Configure

3

Methods

4

Process



## Processing Video

sample\_video.mp4

Method: SAM2 + OpenPose Body 25B

PROCESSING

Overall Progress

67%

Estimated time remaining: ~45 seconds



Pose Detection Completed

YOLOv11 inference on first frame  
Duration: 2.3s



Segmentation Initialized

SAM2 prompt built from pose data  
Duration: 1.8s



Frame-by-Frame Processing

Segmenting and masking video frames  
Frame 302 / 450

67%



Pose Landmark Extraction

Pending



Final Rendering

Pending

Cancel Processing

Processing...

### Processing Info

|              |           |
|--------------|-----------|
| Worker       | Worker 1  |
| GPU          | GPU 0     |
| Method       | SAM2      |
| Resolution   | 1920x1080 |
| Total Frames | 450       |
| Start Time   | 14:23:45  |

### Performance

|                                   |            |
|-----------------------------------|------------|
| GPU Usage                         | 87%        |
| Memory Usage                      | 6.2 / 8 GB |
| Processing: ~4.8 min/min of video |            |